

Education, Globalization, and Social Change

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J. King

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ED - TECH PRESS
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Published by ED-Tech Press,
71-75 Shelton Street, Covent Garden
London WC2H9JQ
United Kingdom

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ISBN: 978-1-83535-142-0

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British Library Cataloguing in Publication Data.
A catalogue record for this book is available from the British Library.

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Preface

The most effective tool for bringing about social change is education. The only way for society to modernise and bring about the desired changes is through education. By giving people the chances and experiences to develop themselves for acclimating to the new demands and philosophies of the changing society, education can change society. Planning is essential for a sound social progress in all spheres of life, including social, cultural, economic, and political. Planning for education must take into account the needs and ambitions of the populace as a whole.

Obtaining social and economic benefits of society requires education. It is now crucial to the economy. Education is currently a significant organisation with a prominent public profile. In order to suit their conception of society's needs, the dominant groups in society now have influence over education. Social and economic transformation, more social mobility, and more trained labour for technologically oriented industries are all conditions of changes in the educational system. The social integration and a better educated labour force and voter may be facilitated through planned educational innovations, policies, and programmes.

The transmission of a way of life to the pupils is not given much priority in contemporary schools, colleges, and universities. The standing of women has dramatically improved because to education. The double standard of morality is no longer acceptable to educated modern women. They have found it easier to get work outside the family. Peter Worsely emphasises "that education reflects society, and educational change follows social change" while discussing the relationship between education and social change and development. Additionally, education plays a separate role in social and economic development, having both planned and unforeseen effects.

Author

Introduction

‘Education’ is one of the intervening variables in the phenomenon of social change. ‘Durkheim’ conceives of education as “the socialisation of the younger generation”. According to James Welton, education consists in “an attempt on the part of the adult members of human society to shape the development of the coming generation with its own ideals of life.” As Samuel Koenig has pointed out, it is a “process whereby the social heritage of a group is passed on from generation to another”.

Education can also be understood as a factor of social change. The role of education as an agent or instrument of social change and development is widely recognised today. Education can initiate social change by bringing about a change in the outlook and attitudes of man. It can bring about a change in the pattern of social relationship and thereby it may cause social changes. One of the purposes of education is to change man and his life and living style. To change man is to change society only.

There was a time when educational institutions and teachers were engaged in transmitting a way of life to the students. During those days, education was more a means of social control than an instrument of social change. Modern schools, colleges and universities do not place much emphasis upon transmitting a way of life to the students. The traditional education was meant for an unchanging, static society, not marked by rapid changes. But today, education aims at imparting empirical knowledge, that is, knowledge, about science, technology and other type of specialised knowledge. Education was associated with religion. It has, however, become secular today. It is an independent institution now. Education today has been chiefly instrumental in preparing the way for the development of science and technology.

Education has brought about phenomenal changes in every aspect of man's life. Francis J. Brown remarks that education is a process which brings about changes in the behaviour of society. It is a process which enables every individual to effectively participate in the activities of society, and to make positive contribution to the progress of society. As Drucker has stated that the "highly educated man has become the central resource of today's society and the supply of such men is the true measure of its economic, military and even its political potential."

Modern education has changed our attitude and outlook. It has affected our customs and traditions, manners and morals, religious beliefs and philosophical principles. It has removed to a great extent the superstitious beliefs and unreasoned fears about the supernatural beings. It has widened our vision and removed our narrow ideals, prejudices and misunderstandings. Higher education has brought about more refined behaviour.

Education has contributed to a radical improvement in the status of women. Educated modern women no more tolerate the double standard of morality. It has helped them to seek employment outside the family. Particularly, mass education in civilised societies has fostered the sense and the feeling of equality. Referring to the relation between education and social change and development, Peter Worsely points out "that education reflects society, and educational change follows social change".

Though education conditions development it itself is a product of prior social and economic changes in society. Further, education is an independent factor in social and economic development producing intended and unintended consequences and conflicts of values and goals. Education is an important means of attaining social and economic rewards of society. It has become essential for the economy. Education has now become a large-scale and a highly visible organisation. Education is now controlled by the dominant groups of society so as to meet their definition of society's needs.

Changes in the educational system condition social and economic changes, greater social mobility and more skilled man-power for technologically based industry. Planned educational innovations, policies and programmes may contribute to the social integration and a more highly educated labour force and electorate. Education has been playing a great role in getting occupations which are key determiners of general social status. Thus the schools are agents in realising the desire for upward social mobility. In many highly industrialised societies the proportion of people in the manual working class has steadily declined. It is so in the case of America, Britain, France, *etc.* The schools have been instrumental in transforming the occupational structure and modifying the class structure as well.

In most developing countries education is regarded as "the gateway to an improved social position." Hence one may find an unsatisfied demand for education in such countries. This is especially true in the case of a developing nation like India. Educational change in these countries can effectively proceed only if corresponding changes take place in the other aspects of their social structure.

Where education is a condition of social and economic change, it is more likely to produce intended consequences. This happens because educational change is following other changes in society. The social context is thus favourable to particular change. For example, educational re-forms, designed to raise educational standards among low-income people have become more successful in Cuba than in Guatemala.

This has been so, because, in Cuba, more than in Guatemala, educational change has followed social and economic changes enabling the low-income people to take an active participation in the development of national society. As far as India is concerned, there is no proper coordination between educational changes and socioeconomic needs.

Education increases political awareness among poor people also. This would bring about wider political changes with the increasingly organised participation of people in national politics. Modern states, particularly the totalitarian ones, have made education an installment for establishing their regime. Under authoritarian principles, the control of the school touches every aspect of education. The teachers are carefully chosen and supervised, and deviations from the party line are severely punished. Students of all ages must be given nothing but the truth as the ruling elite see it.

The principal of a school in Moscow once said: “the prime duty of the Soviet teacher is to train our younger generation for the work of building communism”. On the contrary, in the democratic countries there is the belief that “The State is for man, not man for the State”. Education is made free and open. Here, education makes a man to become more conscious of his rights and also of his duty to provide and guard similar rights for others. Education is expected to contribute to ‘progress’, to modify the cultural heritage as well as to preserve and transmit it. In modern industrial societies educational organisations have become innovators. They are gathering and storing new knowledge and are promoting change in the process of transmitting that knowledge.

It is now widely held that educational system should dedicate itself to the task of bringing about desirable changes. The emphasis upon research in universities reflects the judgment that discovery itself is good. For the first time in history, societies are marshalling their huge resources and talents to make advances in knowledge through educational organisations. Changes do not take place with equal rate of speed in all areas of life. Generally, there is more enthusiasm for change in areas of material culture than in non-material culture. Through educational researches any kind of innovation can be made to maximise production and minimise cost.

When education challenges cherished traditions, it becomes the object of some hostility. Education cannot be used as an instrument to bring about any kind of change. Because education operates in the context of other institutions and is constrained by them. As Alex Inkeles has pointed out, different levels of education have different levels of effects. In the developing countries primary school education is enabling whole population to do things they would never have been able to do before. Literacy helps them to read labels on cans, bottles, tins, to read sign boards, newspapers, birth-control leaflets, to move around the strange city, *etc.* These events are social changes. In the

developing countries primary school education is more important than higher education. Even though widespread primary education can have a great impact upon people in the developing countries the ideological content of primary school education remains almost conservative. Because, governments organise school systems in a stereotyped way. There is less or no scope for the teachers to make researches or to become revolutionary leaders.

It can be said that basic literacy brings a society into the modern world. But only higher education provokes persons to question the values of everyday life. The high school or primary school teacher is not as free to speak critically as the university professor is. At elementary level of education students normally live with their parents at home and hence not free to entertain ideas which their parents may dislike. On the other hand, at the university level, the intellectual work requires students to do more critical thinking than they might do at a lower level.

University student movements have often been the major force demanding social change in many societies. A decade between 1960 and 1970 witnessed a large number of student upsurges resulting in social and political changes. For example, in China, India, Japan, America, Germany, France, Italy, England, Indonesia and in many other countries students agitated for various reasons causing vast changes. In some cases, the student movements stood with the establishment and in majority of the cases they tried to discredit, transform or topple governments. The students are today a new social force of incalculable significance. But student movements have been far less active in the late 1970s than a decade ago.

It is true that college educated persons are still the most progressive group in society whether they are quiet or vocal in calling for social reform or change. More and more persons are receiving higher education. Majority will attain a degree. If that is so, it means that society will contain a “built in” engine for social change. As long as universities continue to occupy an increasingly important place in society, so long changes are bound to be initiated through education in some way or the other.

THE IMPORTANCE OF EDUCATION

Education is very important for an individual's success in life. Education provides pupils teaching skills that prepare them physically, mentally and socially for the world of work in later life. Education is generally seen as the foundation of society which brings economic wealth, social prosperity and political stability. Higher education helps in maintaining a healthy society which prepares health care professionals, educated health care consumers and maintaining healthy population. Education is major aspect of development of any modern society since if there is a deficit of educated people then society will stop its further progress. Government should pay serious attention to education and support it economically and morally all over the country. Education is the best investment for the people because well educated people have more opportunities to get a job which gives them satisfaction.

Educated individuals enjoy respect among their colleagues and they can effectively contribute to the development of their country and society by inventing new devices and discoveries. Today's ever growing numbers of people mostly are not satisfied with their basic education and try to get secondary or tertiary education in order to meet the demands of contemporary society.

Some of them enter higher educational institutions and some search additional information on the internet. Good People sacrifices their time and money and sometimes even their health to raise educational level because they realise that education is their passport to the future and for tomorrow. Main purpose of education is to educate individuals within society, to prepare and qualify them for work in economy as well as to integrate people into society and teach them values and morals of society.

Role of education is means of socializing individuals and to keep society smoothing and remain stable. Education in society prepares youngsters for adulthood so that they may form the next generation of leaders. It will yield strong families and strong communities. Indeed, parents taking an active role in their child education produce a willingness in children to learn. Education and society provides a forum where teachers and scholars all over the world are able to evaluate problems in education and society from a balanced and comparative social and economic perspective. Education is an important aspect of the work of society and it will raise the countryside issues and promote knowledge and understanding of rural communities.

One of the education essential tasks is to enable people to understand themselves. Students must be equipped with knowledge and skills which are needed to participate effectively as member of society and contribute towards the development of shared values and common identity. Education has a vital role to play in assisting students to understand their cultural identity. Education acts as the distribution mechanism of the cultural values such as it more layered the society and participate in society that carries the culture. Education and society provides a forum where teachers and scholars all over the world are able to evaluate problems in education and society from a balanced and comparative social and economic perspective.

Education is an important aspect of the work of society and it will raise the countryside issues and promote knowledge and understanding of rural communities. One of the education essential tasks is to enable people to understand themselves. Students must be equipped with knowledge and skills which are needed to participate effectively as member of society and contribute towards the development of shared values and common identity. Education has a vital role to play in assisting students to understand their cultural identity. Education acts as the distribution mechanism of the cultural values such as it more layered the society and participate in society that carries the culture. In our culture today, there is a great emphasis on higher education.

In a society, more educated you are, better off you are. Every society has specialized individuals that require extended education to fulfill certain main positions. These persons are normally known as professors, priests, doctors, mechanics or artists.

Education has been a higher part of every culture on earth and education is a systemic project. Whole society should care for and support the education patriotism, cause and socialism among the young people. Everyone must do work hard to cultivate moral conduct. Education mainly begins at home; one does not acquire knowledge from a teacher, one can learn and get knowledge from a parent or a family member.

In almost all societies, receiving education and attending school is very necessary is one wants to achieve success. Education is the key to move in the world, seek better jobs and ultimately succeed in life. Schools play a vital role in preparing our children and young people for effective participation and responsible citizenship in society. The development of education and educational opportunities is built on creativity tempered by knowledge and wisdom gain through the experience of learning.

Investment in human capital, life long learning and quality education help in the development of society. Teachers are the most important factors for an innovative society because teachers' knowledge and skills not only enhance the quality and efficiency of education, but also improve the prerequisites of research and innovation. Many members of our society are not provided with a safe and secure environment in which children can develop, child abuse, violence against women and interpersonal violence cause a cancer on our society. Society play a key role in the realization of life long learning. The improvement of social education facilities such as libraries and the learning opportunities are implemented by the local governments. Students today are exposed to loads of technology and information at everywhere.

MEANING OF PRE-PRIMARY EDUCATION

It is one of the many consequences of the general lack of educational planning in India that a very impressionable, plastic and educationally potent period of a child's life has received so little attention. In most countries in Europe and America it has now been clearly recognised that the nursery or infants School has an important part to play in every school System. Russia, where an extremely efficient system of kindergartens, creches and nurseries has been evolved, deserves Special mention in this connection. Outside India the nursery school has come into its own and has taken a well-defined place in the fabric of public instruction, but the importance of looking after the physical and mental welfare, of future citizens from their earliest years has still to be brought home to the responsible authorities, in this country.

Official neglect, however, is far from being the only obstacle; ignorance and indifference on the part of the people most closely concerned has also contributed to the present sorry state of affairs. Even if proper facilities were provided, it would be by no means an easy matter to persuade the Indian mother to subordinate her natural affections in the interest of a more healthy physical and mental environment for her children. A great deal of propaganda and training of public opinion will be necessary before a system of pre-primary education can be successfully introduced. It is necessary first of all to define what is meant by pre- primary education as applied to India's needs.

In Great Britain and other Western countries, where the age for compulsory attendance is five or even lower, it is usual to distinguish between Infants Schools or classes, which cater for children between the ages of five and seven or eight and Nursery Schools, or classes, which are designed for those below the compulsory school, age. Nursery and Infants classes, as distinct from Nursery and Infants Schools, mean special classes attached to schools for older children. In the Indian system, so long as the age for compulsion is fixed at six, there will hardly be room for the Infants School as understood in the West. All pre-primary instruction, therefore, whether provided in Separate schools or in classes attached to Junior Basic Schools, may conveniently be called 'Nursery', because this word clearly denotes the character of the environment and teaching that will be required.

Of the reasons for providing special educational facilities for children below a compulsory age of six, which many people regard as too late an age for the great majority of children to start their schooling, the most urgent concerns the physical aspect. It has been established that many defects, which ought to be remedied quickly, reveal themselves before this age and it is therefore necessary, especially in town or other areas where housing conditions are unsatisfactory, that primary education should be supplemented by a generous provision of Nursery Schools and classes for children below the minimum age for compulsory attendance.

The Central Advisory Board's second Committee on Basic Education recognise this necessity, though it is possible that they under- emphasised the State's responsibility in the matter. It may be doubted whether private enterprise can be relied upon to fill the gap. Apart, from consideration of the children's health the case of the working mother has to be borne in mind. It would be regrettable if women were hindered from taking their proper position in society and contributing their share to its economic and cultural progress on account of domestic liabilities, but it would be still more regrettable if they were able to do so only at the expense of their children.

It is the duty of the State to come to the rescue, for the sake both of its future citizens and of those that bear them, by providing bright, well-equipped and well-staffed Nursery Schools, where the children can be properly looked after while their mothers are at work. Educationally the Nursery School aims at providing a carefully controlled environment which should cater in the healthiest and wisest manner possible for the mental, physical and social needs of the growing child. If it is conceded that at any stage of development the influence of environment is of importance, then it stands to reason that in the very early years of childhood the provision of a good environment is of paramount importance. Where the home cannot provide this, it is the business of the State to intervene.

The question then arises as to what areas would be most suitable for the establishment of Nursery Schools. In this connection it is necessary to take into consideration, besides the actual needs of a given area, its economic aspect and the problem of sufficient numbers. It, has already been pointed out that due to congestion, bad housing and greater

vulnerability to disease, children in town areas need most the facilities that a Nursery School can offer and there are usually enough of them in a given area to make such provision a practicable proposition.

Village children also, whose mothers in many cases have to work in the fields and whose homes are often hardly worthy of the name, have a strong claim for similar amenities. But it is doubtful whether in a radius of about a mile and a half, the maximum distance which small children can be expected to travel, and especially in rural districts where there are hardly any means of conveyance, enough children can be mustered to form a reasonably economical school.

For this reason, therefore, it is suggested that instead of attempting to establish fullfledged Nursery Schools in village areas, a Nursery class should be added to the ordinary Junior Basic School. In fact it is desirable that the lowest class in all village Junior Basic Schools should be known as the Nursery or Infants 'clasp and should be specially staffed and equipped for that purpose. It is hardly necessary to say that it should be in the charge of a woman teacher who have received the appropriate training.

The regular Nursery School may then be confined for the present to towns. It may conveniently be housed on the same site as, or even form a department of, the Junior Basic School, since that will enable older children to look after their younger brothers and sisters on the way to and, from school.

The question of staff for Nursery Schools and classes is an all-important one. A very specialised kind of training is required for dealing intelligently with the very young. That teachers for infants should be women goes without saying: they alone Possess the essential sympathy and knowledge of child mature. At the present time the supply of such teachers in India is negligible and one of the first tasks of any re-organisation will be to enlist the services of suitable women and to see that they are properly trained.

The Second Education Committee of the Central Advisory Board of Education, 1939, laid down as one of their main decisions:

- “ That..... the provision of ‘pre-basic’ education in Nursery and Infants Schools and Classes is highly desirable..... Provincial Governments should aim in the first place;
 - At providing model Infants and Nursery Schools in suitable centres,
 - At increasing the supply of properly trained infants teachers, who should be women,
 - At encouraging the enrolment in ‘basic’ schools of children below the minimum age for compulsory attendance and
 - At stimulating the provision by voluntary agencies of efficient ‘pre-basic’ schools”.

As has been already pointed out, the words ‘highly desirable’ are an understatement. The provision of an adequate number of Nursery Schools and classes is essential and although it may hardly be practicable, at any rate in the beginning to make attendance compulsory. the schools should be free and no pains should be, spared to make them as

attractive as possible with the object of persuading parents to send their children voluntarily. Apart from the provision made by the State there should always be scope in education for private enterprise. The fact that Government has hitherto done so little in this sphere makes the private ventures that have been launched all the more valuable and praiseworthy. Among the best known of these are the Jitendra Narayan Ray, Infant and Nursery School for Indian children in Bengal the Raj- Ghat School and Children's Hostel, Benares, and the Infant's Part of the Besant Theosophical School, Adyar, in which the Monterey system of education is followed.

A word about the methods of education in the Nursery School is perhaps necessary. A young child, as Abbott and Wood remark in their report, "needs experience more than instruction" and this is precisely what the Nursery School should aim at providing. They go on to say "that the education of the young children should provide for their physical care, for training them in good habits and for widening their experience through interesting activities. We envisage such activities as the following: acting and singing, physical exercises, games and dancing, care of flowers and animals, drawing and making things.

These activities minister to one or other of the characteristic needs of children and provide them with experience which gives them confidence in their growing powers. The infant schools must be sensible, happy institutions which patently do something for the children which the home does not do but yet which the parents appreciate when it is done". There is hardly any room for any formal instruction in the 3 Rs in the daily programme of a Nursery School. But through sensory-training, through the promotion of self-expression, through community living and companionship in an educationally controlled environment, the all round development— mental, social and physical—of the child is fostered. The foundations are laid which cover all these aspects of growth and development and the young child is then ready to absorb and assimilate that education for citizenship which a well planned Basic School should impart.

The total number of children to be provided for on these lines can hardly be calculated accurately. This will depend entirely on the nature of the areas under question and on the success of "nursery" propaganda. Then too the age range cannot be rigidly fixed. The two and three year old should be as welcome in a Pre-Primary School as the fours and fives. But the ages three to six may be taken as the normal range for the nursery school child and it would appear from the latest figures available for British India that there are 2,13,08,000 such children in the rural areas, and 31,00,000 in the towns.

In England, about 1 child in 7 attends school voluntarily before the minimum compulsory age. 'If the same proportion were taken for this country the number between three and six would be 35,00,000. For the present, however, it may be sufficient to provide places for roughly, a third of these in a Nursery Schools and classes. On the same basis of calculation as for the Junior Basic Schools, *i.e.*, 30 children per teacher at an average salary of ₹. 42 5 p.m. the annual cost per capita comes to ₹. 31.84. In actual practice, this will probably work out somewhat higher, since the great majority of nursery

children will be educated in town areas where the average salary of a teacher will have to be raised in order to meet the higher cost of living. Moreover, Nursery Schools require more space and more equipment. On the other hand relief may be given so far as staffing is concerned if older girls from High schools and Senior Basic Schools are attached to Nursery Schools for definite periods for training in child welfare. For the reasons above both the figure of 10,00,000 and the estimated cost must be regarded as tentative.

HIGHER EDUCATION

Higher education—as distinct from professional training—is abstract and often has no direct relation to the practical problems of life. Its abstractness is however its saving virtue, for it gives to such education a universal aspect. Not restricted to any particular field, it enables the mind to rise to those general truths which give us our first glimpse of the world of spirituality. The discovery of these universal truths is also the basis of all human progress. There can of course be no simple answer to the question as to whether education should be confined to matters of common concern to all human beings or devoted to training of specific abilities of different individuals.

A popular way of expressing this opposition is to describe it as education for life versus education for a profession. Doubts, however, arise the moment the question is formulated in this form. What exactly do we mean by the term education for life? There is no such thing as life in general. Each individual is a member of a particular society and performs certain specific functions associated with the station to which he is called. Insistence on the specific duties of a particular station in life would tend to obliterate the difference between education for life and education for a profession.

The same result follows if we work out the implications of what is meant by education for a profession. A profession itself is a way of life, and hence education for a profession is education for a particular way of life. We must also be careful that the concept of one's station in life is not overworked. Insistence on station may divide society into rigid strata and lead to some kind of caste or class structure in which the individual is predetermined to perform a specific function. If this happens, the distinction between education for life and education for a profession again disappears. It may also be pointed out that insistence on one's station is likely to lead to a static society. One of the main lines of human progress has been to move away from rigid class or caste structure and give the individual a greater liberty in choosing his future vocation.

The failure to maintain an opposition so commonly accepted underlines the need to redefine from time to time the basic aims of education. Only if these aims are clearly realized can we attempt to relate what may be called general education to specialized education. No comprehensive analysis of these aims can be offered in a short essay. A proper understanding of the nature of education would have to be based on an analysis of the nature of the mind and its operations and the relation of the human mind to society and the world. Without entering into the far-flung speculation to which such an

analysis would lead, we can distinguish four related but distinct purposes which together constitute the end of education. Its first purpose is to develop the personality of the individual. It also seeks to give him knowledge of the world in which he lives. A third purpose is to develop skills needed to sustain and advance social life so that he can be a creative member of society. Connected with all the three but at the same time distinct from them is the fourth purpose which is to satisfy the individual's search after values.

Each of these four aims requires more detailed and specific formulation than can be attempted here. When we talk of the development of personality, we refer to the growth and maturity of the individual's physical, mental, emotional and spiritual abilities. A retarded body is not merely an absence of physical growth but has certain positively undesirable characteristics. Similarly an undeveloped mind not only lacks intellectual freedom, but is a source of superstition, obscurantism, fear and hatred. It is common experience that to deny a child affection means not only that the child is emotionally starved but that he becomes a source of danger and infection to others. The spiritual flowering of a personality may be more difficult to define but there is little doubt that, as in the other cases, here also the failure to grow is not a mere privation but a negation which has an adverse effect on the individual as well as others. One is in fact reminded of Alice's predicament in Wonderland. She finds that she has to run for all she is worth merely to stay where she is. The failure to grow does not leave one where one was: it sweeps one back to a depth below one's original starting-point.

The second aim of education is to secure knowledge of the physical world as well as of the ideas and ideals of society. Without such knowledge, the individual cannot survive, let alone develop his personality. In fact, possession of such knowledge is a condition of both personal development and service to society. There may at times be a tendency to equate education with the acquisition of mere information. We must also guard against placing an undue emphasis on knowledge of the physical or the ideational environment severally, for both are equally important for our understanding of the world. Nevertheless, the acquisition of knowledge about the world in which we live is basic to all programmes of education.

The development of personality and the acquisition of knowledge of the environment are necessary for achieving the third aim of education. The individual can function only in a social context. If this milieu does not change and grow with the individual's growth, tensions are bound to develop. In fact the individual's growth is fostered by a friendly and retarded by an uncongenial environment. On the other hand, progress or deterioration of the different individuals leads to social progress or the reverse. If one is to be a creative member of society, one must not only sustain one's own growth, but contribute something to the growth of society.

In the social conditions of today, the average man needs all his limited energy to keep society functioning. Progress, on the other hand, demands an addition to what society has already achieved. Since in existing circumstances, a major part of our energy is needed merely to keep up to the norm, it requires an uncommon effort to go beyond

it. Nevertheless, it is an obligation which each individual must in some measure fulfil. We cannot take unless we give, and vice versa.

The position is further complicated by the fact that apart from his role as a member of society, each individual has an inviolate identity that is unique. He can find satisfaction only by fulfilling the demands of his inner nature. This may be described as a quest for values or a voyage of self-realization, but however we describe it, each individual has in him an element where he transcends his social needs. He cannot achieve self-realization by merely satisfying the demands of his society or performing the functions of the profession he may follow.

Reference has already been made to the paradox that education for life may in fact mean education for a specific function in life. In other words, education for life if interpreted in a narrow sense is hardly distinguishable from education for a profession. Nevertheless, a distinction has to be drawn for the reasons already mentioned. On the one hand, the progress of society is from status to contract, from rigidity to fluidity. On the other, the present organization of society demands from the majority of its members almost their entire energy merely to sustain the standard of civilization already achieved. It may be that with progress in social development, we will discover that the creative spark in man is more widely spread than we at present imagine but, even in a happier society, there will be differences in the brilliance of the spark. In future all members of society may contribute more to general social welfare, but the really significant advances are likely to take place, then as now, through the efforts of individuals of genius. Before, however, such an effort can be made, it has to be ensured that the individual or group making it has sustained the standard common to most members of the community. It is only from the achieved successes of today that we can launch into new experiments of the future.

The first two aims of education are to develop the various faculties of the individual and give him, some knowledge of the world. So far as the first is concerned, there is no need to think of any special functions which he may be called upon to perform in later life. Specialization begins to show in the second aim and determines the aspects of reality of which he requires greater and more intimate knowledge.

Specialization shows more clearly as we consider the third and fourth purposes of education. An individual can contribute to the sustenance and advancement of social purposes and thus be a creative member of society only in his individual way. Similarly, his contribution to the realization of old or discovery of new values must be based on his individual capacity. Self-realization is from the nature of the case essentially an individual function. It is said of religion that it is what a man does with his solitariness. The values of science, philosophy and art no doubt contribute to the enrichment of the life of the community, but like religion, they also are essentially the result of what a man does in his solitude.

Prima facie, it therefore appears that the first aim of education leads to a theory of general education while the third suggests training of special faculties and aptitudes.

The second aim includes both general knowledge of the world and special knowledge suited to particular avocations. The fourth does not fit into either category and stands by itself. Analysis will, however, show that such clear-cut distinctions are not tenable. Each aim supports and is supported by the others. A striking example of this is furnished by the advance in material standards of life achieved as a result of abstract research which has at first sight no relation to any practical problem.

We may now try to see if these purposes of education may be related to different stages of education. One word of caution is, however, necessary. To relate a particular purpose with a particular stage does not imply that it does not operate in the other stages. All that is meant is that a particular purpose is more clearly seen in a particular stage. It is then obvious that the development of the personality of the individual and acquisition of general knowledge of the world must begin at the primary or elementary level. This covers early childhood and continues up to or a little beyond the advent of adolescence.

There can be no question of developing any specific skills or abilities at this stage. The only aim of education at this level is to give the child a corpus of knowledge which he can share with all members of his society and to develop in him physical, intellectual, social and moral habits necessary for his survival and progress. Besides, his aptitudes are largely undifferentiated at this early stage of life. That is why we can and often do speak in terms of generalized education for children.

While elementary education is the most general in purpose, it is simultaneously the most concrete of all the stages. We can give the child knowledge of the world in which he lives only by starting from his immediate surroundings. Any skills we want to develop in him have also to be based on the immediate requirements of his environment. If elementary education is to fulfil its real objective, it must be embedded in the local experience of the community. Since such experience is specific in nature, elementary education, in spite of the generality of its purpose and perhaps also of its pattern, is more specific in content than education at any other stage.

The concept of Basic education which has in recent times developed in India is a recognition of this truth. Basic education seeks to develop the child's personality by giving him knowledge correlated to a craft with which he is familiar. In its insistence upon a craft which is prevalent, Basic education has fastened upon an important truth. It recognizes that for a child, abstract teaching is not merely a strain but remains unreal.

By emphasizing craft, the element of activity is immediately brought into the pattern of education. Further, craft means socially useful activity so that the child is from the very beginning taught to recognize his function as a member of society. Insistence on a local craft is also a recognition of the educational principle that learning must proceed from the familiar to the unfamiliar. Finally, it seeks to develop habits of citizenship in children, not through abstract principles which have little meaning for a child, but by making them practise it in their daily life. The very virtues of Basic education, however, suggest that the content of elementary education must differ from locality to locality.

The pattern may no doubt be the same, but the specific details may differ so much that the underlying unity may be missed by all but the careful observer. The purposes of education are common to all but the way in which such purposes can be fulfilled depends at this stage on making the education as concrete and specific as possible.

Psychologically, there is a justification why elementary education is general in a specific environment. It is general because it caters to the wide-awake curiosity of a child. The child wants to know the why of everything that comes within its ken. It seeks to unite in a common world the diverse experiences that come to it with all their novelty. Elementary education has to be specific in respect of the environment but general in respect of its purpose. It is specific because it starts with the known environment. It is general because it aims at the integration of the life of the child with that of the community. As we proceed from elementary to secondary education, a new principle comes into operation. What distinguishes man from all other animals is his power of abstraction. In spite of deficiency of physical power and weakness of the senses, man has triumphed over all other animals because of his capacity to derive general laws from particular instances.

The essence of such generalization is the isolation of the significant from the irrelevant. Our experience at any moment is full of an infinite number of isolated items. We may focus our attention on a particular object, but we cannot altogether shut out impressions of odour, sound and sight on the fringes of our consciousness. If, however, we allowed this multitudinous world of senses to distract us, we could never find any unity in our experience. It is only by selecting the relevant items and combining them into significant objects that our experience becomes coherent and intelligible.

It may be said that, in a sense, even animals separate the significant from the irrelevant in their experience. A tiger stalking its prey excludes from its attention almost everything else. There is nevertheless a distinction between such selection and the selective process which governs human thought. For the animal, the selection is invariably an instinctive operation. It is therefore repetitive and at the same time unilinear. Given the same sort of situation, it is almost infallible.

If the situation changes, the instinctive response may and often does lead to disaster. In the case of man, selection is an act of the intellect, not of the instincts. It is much more complex and rarely repetitive. Because such selection is based on abstraction, it derives from man's power to break up his experience into its constituent units and determine the relations which obtain among them. This power enables man to combine great complexity of content with a unity of structure and is the basis of his superiority over all other animals. This may be seen with reference to his capacity for communicating his ideas. Birds and animals produce noises that have a limited significance, but it is man alone that has created language out of articulate sounds.

While the power of analysis gives man ultimate superiority over all other animals, it makes him comparatively helpless till these powers have developed adequately. In the case of most animals, the skills needed for the sustenance of life are acquired before the

end of adolescence. The human adolescent, particularly in modern society, cannot look after itself adequately. In addition, the duration of adolescence is longer in the case of man. His preparation for life therefore continues throughout childhood and adolescence and even beyond.

EDUCATION AS A SOCIAL FUNCTION

We have seen that a community or social group sustains itself through continuous self-renewal, and that this renewal takes place by means of the educational growth of the immature members of the group. By various agencies, unintentional and designed, a society transforms uninitiated and seemingly alien beings into robust trustees of its own resources and ideals. Education is thus a fostering, a nurturing, a cultivating, process. All of these words mean that it implies attention to the conditions of growth. We also speak of rearing, raising, bringing up — words which express the difference of level which education aims to cover.

Etymologically, the word education means just a process of leading or bringing up. When we have the outcome of the process in mind, we speak of education as shaping, forming, molding activity — that is, a shaping into the standard form of social activity. In this chapter we are concerned with the general features of the way in which a social group brings up its immature members into its own social form. Since what is required is a transformation of the quality of experience till it partakes in the interests, purposes, and ideas current in the social group, the problem is evidently not one of mere physical forming.

Things can be physically transported in space; they may be bodily conveyed. Beliefs and aspirations cannot be physically extracted and inserted. How then are they communicated? Given the impossibility of direct contagion or literal inculcation, our problem is to discover the method by which the young assimilate the point of view of the old, or the older bring the young into like-mindedness with themselves. The answer, in general formulation, is: By means of the action of the environment in calling out certain responses. The required beliefs cannot be hammered in; the needed attitudes cannot be plastered on. But the particular medium in which an individual exists leads him to see and feel one thing rather than another; it leads him to have certain plans in order that he may act successfully with others; it strengthens some beliefs and weakens others as a condition of winning the approval of others.

Thus it gradually produces in him a certain system of behaviour, a certain disposition of action. The words “environment,” “medium” denote something more than surroundings which encompass an individual. They denote the specific continuity of the surroundings with his own active tendencies. An inanimate being is, of course, continuous with its surroundings; but the envining circumstances do not, save metaphorically, constitute an environment. For the inorganic being is not concerned in the influences which affect it. On the other hand, some things which are remote in space and time from a living creature, especially a human creature, may form his

environment even more truly than some of the things close to him. The things with which a man varies are his genuine environment. Thus the activities of the astronomer vary with the stars at which he gazes or about which he calculates. Of his immediate surroundings, his telescope is most intimately his environment.

The environment of an antiquarian, as an antiquarian, consists of the remote epoch of human life with which he is concerned, and the relics, inscriptions, *etc.*, by which he establishes connections with that period. In brief, the environment consists of those conditions that promote or hinder, stimulate or inhibit, the characteristic activities of a living being. Water is the environment of a fish because it is necessary to the fish's activities — to its life. The north pole is a significant element in the environment of an arctic explorer, whether he succeeds in reaching it or not, because it defines his activities, makes them what they distinctively are. Just because life signifies not bare passive existence but a way of acting, environment or medium signifies what enters into this activity as a sustaining or frustrating condition.

SOCIAL CHANGE

Change means differentiation in anything observation over sometime. If we feel that there's come alteration we call it changes. If this change is in contest to social structure, institution *etc.*, *i.e.*, social context then it is social change.

According Fictor "*Change means variations from previous state or mode of existence*".

Change is an universal phenomena *i.e.*, it is a law of nature. There's always a change in nature. Society is a part of nature and so society also changes and static society is unthinkable. Society is on the wheel of change, which may occur due to various factors (like demography, ideas *etc.* If there is any change in Technology *etc.* there's change in society) out the change varies in speed and form. In some places the change is rapid whereas in other places it may be slow. These days due to industrialization and urbanization the change is rapid as compared to earlier times. The form may be economic, political, social (industrialization) religious (industrialization), change in any part of society affects all the other parts of society. *Eg.* An individual is the fundamental unit of society and there's change in the life of the individual which is called evolutionary process of social change (birth to death). This is a slow process.

DEFINITION OF SOCIAL CHANGE

Ginsberg (By social change I understand a change in the social structure). Kingsley Doris "By social change is meant only such alternations as occur in social organization *i.e.*, the structure and functions of society". Merrill and Elbridge "*Social change means, that large no. of persons are engaging in activities that differ from those which they or their immediate fore-fathers engaged in some time before.*"

Gillin & Gillin "Social changes are variations from the accepted mode of life, whether due to alteration in geographical condition, in cultural equipment, composition of the

population. Or ideologies & whether brought about by diffusion or inventions within the group.”

Jones’ *“Social change is a term used to describe variations in or modification of any aspect of social process, social patterns, social interaction or social organization.”*

M.D. Jenson – Describes –Social change as *“modification in ways of doing and thinking of people.”*

Characteristics of Social Change

1. Social change is universal or it is an essential law.
2. Change with diff. in speed and form simple society ... change was slower.
3. Change is unpredictable in general Revol is a process of social change. What speed and in what form the change takes place is not easily predictable.
4. Social change is change in community
5. Social change generally changes in direction. There are 3 patterns of social change.
 - i. linear failure change generally leads to progress (change for good) can’t cycle –car – train –plain
 - ii. Fluctuating change – the change may be upward and downward. The demographic change is such also economic change,
 - iii. Cyclical change – the change is in a cycle. Fashion, sometimes also in economical aspect (Karl max gave this idea. He says earlier there was no private property and we may go back to it).

FACTORS OF SOCIAL CHANGE

Biological Factors

Demographic factors – Population plays an important role in society if there is change in the composition of pop there is change in society by composition we mean the structure *i.e.*, sex ratio. For balance in society the sex ratio should be 1:1 and if there is change in the ratio there is change in society if there are more females than the status and position goes down (because in Polygamy more wives and the hubby now their status goes down). In the other case the females position rises. The bride –price increases (in the tribal society).

Age group – childhood, adulthood, old age. If the population of children is most then increase of population will be slower. If adults more than there will be rapid change in society cause they are the most regenerative. In case of old more there is conflict in society they don’t wish for change.

Marital status in production of children. If girls are married young there will be over population and he health is also in danger. Status of women becomes lower. And if at too late a stage – a girl is married fertility is less. Changes in demography – Birth rate and Death rate. Higher birth rate creates a lot of problems. Malthus theme of population

– Economics. Over population-poverty unemployment increases. Death – rate – man – power decreases.

Immigration and Emigration – 1 is coming into country, 2 – going out of the country. Causes cultural problems leads to over population. 2 – Brain – drain is the problem.

Natural factors – now native affect society – National calamities, floods, epidemics affects society in its social relationships (i) structure. People become selfish as during scarcities they are more bothered feeding themselves.

Technological Factors

- Mechanization and social change – machines bring about this gave women the chance to work gave rise to women's lib.
- Unemployment and such problems arose these affected cottage industries.
- Urbanization – changed job opportunities.

Transport gave rise to social contacts. Communication gives rise to greater awareness and is means of recreation too.

Cultural Factors

Write about concept of cultural lag by W.F. Ogburn book – social change brings change. He says material and non – material change. Usually non-material can't cope up with material changed and gives rise to cultural lag. Change in values ideas and custom's changes society (Habits). In handbook of Sociology, he said it may so happen that material behind education, urbanization *etc.*, too brings change in marriage system *etc.* Marxian theory of social change *i.e.*, Technological Deterministic theory.

On interpretative theory – change according to him is inevitable and a continuous process. He has given more importance to the economical factors. He says if there is change in economy the only tractor may (changes of demography *etc.* affect the individuals) there is change in society – change in the production system *i.e.*, change in technology because it is due to change in technology that there's change in production that's why his theme is called technological data. Two change in production system. Has two aspects productive forces and productive relations – this is due to change in technology productive apparatuses, labour and production experience and labour still ? productive forces. Productive relations ? Capitalists and labourers (master and slaves).

MEANING AND NATURE OF SOCIAL CHANGE

MEANING

Any change that alters basically the established forms of social relationships, thus transforming in some measure the social structure, is social change. It is a change that reflects in all aspects of social life, *i.e.*, family, neighbourhood, play group, work group (industry or hospital *etc.*), religion, economic and political spheres, leisure activities *etc.* Social change may be partial or total, though mostly it is partial.

A change resulting from enacting a law prohibiting untouchability might be called a partial change in society. Total change hardly happens. We always find partial change in caste system, family or marriage systems. Societies do change, it is inevitable but total structure hardly ever changes. Changes are not necessarily good or bad. Sociologists do not necessarily consider social change *per se* to be good or bad.

According to Kingsley Davis (1949), “By social change is meant only such alterations that occur in social organisation, that is, structure and function of society.” M.E. Jones (1962) states: “Social change is a term used to describe variations in, or modifications of, any aspect of social processes, social patterns, social interaction or social organisation.” Horton and Hunt (1968) defined it as “changes in the social structure and social relationships of the society”. As sociologists we are not concerned with all types of changes that occur in society all the time. Our direct concern is with social relationships. It is the change which alone we shall regard as social change. Thus, any modification that occurs in the web of social relationships, which alters the long established pattern of social structure and social institutions, may be termed as social change. For sociologist, alterations in the properties of social structure (or social systems) through time constitute the focus of the subject of social change or social dynamics.

Nature of Social Change

Social change is not a uniquely modern phenomenon. Some kinds and degrees of change are universal in human experience. No society is static whether it is primitive or modern. The speed of change may differ from society to society. This speed may be so slow that it may appear that the society is static but it is not so. The so-called modern contemporary society is changing rapidly and constantly. These changes are neither temporally nor spatially isolated, that is, changes occur in sequential chains rather than as ‘temporary’ crises. Change takes place in chain-reaction sequence, *i.e.*, change in one part of society brings changes in other parts also. But, the rate or speed and direction of change are not always uniform in all aspects of society.

Some aspects change faster than the other. Material aspect of society (civilization) changes faster than the non-material aspects (ideology, thinking pattern or social relationships). Causes and effects of change may also differ from society to society. The direction or patterns of social change may be linear, circular or fluctuating up and down. It is such a complex phenomenon that no prediction or measurement about its speed, direction and causes can be made in accurate and precise terms as we see in natural sciences. As an astronomer can predict the orderly relations among planets in a solar system, sociologist cannot precisely predict about human relations. Whether a marital relation between a woman and a man will last for life long or will just end in divorce after sometime.

The proportion of contemporary change (planned or unplanned—latent consequences of deliberate innovations such as information technology) is much higher than in former times. The range of material technology and social strategies is expanding rapidly and

its net effect is additive or cumulative. Contemporary change is probably everywhere and its consequences may also be everywhere. Besides, increasing the expectations of life, it has brought many social problems in its train to humanity. In the modern world social change has taken on some special qualities and magnitudes. The modes of life and social institutions characteristic of the modern world are radically different from those of even the recent past. During a period of only two or three centuries human social life has been wrenched away from the types of social order in which people lived for thousands of years. But though we are largely immune from the natural disasters such as flood, famine and diseases like plague in the technologically developed countries today, we have to deal now with the social forces we ourselves have created. These forces bring social change into our lives in a continuous way.

THEORIES OF SOCIAL CHANGE: MEANING, NATURE AND PROCESSES

Change is the law of nature. What is today shall be different from what it would be tomorrow. The social structure is subject to incessant change. Forty years hence government is due to make important changes. Family and religion will not remain the same during this period because institutions are changing. Individuals may strive for stability, societies may create the illusion of permanence, the quest for certainty may continue unabated, yet the fact remains that society is an ever-changing phenomenon, growing, decaying, renewing and accommodating itself to changing conditions and suffering vast modifications in the course of time. Our understanding of it will not be complete unless we take into consideration this changeable nature of society, study how differences emerge and discover the direction of change.

The Meaning of Social Change

The word “change” denotes a difference in anything observed over some period of time. Social change, therefore, would mean observable differences in any social phenomena over any period of time.

The Following are Some of its Definitions:

- (i) Jones. “Social change is a term used to describe variations in, or modifications of, any aspect of social processes, social patterns, social interaction or social organisation.”
- (ii) Mazumdar, H. T. “Social change may be defined as a new fashion or mode, either modifying or replacing the old, in the life of a people, or in the operation of a society.”
- (iii) Gillin and Gillin. “Social changes are variations from the accepted modes of life; whether due to alteration in geographical conditions, in cultural equipment, composition of the population or ideologies and whether brought about by diffusion or inventions within the group.”
- (iv) Davis. By “Social change is meant only such alterations as occur in social organisation, that is, structure and functions of society.”

- (v) Merrill and Eldredge. "Social change means that large number of persons are engaging in activities that differ from those which they or their immediate forefathers engaged in some time before."
- (vi) MacIver and Page. "...Our direct concern as sociologists is with social relationships. It is the change in these relationships which alone we shall regard as social change."
- (vii) M. D. Jenson. "Social change may be defined as modification in ways of doing and thinking of people."
- (viii) Koenig, S. "Social change refers to the modifications which occur in the life patterns of a people."
- (ix) Lundberg and others. "Social change refers to any modification in established patterns of inter human relationships and standards of conduct."
- (x) Anderson and Parker. "Social change involves alteration in the structure or functioning of social forms or processes themselves."
- (xi) Ginsberg, M. "By social change, I understand a change in social structure *e.g.*, the size of a society, the composition or balance of its parts or the type of its organisation."

On the basis of these definitions it may be concluded that social change refers to the modifications which take place in the life patterns of people. It does not refer to all the changes going on in the society. The changes in art, language, technology; philosophy *etc.*, may not be included in the term 'Social change' which should be interpreted in a narrow sense to mean alterations in the field of social relationships. Social relationships are social processes, social patterns and social interactions. Thus social change will mean variations of any aspect of social processes, social patterns, social interactions or social organisation. It is a change in the institutional and normative structure of society.

NATURE OF SOCIAL CHANGE

The Main Characteristics of the Nature of Social Change are as follows:

Social Change is a Universal Phenomenon

Social change occurs in all societies. No society remains completely static. This is true of all societies, primitive as well as civilized. Society exists in a universe of dynamic influences. The population changes, technologies expand, material equipment changes, ideologies and values take on new components and institutional structures and functions undergo reshaping. The speed and extent of change may differ from society to society. Some change rapidly, others change slowly.

Social Change is Community Change

Social change does not refer to the change in the life of an individual or the life patterns of several individuals. It is a change which occurs in the life of the entire

community. In other words, only that change can be called social change whose influence can be felt in a community form. Social change is social and not individual.

Speed of Social Change is not Uniform

While social change occurs in all societies, its speed is not uniform in every society. In most societies it occurs so slowly that it is often not noticed by those who live in them. Even in modern societies there seems to be little or no change in many areas. Social change in urban areas is faster than in rural areas.

Nature and Speed of Social Change is Affected by and Related to Time Factor

The speed of social change is not uniform in each age or period in the same society. In modern times the speed of social change is faster today than before 1947. Thus, the speed of social change differs from age to age. The reason is that the factors which cause social change do not remain uniform with the change in times. Before 1947 there was less industrialization in India, after 1947 India has become more industrialized. Therefore, the speed of social change after 1947 is faster than before 1947.

Social Change Occurs as an Essential Law

Change is the law of nature. Social change also is natural. It may occur either in the natural course or as a result of planned efforts. By nature we desire change. Our needs keep on changing. To satisfy our desire for change and our changing needs social change becomes a necessity. The truth is that we are anxiously waiting for a change. According to Green, "The enthusiastic response of change has become almost a way of life."

Definite Prediction of Social Change is not Possible

It is difficult to make any prediction about the exact forms of social change. There is no inherent law of social change according to which it would assume definite forms. We may say that on account of the social reform movement untouchability will be abolished from the Indian society; that the basis and ideals of marriage will change due to the marriage laws passed by the government; that industrialization will increase the speed of urbanisation but we cannot predict the exact forms which social relationships will assume in future. Likewise it cannot be predicted as to what shall be our attitudes, ideas, norms and values in future.

Social Change Shows Chain-Reaction Sequence

A society's pattern of living is a dynamic system of inter-related parts. Therefore, change in one of these parts usually reacts on others and those on additional ones until they bring a change in the whole mode of life of many people. For example, industrialism has destroyed the domestic system of production. The destruction of domestic system of production brought women from the home to the factory and the office. The employment of women meant their independence from the bondage of man. It brought

a change in their attitudes and idea. It meant a new social life for women. It consequent affected every part of the family life.

Social Change Results from the Interaction of a Number of Factors

Generally, it is thought that a particular factor like changes in technology, economic development or climatic conditions causes social change. This is called monistic theory which seeks to interpret social change in terms of one single factor. But the monistic theory does not provide an adequate explanation of the complex phenomenon of social change. As a matter of fact, social change is the consequence of a number of factors. A special factor may trigger a change but it is always associated with other factors that make the triggering possible. The reason is that social phenomena are mutually interdependent. None stand out as isolated forces that bring about change of themselves. Rather each is an element in a system. Modification of one part influences the other parts and these influence the rest, until the whole is involved.

Social Changes are Chiefly those of Modification or of Replacement

Social changes may be broadly categorised as modifications or replacements. It may be modification of physical goods or social relationships. For example, the form of our breakfast food has changed. Though we eat the same basic materials which we ate earlier, wheat, eggs, corn, but their form is changed. Ready-to-eat-cornflakes, breads, omelets are substituted for the form in which these same materials were consumed in yester years. There may also be modifications of social relationships. The old authoritarian family has become the small equalitarian family, the one room school has become a centralized school. Our ideas about women's rights, religion, government and co-education stand modified today.

Change also takes the form of replacement. A new material or non-material form supplants an old one Horses have been replaced by automobiles. Similarly, old ideas have been replaced by new ideas. The germ theory of medicine has replaced older views of the cause of disease. Democracy has replaced aristocracy.

THEORIES OF SOCIAL CHANGE

Among the theories of social change we shall study the theories regarding:

- The direction of social change and
- The causes of social change.

The Direction of Social Change

Early sociologists viewed the culture of primitive peoples as completely static, but this was abandoned with the appearance of scientific studies of preliterate cultures. Anthropologists now agree that primitive cultures have undergone changes although at such a slow pace as to give the impression of being stationary. In recent years the social change has proceeded at a very rapid rate. Since World War I numerous countries have

passed through profound changes not only in their political institutions but in their class structures, their economic systems, their modes of living. Various theories have been advanced to explain the direction of social change. We take a brief consideration of each of them.

Theory of Deterioration

Some thinkers have identified social change with deterioration. According to them, man originally lived in a perfect state of happiness in a golden age. Subsequently, however, deterioration began to take place with the result that man reached an age of comparative degeneration. This was the notion in the ancient Orient. It was expressed in the epic poems of India, Persia and Sumeria. Thus, according to Indian mythology man has passed through four ages—Satyug, Treta, Dwapar and Kaliyug. The Satyug was the best age in which man was honest, truthful and perfectly happy. Thereafter degeneration began to take place. The modern age is the age of Kaliyug wherein man is deceitful, treacherous, false, dishonest, selfish and consequently unhappy. That such should be the concept of history in early times is understandable, since we observe deterioration in every walk of life today.

Cyclic Theory

Another ancient notion of social change found side by side with the afore-mentioned one, is that human society goes through certain cycles. Looking to the cyclic changes of days and nights and of climates some sociologists like Spengler believe that society has a predetermined life cycle and has birth, growth, maturity, and decline.

Modern society is in the last stage. It is in its old age. But since history repeats itself, society after passing through all the stages, returns to the original stage, whence the cycle again begins. This concept is found in Hindu mythology, according to which Satyug will again start after Kaliyug is over. J.B. Bury in his *The Idea of Progress*, pointed out that this concept is also found in the teachings of stoic philosophers of Greece as well as in those of some of the Roman philosophers, particularly Marcus Aurelius. The view that change takes place in a cyclical way has been accepted by some modern thinkers also who have given different versions of the cyclical theory. The French anthropologist and biologist Vacher de Lapouge held that race is the most important determinant of culture. Civilization, he maintained, develops and progresses when a society is composed of individuals belonging to superior races and declines when racially inferior people are absorbed into it.

Western civilization, according to him, is doomed to extinction because of the constant infiltration of foreign inferior elements and their increasing control over it. The German anthropologist Otto Ammon, the Englishman Houston Stewart Chamberlain and American Madison Grant and Lothrop Stoddard also agreed with the view of Lapouge which may be called the theory of biological cycle. Spengler developed another version of cyclical theory of social change. He analysed the history of various civilizations

including the Egyptian, Greek and Roman and concluded that all civilizations pass through a similar cycle of birth, maturity and death. The western civilization is now on its decline which is unavoidable.

Vilfredo Pareto propounded the theory that societies pass through the periods of political vigour and decline which repeat themselves in cyclical fashion. The society according to him, consists of two types of people—one, who like to follow traditional ways whom he called rentiers, and those who like to take chances for attaining their ends whom he called as Speculators. Political change is initiated by a strong aristocracy, the speculators who later lose their energy and become incapable of vigorous role. Thus ruling class eventually resort to tricks or to clever manipulations and they come to possess individuals characterized by the rentier mentality. The society declines, but at the same time speculators arise from among the subjugated to become the new ruling class and overthrow the old group. Then the cycle begins.

F. Stuart Chapin gave another version of cyclical change. He made the concept of accumulation the basis for his theory of social change. According to him, cultural change is “selectively accumulative in time.” He wrote, “The most hopeful approach to the concept of cultural change would seem to be to regard the process as selectively accumulative in time and cyclical or oscillatory in character.” Thus, according to Chapin, cultural change is both selectively accumulative and cyclical in character. He postulated a hypothesis of synchronous cyclical change. According to him, the different parts of culture go through a cycle of growth, vigour and decay.

If the cycles of the major parts, such as government and the family, coincide or synchronize, the whole culture will be in a state of integration. If they do not synchronize, the culture will be in a disintegrated condition. Growth and decay, according to Chapin, in cultural forms are as inescapable as they are in all living things. Relying upon data drawn from the history of various civilizations, Sorokin concluded that civilizations fall into three major types namely, the ideational, the idealistic and the sensate. In the ideational type of civilization, reality and value are conceived of in terms of a “supersensory and super-rational God”, while the sensory world appears as illusory.

In a word, ideational culture is god-ridden. In the idealistic type of culture, reality and value are regarded sensory as well as supersensory. This is a synthesis of ideational and the sensate. The thought and behaviour of man are partly anchored in the materialistic and are partly anchored with the other world. In the sensate type of culture the whole way of life is characterized by a positivistic, materialistic outlook. Reality and value are merely what the senses perceive and beyond sense perception there is no reality. The western civilization, according to Sorokin, is now in an “overripe” sensate phase that must be supplanted by a new ideational system.

In recent times Arnold J. Toynbee, the noted English historian, has also propounded a cyclical theory of the history of world civilization. He maintained that civilizations pass through three stages, corresponding to youth, maturity and decline. The first is

marked by a “response to challenge”, the second is a “time of troubles,” and the third is characterized by gradual degeneration. He was also of the view that our civilization, although in the state of final downfall, can still ‘be saved by means of proper guidance by the “creative minority” by which he meant a select group of leaders who withdraw from the corrupting influences, commune with God, become spiritually regenerated and then return to inspire the masses.

The above concepts of the cyclical nature of social change may be called theories of cultural cycles. They are as a matter of fact the result of philosophical rather than scientific studies. The authors of these concepts begin with presumptions which they try to substantiate by marshalling a mass of data from history. They are philosophical doctrines, spun from the whole cloth, however heavily documented and illustrated by distorted historical evidences. Barnes, while appraising Toynbee’s work, wrote, “It is not objective or even interpretative history.

It is theology, employing selected facts of history to illustrate the will of God as the medieval bestiaries utilized biological fantasies to achieve the same results... Toynbee’s vast materials throw far more light upon the processes of Toynbee’s mind than upon the actual process of history... He writes history as he thinks it should be to further the cause of salvation, rather than as it has really been.”

Linear Theory

Some thinkers subscribe to the linear theory of social change. According to them, society gradually moves to an even higher state of civilization and that it advances in a linear fashion and in the direction of improvement. Auguste Comte postulated three stages of social change: the Theological, the Metaphysical and the Positive. Man has passed through the first two stages, even though in some aspects of life they still prevail, and is gradually reaching the Positive stage. In the first stage man believed that supernatural powers controlled and designed the world. He advanced gradually from belief in fetishes and deities to monotheism.

This stage gave way to the Metaphysical stage, during which man tries to explain phenomena by resorting to abstractions. On the positive stage man considers the search for ultimate causes hopeless and seeks the explanatory facts that can be empirically observed. This implies progress which according to Comte will be assured if man adopts a positive attitude in the understanding of natural and social phenomena.

Herbert Spencer, who likened society to an organism, maintained that human society has been gradually progressing towards a better state. In its primitive state, the state of militarism, society was characterized by warring groups, by a merciless struggle for existence. From militarism society moved towards a state of industrialism. Society in the stage of industrialism is marked by greater differentiation and integration of its parts. The establishment of an integrated system makes it possible for the different groups—social, economic and racial, to live in peace. Some Russian sociologists also subscribed to the linear theory of social change

Nikolai K. Mikhailovsky opined that human society passes through three stages;

1. The objective anthropocentric,
2. The eccentric, and;
3. The subjective anthropocentric.

In the first stage, man considers himself the centre of the universe and is preoccupied with mystic beliefs in the supernatural. In the second stage, man is given over to abstractions; the abstract is more “real” to him than the actual. In the third stage, man comes to rely upon empirical knowledge by means of which he exercises more and more control over nature for his own benefit. Solo-view conceived of the three stages as the tribal, the national governmental, and the period of universal brotherhood.

Pritirim Sorokin in his concept of variable recurrence has attempted to include both cyclical and linear change. In his view culture may proceed in a given direction for a time and thus appear to conform to a linear formula. But eventually, as a result of forces that are internal within the culture itself, there will be a shift of direction and a new period of development will be ushered in. Perhaps the new trend is also linear, perhaps it is oscillating, perhaps it conforms to some particular type of curve. At any rate, it also reaches limits and still another trend takes its place.

The description given by Sorokin makes room for almost any possibility, deterioration, progress or cyclical change and, therefore, sociologists find little quarrel with his description. But at any rate, Sorokin’s variable occurrence is an admission that the present state of sociological knowledge does not warrant the construction of theories regarding the long-run trend or character of social change. Whether contemporary civilization is headed for the scrap-heap via internal disintegration or atomic warfare, or is destined to be replaced by some stabler and idealistic system of social relationships cannot be predicted on other than grounds of faith. The factual evidence which is available to us can only lead us to remark that whatever direction social change takes in future, that direction will be determined by man himself.

The Causes of Social Change

The direction in which social change has taken place according to the writers. But none of the above theories strikes the central question of causation of change. Among the causal theories of social change the deterministic theory is the most popular. Now we take a brief review of this theory.

Deterministic Theories of Social Change

The deterministic theory of social change is a widely accepted theory of social change among contemporary sociologists. According to this theory there are certain forces, social or natural or both, which bring about social change. It is not reason or intellect but the presence of certain forces and circumstances which determine the course of social change. Sumner and Keller insisted that social change is automatically determined by economic factors. Keller maintained that conscious effort and rational planning have

very little chance to effect change unless and until the folkways and mores are ready for it. Social change is an essentially irrational and unconscious process. Variation in the folkways which occurs in response to a need is not planned. Man can at most only assist or retard the change that is under way. It was Karl Marx who, deeply impressed by the German philosopher Hegel's metaphysical idealism, held that material conditions of life are the determining factors of social change. His theory is known as the theory of economic determinism or "the materialist interpretation of history".

Briefly put Marx held that human society passes through various stages, each with its own well-defined organisational system. Each successive stage comes into existence as a result of conflict with the one preceding it. Change from one stage to another is due to changes in the economic factors, namely, the methods of production and distribution. The material forces of production are subject to change, and thus a rift arises between the underlying factors and the relationships built upon them. A change in the material conditions of life brings changes in all social institutions, such as state, religion and family. It alters the primary socio-economic relationships. To put in his own words, "Legal relations as well as forms of state could neither be understood by themselves, nor explained by the so-called general progress of the human mind, but they are rooted in the material conditions of life... The mode of production in material life determines the general character of the social, political and spiritual process of life.

It is not the consciousness of man that determines their existence, but on the contrary, their social existence determines their consciousness." Thus the economic factor is a primary one in society, for all social phases of life are dependent upon it and are almost entirely determined by it. According to Engels, a close associate of Marx, "The ultimate causes of all social changes and political revolutions are to be sought not in the minds of men, in their increasing insight into the eternal truth and justice, but in changes in the mode of production and exchange." According to Marx, the social order has passed through five phases called the oriental, the ancient, the feudal, the capitalistic, and the communistic. The modern capitalistic system has been moving towards its doom because the conditions it produced and the forces it unloosed make its disintegration inevitable. In it the class struggle is simplified, revealing itself more and more into the clear-cut conflict of two great classes, the bourgeoisie and the proletariat.

As Marx puts... "The weapons with which the bourgeoisie felled feudalism to the ground are now turned against the bourgeoisie itself. But not only has the bourgeoisie forged, the weapons that bring death to itself, it has called into existence the men who are to wield those weapons—the modern working class, the proletariat." Coker has beautifully summed up the tendencies of capitalism in the following words.

"Thus the capitalist system enlarges the number of workers, orings them together into compact groups, makes them class conscious, supplies them with means of inter-communication and co-operation on a worldwide scale, reduces their purchasing power, and by increasingly exploiting them arouses them to organised resistance. Capitalists acting persistently in pursuit of their natural needs and in vindication of a system

dependent upon the maintenance of profits, are all the time creating conditions which stimulate and strengthen the natural efforts of workers in preparing for a system that will fit the needs of working men's society."

The resulting social order will not reach its full development at once but will go through two stages. In the first, there will be a dictatorship of the proletariat during which the proletariat will rule despotically and crush out all the remnants of capitalism. In the second, there will be real communism, during which there shall be no state, no class, no conflict, and no exploitation. Marx visualized a society in which the social order will have reached a state of perfection. In that society the prevailing principle will be "from each according to his capacities, to each according to his needs."

Marx's theory of determinism contains a great element of truth but it cannot be said to contain the whole truth. Few deny that economic factors influence social conditions of life but few hold that economic factors are the only activating forces in human history. There are other causes obviously also at work. There is no scientific proof that human society is going through the stages visualized by Marx. His claim that man is destined to attain an ideal stage of existence is little more than visionary. His theory of value and its corollary of surplus value, his theory of the sole productivity of labour as such, and his law of the accumulation of capital are derived from an outmoded, abstract and narrow doctrine of the equivalence of price and cost which has been now rejected by modern economists.

Moreover, Marx's thesis of the relation between social change and economic process is based upon an inadequate psychology. In a way it may be said that an inadequate psychology is perhaps the fatal weakness of all determinisms. He does not tell us as how change is reproduced in the modes of production. He speaks as though the changing technique of production explained itself and was a first cause.

He gives a simple explanation of social change and ignores the complexities of habituation on the one hand and of revulsion on the other. He simplifies the attitudes that gather around institution; the solidarities and loyalties of family, occupation and nation are subjected to those of economic class. He as a matter of fact has not squarely faced the intricate question of social causation. That the economic changes and social changes are correlated, none may deny. But to say that the superstructure of social relationships is determined by the economic structure is going too far.

Russell writes, "Men desire power, they desire satisfactions for their pride and their self-respect. They desire victory over rivals so profoundly that they will invent a rivalry for the unconscious purpose of making a victory possible. All these motives cut across the pure economic motive in ways that are practically important." The deterministic interpretation of social change is too simple.

A number of social thinkers opposed to the theory of economic determinism consider non-material elements of culture the basic sources of social change. They regard ideas as the prime movers in social life. The economic or material phenomena are conceived to be subordinate to the non-material. Gustave Le Bon, George Sorel, James G. Frazer

and Max Weber held that religion is the chief initiator of social changes. Thus Hinduism, Buddhism and Judaism have had a determining influence upon the economics of their adherents. The theory of religious determinism has been criticised by Sorokin in his *Contemporary Sociological Theories*. He posed the question; "If all social institutions change under the influence of the changes in religion, how, when and why does religion change itself"? According to Sorokin change is caused by the interaction of the various parts of a culture, none of which may be considered primary.

It means that change is pluralistic rather than monistic in origin. But this pluralistic theory of social change is initiated in the material culture and thence spreads to other spheres. Change is caused not only by economic factors but is also largely automatic in nature. A number of sociologists have held that social change can be brought about by means of conscious and systematic efforts. Thus, Lester F. Ward asserted that progress can be achieved by means of purposive efforts of conscious planning. Through education and knowledge intellect can assert itself over the emotions so that effective planning is made possible.

Natural evolution, according to Ward, is a slow process, whereas intelligent planning accelerates the processes of nature. Charles A Ellwood agreed with Ward that progress is promoted by education and knowledge. Lund-wig Stein, a German sociologist and philosopher, and L.T. Hobhouse, an English sociologist, also expounded theories closely resembling Ward's. They expressed the view that progress can be achieved through the control of material factors by the mind. Human affairs are amenable to control by reason and, therefore, rational element in our nature must be developed so that it may be utilized as a factor in the evolutionary process.

THE SOCIAL ENVIRONMENT

A being whose activities are associated with others has a social environment. What he does and what he can do depend upon the expectations, demands, approvals, and condemnations of others. A being connected with other beings cannot perform his own activities without taking the activities of others into account. For they are the indispensable conditions of the realization of his tendencies. When he moves he stirs them and reciprocally. We might as well try to imagine a business man doing business, buying and selling, all by himself, as to conceive it possible to define the activities of an individual in terms of his isolated actions. The manufacturer moreover is as truly socially guided in his activities when he is laying plans in the privacy of his own counting house as when he is buying his raw material or selling his finished goods.

Thinking and feeling that have to do with action in association with others is as much a social mode of behaviour as is the most overt cooperative or hostile act. What we have more especially to indicate is how the social medium nurtures its immature members. There is no great difficulty in seeing how it shapes the external habits of action. Even dogs and horses have their actions modified by association with human beings; they form different habits because human beings are concerned with what they

do. Human beings control animals by controlling the natural stimuli which influence them; by creating a certain environment in other words. Food, bits and bridles, noises, vehicles, are used to direct the ways in which the natural or instinctive responses of horses occur. By operating steadily to call out certain acts, habits are formed which function with the same uniformity as the original stimuli.

If a rat is put in a maze and finds food only by making a given number of turns in a given sequence, his activity is gradually modified till he habitually takes that course rather than another when he is hungry. Human actions are modified in a like fashion. A burnt child dreads the fire; if a parent arranged conditions so that every time a child touched a certain toy he got burned, the child would learn to avoid that toy as automatically as he avoids touching fire. So far, however, we are dealing with what may be called training in distinction from educative teaching. The changes considered are in outer action rather than in mental and emotional dispositions of behaviour. The distinction is not, however, a sharp one. The child might conceivably generate in time a violent antipathy, not only to that particular toy, but to the class of toys resembling it. The aversion might even persist after he had forgotten about the original burns; later on he might even invent some reason to account for his seemingly irrational antipathy.

In some cases, altering the external habit of action by changing the environment to affect the stimuli to action will also alter the mental disposition concerned in the action. Yet this does not always happen; a person trained to dodge a threatening blow, dodges automatically with no corresponding thought or emotion. We have to find, then, some differentia of training from education. A clew may be found in the fact that the horse does not really share in the social use to which his action is put. Some one else uses the horse to secure a result which is advantageous by making it advantageous to the horse to perform the act — he gets food, *etc.* But the horse, presumably, does not get any new interest. He remains interested in food, not in the service he is rendering. He is not a partner in a shared activity.

Were he to become a copartner, he would, in engaging in the conjoint activity, have the same interest in its accomplishment which others have. He would share their ideas and emotions. Now in many cases — too many cases — the activity of the immature human being is simply played upon to secure habits which are useful. He is trained like an animal rather than educated like a human being. His instincts remain attached to their original objects of pain or pleasure. But to get happiness or to avoid the pain of failure he has to act in a way agreeable to others. In other cases, he really shares or participates in the common activity. In this case, his original impulse is modified. He not merely acts in a way agreeing with the actions of others, but, in so acting, the same ideas and emotions are aroused in him that animate the others. A tribe, let us say, is warlike.

The successes for which it strives, the achievements upon which it sets store, are connected with fighting and victory. The presence of this medium incites bellicose exhibitions in a boy, first in games, then in fact when he is strong enough. As he fights

he wins approval and advancement; as he refrains, he is disliked, ridiculed, shut out from favourable recognition. It is not surprising that his original belligerent tendencies and emotions are strengthened at the expense of others, and that his ideas turn to things connected with war. Only in this way can he become fully a recognized member of his group. Thus his mental habitudes are gradually assimilated to those of his group. If we formulate the principle involved in this illustration, we shall perceive that the social medium neither implants certain desires and ideas directly, nor yet merely establishes certain purely muscular habits of action, like “instinctively” winking or dodging a blow. Setting up conditions which stimulate certain visible and tangible ways of acting is the first step.

Making the individual a sharer or partner in the associated activity so that he feels its success as his success, its failure as his failure, is the completing step. As soon as he is possessed by the emotional attitude of the group, he will be alert to recognize the special ends at which it aims and the means employed to secure success. His beliefs and ideas, in other words, will take a form similar to those of others in the group. He will also achieve pretty much the same stock of knowledge since that knowledge is an ingredient of his habitual pursuits. The importance of language in gaining knowledge is doubtless the chief cause of the common notion that knowledge may be passed directly from one to another.

It almost seems as if all we have to do to convey an idea into the mind of another is to convey a sound into his ear. Thus imparting knowledge gets assimilated to a purely physical process. But learning from language will be found, when analyzed, to confirm the principle just laid down. It would probably be admitted with little hesitation that a child gets the idea of, say, a hat by using it as other persons do; by covering the head with it, giving it to others to wear, having it put on by others when going out, *etc.* But it may be asked how this principle of shared activity applies to getting through speech or reading the idea of, say, a Greek helmet, where no direct use of any kind enters in. What shared activity is there in learning from books about the discovery of America? Since language tends to become the chief instrument of learning about many things, let us see how it works.

The baby begins of course with mere sounds, noises, and tones having no meaning, expressing, that is, no idea. Sounds are just one kind of stimulus to direct response, some having a soothing effect, others tending to make one jump, and so on. The sound h-a-t would remain as meaningless as a sound in Choctaw, a seemingly inarticulate grunt, if it were not uttered in connection with an action which is participated in by a number of people. When the mother is taking the infant out of doors, she says “hat” as she puts something on the baby’s head. Being taken out becomes an interest to the child; mother and child not only go out with each other physically, but both are concerned in the going out; they enjoy it in common. By conjunction with the other factors in activity the sound “hat” soon gets the same meaning for the child that it has for the parent; it becomes a sign of the activity into which it enters.

The bare fact that language consists of sounds which are mutually intelligible is enough of itself to show that its meaning depends upon connection with a shared experience. In short, the sound h-a-t gains meaning in precisely the same way that the thing “hat” gains it, by being used in a given way. And they acquire the same meaning with the child which they have with the adult because they are used in a common experience by both. The guarantee for the same manner of use is found in the fact that the thing and the sound are first employed in a joint activity, as a means of setting up an active connection between the child and a grownup. Similar ideas or meanings spring up because both persons are engaged as partners in an action where what each does depends upon and influences what the other does.

If two savages were engaged in a joint hunt for game, and a certain signal meant “move to the right” to the one who uttered it, and “move to the left” to the one who heard it, they obviously could not successfully carry on their hunt together. Understanding one another means that objects, including sounds, have the same value for both with respect to carrying on a common pursuit. After sounds have got meaning through connection with other things employed in a joint undertaking, they can be used in connection with other like sounds to develop new meanings, precisely as the things for which they stand are combined. Thus the words in which a child learns about, say, the Greek helmet originally got a meaning by use in an action having a common interest and end.

They now arouse a new meaning by inciting the one who hears or reads to rehearse imaginatively the activities in which the helmet has its use. For the time being, the one who understands the words “Greek helmet” becomes mentally a partner with those who used the helmet. He engages, through his imagination, in a shared activity. It is not easy to get the full meaning of words. Most persons probably stop with the idea that “helmet” denotes a queer kind of headgear a people called the Greeks once wore. We conclude, accordingly, that the use of language to convey and acquire ideas is an extension and refinement of the principle that things gain meaning by being used in a shared experience or joint action; in no sense does it contravene that principle.

When words do not enter as factors into a shared situation, either overtly or imaginatively, they operate as pure physical stimuli, not as having a meaning or intellectual value. They set activity running in a given groove, but there is no accompanying conscious purpose or meaning. Thus, for example, the plus sign may be a stimulus to perform the act of writing one number under another and adding the numbers, but the person performing the act will operate much as an automaton would unless he realizes the meaning of what he does.

THE SOCIAL MEDIUM AS EDUCATIVE

Our net result thus far is that social environment forms the mental and emotional disposition of behaviour in individuals by engaging them in activities that arouse and strengthen certain impulses, that have certain purposes and entail certain consequences.

A child growing up in a family of musicians will inevitably have whatever capacities he has in music stimulated, and, relatively, stimulated more than other impulses which might have been awakened in another environment. Save as he takes an interest in music and gains a certain competency in it, he is “out of it”; he is unable to share in the life of the group to which he belongs. Some kinds of participation in the life of those with whom the individual is connected are inevitable; with respect to them, the social environment exercises an educative or formative influence unconsciously and apart from any set purpose.

In savage and barbarian communities, such direct participation furnishes almost the sole influence for rearing the young into the practices and beliefs of the group. Even in present-day societies, it furnishes the basic nurture of even the most insistently schooled youth. In accord with the interests and occupations of the group, certain things become objects of high esteem; others of aversion. Association does not create impulses or affection and dislike, but it furnishes the objects to which they attach themselves. The way our group or class does things tends to determine the proper objects of attention, and thus to prescribe the directions and limits of observation and memory.

What is strange or foreign tends to be morally forbidden and intellectually suspect. It seems almost incredible to us, for example, that things which we know very well could have escaped recognition in past ages. We incline to account for it by attributing congenital stupidity to our forerunners and by assuming superior native intelligence on our own part. But the explanation is that their modes of life did not call for attention to such facts, but held their minds riveted to other things. Just as the senses require sensible objects to stimulate them, so our powers of observation, recollection, and imagination do not work spontaneously, but are set in motion by the demands set up by current social occupations.

The main texture of disposition is formed, independently of schooling, by such influences. What conscious, deliberate teaching can do is at most to free the capacities thus formed for fuller exercise, to purge them of some of their grossness, and to furnish objects which make their activity more productive of meaning. While this “unconscious influence of the environment” is so subtle and pervasive that it affects every fibre of character and mind, it may be worth while to specify a few directions in which its effect is most marked. First, the habits of language. Fundamental modes of speech, the bulk of the vocabulary, are formed in the ordinary intercourse of life, carried on not as a set means of instruction but as a social necessity. The babe acquires, as we well say, the mother tongue. While speech habits thus contracted may be corrected or even displaced by conscious teaching, yet, in times of excitement, intentionally acquired modes of speech often fall away, and individuals relapse into their really native tongue. Secondly, manners. Example is notoriously more potent than precept.

Good manners come, as we say, from good breeding or rather are good breeding; and breeding is acquired by habitual action, in response to habitual stimuli, not by conveying information. Despite the never ending play of conscious correction and

instruction, the surrounding atmosphere and spirit is in the end the chief agent in forming manners. And manners are but minor morals. Moreover, in major morals, conscious instruction is likely to be efficacious only in the degree in which it falls in with the general “walk and conversation” of those who constitute the child’s social environment. Thirdly, good taste and esthetic appreciation. If the eye is constantly greeted by harmonious objects, having elegance of form and colour, a standard of taste naturally grows up. The effect of a tawdry, unarranged, and over-decorated environment works for the deterioration of taste, just as meagre and barren surroundings starve out the desire for beauty. Against such odds, conscious teaching can hardly do more than convey second-hand information as to what others think.

Such taste never becomes spontaneous and personally engrained, but remains a labored reminder of what those think to whom one has been taught to look up. To say that the deeper standards of judgements of value are framed by the situations into which a person habitually enters is not so much to mention a fourth point, as it is to point out a fusion of those already mentioned. We rarely recognize the extent in which our conscious estimates of what is worth while and what is not, are due to standards of which we are not conscious at all. But in general it may be said that the things which we take for granted without enquiry or reflection are just the things which determine our conscious thinking and decide our conclusions. And these habitudes which lie below the level of reflection are just those which have been formed in the constant give and take of relationship with others.

THE SCHOOL AS A SPECIAL ENVIRONMENT

The chief importance of this foregoing statement of the educative process which goes on willy-nilly is to lead us to note that the only way in which adults consciously control the kind of education which the immature get is by controlling the environment in which they act, and hence think and feel. We never educate directly, but indirectly by means of the environment. Whether we permit chance environments to do the work, or whether we design environments for the purpose makes a great difference. And any environment is a chance environment so far as its educative influence is concerned unless it has been deliberately regulated with reference to its educative effect. An intelligent home differs from an unintelligent one chiefly in that the habits of life and intercourse which prevail are chosen, or at least coloured, by the thought of their bearing upon the development of children. But schools remain, of course, the typical instance of environments framed with express reference to influencing the mental and moral disposition of their members. Roughly speaking, they come into existence when social traditions are so complex that a considerable part of the social store is committed to writing and transmitted through written symbols.

Written symbols are even more artificial or conventional than spoken; they cannot be picked up in accidental intercourse with others. In addition, the written form tends to select and record matters which are comparatively foreign to everyday life. The

achievements accumulated from generation to generation are deposited in it even though some of them have fallen temporarily out of use. Consequently as soon as a community depends to any considerable extent upon what lies beyond its own territory and its own immediate generation, it must rely upon the set agency of schools to insure adequate transmission of all its resources. To take an obvious illustration: The life of the ancient Greeks and Romans has profoundly influenced our own, and yet the ways in which they affect us do not present themselves on the surface of our ordinary experiences.

In similar fashion, peoples still existing, but remote in space, British, Germans, Italians, directly concern our own social affairs, but the nature of the interaction cannot be understood without explicit statement and attention. In precisely similar fashion, our daily associations cannot be trusted to make clear to the young the part played in our activities by remote physical energies, and by invisible structures. Hence a special mode of social intercourse is instituted, the school, to care for such matters. This mode of association has three functions sufficiently specific, as compared with ordinary associations of life, to be noted. First, a complex civilization is too complex to be assimilated in toto.

It has to be broken up into portions, as it were, and assimilated piecemeal, in a gradual and graded way. The relationships of our present social life are so numerous and so interwoven that a child placed in the most favourable position could not readily share in many of the most important of them. Not sharing in them, their meaning would not be communicated to him, would not become a part of his own mental disposition. There would be no seeing the trees because of the forest. Business, politics, art, science, religion, would make all at once a clamor for attention; confusion would be the outcome. The first office of the social organ we call the school is to provide a simplified environment.

It selects the features which are fairly fundamental and capable of being responded to by the young. Then it establishes a progressive order, using the factors first acquired as means of gaining insight into what is more complicated. In the second place, it is the business of the school environment to eliminate, so far as possible, the unworthy features of the existing environment from influence upon mental habitudes. It establishes a purified medium of action. Selection aims not only at simplifying but at weeding out what is undesirable. Every society gets encumbered with what is trivial, with dead wood from the past, and with what is positively perverse. The school has the duty of omitting such things from the environment which it supplies, and thereby doing what it can to counteract their influence in the ordinary social environment. By selecting the best for its exclusive use, it strives to reinforce the power of this best.

As a society becomes more enlightened, it realizes that it is responsible not to transmit and conserve the whole of its existing achievements, but only such as make for a better future society. The school is its chief agency for the accomplishment of this end. In the third place, it is the office of the school environment to balance the various elements in the social environment, and to see to it that each individual gets an opportunity to

escape from the limitations of the social group in which he was born, and to come into living contact with a broader environment. Such words as “society” and “community” are likely to be misleading, for they have a tendency to make us think there is a single thing corresponding to the single word.

As a matter of fact, a modern society is many societies more or less loosely connected. Each household with its immediate extension of friends makes a society; the village or street group of playmates is a community; each business group, each club, is another. Passing beyond these more intimate groups, there is in a country like our own a variety of races, religious affiliations, economic divisions. Inside the modern city, in spite of its nominal political unity, there are probably more communities, more differing customs, traditions, aspirations, and forms of government or control, than existed in an entire continent at an earlier epoch. Each such group exercises a formative influence on the active dispositions of its members.

A clique, a club, a gang, a Fagin’s household of thieves, the prisoners in a jail, provide educative environments for those who enter into their collective or conjoint activities, as truly as a church, a labour union, a business partnership, or a political party. Each of them is a mode of associated or community life, quite as much as is a family, a town, or a state. There are also communities whose members have little or no direct contact with one another, like the guild of artists, the republic of letters, the members of the professional learned class scattered over the face of the earth. For they have aims in common, and the activity of each member is directly modified by knowledge of what others are doing. In the olden times, the diversity of groups was largely a geographical matter.

There were many societies, but each, within its own territory, was comparatively homogeneous. But with the development of commerce, transportation, intercommunication, and emigration, countries like the United States are composed of a combination of different groups with different traditional customs. It is this situation which has, perhaps more than any other one cause, forced the demand for an educational institution which shall provide something like a homogeneous and balanced environment for the young. Only in this way can the centrifugal forces set up by juxtaposition of different groups within one and the same political unit be counteracted. The intermingling in the school of youth of different races, differing religions, and unlike customs creates for all a new and broader environment. Common subject matter accustoms all to a unity of outlook upon a broader horizon than is visible to the members of any group while it is isolated. The assimilative force of the American public school is eloquent testimony to the efficacy of the common and balanced appeal.

The school has the function also of coordinating within the disposition of each individual the diverse influences of the various social environments into which he enters. One code prevails in the family; another, on the street; a third, in the workshop or store; a fourth, in the religious association. As a person passes from one of the environments to another, he is subjected to antagonistic pulls, and is in danger of being split into a

being having different standards of judgement and emotion for different occasions. This danger imposes upon the school a steady and integrating office.

SOCIAL AND NATIONAL SERVICE

The present educational system is also responsible for increasing the gulf between the educated and the uneducated classes, between the intelligentsia and the masses. The intelligentsia should try to become a real service-group striving to uplift the masses and should resist the temptation to become a parasitical group living for itself and perpetuating its own privileged position. Our traditional 'elite' as a whole-with some noble exceptions-had no close ties with the masses, and the new 'elite' created by modern education also remained largely aloof from the people, except during the struggle for freedom under Mahatma Gandhi when he was able to inspire large numbers of educated and even well-to-do persons to identify themselves with the interests of the masses and the country as a whole. But from the time the struggle for freedom came to an end with the attainment of independence, they have again tended to move away from the people.

This is a great danger and, with a view to meeting it, suitable programmes to help in the evolution of a well-knit and united nation have to be devised. For this purpose, we recommend that some form of social and national service should be made obligatory for all students and should form an integral part of education at all stages. This can become an instrument to build character, improve discipline, inculcate a faith in the dignity of manual labour and develop a sense of social responsibility. Programmes of social and national service can be organized in one of two ways: they can either be carried out occasionally on a part-time basis during the period of education, or organized as a period of full-time service in addition to the present period of education. The second approach was recommended by the National Service Committee appointed under the chairmanship of Dr. C. D. Deshmukh.

It was suggested that national service should be a full-time programme of nine to twelve months for all young persons after passing out of higher secondary schools or the pre-university course and before entering employment or the university. The programme was to include some military training, social service, manual labour and general education. These proposals have not been adopted so far; nor have the public reactions to them been very favourable. Instead, at the time of the national emergency, the scheme of compulsory *National Cadet Corps* (NCC) was introduced at the university stage. About the same time, the Ministry of Education, sponsored a study of national service in several countries including Yugoslavia, Czechoslovakia, Germany, Norway, Sweden, the UK, the USA, Japan and the Philippines. The report of this study has been published under the title *National Service for Youth*.

Apart from discussing the salient features of such service in different countries, it made a number of recommendations about what could be done in India to develop a practicable scheme. It did not favour the 'introduction of compulsion, which has not

been tried or favoured in any of these countries, but recommended that national or social service should be introduced, at least to begin with, on a voluntary basis and extended as widely as possible and that it should provide a rich and varied programme of activities which will make an imaginative approach to youth. We believe that it would be more realistic and practicable to introduce the idea of social or national service broadly on the lines recommended in this study. Instead of adding one year to the total span of education for the purpose, a more feasible plan may be to develop, as an integral part of education, a programme which would run concurrently with academic studies in school and college. The programme should begin from the upper primary stage and continue up to the university so that the right attitudes are developed from an early age and every young person is ultimately brought within its ambit.

There are two main forms in which such service can be organized:

1. Encouraging and enabling students to participate in community living on the school or college campus; and
2. Providing opportunities of participation in programmes of community development and national service.

THE COMMON SCHOOL

In a situation of the type we have in India, it is the responsibility of the educational system to bring the different social classes and groups together and thus promote the emergence of an egalitarian and integrated society. But at present instead of doing so, education itself is tending to increase social segregation and to perpetuate and widen class distinctions. At the primary stage, the free schools to which the masses send their children are maintained by the government and local authorities and are generally of poor quality. Some of the private schools are definitely better; but since many of them charge high fees, they are availed of only by the middle and the higher classes. At the secondary stage, a large proportion of the good schools are private but many of them also charge high fees which are normally beyond the means of any but the top ten per cent of the people, though some of the middle class parents make great sacrifices to send their children to them. There is thus segregation in education itself-the minority of private, fee-charging, better schools meeting the needs of the upper classes and the vast bulk of free, publicly maintained, but poor schools being utilized by the rest. What is worse, this segregation is 'increasing and tending to widen the gulf between the classes and the masses.

This is one of the major weaknesses of the existing educational system. Good education, instead of being available to all children, or at least to all the able children from every stratum of society, is available only to a small minority which is usually selected not on the basis of talent but on the basis of its capacity to pay fees. The identification and development of the total national pool of ability is greatly hampered. The position is thus undemocratic and inconsistent with the ideal of an egalitarian society. The children of the masses are compelled to receive sub-standard education

and, as the programme of scholarships is not very large sometimes even the ablest among them are unable to find access to such good schools as exist, while the economically privileged parents are able to 'buy' good education for their children.

This is bad not only for the children of the poor but also for the children from the rich and privileged groups. It gives them a short-term advantage in so far as it enables them to perpetuate and consolidate their position. But it must be realized that, in the long run, their self-interest lies in identifying themselves with the masses. By segregating their children, such privileged parents prevent them from sharing the life and experiences of the children of the poor and coming into contact with the realities of life. In addition to weakening social cohesion, they also render the education of their own children anemic and incomplete. If these evils are to be eliminated and the educational system is to become a powerful instrument of national development in general, and social and national integration in particular, we must move towards the goal of a common school system of public education.

- Which will be open to all children, irrespective of caste, creed, community, religion, economic conditions or social status;
- Here access to good education will depend not on wealth or class but on talent;
- Which will maintain adequate standards in all schools and provide at least a reasonable proportion of quality institutions;
- In which no tuition fee will be charged; and
- Which would meet the needs of the average parent so that he would not ordinarily feel the need to send his children to expensive schools outside the system.

Such an educational system has, for instance, been built up in the USSR and is one of the major factors which have contributed to its progress. It has also been developed, in different forms and to varying degrees, in other nations like the USA, France and the Scandinavian countries. The traditional English system has been different and has allowed good education, under private management, to be largely reserved for those who have the capacity to pay the necessary fees. But recently, the so-called Public Schools have come in for strong criticism in England itself and it is not unlikely that a radical change may be initiated to make them more democratic. A somewhat similar system was transplanted in India by British administrators and we have clung to it so long because it happened to be in tune with the traditional hierarchical structure of our society. Whatever its past history may be, such a system has no valid place in the new democratic and socialistic society we desire to create. We examine more fully elsewhere how this programme can be implemented.

Social and Cultural Development

SOCIAL DEVELOPMENT

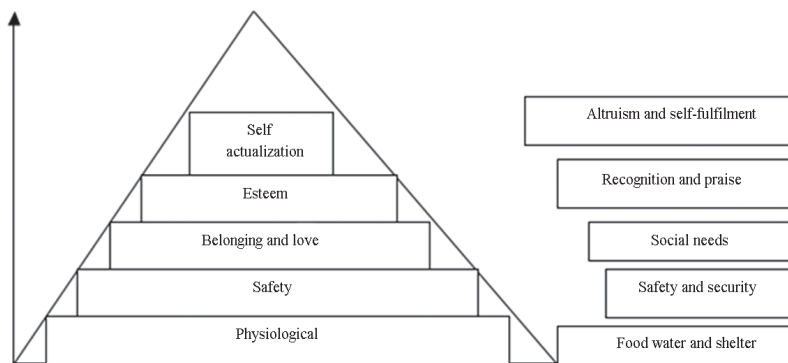
Social Development encompasses a commitment to individual and societal well-being, and the opportunity for citizens to determine their own and their society's needs and to influence decisions that affect these. Social change incorporates public concerns in developing social policy and economic initiatives. Until relatively recently, social development was conceived in terms of a set of desirable results - higher incomes, longer life expectancy, lower infant mortality, more and better education *etc.* Recently emphasis has shifted from the results to the enabling conditions, strategies and public policies for achieving those results. But still little attention has been placed on the underlying social process of development that determines how society formulates, adopts, initiates, and organizes; and few attempts have been made to formulate such a framework. However, there are some recognized theories and principles, which will be examined briefly.

KEY SOCIAL DEVELOPMENT PRINCIPLES

Social development is defined in the broadest social terms as an upward directional movement of society from lesser to greater levels of energy, efficiency, quality, productivity, complexity, comprehension, creativity, choice, mastery, enjoyment and accomplishment. Growth and development usually go together, but they are different phenomena subject to different laws. Growth involves an expansion of existing types and forms of activities. Development involves a qualitative enhancement. Social

development is driven by the subconscious aspirations of society for advancement or progress. Society (and individuals) will seek the progressive fulfillment of a prioritized hierarchy of needs.

Motivation is complex and highly individual. The motivation to work can be physical (earning money for food or shelter), psychological (seeking social satisfaction or security) or more unconscious and instinctive – which applies particularly to altruistic and self-fulfillment reasons. One of the most popular theories explaining motivation is Maslow's 'hierarchy of needs', which categorizes human motivations as follows:



The theory works on the basis that needs are only motivators when they are unsatisfied. The lower order needs (physiology and safety) are dominant until satisfied, when the higher needs come into being. This theory is important in terms of social development theory, as it helps to explain why more altruistic concerns (such as animal welfare activity) are often not burning social issues until society has developed to a level that meets individuals' lower order needs (personal, shelter, security *etc.*).

As can be seen, in the course of social development, society is moved by a range of different psychological motives. Self actualizing motives (wanting self-development and social progress for altruistic and ethical reasons) are normally the last to be fulfilled. However, the globalization of culture and information is bringing such concerns more rapidly into public consciousness. The revolution of rising expectations represents a new and more powerful motivating force for development; which by its nature is not limited, as all the others have been, to a specific class or section of society.

Development of society occurs only in fields where the collective will is sufficiently strong and seeking expression. Development strategies will be most effective when they focus on identifying areas where the social will is mature and can provide better means for the awakened social energy to express itself. Only those initiatives that are in concordance with this subconscious urge will be likely to succeed and gain 'critical-mass'. Every society possesses a huge reservoir of potential human energy that is absorbed and held static in its organized foundations - its cultural values, physical security, social beliefs, and political structures. At times of transition, crisis, and opportunities, those energies are released and expressed in action. Policies, strategies, and programmes that tap this latent energy and channel it into constructive activities

can stir an entire nation to action and rapid advancement. The theory of critical mass was expounded by Nobel Prize winner Thomas Schelling (back in 1969-1971), and is explained in further detail in a new book of his 'Game Theory' by Dodge (2012). It is also known as the 'bandwagon effect'. The success or failure of many things is determined by whether 'critical mass' is achieved; so increasing numbers are encouraged to join movements as they gain attention and impact. Schelling also introduced the concept of the 'tipping point', which is the exact point at which a movement becomes self-sustaining. However, social change requires an enormous investment of energy to break existing patterns of social behaviour and form new ones. Change is triggered when societal energies accumulate beyond the level required for functioning at the present level. The social energy may be released in response to the opening up of a new opportunity or confrontation by a severe challenge. Where different cultures meet and blend, explosive energies for social evolution are released.

The rate and extent of development is determined by prevalent social attitudes, which control the flow of social energies. Where attitudes are not conducive, development strategies will not yield results. In this case the emphasis should be placed on strategies to bring about a change in societal attitudes - such as public education and awareness, demonstration, and encouragement of successful pioneers. The implications of this for the animal welfare movement can be that if the 'time is not ripe' for the animal movement to 'take off' as a social movement in its own right in certain countries, then tapping into other burgeoning social change concerns may be necessary (whilst always building the capacity, models, awareness and moral force needed to grow the animal welfare movement).

Bringing About Social Change

There are various social change models. These all require public awareness and learning, and consensus building (building critical mass), before policy change can effectively be implemented and enforced.

We found the following social change model applicable and relevant to animal welfare. It is a model of the stages needed to institutionalize social change:

- Official structures – the development of departments or individuals dealing with the issue (when there is a need to put the issue on the agenda)
- Legislation – when the need for official policy change is accepted
- Enforcement – when the authorities accept the need to enforce change
- Transmission by education – when it is accepted that this needs to be an issue for society
- Cultural transmission by family – when the issue is generally accepted, and grassroots education takes place within families

The last two stages can be grouped together as 'public awareness and support', which is a long-term process beginning with education and ending with final acceptance in the family unit.

It is important to remember that social change does not occur until all stages of the social change model have been accomplished. It is a common mistake to view policy change in terms of obtaining policies/laws alone, without considering the structures and enforcement needed to implement these. A policy or law is not worth the paper it is written on if it is not put into practical effect. In fact, policy change objectives may be seen as an intermediate aim, with change in practice being the end point that brings real improvements in the situation of the animals. The problem for various social change movements is how to press their movement's concerns up the ladder of people's own 'hierarchy of needs'. That is: 'how to make the issue become a priority to people?' This is a real problem for animal welfare, as the suffering does not impact directly upon individual humans (*i.e.*, it is not a personal problem, and requires altruism and empathy to be considered important). Increased awareness can accelerate this process. But to become a burning issue, there needs to be emotional engagement, intellectual challenge, and/or a real sense of justice (a strong ethical perspective).

The process of discovery expands human consciousness. The process of application enhances social organization. This explains why 'best practice pilot projects' are important – as they are a practical demonstration of how positive change can take place (and both national and international 'best practice' case studies can be used). It also explains why capacity building is necessary – providing not only necessary skills, but also the mindset that positive change is achievable.

Development occurs when pioneering individual initiatives are imitated by others, multiplied and actively supported by the society. Society then actively organizes the new activity by establishing supportive laws, systems and institutions. At the next stage, it integrates the new activity with other fields of activity and assimilates it into its educational system. The activity has become fully assimilated as part of the culture when it is passed on to the next generation as values through the family.

There are various causes of social change. Culture can be a cause, and there are said to be three main sources of cultural change: The first source is invention. Inventions produce new products, ideas, and social patterns. The invention of rocket propulsion led to space travel, which in the future may lead to inhabitation of other planets. The second source is discovery. Discovery is finding something that has never been found before, or finding something new in something that already exists. The third source is diffusion. Diffusion is the spreading of ideas and objects to other societies. This would involve trading, migration, and mass communication. The 'mass media' is a vital factor in the speed of social change. It permits rapid diffusion of ideas, by delivering these in the private and relaxing environs of the home, where audiences are at their most susceptible.

The Role of Civil Society

Social development takes place within a larger evolutionary context. Social learning is a subconscious seeking (for ethics and values, or even reason and purpose) that

ultimately leads to conscious knowledge. We experience first and understand later. Our mental comprehension perpetually lags behind physical experience and struggles to catch up with it. But as society advances, development becomes more conscious and more rapid.

Social movements can also play a pivotal role in promoting change. Civil society or civil institutions refer to the totality of voluntary civic and social organizations or institutions that form the basis of a functioning society as opposed to the force backed structures of a state (regardless of that state's political system).

While there are myriad definitions of civil society, the London School of Economics Centre for Civil Society working definition is illustrative:

“Civil society refers to the arena of un-coerced collective action around shared interests, purposes, and values. In theory, its institutional forms are distinct from those of the state, family, and market, though in practice, the boundaries between state, civil society, family, and market are often complex, blurred and negotiated. Civil society commonly embraces a diversity of spaces, actors, and institutional forms, varying in their degree of formality, autonomy, and power. Civil societies are often populated by organizations such as registered charities, development non-governmental organizations, community groups, women's organizations, faith-based organizations, professional associations, trades unions, self-help groups, social movements, business associations, coalitions and advocacy group.”

Civil society organizations, particularly those in the social change sector, are strong proponents of the public sphere, and frequently make public policy discussion and public education major parts of their missions. They seek to effect change through dialogue with others sharing an interest in a social concern. In recent years, the rise of the new communications technologies and the Internet has had a significant effect on public sphere communications. The rapid evolution of the Internet has led many civil society organizations to adopt different software tools and information dissemination techniques to enhance their strategic effectiveness for social change.

Role of the Individual in Social Change

Society has no direct means to give conscious expression to its subconscious collective aspirations and urges. That essential role is played by pioneering conscious individuals - visionary intellectuals, political leaders, entrepreneurs, artists and spiritual seekers who are inspired to express and achieve what the collective subconsciously aspires to and is prepared for. Where the aspiration and action of the leader do not reflect the will of the collective, it is ignored or rejected. Where it gives expression to a deeply felt collective urge, it is endorsed, imitated, supported, and systematically propagated. This is most evident at times of war, social revolution, or communal conflict.

For example, India's early freedom fighters consciously advocated the goal of freedom from British rule long before that goal had become a felt aspiration of the masses. The leaders spent decades urging a reluctant population to conceive of itself as a free nation

and to aspire to achieve that dream. When finally the collective endorsed this conception, no foreign nation had the power to impose its will on the Indian people.

All human creative processes release and harness human energy and convert it into results. The process of skill formation involves acquiring mastery over our physical-nervous energies so that we can direct our physical movements in a precisely controlled manner.

In the absence of skill, physical movements are clumsy, inefficient, and unproductive, like the stumbling efforts of a child learning to walk. Whilst the energies are the motivating force, it is strategic ability and professional skills that turn energy into effective action. Often the high emotions of social change movements are a facet of this undirected energy. Strategic advocacy helps to direct this energy, focusing it in directions where change can most effectively be triggered.

Development occurs when the subconscious preparedness of society leads to the generation of new ideas and conscious initiatives by individuals. The accumulated surplus energy of society releases the initiative of pioneers who apply new ideas, acquire new skills and introduce new types of activities. Imitation of successful pioneers eventually attracts the attention and overcomes the resistance of conservative forces in society, leading the society to accept and embrace the new activity. The potentials for development always far exceed the initiative of society to exploit them. The actual achievements of society depend on the measure that it is ready to actively respond to new opportunities and challenges. That response is the real determinant of development. Three fundamental conditions determine a society's level of preparedness: energy, awareness and aspiration.

Role of the Pioneer in Social Change

Social progress is stimulated by pioneering individuals who become conscious of new opportunities and initiate new behaviours and activities to take advantage of them. Pioneers are the lever or spearhead for collective advancement. Pioneers give conscious expression to the subconscious urges and readiness of society.

When society is subconsciously prepared for change, it still needs an agent through which to express this preparedness in action. In natural development, that is the role of pioneering individuals. Once society is prepared, sooner or later it gives rise to the initiative of one or more pioneering individuals who break out from the existing mold and attempt something new. Although exceptional and eccentric individuals may initiate new activities in any society, these activities usually disappear with the passing of their founder or give rise to isolated imitation that never acquires significant momentum.

The social change pioneer is a conscious product of the society whose aspiration and initiative give expression to the subconscious aspiration of the society in which he lives. Every new developmental activity is initially conceived and introduced by one or a few pioneers. The pioneer is one who sees, believes in and acts upon an opportunity that others fail to see or believe in; or lack the energy or courage to pursue. The pioneer

exhibits a new understanding, new attitudes, new skills and behaviours different from those prevalent in the community at the time. If the pioneer's initiative is in tune with the social aspiration and social preparedness, it inspires and encourages other dynamic individuals to imitate or improve upon the new initiative.

The role of the pioneer is vital to social change, because the next stage of social progress almost always remains unseen by the collective. It is the free thinking, far seeking individual who dares to imagine or conceive what the popular mind is unaware of and then translates that vague possibility into a reality that all can see. By acquiring one new or different attribute or behaviour, he charts a new course and reveals a new possibility, all the time basing him or herself on the society's present accomplishments and in most cases moving in a direction that the society has already indicated. Max Weber thought that the expression of ideas by charismatic individuals could change the world. Here are some examples of influential people who caused changes in the world (good and bad): Martin Luther King, Jr.; Adolf Hitler; Mao Tseng Tug; Mohandas Gandhi and Nelson Mandela.

Multiplier Effect

It does not really matter whether pioneers come forward on their own internal prompting or in response to an opportunity or demonstration created by government. In either case, the individual embodies and represents the social initiative. What does matter is the response of the society to the pioneer. Often the early pioneer meets with a response of indifference, resistance, contempt or hostility from the community around him, especially when his actions represent a radical departure from the status quo. This usually occurs when the pioneer comes too much before his time, before society is fully ready to act on its urge for something new.

At other times the successful pioneer is actively admired and respected, yet no one else comes forward to imitate his success. In either case, the pioneer's initiative fails to catch on. If the pioneer pushes through change before the society is fully prepared, the change comes abruptly in the form of a revolution. If society is fully prepared to accept and follow the pioneer, then the change occurs by a smooth evolution. Revolution is premature evolution.

CULTURAL FACTORS OF SOCIAL CHANGE

In sociology the word 'Culture' denotes acquired behaviour which are shared by and transmitted among the members of the society. Man learns his behaviour and behaviour which is learnt is called culture. Singing, dancing, eating, playing belong to the category of culture. It includes all that man has acquired in the mental and intellectual sphere of his individual and social life. It is the expression of our nature, in our modes of living and thinking, in our everyday intercourse, in art, in literature, in recreation and enjoyment. For the clear understanding of the term 'Culture' here, it is necessary to discuss some of the important definitions of culture.

DEFINITIONS

According to White, *“Culture is a symbolic, continuous, cumulative and progressive process.”*

Malinowski defines culture, *“as the handiwork of man and as the medium through which he achieves his end.”*

MacIver and Page defines, *“Culture is the realm of styles, of values, of emotional attachments, of intellectual adventures.”*

In the words of Green, *“Culture is the socially transmitted system of idealized ways in knowledge, practices and beliefs, along with the artifacts that knowledge and practice produce and maintain as they change in time.”*

Culture, as is defined by Tylor, refers to, *“that complex whole which includes knowledge, belief, art, morals, law, customs and any other capabilities and habits acquired by man as a member of society.”*

From the above definitions of culture, the features of culture may be summarized below. They are:

1. Culture is social, but not individual.
2. Culture is idealistic.
3. Culture is communicative.
4. Culture is an acquired quality.
5. Culture meets the recurring demand of mankind.
6. Culture has the characteristics of adaptation.
7. Culture has the quality of becoming integrated.

Culture plays a very significant role in social change. No culture even remains constant. It always keeps on changing. So it is treated as an important source or instrument of social change. The very nature of culture makes it dynamic. The terms like continuous, cumulative adoptive, transmissive *etc.* signify the elements of dynamism within the culture. Besides, there are host of other internal and external variables that interact upon the culture and cause changes within it.

Change in culture takes place by three important factors. They are discovery, invention and diffusion. A discovery may be a material item or a non material item. But any discovery brings about social change only when it is widely used. An invention generates pressure for change in different parts of social life. Diffusion is the process by which cultural traits spread from one culture to another or from one part of culture to another. It is a two-way process. Because in this process one culture gives something to the other and at the same time takes something from it.

Several sociologists have highlighted the determining role of culture effecting social change. Some of the important effects are given below. They are:

- Culture gives speed and direction to social change: If the culture is too much conservative, then its rate of change becomes too low and vice versa. People whether accept change or not depends upon their attitudes and values which are the products of the culture.

- Culture influences the direction and character of technological change: Our belief and social institutions not only correspond to the changes in technology but these determine the use to which the technological inventions will be put. It is the culture that decides the purpose to which a technical invention must be put.
- Culture shapes economy and is effective towards economic growth: Culture not only gives direction to technology but it shapes the economy which is too much effective towards economic growth.
- It keeps the social relationship intact: It makes people think not of their own but also of the others. By regulating the behaviour of the people and satisfying their primary drives pertaining to hunger, shelter and sex, it has been able to maintain group life.

THE CULTURAL FACTORS THAT INFLUENCE SOCIAL CHANGE

Cultural factors constitute yet another source of social change. Cultural factors consist of our values, and beliefs, ideas and ideologies, morals and manners, customs and traditions and various institutions. Not only social values direct social changes but they themselves are subject to change. Ideas and ideals, ideologies and philosophies are inherently changeful. They change with time, and in turn, initiate change in the social order. In some periods ideas of liberty, equality and democracy may be found, in some others, ideas of strict discipline and centralised order may be observed, and still in some other period's religious orthodoxy or religious non-conformity may prevail.

There is an intimate connection between our beliefs and institutions, our valuations and social relationships. Certainly cultural change involves social change (in so far as it affects human relations) as the social and cultural changes are closely related. "What people think, in short, determines in every measure... what they do and what they want"—Robert Bierstedt. Cultural factors influence social changes. There is an intimate relationship between society and culture. Hence cultural changes involve social change. Cultural factors such as values, attitudes, ideologies, ideas of greatmen *etc.* play a vital role in influencing social change. W. F. Ogburn's concept of cultural lag explains the role of culture in social change.

Role of Values

Values play a major role in acceptance or rejection of any social change. In our country some communities do not approve of the use of contraceptive as they believe that it goes against their religious values as well as the economic values of having large family.

Role of Attitude

The attitude of a society towards scientific enquiry may encourage or discourage change. In the medieval Europe, Galileo was tortured because of the speculative attitude of the church leaders. In contrast modern countries at present encourage science. The

successful scientists in our country are rewarded with honour, prestige, and sometimes with money. High value placed on science has encouraged many youngmen to take up science as a career.

Role of Greatmen

Greatmen like the Buddha, Mahavira, Sankaracharya, Jagannatha Das, Aurovinda, Vivekananda, Mahammad, Jesus Christ, Mahatma Gandhi and others have introduced revolutionary changes in the spheres of religion. Napoleon, Hitler, Karl Marx, Gandhi and others have caused radical changes in the sphere of politics. Similarly writers, scientists, philosophers have also contributed to social changes.

Role of Ideology

Ideology means a set of beliefs justifying the interest of those who held it. New ideas and ideologies also bring about significant changes in the world's social and economic order. Nobody can deny the influences of the ideas like liberty, equality and fraternity in giving rise to the French revolution. Similarly new ideologies such as socialism, Gandhisim and democracy have exerted profound influence on social organisation. Recent change in ideologies has caused disintegration of the U.S.S.R.

Concept of Cultural Lag

The term cultural lag was first used by W. F. Ogburn in his book 'Social Change'. The concept of cultural lag is a necessary concept in the explanation of social change.

Ogburn classified culture into two categories:

1. Material culture an
2. Non-material culture.

By material culture, he means things like books, utensils, machines, houses and such other forms of technology. Non-material aspect of culture include beliefs, attitudes, values, norms, religion, education *etc.* According to Ogburn changes come first into the material aspects of culture. But the non-material aspects of culture take time to adjust to the changes taking place in material aspects. As a result, non-material culture falls behind material culture, which is known as 'cultural lag'.

Ogburn has cited number of examples of cultural lag. According to him the number of policemen in towns remains constant whereas the population of town is fast increasing. Secondly family planning technologies have advanced but people take time to accept them. Even some communities do not accept family planning measures due to traditional religious values and social pressure. As a result cultural lag occurs.

From the above discussion the cause of cultural lag is that the various parts of culture change at different rates. Material culture changes more rapidly than non-material culture. But cultural lag is due to man's ideologies, dogmatism and conservatism. Because of his inclination to tradition, man cannot change social institutions. The failure of social institutions to adapt to the changes in technology leads to cultural lag.

CONCEPT OF CULTURAL LAG

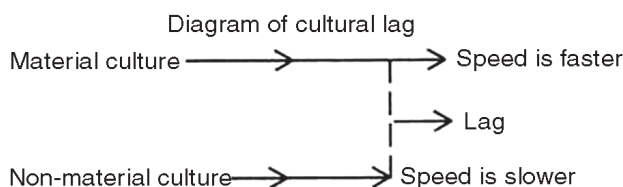
To examine the role of cultural factors in social change, it is necessary to discuss the concept of “Cultural Lag”. W.F. Ogburn, an American sociologist introduced the concept of ‘Cultural Lag’ in his book “social change: which was published in the year 1920. The word ‘lag’ connotes crippled movement. Hence cultural lag means the flatering of one aspect of culture behind another. According to Ogburn there are two aspects of culture. One is the material and another one is the non-material.

Material Culture

It includes those elements or things which are tangible, visible and touchable like goods, tools, machines and furnitures *etc.*

Non-material Culture

It includes those elements which are neither tangible nor touchable rather these are experienced by men like customs, values, ideologies, religion and behaviour pattern *etc.* It is found that technological innovations and discoveries stimulate the material culture to a great extent. As a result, the changes are quick 4n material culture. But non-material culture responds very slowly to such changes in material culture. So the material culture goes ahead leaving behind the non-material culture. This is called the “cultural lag”. For example-The development in the field of industry requires a corresponding change in the system of education. The failure of education to meet the needs of modern industrial development leads to cultural lag.



CAUSES OF CULTURAL LAG

What is the Cause of this Cultural Lag

Regarding the answer we find many factors causing cultural lag.

They are:

1. Difference in the degrees of changeability of various elements of culture.
2. Changes in law.
3. Man's psychological dogmatism.

Criticisms

The theory of ‘cultural lag’ of Ogburn is not free from the criticisms.

1. There is no clear-cut distinction between material and non- material culture.

Again, it is not necessary that non-material culture should invariably lag behind material culture.

2. A major defect in Ogburn's theory is that he uses the "cultural lag" for expanding all gaps in the process of social change. Maclver has suggested the use of different terms for the various types of disequilibrium and conflicts, such as technological lag, technological restraint and cultural clash *etc.*
3. According to the theory of cultural lag, while one thing progresses forward, another lags. Hence this word should not be used in the context of those objects which are the encouraging as well as the restraining and are similar and possess a common standard of evaluation.
4. Finally Ogburn has not provided any standard or scale to measure the units of material and non-material culture. So it is difficult to find out whether one aspect of culture changes faster than the other or not.

In spite of the above shortcomings, the theory of "cultural lag" of Ogburn has been proved to be beneficial for understanding the cultural factors in bringing about social change.

CULTURE AND THE SOCIOLOGICAL PERSPECTIVE

As this evidence on kissing suggests, what seems to us a very natural, even instinctual act turns out not to be so natural and biological after all. Instead, kissing seems best understood as something we learn to enjoy from our culture. The artifacts and ways of thinking, feeling, and acting that are part of any society, or the symbols, language, beliefs, values, and artifacts that are part of a society. Because Society, Sociology and the Sociological Perspective, refers to a group of people who live in a defined territory and who share a culture, it is obvious that culture is a critical component of any society.

If the culture we learn influences our beliefs and behaviours, then culture is a key concept to the sociological perspective. Someone who grows up in the United States differs in many ways, some of them obvious and some of them not so obvious, from someone growing up in China, Sweden, South Korea, Peru, or Nigeria. Culture influences not only language but the gestures we use when we interact, how far apart we stand from each other when we talk, and the values we consider most important for our children to learn, to name just a few. Without culture, we could not have a society.

The profound impact of culture becomes most evident when we examine behaviours or conditions that, like kissing, are normally considered biological in nature. Consider morning sickness and labour pains, both very familiar to pregnant women before and during childbirth, respectively. These two types of discomfort have known biological causes, and we are not surprised that so many pregnant women experience them. But we *would* be surprised if the husbands of pregnant women woke up sick in the morning during their wives' pregnancies or experienced severe abdominal pains while their wives gave birth.

These men are neither carrying nor delivering a baby, and there is no logical—that is, biological—reason for them to suffer either type of discomfort. And yet anthropologists have discovered several societies in which men about to become fathers experience precisely these symptoms. They are nauseous during their wives' pregnancies, and they experience labour pains while their wives give birth. The term *couvade* refers to these symptoms, which do not have any known biological origin. Yet the men feel them nonetheless, because they have learned from their culture that they *should* feel these types of discomfort. And because they should feel these symptoms, they actually do so.

Perhaps their minds are playing tricks on them, but that is often the point of culture. As sociologists William I. and Dorothy Swaine Thomas once pointed out, if things are perceived as real, then they are real in their consequences. These men learn how they should feel as budding fathers, and thus they feel this way. Unfortunately for them, the perceptions they learn from their culture are real in their consequences. The example of drunkenness further shows how cultural expectations influence a behaviour that is commonly thought to have biological causes. In the United States, when people drink too much alcohol, they become intoxicated and their behaviour changes.

Most typically, their inhibitions lower and they become loud, boisterous, and even rowdy. We attribute these changes to alcohol's biological effect as a drug on our central nervous system, and scientists have documented how alcohol breaks down in our body to achieve this effect. This explanation of alcohol's effect is OK as far as it goes, but it turns out that *how* alcohol affects our behaviour depends on our culture. In some societies anthropologists have studied, people drink alcohol until they pass out, but they never get loud or boisterous; they might not even appear to be enjoying themselves. In other societies, they drink lots of alcohol and get loud but not rowdy. In some societies, including our own, people lose sexual inhibitions as they drink, but in other societies they do not become more aroused. The anthropological evidence is very clear: alcohol as a drug does affect human behaviour, but culture influences the types of effects that occur. We learn from our culture how to behave when drunk just as we learn how to behave when sober.

CULTURE VERSUS BIOLOGY

These examples suggest that human behaviour is more the result of culture than it is of biology. This is not to say that biology is entirely unimportant. As just one example, humans have a biological need to eat, and so they do. But humans are much less under the control of biology than any other animal species, including other primates such as monkeys and chimpanzees. These and other animals are governed largely by biological instincts that control them totally. A dog chases any squirrel it sees because of instinct, and a cat chases a mouse for the same reason. Different breeds of dogs do have different personalities, but even these stem from the biological differences among breeds passed down from one generation to another. Instinct prompts many dogs to turn around before

they lie down, and it prompts most dogs to defend their territory. When the doorbell rings and a dog begins barking, it is responding to ancient biological instinct. Because humans have such a large, complex central nervous system, we are less controlled by biology.

The critical question then becomes, how much does biology influence our behaviour? Predictably, scholars in different disciplines answer this question in different ways. Most sociologists and anthropologists would probably say that culture affects behaviour much more than biology does. In contrast, many biologists and psychologists would give much more weight to biology. Advocating a view called socio-biology. The view that genes and other aspects of human biology influence human behaviour and values., some scholars say that several important human behaviours and emotions, such as competition, aggression, and altruism, stem from our biological makeup. Sociobiology has been roundly criticized and just as staunchly defended, and respected scholars continue to debate its premises.

Why do sociologists generally favour culture over biology? Two reasons stand out. First, and as kissing and the other examples show, many behaviours differ dramatically among societies in ways that show the strong impact of culture. Second, biology cannot easily account for why groups and locations differ in their rates of committing certain behaviours. For example, what biological reason could explain why suicide rates west of the Mississippi River are higher than those east of it or why the U. S. homicide rate is so much higher than Canada's? Various aspects of culture and social structure seem much better able than biology to explain these differences. Many sociologists also warn of certain implications of biological explanations.

First, they say, these explanations implicitly support the status quo. Because it is difficult to change biology, any problem with biological causes cannot be easily fixed. Consider evidence that women do worse than men on the math SAT exam and are less likely to be mathematically gifted. Some researchers attribute this difference to women's lower testosterone levels or to their brain structures. Suppose either explanation is true. What, then, can we do to improve women's math SAT scores? Operate on their brains? Give them more testosterone?

Obviously either option is morally unethical and practically impossible. If these are the only options, then there is little hope for improving women's math ability, and gender inequality in math will continue. Suppose instead, as many educators think, that the gender math difference stems from social and cultural factors, including the way girls and boys are brought up, the amount of attention teachers pay to them, and gender stereotyping in children's books. None of these factors will be easy to change, but at least it is more possible to change them than to change biological conditions. Sociology's perspective on gender and math performance thus promises at least some hope in reducing gender inequality in math performance.

A second possible implication of biological explanations that concerns some sociologists harkens back to an earlier time. This was a time when perceived biological

differences among races and religions were used to justify forced sterilization and mass violence, including genocide, against certain groups. As just one example, in the early 1900s, some 70,000 people, most of them poor and many of them immigrants or African Americans, were involuntarily sterilized in the United States as part of the eugenics.

The view, popular in the early 20th century, that certain categories of people were biologically inferior and hence should be sterilized movement, which said that certain kinds of people were biologically inferior and must not be allowed to reproduce. The Nazi Holocaust a few decades later used a similar eugenics argument to justify its genocide against Jews, Catholics, gypsies, and gays. With this history in mind, some scholars fear that biological explanations of human behaviour might still be used to support views of biological inferiority.

THE ELEMENTS OF CULTURE

Culture was defined earlier as the symbols, language, beliefs, values, and artifacts that are part of any society. As this definition suggests, there are two basic components of culture: ideas and symbols on the one hand and artifacts on the other. The first type, called non-material culture. The symbols, language, norms, and values that constitute a major part of a society's culture., includes the values, beliefs, symbols, and language that define a society. The second type, called material. An element of culture consisting of society's material objects, or artifacts., includes all the society's physical objects, such as its tools and technology, clothing, eating utensils, and means of transportation. These elements of culture are discussed next.

SYMBOLS

Every culture is filled with symbols. Things that stand for something else and that often evoke various reactions and emotions., or things that stand for something else and that often evoke various reactions and emotions. Some symbols are actually types of nonverbal communication, while other symbols are in fact material objects. Sociology and the Sociological Perspective emphasizes, shared symbols make social interaction possible. Let's look at nonverbal symbols first. A common one is shaking hands, which is done in some societies but not in others.

It commonly conveys friendship and is used as a sign of both greeting and departure. Probably all societies have nonverbal symbols we call gestures. Movements of the hand, arm, head, and other parts of the body that are meant to convey ideas or emotions non-verbally., movements of the hand, arm, or other parts of the body that are meant to convey certain ideas or emotions. However, the same gesture can mean one thing in one society and something quite different in another society. In the United States, for example, if we nod our head up and down, we mean yes, and if we shake it back and forth, we mean no. In Bulgaria, however, nodding means no, while shaking our head back and forth means yes! In the United States, if we make an "O" by putting our thumb

and forefinger together, we mean “OK,” but the same gesture in certain parts of Europe signifies an obscenity.

“Thumbs up” in the United States means “great” or “wonderful,” but in Australia it means the same thing as extending the middle finger in the United States. Certain parts of the Middle East and Asia would be offended if they saw you using your left hand to eat, because they use their left hand for bathroom hygiene. Some of our most important symbols are objects. Here the U. S. flag is a prime example. For most Americans, the flag is not just a piece of cloth with red and white stripes and white stars against a field of blue. Instead, it is a symbol of freedom, democracy, and other American values and inspires pride and patriotism. During the Vietnam War, however, the flag became to many Americans a symbol of war and imperialism. Some burned the flag in protest, prompting angry attacks by bystanders and negative coverage by the news media. Other objects have symbolic value for religious reasons.

Three of the most familiar religious symbols in many nations are the cross, the Star of David, and the crescent moon, which stand for Christianity, Judaism, and Islam, respectively. Whereas many cultures attach no religious significance to these shapes, for many people across the world they evoke very strong feelings of religious faith. Recognizing this, hate groups have often desecrated these symbols. As these examples indicate, shared symbols, both nonverbal communication and tangible objects, are an important part of any culture but also can lead to misunderstandings and even hostility. These problems underscore the significance of symbols for social interaction and meaning.

LANGUAGE

Perhaps our most important set of symbols is language. In English, the word *chair* means something we sit on. In French, the word *chaise* means the same thing. As long as we agree how to interpret these words, a shared language and thus society are possible. By the same token, differences in languages can make it quite difficult to communicate. For example, imagine you are in a foreign country where you do not know their language and they do not know yours. Worse yet, you forgot to bring your dictionary that translates their language into yours, and vice versa, and your iPhone battery has died. You become lost.

How will you get help? What will you do? Is there any way to communicate your plight? As this scenario suggests, language is crucial to communication and thus to any society’s culture. Children learn language from their culture just as they learn about shaking hands, about gestures, and about the significance of the flag and other symbols. Humans have a capacity for language that no other animal species possesses. Our capacity for language in turn helps make our complex culture possible. In the United States, some people consider a common language so important that they advocate making English the official language of certain cities or states or even the whole country and banning bilingual education in the public schools. Critics acknowledge the importance

of English but allege that this movement smacks of anti-immigrant prejudice and would help destroy ethnic subcultures.

In 2009, voters in Nashville, Tennessee, rejected a proposal that would have made English the city's official language and required all city workers to speak in English rather than their native language. Language, of course, can be spoken or written. One of the most important developments in the evolution of society was the creation of written language. Some of the preindustrial societies that anthropologists have studied have written language, while others do not, and in the remaining societies the "written" language consists mainly of pictures, not words.

This variation with data from 186 preindustrial societies called the Standard Cross-Cultural Sample, a famous data set compiled several decades ago by anthropologist George Murdock and colleagues from information that had been gathered on hundreds of preindustrial societies around the world. In Figure, "The Presence of Written Language", we see that only about one-fourth of the SCCS societies have a written language, while about equal proportions have no language at all or only pictures.

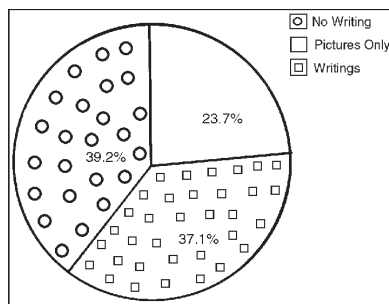


Fig. The Presence of Written Language (Percentage of Societies)

To what extent does language influence how we think and how we perceive the social and physical worlds? The famous but controversial Sapir-Whorf hypothesis Sapir-Whorf hypothesis. The view that language influences the thoughts and perceptions of people in a society., named after two linguistic anthropologists, Edward Sapir and Benjamin Lee Whorf, argues that people cannot easily understand concepts and objects unless their language contains words for these items. Language thus influences how we understand the world around us. For example, people in a country such as the United States that has many terms for different types of kisses are better able to appreciate these different types than people in a country such as Japan, which, as we saw earlier, only fairly recently developed the word *kissu* for kiss.

Another illustration of the Sapir-Whorf hypothesis is seen in sexist language, in which the use of male nouns and pronouns shapes how we think about the world. In older children's books, words like *fireman* and *mailman* are common, along with pictures of men in these jobs, and critics say they send a message to children that these are male jobs, not female jobs. If a teacher tells a second-grade class, "Every student should put his books under his desk," the teacher obviously means students of both sexes but may

be sending a subtle message that boys matter more than girls. For these reasons, several guidebooks promote the use of nonsexist language. Table, “Examples of Sexist Terms and Nonsexist Alternatives” provides examples of sexist language and nonsexist alternatives.

Table. Examples of Sexist Terms and Nonsexist Alternatives.

Term	Alternative
Businessman	Business person, executive
Fireman	Fire fighter
Chairman	Chair, chairperson
Policeman	Police officer
Mailman	Letter carrier, postal worker
Mankind	Humankind, people
Man-made	Artificial, synthetic
Waitress	Server
He (as generic pronoun)	He or she; he/she; s/he
“A professor should be devoted	“Professors should be devoted to his students” to their students”

The use of racist language also shows the Sapir-Whorf hypothesis. An old saying goes, “Sticks and stones may break my bones, but names will never hurt me.” That may be true in theory but not in reality. Names can hurt, especially names that are racial slurs, which African Americans growing up before the era of the civil rights movement routinely heard. The Sapir-Whorf hypothesis, the use of these words would have affected how whites perceived African Americans. More generally, the use of racist terms may reinforce racial prejudice and racial stereotypes.

NORMS

Cultures differ widely in their norms. Socially acceptable ways of behaving., or standards and expectations for behaving. We already saw that how people behave when drunk stems from society’s expectations of how they should behave when drunk. Norms of drunken behaviour influence how we behave when we drink too much. Norms are often divided into two types, formal norms. Norms that are very important and usually written down; also called laws and mores and informal. Relatively unimportant norms, often unwritten, that still affect people’s behaviour.. Formal norms, also called *mores* and *laws*, refer to the standards of behaviour considered the most important in any society.

Examples in the United States include traffic laws, criminal codes, and, in a college context, student behaviour codes addressing such things as cheating and hate speech. Informal norms, also called *folkways* and *customs*, refer to standards of behaviour that are considered less important but still influence how we behave. Table manners are a common example of informal norms, as are such everyday behaviours as how we interact with a cashier and how we ride in an elevator. Many norms differ dramatically from one culture to the next. Some of the best evidence for cultural variation in norms comes

from the study of sexual behaviour. Among the Pokot of East Africa, for example, women are expected to enjoy sex, while among the Gusii a few hundred miles away, women who enjoy sex are considered deviant. In Beag, a small island off the coast of Ireland, sex is considered embarrassing and even disgusting; men feel that intercourse drains their strength, while women consider it a burden. Even nudity is considered terrible, and people on Beag keep their clothes on while they bathe. The situation is quite different in Mangaia, a small island in the South Pacific. Here sex is considered very enjoyable, and it is the major subject of songs and stories. While many societies frown on homosexuality, others accept it.

Among the Azande of East Africa, for example, young warriors live with each other and are not allowed to marry. During this time, they often have sex with younger boys, and this homosexuality is approved by their culture. Among the Sambia of New Guinea, young males live separately from females and engage in homosexual behaviour for at least a decade. It is felt that the boys would be less masculine if they continued to live with their mothers and that the semen of older males helps young boys become strong and fierce. Other evidence for cultural variation in norms comes from the study of how men and women are expected to behave in various societies.

For example, many preindustrial societies are simple hunting and gathering societies. In most of these, men tend to hunt and women tend to gather. Many observers attribute this gender difference to at least two biological differences between the sexes. First, men tend to be bigger and stronger than women and are thus better suited for hunting. Second, women become pregnant and bear children and are less able to hunt. Yet a different pattern emerges in some hunting and gathering societies. Among a group of Australian aborigines called the Tiwi and a tribal society in the Philippines called the Agta, both sexes hunt.

After becoming pregnant, Agta women continue to hunt for most of their pregnancy and resume hunting after their child is born. Some of the most interesting norms that differ by culture govern how people stand apart when they talk with each other. In the United States, people who are not intimates usually stand about three to four feet apart when they talk. If someone stands more closely to us, especially if we are of northern European heritage, we feel uncomfortable. Yet people in other countries—especially Italy, France, Spain, and many of the nations of Latin America and the Middle East—would feel uncomfortable if they were standing three to four feet apart. To them, this distance is too great and indicates that the people talking dislike each other. If a U. S. native of British or Scandinavian heritage were talking with a member of one of these societies, they might well have trouble interacting, because at least one of them will be uncomfortable with the physical distance separating them.

Rituals

Different cultures also have different rituals. Established procedures and ceremonies that often mark transitions in the life course, or established procedures and

ceremonies that often mark transitions in the life course. As such, rituals both reflect and transmit a culture's norms and other elements from one generation to the next. Initiation and commencement ceremonies in colleges and universities are familiar examples of time-honoured rituals. In many societies, rituals help signify one's gender identity.

For example, girls around the world undergo various types of initiation ceremonies to mark their transition to adulthood. Among the Bemba of Zambia, girls undergo a month-long initiation ceremony called the *chisungu*, in which girls learn songs, dances, and secret terms that only women know. In some cultures, special ceremonies also mark a girl's first menstrual period. Such ceremonies are largely absent in the United States, where a girl's first period is a private matter. But in other cultures the first period is a cause for celebration involving gifts, music, and food.

Boys have their own initiation ceremonies, some of them involving circumcision. That said, the ways in which circumcisions are done and the ceremonies accompanying them differ widely. In the United States, boys who are circumcised usually undergo a quick procedure in the hospital. If their parents are observant Jews, circumcision will be part of a religious ceremony, and a religious figure called a *moyel* will perform the circumcision. In contrast, circumcision among the Maasai of East Africa is used as a test of manhood. If a boy being circumcised shows signs of fear, he might well be ridiculed. Are rituals more common in preindustrial societies than in modern ones such as the United States? Consider the Nacirema, studied by anthropologist Horace Miner more than 50 years ago.

In this society, many rituals have been developed to deal with the culture's fundamental belief that the human body is ugly and in danger of suffering many diseases. Reflecting this belief, every household has at least one shrine in which various rituals are performed to cleanse the body. Often these shrines contain magic potions acquired from medicine men. The Nacirema are especially concerned about diseases of the mouth. Miner writes, "Were it not for the rituals of the mouth, they believe that their teeth would fall out, their gums bleed, their jaws shrink, their friends desert them, and their lovers reject them".

Many Nacirema engage in "mouth-rites" and see a "holy-mouth-man" once or twice yearly. Spell Nacirema backward and you will see that Miner was describing American culture. As his satire suggests, rituals are not limited to preindustrial societies. Instead, they function in many kinds of societies to mark transitions in the life course and to transmit the norms of the culture from one generation to the next.

Changing Norms and Beliefs

Our examples show that different cultures have different norms, even if they share other types of practices and beliefs. It is also true that norms change over time within a given culture. Two obvious examples here are hairstyles and clothing styles. When the Beatles first became popular in the early 1960s, their hair barely covered their ears, but

parents of teenagers back then were aghast at how they looked. If anything, clothing styles change even more often than hairstyles.

Hemlines go up, hemlines go down. Lapels become wider, lapels become narrower. This colour is in, that colour is out. Hold on to your out-of-style clothes long enough, and eventually they may well end up back in style. A more important topic on which norms have changed is abortion and birth control. Despite the controversy surrounding abortion today, it was very common in the ancient world. Much later, medieval theologians generally felt that abortion was not murder if it occurred within the first several weeks after conception.

This distinction was eliminated in 1869, when Pope Pius IX declared abortion at any time to be murder. In the United States, abortion was not illegal until 1828, when New York state banned it to protect women from unskilled abortionists, and most other states followed suit by the end of the century. However, the sheer number of unsafe, illegal abortions over the next several decades helped fuel a demand for repeal of abortion laws that in turn helped lead to the *Roe v. Wade* Supreme Court decision in 1973 that generally legalized abortion during the first two trimesters.

Contraception was also practiced in ancient times, only to be opposed by early Christianity. Over the centuries, scientific discoveries of the nature of the reproductive process led to more effective means of contraception and to greater calls for its use, despite legal bans on the distribution of information about contraception. In the early 1900s, Margaret Sanger, an American nurse, spearheaded the growing birth-control movement and helped open a birth-control clinic in Brooklyn in 1916. She and two other women were arrested within 10 days, and Sanger and one other defendant were sentenced to 30 days in jail.

Efforts by Sanger and other activists helped to change views on contraception over time, and finally, in 1965, the U. S. Supreme Court ruled in *Griswold v. Connecticut* that contraception information could not be banned. As this brief summary shows, norms about contraception changed dramatically during the last century. Other types of cultural beliefs also change over time. Since the 1960s, the U. S. public has changed its views about some important racial and gender issues. The percentage of Americans who would vote for a qualified black person as president rose almost 20 points from the early 1970s to the middle of 1996, when the GSS stopped asking the question. If beliefs about voting for an African American had not changed, Barack Obama would almost certainly not have been elected in 2008. Several years of the GSS, shows that the percentage saying that women should take care of running their homes and leave running the country to men declined from almost 36% in the early 1970s to only about 15% in 1998, again, when the GSS stopped asking the question.

Values of what is desirable or undesirable and right or wrong are another important element of culture and involve judgments of what is good or bad and desirable or undesirable. A culture's values shape its norms. In Japan, for example, a central value is group harmony. The Japanese place great emphasis on harmonious social relationships

and dislike interpersonal conflict. Individuals are fairly unassertive by American standards, lest they be perceived as trying to force their will upon others. When interpersonal disputes do arise, Japanese do their best to minimize conflict by trying to resolve the disputes amicably. Lawsuits are thus uncommon; in one case involving disease and death from a mercury-polluted river, some Japanese who dared to sue the company responsible for the mercury poisoning were considered bad citizens.

Individualism in the United States

In the United States, of course, the situation is quite different. The American culture extols the rights of the individual and promotes competition in the business and sports worlds and in other areas of life. Lawsuits over the most frivolous of issues are quite common and even expected. Phrases like “Look out for number one!” abound. If the Japanese value harmony and group feeling, Americans value competition and individualism. Because the Japanese value harmony, their norms frown on self-assertion in interpersonal relationships and on lawsuits to correct perceived wrongs. Because Americans value and even thrive on competition, our norms promote assertion in relationships and certainly promote the use of the law to address all kinds of problems.

This difference between the two nations’ cultures with data from the 2002 World Values Survey which was administered to random samples of the adult populations of more than 80 nations around the world. One question asked in these nations was, “On a scale of one to ten, please indicate your views on competition.” The percentages of Americans and Japanese who responded with a “one” or “two” to this question, indicating they think competition is very beneficial. Americans are about three times as likely as Japanese to favour competition.

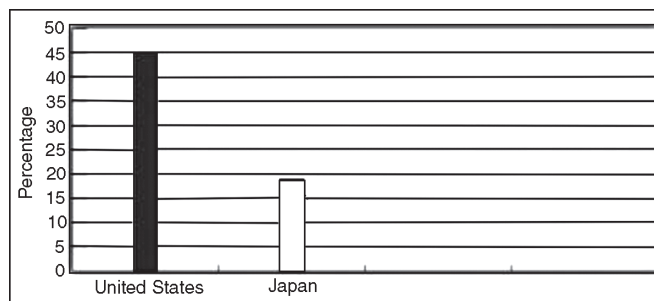


Fig. Percentage That Thinks Competition is Very Beneficial.

The Japanese value system is a bit of an anomaly, because Japan is a modern nation with very traditional influences. Its emphasis on group harmony and community is more usually thought of as a value found in preindustrial societies, while the U. S. emphasis on individuality is more usually thought of as a value found in modern cultures. Anthropologist David Maybury-Lewis describes this difference as follows: “The heart of the difference between the modern world and the traditional one is that in traditional societies people are a valuable resource and the interrelations between them are carefully

tended; in modern society things are the valuables and people are all too often treated as disposable.”

In modern societies, continues Maybury-Lewis, individualism and the rights of the individual are celebrated and any one person’s obligations to the larger community are weakened. Individual achievement becomes more important than values such as kindness, compassion, and generosity. Other scholars take a less bleak view of modern society, where they say the spirit of community still lives even as individualism is extolled. In American society, these two simultaneous values sometimes create tension. In Appalachia, for example, people view themselves as rugged individuals who want to control their own fate. At the same time, they have strong ties to families, relatives, and their neighbours. Thus their sense of independence conflicts with their need for dependence on others.

The Work Ethic

Another important value in the American culture is the work ethic. By the 19th century, Americans had come to view hard work not just as something that had to be done but as something that was morally good to do. The commitment to the work ethic remains strong today: in the 2008 General Social Survey, 72% of respondents said they would continue to work even if they got enough money to live as comfortably as they would like for the rest of their lives.

Cross-cultural evidence supports the importance of the work ethic in the United States. Using earlier World Values Survey data, The percentage of people in United States and three other nations from different parts of the world—Mexico, Poland, and Japan—who take “a great deal of pride” in their work. More than 85% of Americans feel this way, compared to much lower proportions of people in the other three nations.

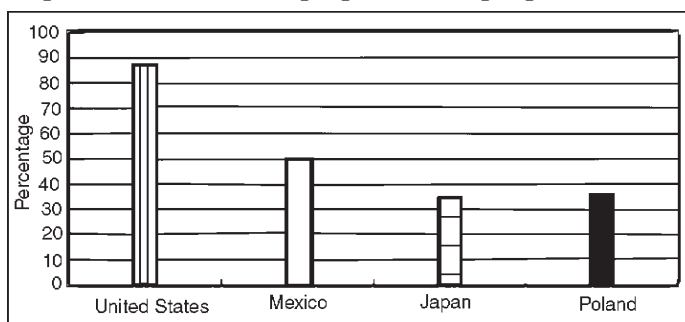


Fig. Percentage of People Who Take a Great Deal of Pride in their Work.

Closely related to the work ethic is the belief that if people work hard enough, they will be successful. Here again the American culture is especially thought to promote the idea that people can pull themselves up by their “bootstraps” if they work hard enough. The WVS asked whether success results from hard work or from luck and connections. The proportions of people in the four nations just examined who most strongly thought that hard work brings success. Once again we see evidence of an important aspect of the

American culture, as U. S. residents were especially likely to think that hard work brings success.

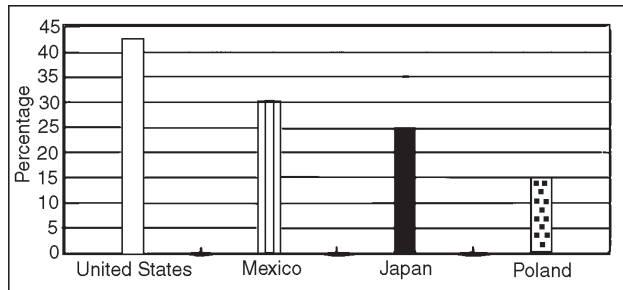


Fig. Percentage of People Who Think Hard Work Brings Success.

If Americans value the work ethic and also believe hard work brings success, then they should be more likely than people in most other nations to believe that poverty stems from not working hard enough. As expected, Americans are much more likely to attribute poverty to not working hard enough.

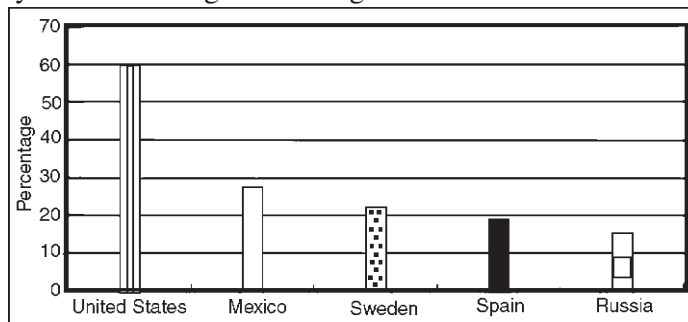


Fig. Percentage of People Who Attribute Poverty to Laziness and Lack of Willpower.

We could discuss many other values, but an important one concerns how much a society values employment of women outside the home. The WVS asked respondents whether they agree or disagree that “when jobs are scarce men should have more right to a job than women.” U. S. residents are more likely than those in nations with more traditional views of women to disagree with this statement.

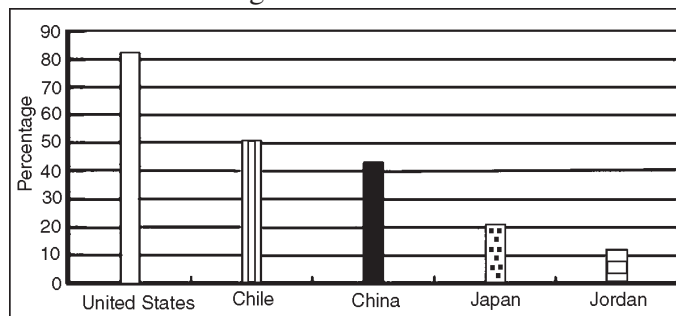


Fig. Percentage of People Who Disagree That Men Have More Right to a Job Than Women When Jobs are Scarce.

ARTIFACTS

The last element of culture is the artifacts. The material objects that constitute a society's material culture., or material objects, that constitute a society's material culture. In the most simple societies, artifacts are largely limited to a few tools, the huts people live in, and the clothing they wear. One of the most important inventions in the evolution of society was the wheel. The very few of the societies in the SCCS use wheels to move heavy loads over land, while the majority use human power and about one-third use pack animals.

Although the wheel was a great invention, artifacts are obviously much more numerous and complex in modern industrial societies. Because of technological advances during the past two decades, many such societies may be said to have a wireless culture, as smartphones, netbooks and laptops, and GPS devices now dominate so much of modern life. The artifacts associated with this culture were unknown a generation ago. Technological development created these artifacts and also new language to describe them and the functions they perform. Today's wireless artifacts in turn help reinforce our own commitment to wireless technology as a way of life, if only because children are now growing up with them, often even before they can read and write. Sometimes people in one society may find it difficult to understand the artifacts that are an important part of another society's culture.

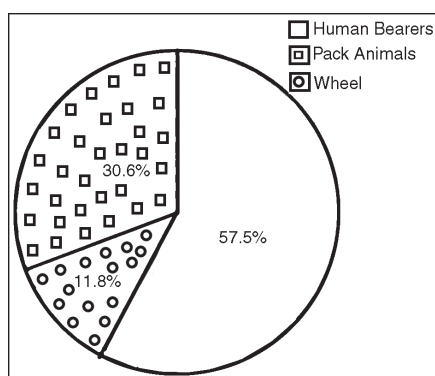


Fig. Primary Means of Moving Heavy Loads.

If a member of a tribal society who had never seen a cell phone, or who had never even used batteries or electricity, were somehow to visit the United States, she or he would obviously have no idea of what a cell phone was or of its importance in almost everything we do these days. Conversely, if we were to visit that person's society, we might not appreciate the importance of some of its artifacts. In this regard, consider once again India's cows. People from India consider cows holy, and they let cows roam the streets of many cities.

In a nation where hunger is so rampant, such cow worship is difficult to understand, at least to Americans, because a ready source of meat is being ignored. Anthropologist Marvin Harris advanced a practical explanation for India's cow worship. Millions of

Indians are peasants who rely on their farms for their food and thus their existence. Oxen and water buffalo, not tractors, are the way they plow their fields. If their ox falls sick or dies, farmers may lose their farms. Because, as Harris observes, oxen are made by cows, it thus becomes essential to preserve cows at all costs. In India, cows also act as an essential source of fertilizer, to the tune of 700 million tons of manure annually, about half of which is used for fertilizer and the other half of which is used as fuel for cooking. Cow manure is also mixed with water and used as flooring material over dirt floors in Indian households. For all of these reasons, cow worship is not so puzzling after all, because it helps preserve animals that are very important for India's economy and other aspects of its way of life.

If Indians exalt cows, many Jews and Muslims feel the opposite about pigs: they refuse to eat any product made from pigs and so obey an injunction from the Old Testament of the Bible and from the Koran. Harris thinks this injunction existed because pig farming in ancient times would have threatened the ecology of the Middle East. Sheep and cattle eat primarily grass while pigs eat foods that people eat, such as nuts, fruits, and especially grains. In another problem, pigs do not provide milk and are much more difficult to herd than sheep or cattle. Next, pigs do not thrive well in the hot, dry climate in which the people of the Old Testament and Koran lived.

Finally, sheep and cattle were a source of food back then because beyond their own meat they provided milk, cheese, and manure, and cattle were also used for plowing. In contrast, pigs would have provided only their own meat. Because sheep and cattle were more "versatile" in all of these ways, and because of the other problems pigs would have posed, it made sense for the eating of pork to be prohibited. In contrast to Jews and Muslims, at least one society, the Maring of the mountains of New Guinea, is characterized by "pig love." Here pigs are held in the highest regard.

INDICATORS OF SOCIAL DEVELOPMENT: SOCIAL, CULTURAL AND POLITICAL ASPECTS

Some of the major indicators of social development are as follows:

1. Social Aspects
2. Cultural Aspects
3. Political Aspects.

SOCIAL ASPECTS

1. The society is more modern and less traditional.
2. The society is more democratic and less authoritarian.
3. The social status is largely determined by achievements and not by birth as in the traditional caste-based society. Social discrimination, if at all exists, is determined by acquired attributes of individuals and not by where they are born.

4. The structure of family is no longer authoritarian and large in size as the traditional joint family used to be. It is small household, nuclear type and democratic in nature as the most urban families seem to be today.
Size of kinship structure in traditional societies used to be large. With social development, it is circumscribed. It is being experienced that the family of procreation and kin people from wife's family of orientation are occupying dominant place in family relationship.
5. There is no religion-based hierarchical division of society.
6. The society is urbanized and the general way of life of people is urban. Rural-to-urban and urban-to-urban migrations are high in a developing society.
7. Social and occupational mobility in society is unrestricted and fast. For example, the traditional Indian caste system had relatively restricted social and occupational mobility. The modern Indian society is now least caste-bound in its social and connubial interactions and occupational choice. However, the Indian society is still largely endogamous.
8. Various governmental and non-governmental agencies are developed to carry out the responsibilities traditionally being carried out by the family. For example, creche, old age homes, home delivery of commodities from the market and the like.
9. The rate of population growth is lower.
10. Rates of mortality, including maternal deaths and infant mortality are also lower.
11. Literacy rates – both male and female – are high.
12. Health facilities are expanded and made available to all – from top to the bottom in the class structure.

Cultural Aspects

1. Attitude of people in developed societies is essentially individualistic, materialistic and profit-oriented. Maximization of achievements is the goal of people.
2. Primeval sentiments are no longer dominant in the social behaviour of individuals. Human behaviour is largely governed by existing situations. Casteism, racism, familialism, fundamentalism, dogmatism and so on are faded and phased out in the wake of social development. People become secular and humanist in their value-orientation.
3. Nationalism and pluralism develop along with social development.
4. The institutions and agencies of human rights grow.
5. The value orientation of a modern person in a developed society is more individual and family-centred and not community-centred. What is to be done or not to be done is determined only by how much satisfaction the individual derives from that action.

6. Customs and traditions become weak. The rate of change in the context of social intercourse, foodstuff, clothes and housing pattern is accelerated. Food habits undergo change to become more metropolitan and continental.
7. Religions and the believers exist but religious practices and rites seem to be on the wane.
8. People become more rational and less superstitious and dogmatic.

Political Aspects

Democracy is the most acceptable form of political system in the developed and developing societies. Except a few countries like Myanmar, Pakistan, Nepal and Bhutan, which do not fall in the category of sound democracies, most of the countries of the world have democracy of one form or other. The kind of social process, which started to evolve with the Renaissance, was named as modernization and development and the first and most radical form of social change that took place in the process was the disjunction of Church from polity and the establishment, in its place, of democracy.

Some Characteristics of Political Development are:

1. Nation and nationhood develop.
2. Liberty to every citizen is ensured. People enjoy freedom of speech, choice of profession, practice of religion, etc.
3. The state is secular. It does not discriminate one citizen from the other on the basis of caste, creed, religion and region.
4. State seeks to guarantee equality among its citizens. Equality does not mean according equal status to everyone, because, an absolutely egalitarian society is perhaps only Utopian. True meaning of equality is to ensure equal opportunity to all. Indian Constitution, for example, guarantees equal opportunity to every citizen to participate in the political, social, economic and cultural activities in the country.
5. Consciousness about the human rights and civil society develops with social development and maturity of democracy. The state and the non-governmental organizations (NGOs) become concerned about these matters and the welfare of citizens gains paramount attention.

CULTURAL CHANGE IN SECULARIZATION

Secularization is the transformation of a society from close identification with religious values and institutions towards non-religious values and secular institutions. *Secularization thesis* refers to the belief that as societies “progress”, particularly through modernization and rationalization, religion loses its authority in all aspects of social life and governance. The term secularization is also used in the context of the lifting of the monastic restrictions from a member of the clergy.

Secularization has many levels of meaning, both as a theory and a historical process. Social theorists such as Karl Marx, Sigmund Freud, Max Weber, and Émile Durkheim,

postulated that the modernization of society would include a decline in levels of religiosity. Study of this process seeks to determine the manner in which, or extent to which religious creeds, practices and institutions are losing social significance. Some theorists argue that the secularization of modern civilization partly results from our inability to adapt broad ethical and spiritual needs of mankind to the increasingly fast advance of the physical sciences.

The term also has additional meanings, primarily historical and religious. Applied to church property, historically it refers to the seizure of monastic lands and buildings by anti-clerical European governments during the 18th and 19th centuries, which resulted in the expulsion and suppression of the religious communities which occupied them. Otherwise, secularization involves the abandonment of goods by the Church where it is sold to purchasers after the government seizes the property, which most commonly happens after reasonable negotiations and arrangements are made. In Catholic canon law, the term can also denote the permission or authorization given for a member of a religious Order to live outside his or her religious community or monastery, either for a fixed or permanent period.

BACKGROUND

Secularization is sometimes credited both to the cultural shifts in society following the emergence of rationality and the development of science as a substitute for superstition — Max Weber called this process, “the disenchantment of the world” — and to the changes made by religious institutions to compensate. At the most basic stages, this begins with a slow transition from oral traditions to a writing culture that diffuses knowledge. This first reduces the authority of clerics as the custodians of revealed knowledge.

As the responsibility for education has moved from the family and community to the state, two consequences have arisen:

1. The *collective conscience* as defined by Durkheim is diminished; and
2. Through the fragmentation of communal activities, religion becomes more a matter of individual choice rather than an observed social obligation.

A major issue in the study of secularization is the extent to which certain trends such as decreased attendance at places of worship indicate a decrease in religiosity or simply a privatization of religious belief, where religious beliefs no longer play a dominant role in public life or in other aspects of decision-making.

The issue of secularization is discussed in various religious traditions. The government of Turkey is an often cited example, following the abolition of the Ottoman Caliphate and foundation of the Turkish republic in 1923. This established popular sovereignty in a secular republican framework, in opposition to a system whose authority is based on religion. As one of many examples of state modernization, this shows secularization and democratization as mutually re-enforcing processes, relying on a separation of religion and state. In expressly secular states like India, it has been argued that the need

was to legislate for toleration and respect between quite different religions, and likewise, the secularization of the West was a response to drastically violent intra-Christian feuds between Catholicism and Protestantism. Some have therefore argued that Western and Indian secularization is radically different in that it deals with autonomy from religious regulation and control. Considerations of both tolerance and autonomy are relevant to any secular state.

DEFINITIONS

John Sommerville outlined six uses of the term secularization in the scientific literature.

The first five are more along the lines of ‘definitions’ while the sixth is more of a ‘clarification of use’:

1. When discussing macro social structures, secularization can refer to *differentiation*: a process in which the various aspects of society, economic, political, legal, and moral, become increasingly specialized and distinct from one another.
2. When discussing individual institutions, secularization can denote the transformation of a religious into a secular institution. Examples would be the evolution of institutions such as Harvard University from a predominantly religious institution into a secular institution.
3. When discussing activities, secularization refers to the transfer of activities from religious to secular institutions, such as a shift in provision of social services from churches to the government.
4. When discussing mentalities, secularization refers to the transition from *ultimate* concerns to *proximate* concerns. *E.g.*, individuals in the West are now more likely to moderate their behaviour in response to more immediately applicable consequences rather than out of concern for *post-mortem* consequences. This is a personal religious decline or movement towards a secular lifestyle.
5. When discussing populations, secularization refers to broad patterns of societal decline in levels of religiosity as opposed to the individual-level secularization of (4) above. This understanding of secularization is also distinct from (1) above in that it refers specifically to religious decline rather than societal differentiation.
6. When discussing religion, secularization can only be used unambiguously to refer to religion in a generic sense. For example, a reference to Christianity is not clear unless one specifies exactly which denominations of Christianity are being discussed.

OPPOSITION

Abdel Wahab Elmessiri outlined two meanings of the secularization term:

1. Partial Secularization: which is the common meaning of the word, and expresses “*The separation between religion and state*”.

2. Complete Secularization: this definition is not limited to the partial definition, but exceeds it to “The separation between all values, and but also to, so that the holiness is removed from the world, and this world is transformed into a usable matter that can be employed for the sake of the strong”.

CULTURAL CHANGE IN SANSKRITIZATION

Sanskritization or Sanskritisation is a particular form of social change found in India and Nepal. It denotes the process by which castes placed lower in the caste hierarchy seek upward mobility by emulating the rituals and practices of the upper or dominant castes. It is a process similar to passing in sociological terms. This term was made popular by Indian sociologist M. N. Srinivas in the 1950s, although earlier references to this process can be found in *Castes in India: Their Mechanism, Genesis and Development* by Dr. B. R. Ambedkar. The earliest references to this process of “descent of imitation from the top to the bottom of social ladder”, in another context, can be found in *The Laws of Imitation* by Gabriel Tarde.

DEFINITION

Srinivas defined sanskritization as a process by which “a *low* or *middle* Hindu caste, or tribal or other group, changes its customs, ritual ideology, and way of life in the direction of a high and frequently *twice-born* caste. Generally such changes are followed by a claim to a higher position in the caste hierarchy than that traditionally conceded to the claimant class by the local community”.

One clear example of sanskritization is the adoption, in emulation of the practice of twice-born castes, of vegetarianism by people belonging to the so-called “low castes” who are traditionally not averse to non-vegetarian food. Sanskritization is not just the adoption of new customs and habits, but also includes exposure to new ideas and values appearing in Sanskrit literature. He says the words *Karma*, *dharma*, *papa*, *maya*, *samsara* and *moksha* are the most common Sanskritic theological ideas which become common in the talk of people who are sanskritized. This phenomenon has also been observed in Nepal among Khas, Magar, Newar and Tharu people.

THEORY

Srinivas first propounded this theory in his D.Phil. thesis at Oxford University. The thesis was later brought out as a book titled *Religion and Society Among the Coorgs of South India*. Published in 1952, the book was an ethnographical study of the Kodava community of Karnataka.

Srinivas writes in the book:

- “The caste system is far from a rigid system in which the position of each component caste is fixed for all time. Movement has always been possible, and especially in the *middle regions of the hierarchy*. A caste was able, in a

generation or two, to rise to a higher position in the hierarchy by adopting vegetarianism and teetotalism, and by Sanskritizing its ritual and pantheon. In short, it took over, as far as possible, the customs, rites, and beliefs of the Brahmins, and adoption of the Brahminic way of life by a low caste seems to have been frequent, though theoretically forbidden. This process has been called 'Sanskritization' in this book, in preference to 'Brahminization', as certain Vedic rites are confined to the Brahmins and the two other 'twice-born' castes."

The book challenged the then prevalent idea that caste was a rigid and unchanging institution. The concept of sanskritization addressed the actual complexity and fluidity of caste relations. It brought into academic focus the dynamics of the renegotiation of status by various castes and communities in India.

CULTURAL CHANGE IN MODERNIZATION

In the social sciences, modernization or modernisation refers to a model of an evolutionary transition from a 'pre-modern' or 'traditional' to a 'modern' society. The teleology of modernization is described in social evolutionism theories, existing as a template that has been generally followed by societies that have achieved modernity. While it may theoretically be possible for some societies to make the transition in entirely different ways, there have been no counterexamples provided by reliable sources.

Historians link modernization to the processes of urbanization and industrialization, as well as to the spread of education. As Kendall notes, "Urbanization accompanied modernization and the rapid process of industrialization. In sociological critical theory, modernization is linked to an overarching process of rationalisation. When modernization increases within a society, the individual becomes that much more important, eventually replacing the family or community as the fundamental unit of society.

Modernization theory and history have been explicitly used as guides for countries eager to develop rapidly, such as China. Indeed, modernization has been proposed as the most useful framework for World history in China, because as one of the developing countries that started late, "China's modernization has to be based on the experiences and lessons of other countries." Instead of being dominated by tradition, societies undergoing the process of modernization typically arrive at governance dictated by abstract principles. Traditional religious beliefs and cultural traits usually becomes less important as modernization takes hold.

THEORY

Theories of modernization, each society can develop from traditionalism to modernity, and that those that make this transition follow similar paths. More modern states are wealthier and more powerful, and their citizens freer, with a higher standard of living. The Social theorist Peter Wagner, modernization can be seen as processes, and as offensives. The former view is commonly projected by politicians and the media, and

suggests that it is developments, such as new data technology or need to update traditional methods, which make modernization necessary or preferable. This view makes critique of modernization difficult, since it implies these developments control the limits of human interaction, and not vice versa.

The view of modernization as offensives argues that both the developments and the altered opportunities made available by these developments are shaped and controlled by human agents. The view of modernization as offensives therefore sees it as a product of human planning and action, an active process capable of being both changed and criticized.

Modernization emerged in the late 19th century and was especially popular among scholars in the mid-20th century. One foremost advocate was Harvard sociologist Talcott Parsons. The theory stressed the importance of societies being open to change and saw reactionary forces as restricting development. Maintaining tradition for tradition's sake was thought to be harmful to progress and development. Proponents of modernization lie in two camps, optimists and pessimist. The former view holds that what a modernizer sees as a setback to the theory are invariably temporary setbacks, with the ability to attain "modernism" still existing. Pessimists argue that such non-modern areas are incapable of becoming modern.

DEMOCRACY

Scholars have long argued that democracy follows modernization, perhaps with a time lag. As Seymour Martin Lipset put it, "All the various aspects of economic development—industrialization, urbanization, wealth, and education—are so closely interrelated as to form one major factor which has the political correlate of democracy. In the 1960s some critics said the link each was too much based on European history, neglecting the Third World. Recent demonstrations of the emergence of democracy in South Korea, Taiwan and South Africa have tended to bolster the thesis.

The historical problem case has always been Germany, in which economic modernization in the 19th century came long before the move to democracy after 1918. Berman, however, concludes that a process of democratization was underway in Imperial Germany, for "during these years Germans developed many of the habits and mores that are now thought by political scientists to augur healthy political development."

Inglehart, and Welzel contend that the realization of democracy is not based solely on an expressed desire for that form of government, but that democracies are born as a result of the admixture of certain social and cultural factors. They argue the ideal social and cultural conditions for the foundation of a democracy are born of significant modernization and economic development that result in mass political participation.

Peerenboom explores the relationships among democracy, the rule of law and their relationship to wealth by pointing to examples of Asian countries, such as Taiwan and South Korea, that have successfully democratized only after economic growth reached relatively high levels and to examples of countries such as the Philippines, Bangladesh,

Cambodia, Thailand, Indonesia and India that sought to democratize at lower levels of wealth but have not done as well. Adam Przeworski and others have challenged Lipset argument. They say political regimes do not transition to democracy as per capita incomes rise. Rather, democratic transitions occur randomly, but once there, countries with higher levels of gross domestic product per capita remain democratic. Epstein retest the modernization hypothesis using new data, new techniques, and a three-way, rather than dichotomous, classification of regimes. Contrary to Przeworski, this study finds that the modernization hypothesis stands up well. Partial democracies emerge as among the most important and least understood regime types. Highly contentious is the idea that modernization implies more human rights, with China in the 21st century being a major test case.

DEVELOPMENT

Development, like modernization, has become the orienting principle of our time. Countries that are seen as modern are also seen as developed, and that means that they are generally more respected by institutions such as the United Nations and even as possible trade partners for other countries. The extent to which a country has modernized or developed dictates its power and importance on the international level.

CRITICISM

Modernization theory has been criticized, mainly because it conflated modernization with Westernization. In this model, the modernization of a society required the destruction of the indigenous culture and its replacement by a more Westernized one. Technically *modernity* simply refers to the present, and any society still in existence is therefore modern. Proponents of modernization typically view only Western society as being truly modern arguing that others are primitive or unevolved by comparison.

This view sees unmodernized societies as inferior even if they have the same standard of living as western societies. Opponents of this view argue that modernity is independent of culture and can be adapted to any society. Japan is cited as an example by both sides. Some see it as proof that a thoroughly modern way of life can exist in a non-western society. Others argue that Japan has become distinctly more western as a result of its modernization.

In addition, this view is accused of being Eurocentric, as modernization began in Europe with the industrial revolution, the French Revolution and the Revolutions of 1848, and has long been regarded as reaching its most advanced stage in Europe and in Europe overseas. Anthropologists typically make their criticism one step further generalized and say that this view is ethnocentric, not being specific to Europe, but Western culture in general. There is a strong criticism of Modernization in Africa due to a major problem with governmental corruption that has ultimately slowed the development of African countries through wars and shady dealings that keep money out of the hands of the people. Modernization in theory, should lead to wealthier more

economically productive countries, however, the corruption that runs rampant through Africa's nations has not allowed for much of this prosperity and growth to take place. The destruction of the tribal identities of Africans into national identities has also helped lead to problems with development bubbles. These development bubbles occur when cities rise into the position of industrialization, but the areas outside of the city do not develop and therefore a big inequality of wealth distribution becomes apparent. Therefore many of the people who once worked the agricultural areas in Africa, move to cities where they are expecting to find jobs that will sustain them.

MODERNIZATION IN INDIA

Everyone nowadays seems obsessed about the question of whether or not Islam can be reconciled with modernization. In discussing this issue, what constitutes modernization is often confused with westernization. Understanding the difference is vital. India's encounter with the West over the past three centuries underscores the distinction between the two processes — modernization and Westernization — that are often assumed to be synonymous. In fact, modernization does not entail Westernization, as the example of contemporary Japan demonstrates. Whereas modernization entails a change in belief about the way the material world operates, Westernization entails a change in cosmological beliefs about the way that one should live.

Like China and unlike Japan, India resisted changes in its ancient beliefs about the way the world works which modernization entails. Instead, like many Islamic countries today, India wrongly believed Gandhi's doctrine that modernization necessarily means Westernization. Fitfully, and under the influence of the British Raj, parts of the economy and society were modernized during the second half of the 19th century of *laissez-faire* and free trade. Some of the traditional literary castes also embraced Westernization.

British policy turned India into a pioneer of Third World industrialization, with an economy increasingly based on domestic capital and entrepreneurship combined with imported technology. But modernization stalled when protectionist pressures from Lancashire and the exigencies of Imperial finance led the British to abandon free trade and *laissez faire*. At the same time, Westernization fueled the rise of a nationalist movement.

The introduction of income taxes and UK labour laws in the late 19th century led to nearly a century of mounting state intervention in the economy, a process that accelerated after independence. This damaged India's growth prospects and hopes of alleviating its ancient scourge of mass poverty. The breakdown of the global economy in the first half of the 20th century in the wake of World War I further eroded India's incipient integration into the world economy during the British Raj. Finally, beginning with the economic reforms of 1991, India at last rejected inward-looking policies, returning to where it left off at the end of the 19th century.

We now have a fairly clear quantitative picture of the performance of the Indian economy throughout this period. During the 130 years from 1868 until 1999-2000, per

capita income more than trebled, as national income increased by a factor of eight while the population grew nearly five-fold. This suggests that the age-old combination of economic stagnation and cultural stability that I call the “Hindu Equilibrium” seems finally to have been broken. But on closer inspection, it turns out that this was largely because of the economic performance of the last two decades.

The sub-period from 1868-1900 saw the beginning of industrialization and India’s partial integration into the world economy under the Raj. National income did not stagnate, as nationalist historians once maintained, but grew modestly, at an average annual rate of 1.1 per cent. Growth was fastest from 1870-1890, followed by large fluctuations in output. A fairly low rate of population growth ensured a modest annual rise in per capita income of about 0.7 per cent during this period.

The second sub-period, 1900-1945, saw the breakdown of the global economy and the start of India’s population explosion. Growth of per capita income decelerated between 1902 and 1930 and declined further in the last fifteen years of British rule until 1947. After 1920, this was due entirely to the rise in population growth, to 1.22 per cent per year, which outpaced fairly high output growth throughout the decade until 1930. Over this entire 45-year sub-period, output grew at just above 1 per cent annually — the same rate as under the Raj — but yearly population growth soared to 0.8 per cent, causing virtual stagnation of per capita income.

The third sub-period, 1950-1980, marked the heyday of economic planning. Infrastructure investment — which had been difficult for the embattled British Raj to finance — suddenly boomed. This, together with high agricultural growth rates, led to a dramatic rise in output, which increased at an annual average rate of 4.5 per cent. But the demographic explosion that began in the 1920s had by now driven the rate of population growth up to 3 per cent per year, so that annual per capita income grew by only 1.5 per cent. A departure from what the late Raj Krishna dubbed the “Hindu rate of growth” came only in the fourth sub-period, from 1980-1999. Partial economic liberalization, undertaken by the Rajiv Gandhi government in the mid-1980s, and the more substantial Narasimha Rao-Manmohan Singh economic reforms in 1991, boosted national income growth to an average annual rate of 6.8 per cent. At the same time, yearly population growth slowed to an average of 2.3 per cent, so that per capita income rose at an impressive annual rate of 4.5 per cent, making a dent in India’s mass structural poverty for the first time in millennia. If India can now complete the unfinished business of fully integrating into the global economy, there is no reason why it cannot improve upon its impressive recent growth performance. If it succeeds, Nehru’s pledge, on the eve of Indian independence, “to wipe every tear from every eye,” will finally be redeemed.

CULTURAL CHANGE IN WESTERNIZATION

Westernization or Westernisation, also occidentalization or occidentalisation is a process whereby societies come under or adopt Western culture in such matters as industry,

technology, law, politics, economics, lifestyle, diet, language, alphabet, religion, philosophy, and/or values. Westernization has been a pervasive and accelerating influence across the world in the last few centuries. It is usually a two-sided

Westernization can also be related to the process of acculturation and/or enculturation. Acculturation refers to the changes that occur within a society or culture when two different groups come into direct continuous contact. After the contact, changes in cultural patterns within either or both cultures are evident. In popular speech, Westernization can also refer to the effects of Western expansion and colonialism on native societies.

For example, natives who have adopted European languages and characteristic Western customs are called acculturated or Westernised. Westernization may be forced or voluntary depending on the situation of the contact.

Different degrees of domination, destruction, resistance, survival, adaptation, and modification of the native culture may follow inter-ethnic contact. In a situation where the native culture experiences destruction as a result of a more powerful outsider, a “shock phase” often is a result from the encounter. This shock phase is especially characteristic during interactions involving expansionist or colonialist eras. During the shock phase, civil repression using military force may lead to a cultural collapse, or ethnocide, which is a culture’s physical extinction. Conrad Phillip Kottak, the Westerners “will attempt to remake the native culture within their own image, ignoring the fact that the models of culture that they have created are inappropriate for settings outside of Western civilization”.

DEFINITION OF THE WEST

Territorial

The “west” was originally defined as Western World. Ancient Romans distinguished between Oriental cultures that inhabited present-day Egypt and Occidental cultures that lived in the West. A thousand years later, the East-West Schism separated the Catholic Church and Eastern Orthodox Church from each other. The definition of Western changed as the West was influenced by and spread to other nations. Islamic and Byzantine scholars added to the Western canon when their stores of Greek and Roman literature jump-started the Renaissance.

The West expanded to include Russia when Peter the Great brought back ideas from Holland. Today, most modern uses of the term refer to the societies of Western and Central Europe and their close genealogical, linguistic, and philosophical descendants, typically included are those countries whose ethnic identity and dominant culture are derived from European culture.

Western civilization can be defined as *at least* North America, Western Europe, Australia and New Zealand. North America includes the U.S.A. and Canada. Widening this definition however invites controversy.

This widened definition can include these countries, or a combination of these countries:

- Other parts of Europe - Due to their membership in the European Union and European Neighbourhood Policy, these countries are included in the definition of the West, as they more or less share a common European identity with Western Europeans. This view has increasingly gained support, especially since the disintegration of the Eastern Block and the current European integration process that is a direct result of that disintegration. Others, however, like the American political scientist Samuel P. Huntington, consider the majority-Orthodox Christian part of Europe as very different from the West. Most who support Huntington's view, believe that Eastern Europe, from the Balkans, European Russia and the Caucasus, did not experience the Renaissance or the full strength of the industrial revolution to be considered as part of "The West". Equally they maintain that Eastern European culture is not shaped by Western Christian traditions of Catholicism or Protestantism, but rather by Eastern Orthodoxy or Islam. From a different angle, this part of the world does not fulfil the economic and standard of living criteria one would associate with the "West".
- Latin-America. Many countries in Latin America are sometimes regarded as Western countries, largely because most of its peoples are racially descended from Europeans. And thus their society operates in a highly Westernised way. Indeed, most countries in Latin America use their official language, either in Spanish or Portuguese. The *CIA -The World Factbook-*, there has also been considerable immigration to Latin America from European nations other than Spain and Portugal. There are also many indigenous cultures with their own languages in Latin America.
- Turkey. Although geographically only 3% of Turkey lies in Europe, Turkey has a similar economic system, has a customs union with the European Union in addition to being an official candidate for membership, and is a member of typical Western organisations such as OECD, Council of Europe, and NATO. It is usually a member of European organisations for sports and cultural events such as UEFA and the Eurovision Song Contest.
- Israel. Many Jewish immigrants to Israel were from Western countries like the United States and Germany. It is a member of the OECD. It is usually a member of European organisations for sports and cultural events such as UEFA and the Eurovision Song Contest.

Personal

A different view on the Western world is not defining it by its territory, but by its people, as these tend to differ in an increasingly globalised world. This view highlights the non-Western population in countries with a Western majority, or vice versa. The Boers for instance can be regarded as Western inhabitants of South Africa.

Differences

It would be incorrect to regard the Western world as a monolithic bloc, as there exist many cultural, linguistic, religious, political, and economical differences between Western countries and populations. The Western world itself is changing over time as it has in the past.

CONSEQUENCES

Due to the colonisation of the Americas and Oceania by Europeans, the cultural, ethnic and linguistic make-up of the Americas and Oceania has been irreversibly changed. This is most visible in settler colonies such as the United States of America, Australia, Canada and New Zealand, where the traditional indigenous population has been overtaken demographically by non-indigenous settlers. This demographic takeover in settler countries has often resulted in the linguistic, social, and cultural marginalisation of indigenous people.

However, even in countries where large populations of indigenous people remain or the indigenous peoples have mixed considerably with European settlers, such as Bolivia, Guatemala, Mexico, Peru and Ecuador, marginalisation still exists. But continued immigration to Chile and eventual “white” majority regions like Costa Rica made these cultures have a castizo or a more Europeanized-mestizo background.

Due to colonisation, the prevalent native languages in the Americas, Australia, New Zealand and part of South Africa, are now usually European languages or creoles based on them:

- Portuguese.
- Spanish (the rest of Latin America (although in Paraguay most people are bilingual with Guaraní and there are more Guaraní speakers than Spanish speakers) and formerly in the Philippines during Spanish colonial rule, which ended in 1898 when the US annexed the islands).
- French (Quebec, New Brunswick and parts of Ontario in Canada and Saint Pierre and Miquelon), French along with French-based creole languages (Haiti, French Guiana, Guadeloupe, Martinique and Saint-Barthelemy).
- Dutch along with creole languages (Suriname, Aruba and the Netherlands Antilles).
- Afrikaans (which contains 90-95% mutual intelligibility with Dutch) along with English (parts of South Africa).
- English (United States, Canada, Australia and New Zealand) or English along with English-based creole languages (Jamaica and most of the West Indies, Guyana).

Many indigenous languages are on the verge of becoming extinct. However, some settler countries have gone to lengths to preserve indigenous languages, for example, in New Zealand the Māori language is the second official language.

WESTERNIZATION OF INDIA

India is considered to be one of the oldest nations in terms of culture and tradition. Epics like ‘Mahabharata’ and ‘Ramayana’ are the real evidences that depict the strong Indian culture. Since ages, it has been a land of sages, saints and various renowned people. Thousands of pious temples built in different parts of the nation reveal that people have been worshipping God and following their culture and tradition since time immemorial.

It was the same going on till few decades back. Most of the people were farmers and survived on their pieces of land. They were satisfied with their earning and hardly had any time to think further. As a result tradition was in touch and people were in accordance with the Indian tradition. As the time passed by and India got independence, the scenario started changing slowly and gradually.

Westernization started attracting the Indian citizens towards it like a magnet. Call it an easy way to earn money or to compete with the rest of the world or to imitate the other nations or for other purposes, but anywhere it was taking us away from our culture at a slow pace. The intoxication of westernization was so strong that people started carrying away with it. The things were new, appealing, attractive and something to try upon, but forgetting that these things might become their habit some day and deviating them from their actual goods and culture.

Seeds of Westernization Sown by the Britishers

India was known as the ‘Golden Bird’ one time. It was known for its abundance in gold, silver, diamonds and various other precious pearls. ‘Kohinoor’ one of the most expensive pearl in the world was mined from Golconda mines. Kings and rulers in India owned these precious things. Scintillations of such commodities attracted the British people towards India and they placed their feet on our motherland in the early 16th century.

Initially their echo chamber behaviour pleased everyone without knowing the hidden motives behind it. It was due to such shrewd techniques that they were able to take away tones of wealth in ships to their places. Eventually they took the authority in their own hands and India became the slave of these English people. It was during this time that they introduced many things in India which the general people were not at all aware of. Guns, cars, cinema, foreign clothes, cigarettes *etc.* were used by them and hence came into existence.

Though we cannot completely say that all the things were bad, most of us were a boon for us, but the extra influence of all these things has hampered our Indian culture. As the years passed by more and more western commodities came into the Indian market and people became aware of them. Thus taking them away from their own things. Mahatma Gandhi, known as the ‘Father of the Nation’ tried to bring back the public from eye catching western things. He started a ‘Swadeshi Movement’, emphasizing on the use of commodities manufactured in the nation itself with its own

raw material. He asked people to make use of 'Khadi clothes' so that British cloth does not stand in market.

It will serve as a boon for the Indian manufacturing unit and will provide more employment. Though it was a great move initiated by Gandhi, but seeds of westernization were so deep sown that after the death of Gandhi the Indian scenario totally changed. India got freedom, and elite group from India started following British trends and implementing them in the nation as well. As a result general public also started following it. Thus drifting the Indian people slowly towards westernization.

Present State of India

If we look into the present state of India, as suggested, find westernization has made a place in each and everything. We cannot even think of running the nation now avoiding these things. These things have now become a necessity. Though culture, tradition, religion is still there and have many followers but its percentage as compared to earlier days as fallen by a big amount. We can have a look at various examples where westernization can be seen at its culmination:

Cinema

Westernization has shown its major effect in the 'Bollywood' industry. Starting from the early 40 movies till 2009 the concept, dresses, fashion and script has totally changed. Earlier main focus was made on the India culture, Indian problems depicting an ordinary man. There was decency in dresses; story of the movie had some moral to the viewers. Actors used to have good theatre experience. Money earning was not the sole concern that time.

These days the idea has totally changed. Vulgarity and obscenity is displayed every now and then. It is not our culture but a western impact. Stories of the movies are copied or idea is taken from western movies. The main aim is always to compete with 'Hollywood'. Cinema influences the youngsters and even the elders a lot. As a result they also start imitating their movie role models. Hence westernization is spreading one or the other way.

Opting Foreign Countries for Profession

Why there is a desire in a young Indian to go abroad after studies and continue there with the profession? Why cannot we work for the country that imparted us with such a magnificent education? The only reason is that western countries have mesmerized us. We are going for momentary things and forgetting the real rich source is with us that is our nation.

Food and Cloth Items

Trendy clothes are in vogue these days. But none could find a new fashion based on Indian work. Jeans, shoes, shirts, jackets and every other accessory is foreign made.

Adidas, Reebok, Wrangler, Woodland, Van Heusen all these are foreign brands and we make use of them. It is not bad but at the same time completely forgetting our tradition and culture is also not good though. One can find very few 'Khadi' showrooms throughout the country. Also various junk food items have also made the hegemony over our traditional food items like rice, pulses, chapattis *etc.* Younger generation is more affected by it. It becomes the duty of the parents to guide them.

Economic Development and Social Change

INTRODUCTION

Economic development, achieved largely through productivity growth, is very important to both developed and developing nations. However, even though we know that higher productivity leads to improved economic outcomes (for example, higher income, more choices to the consumers, better quality products, *etc.*), there has been no consensus among researchers about either the desired path of development or the role of state in economic development.

Concerning the path of development, Lall (2001) says that the appropriate strategy for any country depends not only on its objective economic situation but also on its government policies and national views regarding the appropriate role of the state. Regarding the appropriate role of the state, it seems that for every argument in favour of a smaller government role one can find a counter argument in favour of a more active government role. The role of the state in economic development began to change dramatically with the advent of the Industrial Revolution.

In the West, the resulting industrialization and economic development were based on the establishment of individual property rights that encouraged the growth of private capital. Competition and individual enterprise thrive in this environment because individuals pursue their self-interest of survival and wealth accumulation. The instinct to survive under competitive pressures yields innovation and productivity increases,

which eventually lead to both increased profits for business and lower prices to consumers.¹ However, the rise and spread of capitalism led a number of thinkers to examine the consequences of the market-based approach to development. Socialists argued that capitalism (or private ownership of capital) can lead to greater inequalities of income and wealth, while developmental economists argued that private decisions may not always lead to socially desirable outcomes (particularly in the case of market imperfections).

Indeed, many policymakers at the time saw market failures as quite common and therefore assumed that only appropriate government interventions could guide an economy to a path of sustained economic development. In the early 20th century, the former Soviet Union attempted a bold experiment of improving individual well-being without sacrificing the objective of greater equality of income and wealth through total ownership of capital by the government.

Initially, the Soviet Government was able to raise productivity through directed industrialization and, within a span of 25 years (by the end of World War II), emerged as a superpower. It was around this time that a substantial number of colonized nations were gaining their independence (for example, India, Pakistan and Burma). Unfortunately, during their time as colonies to the Western nations, these countries, for the most part, had been deprived of the industrialization that had engulfed those same Western nations.

Based on the successful experience of the former Soviet Union, many economists and policymakers concluded that, particularly in a poor country, planning was essential for the efficient allocation of an economy's resources. The governments in these newly independent nations assumed a significant role in economic development. They sought to quickly and substantially raise the standard of living through directed and controlled economic development. Apart from everything else, these developing countries invested heavily in education to promote literacy and to ensure an adequate supply of technical manpower to meet their growing needs. Further, these previously colonized nations did not want to subject their poor and weak economies to international economic fluctuations and thus sought to industrialize through import substituting industrialization, where imports were expected to be increasingly replaced by domestic production.

We examine economic development in India, a former British colony that became one of the most closed economies in the world, to contrast the roles of government intervention and individual enterprise in that country's economic growth. In particular, we demonstrate that, given recent economic reforms in India, along with the evidence for the role that individual enterprise can play in a country's economic growth, the Indian government should devise policies that rely more on individual enterprise, with its emphasis upon individual initiative and self-interest, to spur economic development. Further, we describe the special role that can be played in the economic development of India by a greater emphasis upon entrepreneurship. The plan of the paper is as follows. The strategy of economic development and the overall economic environment that has

prevailed in India since its independence from the United Kingdom. The consequences of regulated economic development in India, with particular emphasis on the implications of the microeconomic aspects of India's approach to its economic environment.

The results of India's economic reforms since the country's economic crisis of 1990, and highlights the role that individual enterprise has played and can continue to play in that country's economic fortunes. The special role that entrepreneurship can play in India's efforts at economic growth. Finally, the main findings and concludes the paper.

INDIA'S STRATEGY OF ECONOMIC DEVELOPMENT

India's economic development strategy immediately after Independence was based primarily on the Mahalanobis model, which gave preference to the investment goods industries sector, with secondary importance accorded to the services and household goods sector. For example, the Mahalanobis model placed strong emphasis on mining and manufacturing (for the production of capital goods) and infrastructural development (including electricity generation and transportation).

The model downplayed the role of the factory goods sector because it was more capital intensive and therefore would not address the problem of high unemployment in India. Any increase in planned investments in India required a higher level of savings than existed in the country. Because of the low average incomes in India, the needed higher levels of savings had to be generated mainly by restrictions on the growth of consumption expenditures.

Therefore, the Indian government implemented a progressive tax system not only to generate the higher levels of savings² but also to restrict increases in income and wealth inequalities.³ Among other things, this strategy involved canalization of resources into their most productive uses.⁴ Investments were carried out both by the government and the private sector, with the government investing in strategic sectors (such as national defence) and also those sectors in which private capital would not be forthcoming because of lags or the size of investment required (such as infrastructure). The private sector was required to contribute to India's economic growth in ways envisaged by the government planners. Not only did the government determine where businesses could invest in terms of location, but it also identified what businesses could produce, what they could sell, and what prices they could charge.

Thus the strategy of economic development in India meant:

- Direct participation of the government in economic activities such as production and selling and
- Regulation of private sector economic activities through a complex system of controls.

In addition, the Indian economy was sheltered from foreign competition through use of both the "infant industry argument" and a binding foreign exchange constraint. Imports

were limited to goods considered essential either to the development of the economy (such as raw materials and machines) or to the maintenance of minimal living standards (such as crude oil and food items). It was further decided that exports should play a limited role in economic development, thereby minimizing the need to compete in the global market place. As a result, India became a relatively closed economy, permitting only limited economic transactions with other countries.

Domestic producers were sheltered from foreign competition not only from abroad but also from within India itself. Over time, India created a large number of government institutions to meet the objective of growth with equity. The size of the government grew substantially as it played an increasingly larger role in the economy in such areas as investment, production, retailing, and regulation of the private sector. For example, in the late 1950s and 1960s, the government established public sector enterprises in such areas as production and distribution of electricity, petroleum products, steel, coal, and engineering goods.

In the late 1960s, it nationalized the banking and insurance sectors. To alleviate the shortages of food and other agricultural outputs, it provided modern agricultural inputs (for example farm machinery, irrigation, high yielding varieties of seeds, chemical fertilizers) to farmers at highly subsidized prices. In 1970, to increase foreign exchange earnings, it designated exports as a priority sector for active government help and established, among other things, a duty drawback system, programmes of assistance for market development, and 100 per cent export-oriented entities to help producers export.

Finally, from the late 1970s through the mid-1980s, India liberalized imports such that those not subject to licensing as a proportion to total imports grew from five per cent in 1980-1981 to about 30 per cent in 1987-1988. However, this partial removal of quantitative restrictions was accompanied by a steep rise in tariff rates. This active and dominant participation by the government in economic activities resulted in the creation of a protected, highly-regulated, public sector-dominated economic environment. Along with this government domination of the economy, India soon faced not only some major problems in its overall approach to development, particularly in the area of industrialization but also a dramatic increase in corruption in its economy. Finally, like any other growing economy, the Indian economy faced a number of serious sectoral imbalances, with shortages in some sectors and surpluses in others.

THE CONSEQUENCES OF INDIA'S REGULATED ECONOMIC DEVELOPMENT

India's environment of regulated economic development led to the formulation of policies that were concerned with both macroeconomic and microeconomic aspects. Whereas much attention in the literature has been devoted to the macroeconomic issues, we focus primarily on the microeconomic aspects of Indian economic policies. In

particular, we examine how individuals guided by their self-interests of survival and wealth accumulation will act in a regulated environment, which in fact discourages the pursuit of those self-interests. To do so, we describe the consequences of India's use of price ceilings, in which prices are set below their equilibrium level to make products and services affordable to relatively poorer parts of the society. Figure shows how price ceilings can influence a nation's economy.

Specifically, when prices are kept artificially low, demand outstrips supply. To alleviate the resulting shortage of products and services, the government can either help to increase the supply or help to decrease demand for those products and services.

Considering the supply side options first, the government had the following choices:

- Increase the price of the product;
- Subsidize production of existing suppliers so they will produce and sell more;
- Encourage new businesses to enter the line of production and selling; or
- Permit imports to reduce or eliminate the shortage.

In India, none of these options was seen as satisfactory.

- First, the government certainly did not wish to increase prices, because price ceilings appealed to a majority of the vote bank.
- Second, although the government did subsidize production in several sectors considered essential, the resulting increased production was not sufficient to eliminate the large shortages.
- Third, the government decided to restrict rather than increase the entry of new producers under the pretext of directing scarce resources into their efficient uses.
- Finally, it allowed only limited recourse to imports, in order to protect Indian producers, unless the shortage reached a stage of crisis. The overall result was that inadequate amounts of products and services were supplied to the market.

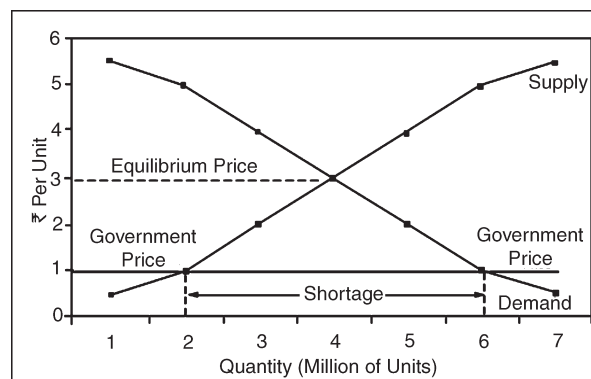


Fig. Price Ceiling.

In contrast to the supply side options just considered, if the government had decided instead to limit demand, it could have done so by increasing taxes or by regulating the

level of demand itself, usually by restricting how much an individual or a family could consume. To ensure the availability of the scarce products and services to Indian consumers, albeit at less than desired levels, the Indian government in fact resorted to large-scale rationing. This rationing was undertaken by government agencies themselves or by licensed private retailers. As might be expected, the rationing regulations required those licensed private retailers to follow government stipulations in their sale of the scarce products and services.

The policy of price ceilings, along with the quantitative restrictions on production and consumption, led to an economic environment ripe for corruption. Specifically, because of the general scarcity of products and services, individuals competed to receive the privilege of economic rights to produce or consume. The implementing authority responsible for allocating these economic rights—politicians, government officials and businesses—enjoyed monopoly power in this situation and, as might be expected, were susceptible to bribes and other illegal favours. The result was an informal and illegal market in which the desired economic rights could be traded.

Also, the lure of higher profits led producers and sellers:

- To have little concern for quality such that many deliberately produced and sold inferior quality products, and
- To resort to the creation of artificial shortages by not releasing to the market all of the products that were available for selling.

Bardhan (1997) defines corruption as the use of public office for private gain, in which an official entrusted with the responsibility for certain public duties engages in malfeasance for personal enrichment that is not easy to monitor. He says that corruption has multiple effects on economic development. For example, it diminishes the efficiency of economic transactions, because corrupt officials will delay or otherwise obstruct those transactions until they receive their expected favours.

Also, the payment of a bribe to receive an investment license tends to reduce the incentive to invest. Honest investors will see the futility of competing with dishonest investors who are guaranteed, through their bribes, to receive the privilege of investment rights. To fully understand the widespread nature of the corruption that existed in India at this time, it is necessary to consider the roles played by the many participants.

For example, business people bribed government officials not only for the right to enter a particular line of business, but also to prevent others from entering that same line of business. Government officials made payoffs to politicians to receive the premium government positions that would allow them to easily contact businesses to seek illegal income and wealth.

Indeed, as Wade (1985) indicates, those officials could earn far more through bribes and other corrupt behaviour than they could earn in salary. Consumers bribed government officials, politicians, and business people to receive a particular amount of a scarce good or a higher quality version of the good. Even individuals and organizations outside India took part in the corruption. Some bribed both officials and politicians, particularly

those connected with the revenue and police departments, to smuggle scarce goods into India at a high profit. The complex system of government controls, including price ceilings, along with the resultant corruption, meant that decision making was arbitrary and the transactions non-transparent. The result was an increase in transaction costs. For example, businesses had to spend more to 'stay connected' with appropriate government officials and politicians.

And consumers, in addition to waiting in line to purchase needed products and services, also made illegal payments for what they should have received at a reasonable price in the first place. It has already been explained how India's government grew in size as it played an increasing role in controlling the economy. It grew even further in trying to be appropriately vigilant in dealing with the increased corruption among government officials, businesses, and other participants. Price controls were only one example of the regulated economic environment.

Another example of a harmful policy was the control of ownership of private capital (both income and wealth) by Indian nationals in India and also by foreign nationals doing business in India. Such policies, coupled with high individual and corporate income tax rates and high customs and excise duties, led to outcomes similar to those resulting from price ceilings namely, increased corruption and higher transaction costs. In conclusion, this part has shown how individuals guided by their selfinterests, will act in a regulated environment. Government controls based on arbitrary and ad hoc administrative decisions lead not only to greater concealed income and wealth but also to diminished productivity, particularly due to the resulting higher transaction costs.

ECONOMIC FACTORS OF SOCIAL CHANGE

CULTURAL AND SOCIAL FACTORS THAT AFFECT DEVELOPMENT

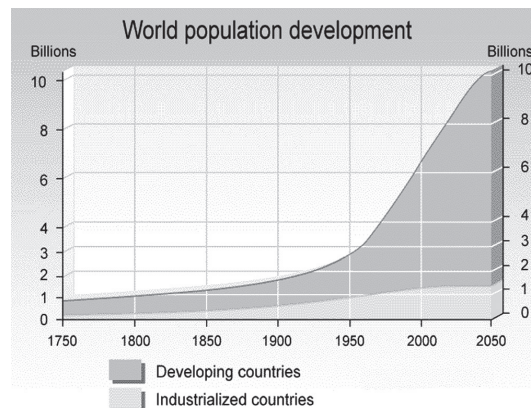
Sometimes there are social or cultural factors that hold back poor countries. Discrimination is one of these. If there are certain people groups that are discriminated against, the country's overall productivity can suffer. This may be a tribe, a caste, a racial category or minority language group. I have already mentioned Cameroon, which has both French speaking and English speaking regions. All the infrastructure happens in the French speaking part. French speakers in Canada complain of the opposite. Welsh speakers in Britain, or Catalans in Spain, have historically faced similar problems. Racial discrimination may be an issue, excluding certain groups from economic activity, either deliberately or not. Racial minorities regularly have poorer exam results and economic prospects than the majority. More serious forms of exclusion would be apartheid South Africa, or the Asian communities driven out of Uganda under Idi Amin, which was disastrous for Uganda's economy.

Another division may be the role of women. Jeffrey Sachs talks about this in *The End of Poverty*: 'Cultural or religious norms may block the role of women... leaving

half the population without economic or political rights and without education, thereby undermining half of the population in its contribution to overall development.’ If you don’t believe that women should work, you have effectively halved the earning potential of your country.

Population

Closely linked to this is the population issue. If women see staying at home and bringing up children as their chief role, they will have more children than those who work. There is nothing wrong with having lots of children, as long as you can provide for them. Jeffrey Sachs again: ‘With fewer children, a poor household can invest more in the health and education of each child, thereby equipping the next generation with the health, nutrition, and education that can lift living standards in future years.’



As Paul has talked about here before, world population has exploded. What is interesting is that the countries where this has happened are often those where women do not play a role in business or society. When women are educated and given a choice, some will stay at home and look after children, and others will pursue careers or start small businesses. This is an important factor, as some countries have seen their population double or triple without their economies keeping pace. That leaves more mouths to feed, and just not enough to go around.

Culture

I’ve already mentioned the role of women, but culture can have hidden effects in business, trade and development. China may be a major power now, but it was the world’s most developed country in the middle ages, and stagnated, or even went backwards, for centuries. Part of this was cultural, a pride and sense of self-sufficiency that led to a closing of China’s borders. ‘China seems to have long been stationary’, Adam Smith wrote in 1776, in his *Wealth of Nations*. ‘A country which neglects or despises foreign commerce... cannot transact the same quantity of business which it might do with different laws and institutions.’ That’s changed, but nationalism, suspicion,

or radical philosophy still has some countries closed down to outside involvement – communism in North Korea, or extremist Islam in Taliban Afghanistan, locking countries out of development. This the far end of the spectrum, but culture works in subtler ways too. Some cultures believe in a greater good, in unity, in the rule of law. They are optimistic, hopeful, ambitious and ready to pull together. Others can be paranoid, fragmented, uncertain of their place in the modern world, angry, resistant to change. Rich countries can be overconfident and brash. Poor countries can see themselves as victims and become despondent. In his *The Wealth and Poverty of Nations*, economic historian David Landes says ‘If we learn anything from the history of economic development it is that culture makes all the difference.’

The Limits of Cultural Interpretations

At the same time, cultural influences on development are notoriously hard to call from the outside. Hinduism was often cited as one of the reasons why India would never develop. Because everyone accepts their place in the world, it was assumed that Hindus would lack the ambition required to innovate and do business on an international stage. The recent growth in India’s economy proves that wrong quite spectacularly. Korean economist Ha-Joon Chang quotes a 1911 travel book that describes Koreans as “sullen, lazy and religionless savages”, something that hardly holds true today. So did Korean culture change, or was the writer simply being superior?

We understand each other better than ever in our globalized world, but our language and traditions are still full of little prejudices that imply we are better than others, and that our neighbours are lazy and dirty and uncouth. I love the fact that not turning up for work is called ‘taking French leave’ in England, and ‘filez a l’Anglaise’ (or ‘doing an English’) in France. A Malagasy friend once joked that in Madagascar, every tribe believes that every other tribe eats cats. In short, culture no doubt plays a role in development, but we have to watch our own biases as we seek to understand why some countries succeed and others fail.

REVIVAL OF ECONOMIC SOCIOLOGY

Despite the efforts of Parsons and Smelser in the mid-1950s and the 1960s to revive economic sociology, it attracted little attention, and by the 1970s the field was somewhat stagnant. A number of works inspired in one way or another by the Marxist tradition—and its general revival in the late 1960s and the early 1970s—made their appearance in this period. Among these were Marxist analyses themselves, dependency theory world systems theory and neo-Marxist analyses of the workplace. In the early 1980s, a few studies suggested a new stirring of interest.

And with the publication in 1985 of a theoretical thesis by Mark Granovetter—“Economic Action and Social Structure: The Problem of Embeddedness”—the new ideas came into focus. The same year Granovetter spoke of “new economic sociology”—yielding a tangible name. Why economic sociology, after decades of neglect, suddenly

would come alive again in the mid- 1980s is not clear. Several factors may have played a role, inside and outside sociology. By the early 1980s, with the coming to power of Reagan and Thatcher, a new neoliberal ideology had become popular, which set the economy—and the economists— at the very center of things. By the mid- 1980s economists had also started to redraw the traditional boundary separating economics and sociology, and to make forays into areas that sociologists by tradition saw as their own territory. It is also during this period that Gary Becker, Oliver Williamson, and others came to the attention of sociologists.

Likewise, sociologists began to reciprocate by taking on economic topics. To some extent this version of what happened resembles Granovetter's version in 1985. He associated "old economic sociology" with the economy and society perspective of Parsons, Smelser, and Wilbert E. Moore, and with industrial sociology— two approaches, he said, that had been full of life in the 1960s but then "suddenly died out". Parsons's attempt to negotiate a truce between economics and sociology had also been replaced by a more militant tone.

According to Granovetter, new economic sociology "attacks neoclassical arguments in fundamental ways," and it wants to take on key economic topics, rather than focus on peripheral ones. Since the mid-1980s new economic sociology has carved out a position for itself in U.S. sociology. It is well represented at a number of universities. Courses are routinely offered in sociology departments.

A part in the American Sociological Association has been formed. A number of high-quality monographs have been produced, such as *The Transformation of Corporate Control* by Neil Fligstein, *Structural Holes* by Ronald Burt, and *The Social Meaning of Money* by Viviana Zelizer. These three works draw on the insights of organization theory, networks theory, and cultural sociology, respectively. The subfield has also seen the appearance of several anthologies, readers, a huge handbook, a textbook, and a general introduction to the field.

GRANOVETTER ON EMBEDDEDNESS

While several attempts have been made to present general theories and paradigms in new economic sociology, the perspective that continues to command most conspicuous attention is Granovetter's theory of embeddedness. Since the mid- 1980s Granovetter has added to his argument and refined it in various writings that are related to his two major projects since the mid-1980s: a general theoretical work in economic sociology entitled *Society and Economy: The Social Construction of Economic Institutions*, and a study of the emergence of the electrical utility industry in the United States.

The most important place in Granovetter's work where embeddedness is discussed is his 1985 article, which operated as a catalyst in the emergence of new economic sociology and which is probably the most cited article in economic sociology since the 1980s. His own definition of embeddedness is quite general and states that economic actions are "embedded in concrete, ongoing systems of social relations". Networks are

central to this concept of embeddedness. An important distinction needs also to be drawn, according to Granovetter, between an actor's immediate connections and the more distant ones— what Granovetter elsewhere calls “relational embeddedness” and “structural embeddedness”.

The most important addition to the 1985 article has been connecting the concept of embeddedness to a theory of institutions. Drawing on Berger and Luckmann Granovetter argues that institutions are “congealed networks”. Interaction between people acquires, after some time, an objective quality that makes people take it for granted. Economic institutions are characterized by “the mobilization of resources for collective action”. Granovetter's argument on embeddedness has been widely discussed and sometimes criticized. An attempt to elaborate it can be found in the work of Brian Uzzi, who argues that a firm can be “underembedded” as well as “overembedded,” and that a firm is most successful when it balances between arm's-length market ties and more solid links. Several other critics have pointed out that Granovetter omits consideration of many aspects of economic action, including a link to the macroeconomic level, culture, and politics. Zukin and DiMaggio suggest that to remedy this lacuna, one should not only talk of “structural embeddedness,” but also of “political,” “cultural,” and “cognitive embeddedness.”

CONTRIBUTIONS USING STRUCTURAL SOCIOLOGY AND NETWORKS

Structural sociology has played a crucial role in promoting and adding to network analysis in sociology, including economic sociology. This approach is centered around the proposition that the relations of persons and positions are crucial to the social process. Its practitioners often use a mathematical approach, focus on social mechanisms, and avoid regression analysis and similar quantitative methods.

Its most prominent scholars are Harrison White and his students, such as Mark Granovetter, Scott Boorman, and Michael Schwartz. White's work in economic sociology has concerned networks, vacancy chains, and markets. He begins his analysis from people's physical dependence on their surroundings but notes that interests are soon embedded in social relations. Network studies have been at the center of the new economic sociology. Many studies have been made of the links between corporations and, more generally, within so-called industrial districts. Burt analyzes competition by drawing on Simmel's idea that you are in a good position if you can play out two competitors against one another. Brian Uzzi's study of embeddedness from 1997 also makes use of networks, as does Granovetter's thesis on business groups. A multitude of other fine studies could be mentioned. One criticism of the network approach is that it has ignored the role in economic life of politics and culture.

CONTRIBUTIONS USING ORGANIZATION THEORY

New economic sociology has been very successful in using organization theory to explore a number of important topics, such as the structure of firms and the links between

corporations and their environments. One fine example is Nicole Woolsey Biggart's *Charismatic Capitalism*, which deals with a very special type of organization: direct selling organizations, such as Tupperware and Mary Kay Cosmetics. Three theoretical approaches in organization theory have been especially important for the development of new economic sociology: resource dependency, population ecology, and new institutionalism.

Resource dependency, as its name suggests, rests on the postulate that organizations are dependent on their environments to survive. An example of this approach is work by Burt, who suggests that three important factors that affect profits are the number of suppliers, competitors, and customers. The more "structural autonomy" a firm has, the higher its profits; that is, a firm with many suppliers, few competitors, and many customers will be in a position to buy cheaply and sell expensively. In population ecology the main driving force of organizations is survival. It has been shown that the diffusion of an organizational form typically passes through several distinct stages: a very slow beginning, then explosive growth, and finally a slow settling down. Individual studies of this process in various industries, such as railroads, banks, and telephone companies, fill a void in economic sociology. New institutionalism is strongly influenced by the ideas of John Meyer and is centered around what may be called cultural and cognitive aspects of organizations.

Meyer argues that organizations seem much more rational than they actually are, and that specific models for organizing activities may be applied widely—including to circumstances they do not fit. It has been argued that the strength of new institutionalism is its exploration of "factors that make actors unlikely to recognize or to act on their interests" and its focus on "circumstances that cause actors who do recognize and try to act on their interests to be unable to do so". The possibility of uniting a more traditional interest analysis with new institutionalism is exemplified by Fligstein's study of the large corporation in the United States. Fligstein notes that the multidivisional form of organization spread for mimetic reasons—but also because this organizational form made it easier for firms to take advantage of new technology and the emerging national market.

CONTRIBUTIONS USING CULTURAL SOCIOLOGY

A group of economic sociologists is committed to a cultural approach, and a substantial number also refer to symbols, meaning structures, and the like in their studies of the economy. Cultural economic sociology owes much to the work of its two most prominent representatives, Viviana Zelizer and Paul DiMaggio. In a programmatic statement Zelizer criticized contemporary economic sociology for its tendency to reduce everything to social relations and networks—"social structural absolutism".

She also rejected the alternative of reducing everything in the economy to culture. The goal should be to take economic and cultural factors into account. DiMaggio has been similarly skeptical of a full-scale cultural analysis of the economy, but argues that

it should include a “‘cultural’ component”—but not more. According to DiMaggio, culture can be either “constitutive,” referring to categories, scripts, and conceptions of agency, or “regulative,” referring to norms, values, and routines. Viviana Zelizer’s work on culture occupies a central position.

Her first major work was a study of life insurance in the United States, with special emphasis on the clash between sacred values and economic values. Over time the economic emphasis came to dominate. Later Zelizer published *Pricing the Priceless Child*, which describes a similar movement, but this time in reverse. Children, who in the nineteenth century had had an economic value, would in the twentieth century increasingly be seen in emotional terms and regarded as “priceless.” In her most recent major study, Zelizer argues that money does not constitute a neutral, nonsocial substance, but appears in a variety of culturally influenced shapes.

CAPITALISM IN ECONOMIC SOCIOLOGY

Capitalism is the dominant economic system in today’s world, and there appears to be no alternatives in sight. Socialism, its main competitor, has been weakened immeasurably by the collapse of the Soviet Union in the early 1990s. Where socialism still prevails, such as in the People’s Republic of China, serious attempts are made to turn the whole economic system in a capitalist direction so that it will operate in a more efficient manner.

“It doesn’t matter if the cat is white or black as long as it catches mice”, to cite Deng Xiaping. While the superiority of capitalism as an economic system and growth machine has fascinated economists for centuries, this has not been the case with sociologists. For sociologists capitalism has mainly been of interest for its *social* effects – how it has led to class struggle, anomie, inequality and social problems in general. Capitalism as an economic system in its own right has been of much less interest. Some of this reaction has probably to do with the unfortunate division of labour that developed between economists and sociologists in the 19th century: economists studied the economy, and the sociologists society minus the economy.

In this respect, as in so many others, sociology has essentially been a “left-over science”. This division of labour between economists and sociologists, however, has not gone unchallenged. In the 1980s sociologists, especially in the United States, turned their attention to the study of the economy itself, asking questions such as the following: Where do markets come from?

How is economic action embedded in social relations? What role do norms and trust play in the economy?. That this set of questions heralded something new soon became clear. A huge amount of research – known as “New Economic Sociology” – soon came into being. By the mid-1990s enough work had been done to put together a handbook in economic sociology, with stages on such topics as “business groups”, “a rational choice perspective on economic sociology” and “networks and economic life”. This trend has continued very strongly, and as of today economic sociology represents one of the

strongholds of American sociology. In all of these writings by sociologists on the economy, the emphasis has primarily been on middle-range phenomena; and little attempt has been made to analyze capitalism. Why this is the case is difficult to say. One answer might be that capitalism is taken for granted; and this would seem to be especially true for the business schools, where a number of economic sociologists are currently to be found. Another may be that new economic sociologists do not seem to have been very interested in politics – and the concept of capitalism is by tradition often associated with a critique of capitalism.

The contributions to the study of capitalism that one can find in Marxist sociology have, for example, not been much explored by contemporary economic sociologists. If we now turn to the economists, these used to stay away from analyses of capitalism as an economic system and instead focus on the workings of the price system and how this led to an efficient allocation of resources. The word “capitalism” was rarely used by economists during the 20th century till a few decades ago when they suddenly embraced it. Since this time, however, the economists have made quick strides forward. As a result, the leading academic scholars on the subject of capitalism are no longer sociologists but economists – from Friedrich Hayek and Milton Friedman, who started the trend, to Douglass North, Oliver Williamson and others who have continued it. In this opening stage an effort will be made to present an agenda for a sociological study of capitalism.

There are two reasons why this type of study may be called *an economic sociology of capitalism*. First, the main emphasis is not on the social effects of capitalism, but on capitalism as an economic system in its own right – on the firms, the banks, the markets and the other economic institutions that make up the core of the economy. This is where the “economic” in “economic sociology” comes in. Second, while we already have several economic theories of capitalism, we do not have one which sufficiently takes the social dimension of the capitalist machinery into account – and this is where the “sociology” in economic sociology comes in.

A study of capitalism as an economic system should consist of two parts. First of all, studies of individual, middle-range phenomena need to be made. Indeed, this constitutes by far the most important task of an economic sociology of capitalism and cannot be replaced by macro level studies of capitalism. But while studies of capitalism itself should not predominate, they have their own distinct *raison d'être*. One of these is that this type of studies outline the basic connections between the different parts of the economy – how the whole economic process hangs together.

They also show, related to this, how the study of the various parts of the economy should be interrelated – something which has hitherto been missing in economic sociology, but which is very important for its attempt to develop a more systematic approach. In studying each individual part of the capitalist system it is also important to be clear what drives the system as a whole. Finally, as already Adam Smith was well aware of, there exists an overall logic to capitalism as an economic system, which the

individual actors are not aware of. Through the logic of unintended consequences capitalism not only produces individual wealth but also social wealth.

A BASIC MODEL OF CAPITALISM

The reference to Adam Smith leads in a natural way to the next step in this argument, namely to the analytical point of departure for an economic sociology of capitalism. This consists of the proposition that *interests* drive the actions of the individuals, and that interests come together in a very specific way in capitalism. The actors in society are driven by a variety of interests – political, economic, legal and so on. It is important to insist here on the plurality of interests since this makes the analysis realistic as well as flexible. Interests of the same type, as well as of different types, may reinforce each other, counterbalance each other, block each other, and so on. Interests, very importantly, is what supplies *the force* in the economic system – what makes millions of people get up in the morning and work all day.

Interests also explain why banks, financial markets and similar institutions are so powerful: they can mobilize and energize masses of people into action through their control over economic resources. At this point it should be noted that sociologists have often tended to ignore interests and focus exclusively on social relations and the impact that these may have. This exclusive emphasis on social relations can to some extent be explained as the professional myopia of the sociologist. It is matched, in the economic profession, by a similar overemphasis on the purely economic side of things – on economic interests and their effects, *minus* social relations as well as other types of interests. A hardhitting economic sociology would attempt to draw on the best of sociology and economics, and to *unite* interests and social relations in one and the same analysis. Our definition of institutions can be used to exemplify this need for drawing on both interests and social relations in the analysis. Institutions are often defined in sociology – especially in the approach that has been developed at Stanford University – in exclusively social terms, that is, as rules, social constructions and so on. Everything, from this perspective, can be an institution, from a handshake and a dance to the state and the firm.

The individuals with their interests is abstracted away, to make room for a visions of institutions as pure and empty structures which are imitated, duplicated and so on in a fairly effortless manner. In contrast to this approach, institutions will here be understood to mean durable lock-ins or amalgamations of interests *and* social relations. This view is currently being developed at Cornell University but has still some way to go. According to this perspective, the interests of individuals as well as of corporate actors need to be taken into account. A business firm, for example, does not exist unless you also take the capital of the firm into account; similarly a family does not exist unless you also take the emotional interests of its members plus the sexual interest between the male and female into account. There is not only a time dimension to institutions – they tend to last for some time – but a normative element as well: they tell you *how* interests should

be realized in society, be it family interests, political interests, economic interests or some other type of interest. Institutions are of crucial importance to the functioning of society, and a sign of this is that they are typically regulated in law. A basic model for capitalism will now be presented which draws on a mixture of sociology and economics. Our general point of departure is the conventional definition of economics as consisting of production, distribution and consumption.

To cite a wellknown textbook: “Economics is the study of how men and society end up *choosing*, with or without the use of money, to employ *scarce* productive resources which could have alternative uses, to produce various commodities and distribute them for consumption, now or in the future, among various people and groups in society”. This definition describes the economy as a process: all economies start with production, continue with distribution, and end with consumption. Now, all economies can be organized in what amounts to two fundamentally different ways.

Weber expressed this fact with the help of his two categories of “householding” and “profitmaking” —you either produce for consumption or for profit. Marx alluded to the same phenomenon when he spoke of “use value” versus “exchange value”. And so did Aristotle, with his famous distinction between “*oekonomia*” and “*chrematistica*”. The key to the different ways of organizing the economy is to be found in the way that the economic product is *distributed* in the sense of being passed on in the economic process. As the reader will notice I am here departing from the way that the term distribution is often used in economics - as the division of what has been produced - and instead focus on the different social mechanisms through which what has been produced is being passed on. To show the fruitfulness of this approach, it is convenient to refer to Polanyi’s three concepts of *redistribution*, *reciprocity* and *exchange*.

Following Polanyi, it is clear that one way of distributing or passing on what has been produced is through *redistribution*. The agent who does the redistribution is typically the state or some other political authority. The modern socialist state is an example of an economic system that is based on redistribution. Other examples can easily be found, for example, in Antiquity. What has been redistributed is then consumed.

Some part of what has been produced is always set aside for future production; and the size of this part is decided by the political authority. An economy which is primarily based on redistribution is capable of growth – but not of the dynamic type of growth that is characteristic of capitalism. The second way of distributing or passing on what has been produced, according to Polanyi, is through *reciprocity*. This means some equitable form of distribution, as is common in a family or in a kin based economy. Again, some part of what is being produced, is always set aside for future production. But, again, the result of this way of distributing what has been produced, is not going to lead to a dynamic economy. An economy which is based on reciprocity tends towards traditionalism and some form of equity. Only the third way of distributing or passing on production – through *exchange* – can lead to a truly dynamic economic system, with an evergrowing economy. The reason for this is that this system is not exclusively driven

by the eternal human interest in consumption but also by the powerful interest of *profit*. The latter activates people in a very different way from redistribution or reciprocity. And on the assumption that the profit is also reinvested in production, a dynamic economic system – *capitalism* – will come into being. What is unique about capitalism, as compared to economic systems based on redistribution and reciprocity, is that it alone operates on the profit motive. The two most important social mechanisms in capitalism are consequently exchange and the feedback of profit into production.

Complexity is added to the capitalist type of economy by the fact that it also contains several sectors based on reciprocity and redistribution. What can be called *the household economy* is, for example, based on a mixture of redistribution and reciprocity; and so is the sector with non-profit organizations. What can be called *the state economy*, which in modern society accounts for 30-50% of GNP, is on the other hand dominated by redistribution.

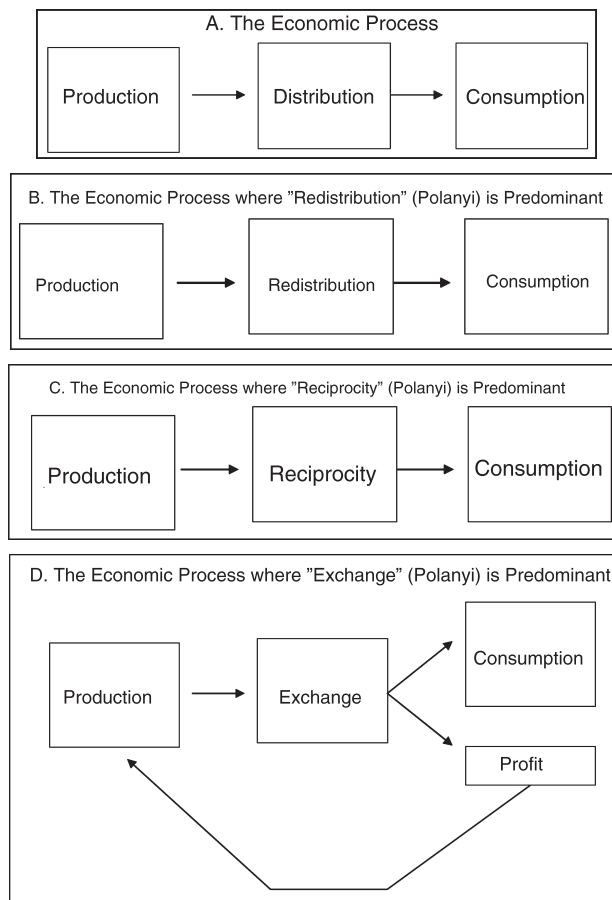


Fig. Capitalism and Other Ways of Organizing the Economic Process and Economic Interests

The rest of this stage is devoted to an attempt to spell out what it would mean for economic sociology to set this model of capitalism at its center. It is clear that this

would have important consequences for what will be seen as the central task of economic. *Comment:* The economic process in any society can be defined as consisting of production, distribution and consumption. The distribution or passing on of what has been produced can be organized in fundamentally three ways; and which of these is chosen will have an enormous impact on the productivity of the economy. Following Polanyi, we may call these *redistribution*, *reciprocity*, and *exchange*.

Exchange characterizes the capitalist organization of the economy; and this type of economy derives its dynamic from the fact that the end goal of the economic process is not exclusively consumption, but also *profit*. The more that this profit is reinvested into production, the more dynamic the economy will be. The two key mechanisms in capitalism, in other words, are *organized exchange* and *the feedback loop of profit into production*. It is the use of these two, it should be stressed, that makes the organization of economic interests in the form of capitalism into such an effective machinery for transforming economic reality. Following this model, the modern capitalist economies can be said to consist of several sectors.

There is, first of all, *the corporate sector*, where exchange dominates. There is also *the non-profit sector*, which is based, among other things, on redistribution. The state accounts for a huge part of GNP and what can be called *the state economy* is primarily based on redistribution. *The household economy* is based on a mixture of redistribution and reciprocity. sociology – namely, to produce studies of *production*, *distribution*, *consumption* and *profit-making*. Added to this are the following three topics: the impact on the economic process by *law*, *politics* and *culture*. For all of these topics it is also imperative to investigate how they can *speed up*, *slow down* or *block* economic growth. Still, our model obviously remains highly simplistic and is in its current shape silent on a number of important economic phenomena, from savings to the dynamics of the business cycle.

Its focus in this stage is on the macroeconomic level of the economic process. The reasoning, on which the model is based, however, may also be used to capture essential aspects of what happens in the economy on the micro- and meso level. Finally, there exist a number of theories of capitalism in social science; and how these can add to the approach that is being presented in this stage will be explored in the next part. Special attention will be paid to the works of Marx, Weber, Schumpeter, Douglass North and advocates of the varieties of capitalism-approach. The stage ends with a discussion of some ways to bring more complexity to the model of capitalism which is advocated here.

THE SOCIOLOGY OF DISTRIBUTION

While the capitalist system consists of three basic processes which are all interdependent and shaped by the fact that they are parts of a dynamic system – production, distribution and consumption – one of these is especially important: *distribution*, in the form of exchange on the market. This represents the reason why it is

preferable to start with distribution rather than with production, which otherwise comes first in the economic process. Once it has been decided to proceed in this manner, it immediately becomes clear that there exists an important precondition for exchange to take place in the first place, namely private property.

From a sociological perspective, Weber explains, property can be conceptualized as a specific form of a closed social relationship. More precisely, it represents a relationship that allows the actor to exclude other actors from the opportunity to use some item or some person. This right is also alienable and can be inherited. Property is typically legally protected, which means that if it is infringed upon, a staff will use coercion to restore it. This view of property is close to the economists' view of property as a collection of enforceable property rights. The main difference is that the element of social relations is given a much more prominent and visible form in the sociological view of property. That there nonetheless exists a basic compatibility between the economic and sociological view of property can be illustrated by the fact that during the last few years a number of sociological studies have appeared, which draw on the notion of property rights. What is crucial about private as opposed to collective property is that the former appeals directly to the individual, and in doing so it activates her in a manner that collective property is unable to do. Some might argue that people should in principle be as motivated by the prospect of acquiring and using collective property as they are by the prospect of acquiring and using private property.

The reason why this is rarely the case, however, has much to do with the free rider problem. It is also very easy for a few individuals to misuse or destroy collective property. Once private property exists, exchange becomes possible. The driving force in an exchange is always that both parties will be better off by trading with each other than by not doing so. Actor A may value her bike at \$50 and Actor B at \$70; and if an exchange takes place both will be better off – and social wealth will have increased by \$20. For an exchange to take place, it is not necessary that one party becomes better off while no-one is worse off. What rather is needed is that both parties become better off by X, without a third party being worse off by more than X, according to the so-called Kaldor-Hicks concept of efficiency.

This latter concept of efficiency is often used in economics because its demands are less stringent than those of Pareto optimality. The reason for referring to it here, however, is that it explains the nature of exchange much better than the idea of Pareto optimality, especially what drives the two parties to engage in an exchange in the first place. Sociologists and economists have developed different approaches to markets – to the role that these play in the economic process, to what is typically regulated in a market, and so on. To economists, markets are primarily processes for price formation, in which the price helps to allocate scarce resources in an efficient manner. By tradition, economists have neglected the institutional dimension of markets, such as rules for exchange, the enforcement machinery and so on. Sociologists, on the other hand, tend to emphasize the role of social relations and institutions in markets. Today's sociologists

will typically analyze the networks which are created by interacting market actors. Weber noted that markets not only consist of repeated acts of exchange, but also of competition among the actors for who will be the one to sell and the one to buy. This idea of “competition for opportunities of exchange” is perfectly compatible with a networks approach, as Ronald Burt has shown in his theory of structural autonomy.

Given the fact that economists and sociologists each hold half of the truth, so to speak, when it comes to markets, it seems natural that they should try to coordinate their efforts. Economists need to better understand the role of the social relations in the market, and sociologists need to better understand how prices are formed and what effect these have on the economy. Prices drive many economic changes in capitalism, as Douglass North has made clear – but they do so via a social structure in which interests are embedded, and where quite a bit else is going on as well. An economic sociology of markets should also study what changes in the exchange mechanism make the capitalist wheel spin faster as well as what slows it down and makes it grind to a halt. According to the theory of transaction costs, lower costs for market deals is a sign of a more efficient exchange mechanism.

This is true indeed, among other reasons since this means that the profit will be larger and that more can be reinvested in production. Lower transaction costs in this context, however, are typically accomplished through changes in social relations and in social mechanisms – and this is where the sociologist can be of help. Take, for example, the clause of *bona fide* or the fact that if the buyer is in good faith it does not matter if the seller did not properly acquire the goods in question. *Bona fide* naturally lowers the transaction costs – but is also a fact of such social complexity that the sociologist, rather than the economist, is best equipped to study it. The same is true for many other forms of trust in economic life. But economic sociology is not only interested in what makes the wheels of capitalism accelerate; there is the equally challenging question of what makes them slow down or grind to a halt. Again, Weber’s work can be used for illustrations.

If bureaucrats in a firm gain power at the expense of the entrepreneur, for example, profit-making will be slowed down since bureaucrats by nature are alien to the idea of profit-making. One important reason for this, Weber says, has to do with the fact that people on a fixed income often find it dishonorable to be swayed exclusively by economic considerations. Similarly, if individual firms and capitalists are not stopped in their attempts to create monopolies, capitalism may wither away because there will be no competition to keep it alive.

Recent scandals in corporate America have also shown how dishonest and false accounting can slow down economic growth and prevent new investments. All in all, the market represents *the* central institution in capitalism. To this should immediately be added that this is only true on condition that most of the production passes through the market. In the great majority of societies throughout history, markets have indeed played a role, but usually a marginal one. It is only since the late 19th century, in

countries such as England and the United States, that the great bulk of production – food, clothes and so on – has been produced in the form of commodities which are exchanged in the market.

By 1790, for example, 80% of all clothing in the United States was made in the home, while a century later 90% was made outside the home. When most of production passes through the market, it can be added, the competition for exchange that Weber speaks about as characteristic of the market will also dominate what happens in the economy *outside* of the market. That is, instead of just bringing a few surplus items to the market, as peasants often did in the Middle Ages, the producers in a modern capitalist economy must start the competition long before they enter the market. When one speaks of a market economy, in other words, what is meant is an economy where the market is not only used for exchange; it also dominates production.

Before leaving the topic of distribution, something also needs to be said about *money* since this is the place in the economic process where money enters into the picture; it is indeed the medium of exchange par excellence. The historical step from barter to exchange with money extended the number of goods that could be exchanged against each other enormously.

Money, more generally, also helps the process exchange to proceed smoothly and lowers the cost of exchange. Many other financial innovations – such as the bill of exchange, certificates of deposit and so on – have similarly helped to lower transition costs and developed in close touch with markets. While money, like any other economic phenomena, has a cultural dimension, it is its place within the overall economic process that is of most interest to economic sociology. In economies based on reciprocity, money often plays a subordinate role since other values than “the cash nexus” decide who should get what. In economies based on redistribution, money is often in use, as recent examples of state socialism are a reminder of.

Political interests, however, dominate the operations of money in this type of societies, and socialist states have usually failed in their attempts to simulate effective market prices. In capitalism, in contrast, money and markets are protected through the existence of “credible commitments” from the political rulers, that is, by assurances of the rulers that they will not intervene in the normal workings of the market. Money, in brief, is allowed to operate “freely” and can therefore help the market to operate more smoothly and cheaply. Money also plays an important role in the capitalist process in the form of capital, that is, as profit generated through exchange. Money and markets, in brief, belong together.

THE SOCIOLOGY OF PRODUCTION

The next major area within the economic sociology of capitalism is that of *production*. No society can live without production, and all production involves social coordination – a sociological element. Nonetheless, an economic sociology of production may want to start from the following wellknown economic premise: that production consists of

combinations of some or all of the traditional factors of production. The sociologist may want to add that all of these factors of production have their distinct sociological profiles - before they enter into production as well as once they interact in the firm. In relation to the basic model of capitalism, the factors of production can be conceptualized as inputs into production.

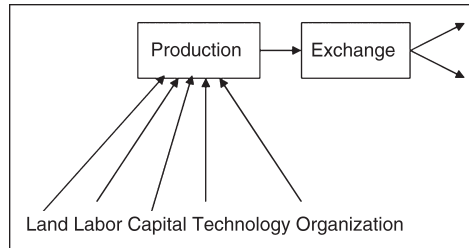


Fig. The Factors of Production in Capitalism.

It should also be emphasized that it is not the organizational form itself that is the sole determinant of productivity. It represents a common error among organization theorists, for example, to think that organizations are what matters the most – just as Marxists think that labour is the key to all production, engineers that technology is the cause of all economic growth, and so on. All of these factors contribute to productivity, individually as well as in combination. This is also where entrepreneurship in its classical Schumpeterian sense comes in. Entrepreneurship is defined by Schumpeter as a new combination for profit making, of already existing factors of production. An innovation may consist of a new commodity or some novel way of lowering the price – and it will result in high profit for the entrepreneur, soon to be followed by competition, overinvestment and lower profit. This way, according to Schumpeter, a business cycle is set off and played out in the economy.

FACTOR OF PRODUCTION NO.1: LABOUR

When factors of production were discussed in the 19th century, land was usually assigned a prominent place. In today's capitalism, however, land is of much less importance, and the average person is not dependent on working the land for her livelihood. Land as a factor of production will therefore be passed over in this stage, and I shall proceed directly to labour, which has retained its importance for the process of production. Labour as a modern factor of production typically passes through two stages. There is the first stage which takes place *before* labour enters into production, and this is followed by the stage once labour is inside the corporation. Two institutions which are central to labour, before it enters into production, are the home and the school. In the home children learn values, discipline and how to interact. They also get to live in a household economy and become influenced by its values.

In school, labour is taught various skills, some of which are of value in production, from reading, writing and elementary mathematics in school to engineering, computer programming and nuclear physics at the university. According to a well-known strand

of sociological theory, when labour is distributed via the market it tends to form into large and loose groups of people with similar economic interests. When there is no market, or the market is being controlled, say, by a profession, labour instead tends to form into small and cohesive groups centered around honour and consumption. Whatever the exact relationship is between class and status, on the one hand, and labour, on the other, it is imperative for economic sociology to attempt to theorize the link between the economic process, on the one hand, and the formation of groups in society, on the other. This is where economic sociology needs to connect with stratification theory – and perhaps also where stratification theory can get some inspiration from economic sociology.

In modern capitalist society labour tends to form into three broad categories: Workers, professionals and managers. Sociology is by tradition skillful in tracing the structure of groups as well as the mentalities of their members, while economists often tend to use a non-differentiated concept of labour and emphasize the crucial role played by the productivity of the worker. Again, it would appear that both sociology and economics can benefit from the insights of the other science. To what extent can labour, before it enters into the process of production, add to, slow down or block the wheels of capitalism? What happens in the home as well as in the school, in terms of creation of values and skills, is clearly of great relevance for an answer to this question, and it will also feed into the formation of status groups and/or classes.

Status groups, Weber argues, are inherently anti-capitalist since they set honour and other non-economic values before profit-making. Status groups are antagonistic to the market since the market disregards the values that its members hold dear. The more that labour feels the impact of the market, on the other hand, the more individual actors will accept its logic: the need for efficiency, profit-making and constant reinvestment.

Factor of Production No. 2: Capital

Economists pay by tradition much attention to the role of capital in the process of production; while sociologists, if they study capital at all, tend to analyze its role outside of production, in the form of so-called wealth. Again, the two approaches may want to draw on each other's insights in order to get a full picture of what is going on. To this can be added that the groups in society who control the economic resources have different attitudes to wealth - how it should be acquired, what it can be used for, and so on. Aristocrats, for example, have traditionally had contempt for merchants and so have warriors.

There is also the fact that some groups of merchants take larger risks than others; and this will have an important impact on the generation of wealth and capital. Merchants furthermore deal in different types of goods, as exemplified by the historical appearance of *the businessman* – a term that was first used in the U.S. in the 1830s to denote a new type of merchant, who not only traded in goods but also in land and whatever could result in a profit. This type of phenomena can be explored with the help of sociology.

Control over capital is often delegated by the owner to some other actor, and this creates a well-known problem for the owner. A flexible type of analysis that economists use to handle this type of situation is agency theory, which is based on the idea that the owner has a different interest than the one to whom she assigns some specific task.

This means that something has to be done about this divergence of interest. One solution is direct observation of the agent, another to give the agent an incentive to act in the interest of the owner. The former is less easy to carry out when it comes to managers, as opposed to workers; there is also the further problem of “who will monitor the monitor”. Agency theory can enrich economic sociology by adding to its analyses of capital, especially when it comes to the question of how the owner can maintain control over her capital, once a manager is in place.

According to Harrison White, for example, the advantage with using agency theory is that “it is intensely social in its mechanisms, since it gets one person to do something for another vis-à-vis a third person but only with heavy reliance on the lay of the social landscape”. James Coleman has a similarly positive view of the sociological potential of agency theory, as is evidenced by the following quote from *Foundations of Social Theory*: “once a transaction has been made, in which the principal satisfies interests of the agent in return for the agent’s using his actions to pursue the principal’s interests, a social system has been created”. A corporation can acquire capital in a variety of ways – from banks, venture capitalists, the capital market, and so on.

Each of these institutions have their own distinct social structure and history which sociology can help to analyze. Pension funds and other mutual funds which have become key players in the modern capital market, are often managed by single individuals; and very little is currently known about these. Agency theory, in combination with economic sociology, represents one way of approaching this type of issues. The way that capital is brought to production will obviously affect the generation of economic value. Risk taking is a crucial factor at this point of the process, and closely related to the profit level. But risk taking is also directly affected by social relations, as the historical emergence of venture capital in the United States a few decades ago illustrates.

What characterizes venture capitalists is an intimate knowledge of the business in which they invest, often in combination with some form of control over the firm that is being targeted. Together, these two measures make risk taking more manageable – and thereby also increase the chance to make a high profit. Again, this is a topic where economic sociology can be of assistance.

Factor of Production No. 3: Technology

Technology is of crucial importance to the capitalist process, primarily because it helps to increase productivity. Exactly how this is done, however, is something that neither economists nor sociologists understand very well. The concept of productivity is in need of much elucidation. Economists realize the great importance of technology in this context, but have difficulty in theorizing it. A sign of this is the discussion about

the role of computers in the economic growth in the United States in the 1990s. “You can see the computer age everywhere but in the productivity statistics”, to quote a famous line by Robert Solow. Another difficulty with the economists’ view of technology is that they often see technology as the one and only reason for growth in productivity. While innovations in technology may well be *the* major reason for growth in productivity in modern capitalism, it is by no means the only one. Social organization, in particular, also affects productivity, a fact that industrial sociology made clear many decades ago. Sociologists differ from economists in that they rarely note that technology is of great importance to productivity as well as for the generation of profits.

Sociologists of science of the old school view science primarily as a public good, which may once have been true but is less so today. Modern sociologists of science, on the other hand, argue that science and technology essentially are to be understood as forms of social construction, which may well be true - but is of limited relevance to an understanding of the role of science and technology in the economic process. From historians of technology we know that economically relevant technology for a long time emerged in a slow, evolutionary manner – as evidenced by the history of boat, the ax, the plow, and so on. At the time of the Industrial Revolution, and even more so during the second half of the 19th century, however, a historical meeting took place between capitalism and science.

This alliance has continued till today and has become ever more important to the dynamic growth of capitalism. It has often been pointed out that social organization can accelerate or impede the emergence of new technology, which in its turn will affect the economy. In his study of religions in India, for example, Weber notes that the caste system blocked innovations by forbidding changes in the tools of the artisans. Since the penalty for a change of this type was religious, Weber’s example also illustrates how a religious interest can be used to block an economic interest. In today’s society, to use a more contemporary example, we are witnessing an important change in the property rights to science, which has helped to speed up production. While science until recently was seen as a common good, ways are now increasingly found to turn it into a private good. The forces that have caused this change are obvious enough. A new pharmaceutical drug can, for example, be worth billions of dollars in profit. It currently also costs around 800 million dollars to develop a new drug.

Factor of Production No.4: Organization (Marshall)

Alfred Marshall sensed the limits to the economics of his days and argued, in *Principles of Economics*, that not only land, labour and capital should be considered factors of production but also “organization”. By organization Marshall meant a number of phenomena, including the individual firm as well as a dynamic collection of firms in the same geographical area, which he termed “industrial district”. The insight that organization is crucial to profit-making is also at the heart of what is known as organizational economics, which draws on a mixture of agency theory, game theory,

transaction cost analysis, and law and economics – but not on sociology. Also sociologists have developed a series of conceptual tools that can be used to analyze the way that the factors of production come together in the profit-making firm.

Sociologists, however, are to a certain extent held back from making the contribution they should be able to make by their fixation on the stance of organization theory that the central unit of analysis is the generic organization and not the corporation. Research on firms is typically translated into knowledge about organizations in general.

The following facts about the modern firm are typically ignored in organization theory, especially as it can be found among sociologists:

- That the firm has its own institutional history;
- That the firm is treated differently from other organizations in laws and regulations; and
- That firms in modern society control more economic resources than any other type of organization, except for the state.

Regardless of this critique, it is clear that several of the tools that have been developed in organization theory can be of considerable help in analyzing corporations; and to some extent they have already been used for this purpose. This, for example, is true for population ecology as well as for resource dependency. Networks also represents a helpful tool in tracing the relations between corporations that emerge as a result of their attempts to make a profit. There is furthermore the insight that work groups can develop norms that go counter to the goals of the corporation, so-called opposition norms.

It is obvious that the way a corporation is organized will speed up, slow down or block profit-making. What was once thought to represent the ideal design for a firm – the huge, bureaucratically organized firm with much of the knowledge concentrated at the top —is today much criticized. It is indeed true that new technologies as well as new ways of appealing to the interests of the employees can replace monitoring with interest alignment, and that this has led to changes in the old type of corporation. The point, however, is not to create a decentralized or a less formal corporation *per se*, as organization theorists are prone to argue, but to do what it takes to make a profit.

THE SOCIOLOGY OF CONSUMPTION

Consumption, to cite *The Wealth of Nations*, represents the end product of production: “consumption is the sole end and purpose of all production”. From the viewpoint of the model of capitalism that has been presented in this stage, however, things are not that simple. For one thing, how the end result of production is divided between consumption and profit is of crucial importance. The more profit that is taken out by the owners and fed back into production, the faster the wheels of capitalism will spin.

There is also the fact that consumption will affect the productivity of labour. We can imagine a line that goes from consumption to production, via labour as a factor of production. Adequate food and some amount of leisure, leading to healthy minds and healthy bodies, are examples of this. Education that is paid by private means would be

another. But even if consumption may have an indirect effect on production, its main contribution to capitalism is obviously that it takes place in the first place. Given that human beings have to satisfy their material needs, this may sound like a triviality. And so it is - except that consumption always has to increase in capitalism, in contrast to economies based on a redistribution or reciprocity. If this does not happen, profit will stagnate and capitalism will lose its vitality.

This means that efforts always have to be made, as part of the process of *production*, to encourage consumption as much as possible. In modern capitalist society whole settings in the form of shopping centers and the like have been created, precisely for this purpose. These have aptly been called “means of consumption”. Consumption can be speeded up, slowed down or blocked through the impact of various forces - and thereby affect the capitalist machinery. The United States, for example, has always been a commercial society, with a population with a strong desire for democratic “comfort”, as opposed to aristocratic “luxury”; and this clearly greases the wheels of capitalism.

After September 11 – to use another example from the United States – shopping was nearly proclaimed a patriotic duty so that the economy would not slump. Examples also exist of societies which have tried to block consumption. One famous example is Florence in the 15th century, when the city was ruled by Savonarola who staged the famous “bonfires of vanities” – that is, public burnings of expensive dresses, sensual paintings, and the like which were judged to detract from a pious life.

THE DEFINITION OF ECONOMIC SOCIOLOGY

Economic sociology—to use a term that Weber and Durkheim introduced—can be defined simply as the sociological perspective applied to economic phenomena. A similar but more elaborate version is the application of the frames of reference, variables, and explanatory models of sociology to that complex of activities which is concerned with the production, distribution, exchange, and consumption of scarce goods and services. One way to make this definition more specific is to indicate the variables, models, and so on, that the economic sociologist employs. When Smelser first put forth that definition, he mentioned the sociological perspectives of personal interaction, groups, social structures, and social controls. Given recent developments, we would add that perspectives of social networks, gender, and cultural contexts have also become central in economic sociology. In addition, the international dimension of economic life has assumed greater salience among economic sociologists, at the same time as that dimension has come to penetrate the actual economies of the contemporary world.

MAINSTREAM ECONOMICS AND ECONOMIC SOCIOLOGY COMPARED

We now compare economic sociology and mainstream economics as a way of further elucidating the sociological perspective on the economy. This is a useful exercise only if qualified by the caution that both bodies of inquiry are much more complex than any

brief comparison would suggest. Any general statement almost immediately yields an exception or qualification.

To illustrate the caution on each side of the comparison:

- In economics the classical and neoclassical traditions have enjoyed a certain dominance—hence the label mainstream—but the basic assumptions of those traditions have been modified and developed in many directions. In a classic statement, Knight stressed that neoclassical economics rested on the premises that actors have complete information and that information is free. Since that time economics has developed traditions of analysis based on assumptions of risk and uncertainty and information as a cost. In addition, numerous versions of economic rationality—for example, Simon’s emphasis on “satisficing” and “bounded rationality”—have appeared. Still other variations on rational behaviour have been developed in behavioural economics, which incorporates many psychological assumptions at variance with the mainstream. Looking in the direction of sociology, some economics now incorporates “norms” and “institutions,” though with meanings different from those found in the sociological tradition.
- Sociology lacks one dominant tradition. Various sociological approaches and schools differ from and compete with one another, and this circumstance has affected economic sociology. For example, Weber was skeptical about the notion of a social “system,” whether applied to economy or society, while Parsons viewed society as a system and economy as one of its subsystems. Furthermore, even if all economic sociologists might accept the definition of economic sociology we have offered, they focus on different kinds of economic behaviour. Some do so following Arrow’s hint that sociologists and economists ask different questions—about consumption, for example. Others, including what is called new economic sociology, argue that sociology should concentrate directly on core economic institutions and problems.

These caveats recorded, a comparison between the central features of mainstream economics and economic sociology will clarify the specific nature of the sociological perspective. The following differences are most salient.

THE CONCEPT OF THE ACTOR

To put the matter baldly, the analytic starting point of economics is the individual; the analytic starting points of economic sociology are typically groups, institutions, and society. In microeconomics, the individualistic approach finds its origins in early British utilitarianism and political economy. This orientation was elucidated systematically by the Austrian economist Carl Menger and given the label methodological individualism by Schumpeter. By contrast, in discussing the individual, the sociologist often focuses on the actor as a socially constructed entity, as “actor-in-interaction,” or “actor-in-society.” Often, moreover, sociologists take the group and the

social-structural levels as phenomena *sui generis*, without reference to the individual actor. Methodological individualism need not be logically incompatible with a sociological approach. In his theoretical stage introductory to *Economy and Society*, Weber constructed his whole sociology on the basis of individual actions. But these actions are of interest to the sociologist only insofar as they are social actions or “take account of the behaviour of other individuals and thereby are oriented in their course”.

This formulation underscores a second difference between microeconomics and economic sociology: the former generally assumes that actors are not connected to one another; the latter assumes that actors are linked with and influence one another. We argue that this difference has implications for how economies function.

THE CONCEPT OF ECONOMIC ACTION IN SOCIOLOGY

In microeconomics the actor is assumed to have a given and stable set of preferences and to choose that alternative line of action which maximizes utility. In economic theory, this way of acting constitutes economically rational action. Sociology, by contrast, encompasses several possible types of economic action. To illustrate from Weber again, economic action can be either rational, traditional, or affectual. Except for residual mention of “habits” and “rules of thumb,” economists give no place to traditional economic action.

Another difference between microeconomics and economic sociology in this context concerns the scope of rational action. The economist traditionally identifies rational action with the efficient use of scarce resources. The sociologist’s view is, once again, broader. Weber referred to the conventional maximization of utility, under conditions of scarcity, as formal rationality. In addition, however, he identified substantive rationality, which refers to allocation within the guidelines of other principles, such as communal loyalties or sacred values.

A further difference lies in the fact that economists regard rationality as an assumption, whereas most sociologists regard it as a variable. For one thing, the actions of some individuals or groups may be more rational than others. Along the same lines, sociologists tend to regard rationality as a phenomenon to be explained, not assumed. Weber dedicated much of his economic sociology to specifying the social conditions under which formal rationality is possible, and Parsons argued that economic rationality was a system of norms—not a psychological universal—associated with specific developmental processes in the West. Another difference emerges in the status of meaning in economic action. Economists tend to regard the meaning of economic action as derivable from the relation between given tastes, on the one hand, and the prices and quantities of goods and services, on the other.

Weber’s conceptualization has a different flavour: “the definition of economic action [in sociology] must . . . bring out the fact that all ‘economic’ processes and objects are

characterized as such entirely by the meaning they have for human action". Meanings are historically constructed and must be investigated empirically, and are not simply to be derived from assumptions and external circumstances. Finally, sociologists tend to give a broader and more salient place to the dimension of power in economic action. Weber insisted that "[it] is essential to include the criterion of power of control and disposal in the sociological concept of economic action," adding that this applies especially in the capitalist economy.

By contrast, microeconomics has tended to regard economic action as an exchange among equals, and has thus had difficulty in incorporating the power dimension. In the tradition of perfect competition, no buyer or seller has the power to influence price or output. It is also true that economists have a tradition of analyzing imperfect competition—in which power to control prices and output is the core ingredient—and that the idea of "market power" is used in labour and industrial economics. Still, the economic conception of power is typically narrower than the sociologist's notion of economic power, which includes its exercise in societal, as well as market, contexts. In a study of the power of the U.S. banking system, for example, Mintz and Schwartz analyze how banks and industries interlock, how certain banks cluster into groups, and how banks sometimes intervene in corporations in order to enforce economic decisions. More generally, sociologists have analyzed and debated the issue of the political implications of wealth inequality and the extent to which corporate leaders constitute a "power elite" in the whole of society.

CONSTRAINTS ON ECONOMIC ACTION

In mainstream economics, actions are constrained by tastes and by the scarcity of resources, including technology. Once these are known, it is in principle possible to predict the actor's behaviour, since he or she will always try to maximize utility or profit. The active influence of other persons and groups, as well as the influence of institutional structures, is set to one side. Knight codified this in the following way: "Every member of society is to act as an individual only, in entire independence of all other persons". Sociologists take such influences directly into account in the analysis of economic action.

Other actors facilitate, deflect, and constrain individuals' action in the market. For example, a friendship between a buyer and a seller may prevent the buyer from deserting the seller just because an item is sold at a lower price elsewhere. Cultural meanings also affect choices that might otherwise be regarded as "rational." In the United States, for example, it is difficult to persuade people to buy cats and dogs for food, even though their meat is as nutritious and cheaper than other kinds.

Moreover, a person's position in the social structure conditions his or her economic choices and activity. Stinchcombe evoked the principle that structural constraints influence career decisions in ways that run counter to considerations of economic payoff. For example, for a person who grows up in a high-crime neighbourhood, the choice

between making a career stealing and getting a job has often less to do with the comparative utility of these two alternatives than with the structure of peer groups and gangs in the neighbourhood.

THE ECONOMY IN RELATION TO SOCIETY

The main foci for the mainstream economist are economic exchange, the market, and the economy. To a large extent, the remainder of society lies beyond where the operative variables of economic change really matter. Economic assumptions typically presuppose stable societal parameters. For example, the long-standing assumption that economic analysis deals with peaceful and lawful transactions, not with force and fraud, involves important presuppositions about the legitimacy and the stability of the state and the legal system. In this way the societal parameters—which would surely affect the economic process if the political legal system were to disintegrate—are frozen by assumption, and thus are omitted from the analysis.

In recent times economists have turned to the analysis of why institutions arise and persist, especially in the new institutional economics and game theory. They have varied the effects of institutional arrangements in various logical experiments. Nevertheless, the contrast with economic sociology remains.

When economists talk about institutions, norms, and the like, their vocabulary is identical to that of sociologists, but they often mean something quite different. It is still very common, for example, for economists to treat the economic arena as lacking norms and institutions. The latter only emerge when markets cannot be constructed or when traditional rational choice analysis fails. Economic sociology, on the other hand, has always regarded the economic process as an organic part of society.

As a consequence, economic sociology has usually concentrated on three main lines of inquiry:

- The sociological analysis of economic process;
- The analysis of the connections and interactions between the economy and the rest of society; and
- The study of changes in the institutional and cultural parameters that constitute the economy's societal context.

GOALS OF ANALYSIS

As social scientists, both economists and sociologists try to explain phenomena encompassed by their respective subject matters. Within this common interest, however, different emphases emerge. Economists tend to be critical of descriptions—they condemn traditional institutional economics as too descriptive and atheoretical. Instead they stress the importance of prediction. Sociologists, by contrast, offer fewer formal predictions, and often find sensitive and telling descriptions both interesting in themselves and essential for explanation. As a result of these differences, sociologists often criticize economists for generating formal and abstract models and ignoring

empirical data, and economists reproach sociologists for their “post factum sociological interpretations”.

Though these differences have become part of the professional cultures of economists and sociologists, it should be noted that the last 10 years have seen a new interest for model building and game theory among sociologists, and a new interest in culture and use of empirical material among economists. It is also possible that the fields of economics and economic sociology may one day agree on some methodological compromise, say along the lines of “analytic narratives”.

MODELS EMPLOYED

The emphasis on prediction constitutes one reason why mainstream economists place such high value on expressing hypotheses and models in mathematical form. Though the advantages of this formal theorizing are readily apparent, economists themselves have at times complained that it tends to become an end in itself. In his presidential address to the American Economic Association in 1970, Wassily Leontief criticized his profession’s “uncritical enthusiasm for mathematical formulation”.

When economists do turn to empirical data, they tend to rely mainly on those generated for them by economic processes themselves. Sample surveys are occasionally used, especially in consumer economics and in labour economics; archival data are seldom consulted, except by economic historians; and ethnographic work is virtually nonexistent. By contrast, sociologists rely heavily on a great variety of methods, including analyses of census data, independent survey analyses, participant observation and fieldwork, and the analysis of qualitative historical and comparative data.

INTELLECTUAL TRADITIONS

Sociologists not only rely on different intellectual traditions that overlap only slightly, but they also regard those traditions differently. Evidently influenced by the natural science model of systematic accumulation of knowledge, economists have shown less interest than sociologists in study and exegesis of their classics. Correspondingly, economics reveals a sharp distinction between current economic theory and the history of economic thought.

In sociology these two facets blend more closely. The classics are very much alive, and are often required reading in theory courses. Despite these differences, and despite the persisting gulf between the traditions of economics and economic sociology, some evidence of synthesis can be identified. Major figures such as Alfred Marshall, Vilfredo Pareto, and Talcott Parsons have attempted theoretical syntheses.

Certain other figures, notably Weber and Schumpeter, have excited interest among both economists and sociologists. In addition, economists and sociologists find it profitable to collaborate in specific problem areas such as poverty and labour markets. Later in the stage as suggested, reraise the issue of intellectual articulation among economists and sociologists.

ECONOMIC SOCIOLOGY IN TRADITION

There exists a large and rich tradition of economic sociology, which roughly begins around the turn of the twentieth century. This tradition has generated both important concepts and ideas and significant research results, which we now present and set in perspective. Economic sociology has peaked twice since its birth: in 1890–1920 with the classic theorists and today, from the early 1980s onwards. A small number of important works in economic sociology—by economists as well as sociologists—were also produced during the period in between. A major thread in the tradition of economic sociology is that investigation must combine the analysis of economic interests with an analysis of social relations.

CLASSICAL ECONOMIC SOCIOLOGY AND ITS PREDECESSORS

The first use of the term economic sociology seems to have been in 1879, when it appears in a work by British economist W. Stanley Jevons. The term was taken over by the sociologists and appears, for example, in the works of Durkheim and Weber during the years 1890–1920. It is also during these decades that classical economic sociology is born, as exemplified by such works as *The Division of Labour in Society* by Durkheim, *The Philosophy of Money* by Simmel, and *Economy and Society* by Weber.

These classics of economic sociology are remarkable for the following characteristics. First, Weber and others shared the sense that they were pioneers, building up a type of analysis that had not existed before. Second, they focused on the most fundamental questions of the field: What is the role of the economy in society? How does the sociological analysis of the economy differ from that of the economists? What is an economic action?

To this should be added that the classical figures were preoccupied with understanding capitalism and its impact on society—"the great transformation" that it had brought about. In hindsight it is clear that several works published before the 1890–1920 period in one way or another prefigure some of the insights of economic sociology. Important reflections on, for example, the role of trade can be found in *The Spirit of the Laws* by Montesquieu, as well as a pioneer comparative analysis of the role of various economic phenomena in republics, monarchies, and despotic states. The role of labour in society is emphasized in the work of Saint-Simon, who also helped to popularize the term industrialism. That the work of Alexis de Tocqueville is full of sharp, sociological observations is something that most sociologists would agree on. That he also made contributions to economic sociology is, however, less known. Of these various precursors as suggested, concentrate only on Karl Marx, a towering figure in nineteenth century thought, even though he was active before the birth of modern sociology.

Karl Marx

Karl Marx was obsessed with the role of the economy in society and developed a theory according to which the economy determined society's general evolution. What

drives people in their everyday lives, Marx also argued, are material interests, and these also determine the structures and processes in society. While Marx wanted to develop a strictly scientific approach to society, his ideas were equally infused by his political desire to change the world. The end result was what we know as “Marxism”—a mixture of social science and political statements, welded into a single doctrine.

For a variety of reasons much of Marxism is erroneous or not relevant to economic sociology. It is far too tendentious and dogmatic to be adopted as a whole. The task that confronts economic sociology today is to extract those aspects of Marxism that are useful. In doing so, it is useful to follow the suggestion of Schumpeter, and distinguish between Marx as a sociologist, Marx as an economist, and Marx as a revolutionary.

We now turn to a preliminary effort to pull out the relevant ingredients for economic sociology. Marx’s point of departure is labour and production. People have to work in order to live, and this fact is universal. Material interests are correspondingly universal. Labour is social rather than individual in nature, since people have to cooperate in order to produce. Marx severely criticized economists for their use of the isolated individual; and he himself sometimes spoke of “social individuals”. The most important interests are also of a collective nature—what Marx calls “class interests.” These interests will, however, only be effective if people become aware that they belong to a certain class. Marx severely criticized Adam Smith’s idea that individual interests merge and further the general interest of society. Rather, according to Marx, classes typically oppress and fight each other with such ferocity that history is as if written with “letters of blood and fire”. Bourgeois society is no exception on this score since it encourages “the most violent, mean and malignant passions of the human heart, the Furies of private interest”.

In various works Marx traced the history of the class struggle, from early times into the future. In a famous formulation from the 1850s, Marx states that at a certain stage the “relations of production” enter into conflict with “the forces of production,” with revolution and passage to a new “mode of production” as a result. In *Capital* Marx writes that he has laid bare “the economic law of motion of modern society” and that this law works “with iron necessity towards inevitable results” of revolutionary change. A positive feature of Marx’s approach is his insight into the extent to which people have been willing to fight for their material interests throughout history.

He also contributed to understanding how large groups of people, with similar economic interests, under certain circumstances can unite and realize their interests. On the negative side, Marx grossly underestimated the role in economic life of interests other than the economic ones. His notion that economic interests in the last hand always determine the rest of society is also impossible to defend; “social structures, types and attitudes are coins that do not readily melt,” to cite a famous quote from Schumpeter.

Max Weber

Among the classics in economic sociology Max Weber occupies a unique place. He proceeded furthest towards developing a distinct economic sociology, laying its

theoretical foundation and carrying out empirical studies. The fact that he had worked as a professor of economics was no doubt helpful in these efforts to build bridges between economics and sociology. Also helpful was the major research task that occupied Weber throughout his career, which was economic as well as social in nature: to understand the origin of modern capitalism. Weber drew heavily on the theoretical work on interests of his time and extended that line of work by making it more sociological. Weber's academic training was broad in nature, and its main emphasis was on law, with the history of law as his specialty.

His two dissertations—one on medieval trading corporations and the other on the sale of land in early Rome—were relevant topics for understanding the rise of capitalism: the emergence of private property in land and of property in the firm. Those works, in combination with a commissioned study of rural workers, earned him a position in economics in the early 1890s. In this capacity he taught economics but published mainly in economic history and in policy questions. Weber wrote, for example, voluminously on the new stock exchange legislation.

Towards the end of the 1890s Weber fell ill, and for the next 20 years he worked as a private scholar. In these years he produced his most celebrated study, *The Protestant Ethic and the Spirit of Capitalism* as well as studies of the economic ethics of the world religions. In 1908 Weber accepted a position as chief editor of a giant handbook of economics. From the very beginning Weber set aside the topic of “economy and society” for himself. The work that today is known as *Economy and Society* consists of a mixture of material that Weber had approved for publication and of manuscripts found after his death. In 1919–20 Weber also taught a course in economic history, which, pieced together a few years later on the basis of students' notes, was published posthumously as *General Economic History*. Though primarily a work in economic history, it contains much interesting material for the economic sociologist. Much of what Weber wrote in economic sociology can be found in *Collected Thesis in the Sociology of Religion and Economy and Society*.

The former contains a revised version *The Protestant Ethic*, “*The Protestant Sects and the Spirit of Capitalism*” and voluminous writings on the economic ethics of the Chinese, Indian, and Judaic world religions and a few other texts. According to Weber, the material in *Collected Thesis* concerns mainly the sociology of religion but is also of interest to economic sociology. The most influential study is *The Protestant Ethic*. This work is centered around Weber's general preoccupation with the articulation of ideal and material interests and ideas.

The believer in ascetic Protestantism is driven by a desire to be saved and acts accordingly. For various paradoxical reasons the individual eventually comes to believe that secular work, carried out in a methodical manner, represents a means to salvation—and when this happens, religious interest is combined with economic interest. The result of this combination is a release of a tremendous force, which shattered the traditional and antieconomic hold of religion over people and introduced a mentality favourable to

capitalist activity. The thesis in *The Protestant Ethic* has led to an enormous debate, with many scholars—probably a majority—arguing against Weber.

While he was writing *The Protestant Ethic* Weber published an thesis, “‘Objectivity’ in Social Science and Social Policy,” that summarized his theoretical views on economic sociology. In this work he argued that the science of economics should be broad and umbrella-like. It should include not only economic theory but also economic history and economic sociology. Weber also proposes that economic analysis should cover not only “economic phenomena” but also “economically relevant phenomena” and “economically conditioned phenomena”.

Economic phenomena consist of economic norms and institutions, often deliberately created for economic ends—for example, banks and stock exchanges. Economically relevant phenomena are noneconomic phenomena that under certain circumstances may have an impact on economic phenomena, as in the case of ascetic Protestantism. Economically conditioned phenomena are those that to some extent are influenced by economic phenomena. The type of religion that a group feels affinity for is, for example, partly dependent on the kind of work that its members do. While economic theory can only handle pure economic phenomena economic history and economic sociology can deal with all three categories of phenomena. A somewhat different approach, both to economic sociology and to interests, can be found in *Economy and Society*. The first stage of this work contains a general sociological analysis.

Two concepts are important building blocks: “social action” and “order”. In the former, “action,” defined as behaviour invested with meaning, is qualified as “social” if it is oriented to some other actor. An “order” is roughly equivalent to an institution, and it comes into being when social actions are repeated over a period, regarded as objective, and surrounded by various sanctions. Economists study pure economic action, which is action exclusively driven by economic interests.

Economic sociologists, however, study social economic action, which is driven not only by economic interest but also by tradition and emotions; furthermore, it is always oriented to some actor. If one disregards single actions, Weber says, and instead focuses on empirical uniformities, it is possible to distinguish three different types: those inspired by “convention,” by “custom” and by “interest”. Most uniform types of action presumably consist of a mixture of all three. Actions that are “determined by interest” are defined by Weber as instrumental in nature and oriented to identical expectations.

An example would be the modern market, where each actor is instrumentally rational and counts on everybody else to be so as well. Weber emphasized that interests are always subjectively perceived; no “objective” interests exist beyond the individual actor. In a typical sentence Weber speaks of “[the] interests of the actors as they themselves are aware of them”.

He also notes that when several individuals behave in an instrumental manner in relation to their individual interests, the typical result is collective patterns of behaviour that are considerably more stable than those driven by norms imposed by an authority.

It is, for example, very difficult to make people do something economic that goes against the individual's interest. A sketch of Weber's economic sociology in *Economy and Society* yields the following main points. Economic actions of two actors who are oriented to one another constitute an economic relationship. These relationships can take various expressions, including conflict, competition, and power.

If two or more actors are held together by a sense of belonging, their relationship is "communal"; and if they are held together by interest, "associative". Economic relationships can also be open or closed. Property represents a special form of closed economic relationship. Economic organizations constitute another important form of closed economic relationships. Some of these organizations are purely economic, while others have some subordinate economic goals or have as their main task the regulation of economic affairs. A trade union is an example. Weber attaches great importance to the role in capitalism of the firm, which he sees as the locus of entrepreneurial activity and as a revolutionary force. A market, like many other economic phenomena, is centered around a conflict of interests—in this case between sellers and buyers. A market involves both exchange and competition.

Competitors must first fight out who will be the final seller and the final buyer and only when this struggle has been settled is the scene set for the exchange itself. Only rational capitalism is centered around the modern type of market. In so-called political capitalism the key to profit making is rather the state or the political power that grants some favour, supplies protection, or the like. Traditional commercial capitalism consists of small-scale trading, in money or merchandise. Rational capitalism has emerged only in the West.

Émile Durkheim

As compared to Weber, Émile Durkheim knew less economics, wrote less about economic topics, and in general made less of a contribution to economic sociology. While none of his major studies can be termed a work in economic sociology, all of them nonetheless touch on economic topics. Durkheim also strongly supported the project of developing a *sociologie économique* by encouraging some of his students to specialize in this area and by routinely including a part on economic sociology in his journal *L'année sociologique*.

At one point he gave the following definition of economic sociology:

- Finally there are the economic institutions: institutions relating to the production of wealth institutions relating to exchange, institutions relating to distribution. They form the subject matter of economic sociology.

Durkheim's first major work, *The Division of Labour in Society* has most direct relevance for economic sociology. Its core consists of the argument that social structure changes as society develops from its undifferentiated state, in primordial times, to a stage characterized by a complex division of labour, in modern times. Economists, Durkheim notes, view the division of labour exclusively as an economic phenomenon,

and its gains in terms of efficiency. What he added was a sociological dimension of the division of labour—how it helps to integrate society by coordinating specialized activities.

As part of society's evolution to a more advanced division of labour, the legal system changes. From being predominantly repressive in nature, and having its center in penal law, it now becomes restitutive and has its center in contractual law. In discussing the contract, Durkheim also described as an illusion the belief, held by Herbert Spencer, that a society can function if all individuals simply follow their private interests and contract accordingly. Spencer also misunderstood the very nature of the contractual relationship.

A contract does not work in situations where self-interest rules supreme, but only where there is a moral or regulative element. "The contract is not sufficient by itself, but is only possible because of the regulation of contracts, which is social in origin". A major concern in *The Division of Labour in Society* is that the recent economic advances in France may destroy society by letting loose individual greed to erode its moral fibre. This problematic is often cast in terms of the private versus the general interest, as when Durkheim notes that "subordination of the particular to the general interest is the very well-spring of all moral activity". Unless the state or some other agency that articulates the general interest steps in to regulate economic life, the result will be "economic anomie," a topic that Durkheim discusses in *Suicide*. People need rules and norms in their economic life, and they react negatively to anarchic situations.

In many of Durkheim's works, one finds a sharp critique of economists; and it was Durkheim's conviction in general that if economics was ever to become scientific, it would have to become a branch of sociology. He attacked the idea of *homo economicus* on the ground that it is impossible to separate out the economic element and disregard the rest of social life. The point is not that economists used an analytical or abstract approach, Durkheim emphasized, but that they had selected the wrong abstractions. Durkheim also attacked the nonempirical tendency of economics and the idea that one can figure out how the economy works through "a simple logical analysis".

Durkheim referred to this as "the ideological tendency of economics". Durkheim's recipe for a harmonious industrial society is as follows: each industry should be organized into a number of corporations, in which the individuals will thrive because of the solidarity and warmth that comes from being a member of a group. He was well aware of the rule that interest plays in economic life, and in *The Elementary Forms of Religious Life* he stresses that "the principal incentive to economic activity has always been the private interest". This does not mean that economic life is purely self-interested and devoid of morality: "We remain [in our economic affairs] in relation with others; the habits, ideas and tendencies which education has impressed upon us and which ordinarily preside over our relations can never be totally absent". But even if this is the case, the social element has another source other than the economy and will eventually be worn down if not renewed.

Georg Simmel

Simmel's works typically lack references to economics as such. Simmel, like Durkheim, usually viewed economic phenomena within some larger, noneconomic setting. Nonetheless, his work still has relevance for economic sociology. Much of Simmel's most important study, *Soziologie*, focuses on the analysis of interests. He suggested what a sociological interest analysis should look like and why it is indispensable to sociology.

Two of his general propositions are that interests drive people to form social relations, and that it is only through these social relations that interests can be expressed:

- Sociation is the form in which individuals grow together into a unity and within which their interests are realized. And it is on the basis of their interests—sensuous or ideal, momentary or lasting, conscious or unconscious, causal or teleological—that individuals form such units.

Another key proposition is that economic interests, like other interests, can take a number of different social expressions. *Soziologie* also contains a number of suggestive analyses of economic phenomena, among them competition. In a stage on the role of the number of actors in social life, Simmel suggests that competition can take the form of *tertius gaudens*. In this situation, which involves three actors, actor A turns to advantage the fact that actors B and C are competing for A's favour—to buy something, to sell something, or the like. Competition is consequently not seen as something that only concerns the competitors; it is in addition related to actor A, the target of the competition.

Simmel also distinguishes competition from conflict. While a conflict typically means a confrontation between two actors, competition rather implies parallel efforts, a circumstance in which society can benefit from the actions of both the actors. Instead of destroying your opponent, as in a conflict, in competition you try to do what your competitor does—but better. *Philosophy of Money*, Simmel's second major sociological work, has always enjoyed a mixed reputation. Durkheim disapproved of it for its mix of genres, and according to Weber economists detested Simmel's way of dealing with economic topics.

Simmel does mix philosophical reflections with sociological observations in an idiosyncratic manner, but *Philosophy of Money* has nonetheless much to give if it is read in its own frame. Simmel's main point is that money and modernity belong together; in today's society there does not exist one exclusive set of dominant values but rather a sense that everything is relative. Simmel's work also contains a myriad of insightful sociological reflections on the connections of money with authority, emotions, trust, and other phenomena. The value of money, Simmel observed, typically extends only as far as the authority that guarantees it.

Money is also surrounded by various “economically important sentiments,” such as “hope and fear, desire and anxiety”. And without trust, Simmel argues, society could simply not exist; and “in the same way, money transactions would collapse without trust”. In relation to money, trust consists of two elements. First, because something

has happened before—for example, that people accept a certain type of money—it is likely to be repeated. Another part of trust, which has no basis in experience and which can be seen as a nonrational belief, Simmel calls “quasi-religious faith,” noting that it is present not only in money but also in credit.

AFTER THE CLASSICS

Despite its foundation in the classics, economic sociology declined after 1920 and would not return to full vigour before the 1980s. Exactly why this happened is still not clear. One reason is probably that neither Weber nor Simmel had any disciples. Durkheim did, however, and the study of Marcel Mauss, *The Gift*, should be singled out. It rests on the argument that a gift typically implies an obligation to reciprocate and should not be mistaken for a one-way act of generosity. *The Gift* also contains a number of interesting observations on credit, the concept of interest, and the emergence of homo economicus. Eventually, however, Durkheimian economic sociology declined. Despite the slowing in economic sociology during the years 1920–80, there were several noteworthy developments, especially the theoretical works of Joseph Schumpeter, Karl Polanyi, and Talcott Parsons. All three produced their most important works while in the United States, but had roots in European social thought.

Joseph Schumpeter

We preface our notes on Schumpeter an economist, by noting some contributions by economists more generally to economic sociology. One example is Alfred Marshall whose analyses of such topics as industries, markets, and preference formation often are profoundly sociological in nature. Vilfredo Pareto is famous for his sociological analyses of rentiers versus speculators, business cycles, and much more.

The work of Thorstein Veblen sometimes appeared in sociological journals, and his analyses include such topics as consumer behaviour why industrialization in England slowed down and the shortcomings of neoclassical economics. Final mention should also be made of Werner Sombart, who wrote on the history of capitalism, on “the economic temper of our time,” and on the need for a “verstehende economics”. The contributions of Schumpeter are especially noteworthy. His life spanned two periods in modern economics—the period around the turn of the century, when modern economics was born, and the period of a few decades later when it was mathematized and secured its place as “mainstream.”

Schumpeter similarly spanned two distinct periods in sociology—from Max Weber in the first decade of the 20th century through Talcott Parsons in the 1930s and 1940s. Schumpeter is also unique among economists for trying to create a place for economic sociology next to economic theory. In this last effort Schumpeter was clearly inspired by Weber and, like the latter, referred to this type of broad economics as *Sozialökonomik*, or “social economics.” Schumpeter defines economic sociology as the study of institutions, within which economic behaviour takes place. Schumpeter produced three

studies in sociology. The first is an article on social classes that is of interest because of his distinction between economists' and sociologists' use of the concept of class. While for the former, he argues, class is a formal category, for the latter it refers to a living reality. The second study is an article about the nature of imperialism that can be compared to the equivalent theories of Hobson, Lenin, and others. Schumpeter's basic idea is that imperialism is precapitalistic and deeply irrational and emotional in nature—essentially an expression for warrior nations of their need to constantly conquer new areas or fall back and lose their power.

The third study is perhaps the most interesting one from the viewpoint of contemporary economic sociology, "The Crisis of the Tax State". Schumpeter characterizes this object as a study in "fiscal sociology"; its main thesis is that the finances of a state represent a privileged position from which to approach the behaviour of the state. As a motto Schumpeter cites the famous line of Rudolf Goldscheid: "The budget is the skeleton of the state stripped of all misleading ideology. Schumpeter did not regard Capitalism, Socialism, and Democracy as a work in sociology, but its main thesis is nonetheless sociological in nature: the motor of capitalism is intact but its institutional structure is weak and damaged, making it likely that socialism will soon replace it. On this point Schumpeter was evidently wrong.

His analysis of the forces that are undermining capitalism may seem idiosyncratic at times. Nonetheless, Schumpeter should be given credit for suggesting that the behaviour of intellectuals, the structure of the modern family, and so on, do affect capitalism. Of special importance are his insights about economic change or, as Schumpeter phrased it with his usual stylistic flair, "creative destruction."

Entrepreneurship is at the heart of Schumpeter's treatment of economic change. He himself saw his theory of entrepreneurship as falling in economic theory, more precisely as an attempt to create a new and more dynamic type of economic theory. Nonetheless, many of his ideas on entrepreneurship are sociological in nature. His central idea—that entrepreneurship consists of an attempt to put together a new combination of already existing elements—can be read sociologically, as can his idea that the main enemy of the entrepreneur is the people who resist innovations.

Karl Polanyi

Trained in law, Polanyi later taught himself Austrian economics as well as economic history and economic anthropology. Though he was interdisciplinary in approach, his main specialty was economic history, with an emphasis on nineteenth-century England and preindustrial economies. Polanyi's most famous work is *The Great Transformation*, conceived and written during World War II. Its main thesis is that a revolutionary attempt was made in nineteenth-century England to introduce a totally new, market-centered type of society.

No outside authority was needed; everything was automatically to be decided by the market. In the 1840s and 1850s a series of laws was introduced to turn this project into

reality, turning land and labour into common commodities. Even the value of money was taken away from the political authorities and handed over to the market. Polanyi, this type of proceeding could only lead to a catastrophe. When the negative effects of the market reforms became obvious in the second half of the nineteenth century, Polanyi continues, countermeasures were set in to rectify them.

These measures, however, only further unbalanced society; and developments such as fascism in the twentieth century were the ultimate results of the illfated attempt in mid-nineteenth-century England to turn everything over to the market. Polanyi also cast his analysis in terms of interests and argued that in all societies, before the nineteenth century, the general interests of groups and societies had been more important than the money interest of the individual. “An all too narrow conception of interest,” Polanyi emphasizes, “must in effect lead to a warped vision of social and political history, and no purely monetary definition of interest can leave room for that vital need for social protection”.

The theoretical part of *The Great Transformation* is centered around Polanyi’s concepts of “embeddedness” and “principles of behaviour”. The fullest elaboration of this line of work is to be found in *Trade and Market in the Early Empires* and especially in Polanyi’s thesis “The Economy as Instituted Process”. Polanyi criticized economic theory for being essentially “formal”—a kind of logic focused on choice, the means-end relationship, and the alleged scarcity of things that people want. There is also “the economistic fallacy,” or the tendency in economics to equate the economy with its market form. To the formal concept of economics Polanyi counterposes a “substantive” concept, grounded in reality and not in logic.

“The substantive meaning of economic derives from man’s dependence for his living upon nature and his fellows”. While the notion of economic interest is directly linked to “the livelihood of man” in substantive economics, it is only an artificial construction in formal economics. The most famous concept associated with Polanyi’s work is “embeddedness,” which, however, he used in a way different from its contemporary use. According to the current use, an economic action is in principle always “embedded” in some form of social structure. According to Polanyi, economic actions become destructive when they are “disembedded,” or not governed by social or noneconomic authorities.

The real problem with capitalism is that instead of society deciding about the economy, it is the economy that decides about society: “instead of the economic system being embedded in social relationships, these relationships were now embedded in the economic system”. Another set of conceptual tools for economic sociology is Polanyi’s “forms of integration.” His general argument is that rational self-interest is too unstable to constitute the foundation for society; an economy must be able to provide people with material sustenance on a continuous basis.

There are three forms of integration, or ways to stabilize the economy and provide it with unity. These are reciprocity, which takes place within symmetrical groups, such as

families, kinship groups, and neighbourhoods; redistribution, in which goods are allocated from a center in the community, such as the state; and exchange, in which goods are distributed via price-making markets. In each economy, Polanyi specifies, there is usually a mixture of these three forms. One of them can be dominant, while the others are subordinate.

Talcott Parsons

Talcott Parsons was educated as an economist in the institutionalist tradition and taught economics for several years before he switched to sociology in the 1930s. At this time he developed the notion that while economics deals with the means-end relationship of social action, sociology deals with its values. In the 1950s Parsons recast his ideas on the relationship of economics to sociology, in a work coauthored with Neil Smelser, *Economy and Society*. This work constitutes Parsons's major contribution to economic sociology, but both before and after its publication Parsons produced a number of studies relevant to economic sociology. In *The Structure of Social Action* Parsons launched a forceful attack on utilitarian social thought, including the idea that interests represent an Archimedean point from which to analyze society. Interest theorists, Parsons notes, cannot handle the Hobbesian problem of order; they try to get out of this dilemma by assuming that everybody's interests harmonize.

What is not understood by the utilitarians is that norms are necessary to integrate society and provide order. Interests are always part of society, but a social order cannot be built on them. In *Economy and Society* Parsons and Smelser suggested that both sociology and economics can be understood as part of the general theory of social systems. The economy is a subsystem, which interchanges with the other three subsystems. The concept of a subsystem is reminiscent of Weber's notion of sphere, but while the latter refers only to values, the economic subsystem also has an adaptive function as well as a distinct institutional structure. It may finally be mentioned that *Economy and Society* got a negative reception by economists and failed to ignite an interest in economic sociology among sociologists. Smelser's attempt to consolidate economic sociology in the next decade helped fix economic sociology as a subfield in the minds of scholars and in the curricula of colleges and universities, but did not spawn distinct new lines of research.

IT, ECONOMICS AND DEVELOPMENT

In abstract, there are two types of potential economic gains from the use of IT. First, there are both static and dynamic efficiency gains. Static gains are one-time, and come from more efficient use of scarce resources, allowing higher consumption in the present. It is useful to distinguish two kinds of static efficiency gains. One kind pertains to increases in operating efficiency, while the other comes from reduced transaction costs. In both cases, the channel for gains is through more effective and lower cost information storage, processing and communication. Dynamic gains come from higher growth,

potentially raising the entire future stream of consumption. The second type of potential benefit comes from reductions in economic inequality, to the extent that such reductions are an agreed-upon social goal, and therefore a social benefit.

The two types of gains may conflict, if growth requires increased inequality, or they may be mutually reinforcing, where broad sharing of the benefits of growth enhances the rate of growth. We can discuss the role of IT in achieving greater economic gains along both dimensions, without having to commit to a particular position on the relationship between inequality and growth. However, a focus on using IT for rural development is, at least on the surface, supportive of reduced inequality along with increased efficiency and growth. Of course, benefits that are measurable as increased market-based economic activity, and hence show up in national accounts statistics, are not the only component of development.

Development can also include improvements in the capabilities of the population, such as education, health and nutrition, independently of any direct or indirect economic impact. The ability to participate in democratic decision-making also falls into this category. Broad-based improvements in capabilities can also have positive impacts on long-run economic well being, but this is not a necessary condition for desiring such improvements. The role of IT in effecting improvements along non-economic dimensions. Turning to specific impacts, note that IT involves the electronic processing, storage and communication of information, where anything that can be represented in digital form is included in the term 'information'.

Thus news, entertainment, personal communications, educational material, blank and filled-out forms, announcements, schedules, and so on are all information. Software programmes that process data (searching, tabulating, and calculating, for example) are also information in this sense, representing a particular kind of intermediate good. We can use standard economic characterizations to classify the different kinds of information. For example, entertainment, personal communications, and sometimes news, are final goods. Educational material, job announcements, or some kinds of news (weather news for farmers, for example) are intermediate goods, typically used for improving income-earning opportunities. Information goods typically have the characteristic that one person's use does not reduce their availability for another person. Thus, a message or weather news can be viewed by many people, simultaneously or sequentially. Depending on the content of the news or message, different people may place different valuations on the information. Only friends and relatives may be interested in a personal message, all farmers in a district may be interested in local weather news, and so on. The ability to share information among users can impact the feasibility of providing it on a commercial basis.

IT dramatically increases shareability of information, and this affects the economics of private provision of information goods and services. The government may provide information goods because they are shareable and non-excludable (pure public goods). The classic example of a pure public good is national defence, but such goods may also

be local in nature, such as public parks or law and order. Of course many local shareable goods can be provided exclusively, in which case private provision is a feasible alternative, in a club-like arrangement.

Here, government provision may be justified more on equity grounds than on the basis of failure of private provision. In some cases, government financing through taxes or statutory user charges can be combined with outsourcing of delivery to private providers to achieve both equity and efficiency goals. Often, private provision is feasible, but neglects the spillover benefits that it creates, in which case government subsidization may be socially beneficial. For example, primary education has private economic benefits that people are willing to pay for, but it can also have substantial non-economic benefits to the individual and to others in the society (improved understanding, ability to make sound judgments, political decisionmaking capacity, and so on). Additional roles of government that are important to bring out are in redistribution to achieve equity objectives, and in regulation of private activities through licensing and certification.

In both cases, the government also uses economic resources, and IT has a potential role in increasing the efficiency of government. For both government and private provision, one of IT's main direct benefits is in increasing efficiency by economizing on resource use in the operations of firms as well as in market transactions. Information that would otherwise be conveyed through face-to-face contact, post, courier, print delivery, telegraph or telephone may instead be communicated in digital electronic form via the Internet. Efficiency gains from Internet use are not automatic: the telephone, in particular, is an efficient means of communication for many types of information.

IT also requires new investment, so the benefits of trips, time and paper saved must be weighed against the costs of installing and maintaining the new infrastructure. Efficiency benefits of IT are not restricted to the communication itself. IT can improve the efficiency of the telephone network, and it can make it possible to track and analyse communications. Word processing, maintaining accounts, inventory management, and other such activities that may not require longdistance communications are also made more efficient by IT. Experience with Internet use in developed countries suggests that information exchange related to the completion of market transactions is especially valuable.

The ability of IT-based communications (combined with storage and processing) to bring together buyers and sellers more effectively represents major potential gains. These gains can come about through lower search costs, better matching of buyers and sellers, and even the creation of new markets. The successes of auction and employment websites in the US show these gains. In the rural Indian context, farmers selling their crops and buying inputs, parents seeking matrimonial alliances for their children, and job seekers are all potential users of Internet-based matching services. Efficiency gains of IT can also come about through the enabling of new goods and services. In many cases, the new good is related to something available earlier, but is presented in a form that reduces costs and expands the size of the market. For example, recorded music is a

mass-consumption item, whereas only a small minority of the population could afford or have access to live performances by the highest quality musicians. Educational material is another example where recording and duplication can replace more expensive, skilled-labour-intensive alternatives for delivery. The possibilities for interactivity with IT-based educational materials show the advantages of IT over older technologies based only on recording and duplication. Interactivity also implies personalization, in that an individual can select the precise content that he or she wishes to see. This feature also distinguishes IT-based content from what was available through previous technologies.

Finally, the sheer volume of information that is accessible through IT is much greater than before: this also allows new kinds of services to be provided at a cost that is affordable to larger segments of the population. We have outlined the potential static efficiency benefits of IT, but the direct dynamic benefits in terms of higher growth are harder to identify. Of course, if IT economizes on current resources, more is available for investment, which can increase growth. If IT increases the efficiency of education delivery to the broader population, this investment in people (human capital acquisition) is also likely to lead to higher growth.

IT may also have positive impacts through impacts on the innovation process. For example, IT can make innovation easier by allowing simulation and low-cost testing of new designs or searching through possible chemical compounds for beneficial drugs. Finally, IT may speed the diffusion of innovations through better communications, which may stimulate further innovation—though this last channel is speculative. An important barrier to realizing the economic benefits of IT is the oftensubstantial up-front cost of investment in new infrastructure—both hardware and software. In developed countries such as the US, large potential customer bases and efficient capital markets help overcome this barrier. Hardware and software designed for developed country markets can easily be adapted to serve higher income consumers in developing countries, but this leaves out the majority of the population in developing countries.

Thus, one potential consequence of IT is an exacerbation of inequality, as only higher income groups enjoy its benefits—this is the so-called ‘digital divide’. On the other hand, because government-provided goods and services, including redistributive transfer payments, are often aimed at lower income groups, to the extent that IT can increase the efficiency and effectiveness of government, the benefits of IT will be more widely spread, partly reducing ‘digital divide’ concerns. However, achieving these benefits requires more than just internal use of IT: beneficiaries of government services (particularly the economically disadvantaged) must be able to access IT resources also. While governments may invest in such front-end interfaces with citizens (and have done so in developed countries), the cost of doing so for governments in developing countries may be prohibitive.

Such governments typically already have difficulties in raising sufficient resources through taxes and user charges. While successful examples of implementation of ‘e-governance’ initiatives exist, there is a conceptual alternative. This comes from

recognizing the fact that citizens typically incur private costs (often substantial) in availing of government-provided services.

If the use of IT can reduce such costs, even low-income individuals may be willing to pay at least some fraction of the cost savings, and there is scope for private provision of intermediate services that reduce the cost of access to government. Of course, this idea is not specific to IT: private intermediaries already help in filling out forms, getting access, and so on. One difference that IT can make is in reducing costs even further, often by an order of magnitude. In broad terms (as is also the case with electronic marketplaces and job-matching boards), IT changes the scope and nature of intermediation. Private providers may therefore have a role in delivering IT-based information services that are complementary to government services, as well as in providing conventional private goods and services. However, the private individual benefits that determine the prices charged by private providers may not reflect the overall social benefits of provision.

As discussed earlier, these may include benefits such as greater awareness and participation in the political process. In such cases, there may be a role for government subsidization of private provision. This assumes that government provision is likely to be less efficient than private provision, which seems to be true in some cases in developing as well as developed countries. In either case, richer information flows increase the transparency with which the government operates, thereby promoting better monitoring, and potentially—depending on whether electoral and legal institutions are effective—greater accountability.

The ultimate payoff is more efficient delivery of government services. Looking at the case of India, in cities and larger towns, cyber kiosks have already begun to proliferate. Urban population densities, income levels, cultural attitudes and telecom infrastructure all seem to be sufficient for the commercial success of these enterprises. The falling cost of hardware and the availability of a variety of English language software have also supported this trend.

Finally, the government's belated opening up of Internet service provision to competitive entrants has been a crucial supporting development. In non-Internet IT-related services, IT education has clearly taken off in cities as well, inspired by India's success in software exports. In rural areas and smaller towns, however, the various demographic and socioeconomic factors such as income levels, cultural attitudes, and geographic and social fragmentation may not be present in configurations that would easily enable the diffusion of commercial access to various IT-enabled services. Furthermore, the market power of traditional rural intermediaries may act as a barrier to partial innovations in how matching of buyers and sellers is conducted. Finally, vernacular language requirements and different demand patterns imply the need for software that is tailored for fragmented rural markets. We examine the technical and organizational issues in implementing widespread rural IT use in India.

Educational Development

SOCIAL AND NATIONAL DEVELOPMENT

Social, aim of education is equally important because an individual lives in society and has his obligations towards his nation. There is a realisation that, “The present education system does not yield required results mainly because it is divorced from the real social content and social goals”. It has, therefore, been suggested that education should be able, to discharge its natural functions and must correspond to its structure, goals and content in the interest of national development and social progress. It has also been suggested in this connection that students from young age should be made aware of the social responsibility cast on them.

At the same time, there are certain constitutional commitments, which are intimately related to this aim.. We as the citizens of the republic, are constitutionally Committed to democracy, social justice, equality of opportunity, secularism and all to a welfare state. It has, therefore, been suggested that, “Educational policy and educational programme should clearly reflect these commitments”.

The objectives of developing a sense of national identity, unity and patriotism are advocated by many associations. It is pointed out that the national objectives of planning and programmes and development with special emphasis on popular participation and the national problems that we face in different fields should. be taught at relevant stages. Individual and social aims of education are not contrary to one another. In fact they are complementary to one another. The following view strikes a balance between individual and social aims of education.

The purpose of education should be the development of the fullest possible capacities and potentialities physical and spiritual of a 'total man'. It should make a man capable of earning his livelihood reasonably well to enjoy a happy and secure life while making effective contributions to the society and national effort of making India strong advanced and prosperous.

SOCIAL TRANSFORMATION

Education should not merely equip an individual to adjust with society to its customs and conventions, but it should enable him to bring desirable changes in the society. It has been, therefore, suggested that, "Every educational institution from secondary school to university college should be developed to become an agency of change...." However, it is essential that we should be quite clear about the purpose of change. It is, therefore, natural to ask the, question, "Reform and change to achieve what"? What type of society we aim at and what type of citizens we wish to produce? The following ideas give an indication of the kind of changes education is expected to bring about.

Modernisation

Modernisation of society in terms of scientific and technological advancement is a view which seems to be quite popular. It is though that education should enable us to move with times and attain excellence in, science and technology. To quote an expression of this kind. Scientific and technological advances are, gaining momentum and conscious efforts are made to incorporate them into the development sectors. This calls for modernisation of education in order to make. it in conformity with the modern times and to keep pace, with the advances in the world.

Modernisation, however, is not interpreted and equated with westernisation. In fact, lot of emphasis is given to 'Indianness' while talking about modernisation. One of the suggestions explicitly points out that, hour education should integrate and unite the people of India, modernise society while preserving what is authentically Indian in our cultural and spiritual heritage". The following suggestion beautifully reconciles the twin objectives of modern technical sophistication and the ancient spirituality. "*New education policy of India should be built on the foundation of ancient spirituality and modern culture and technical sophistication. It should develop scientific temper and spirit of enquiry in the students*".

Productivity

Some documents have insisted on linking education with productivity and thus making individuals as productivie citizens to build a productive society. One of the suggestions, in a memorandum, for example, says. "It should bring about a social transformation, and enhance greater efficiency and productivity in all sectors: agricultural, industrial and service". It is in this context that Mahatma Gandhi's system of basic education is still considered as a basically sound system and a suggestion has been made that with

necessary modification elements of basic education may form part of education not only at the primary stage but at all stages in our national system of education.

These elements are:

- Productive activity in education.
- Correlation of the curriculum with productive activity and physical and social environment.
- Intimate contact between the school and the local community.

Community Participation

In a democracy education without community participation is barren. This aim of education is, therefore voiced by a number of groups and organisations. The change that is envisaged on this front is that of Integrating education with community in all respects. To quote a suggestion in this regard: The education system in all its branches and sectors should get itself involved in activities related to problems of local Community life and shall thus endeavour through the desirable community participation community involvement in the educational field to bring all education of its rightful place in community life.

ACQUISITION OF VALUES

Moral, cultural and spiritual values in education have been given immense importance in the Memoranda documents. One of the expressions emphatically point out that, “certain basic values as respect for others, responsibility, solidarity, creativity and integrity must be fostered in our children”. It is interesting that a number of specific values have been suggested in the documents. Emphasis should be given in cultivating good qualities like cooperation, good will, forgiveness, tolerance, honesty, patience *etc.* in order to encourage universal brother-hood and to prepare students worthy citizens of the country. Values of optimism and secularism, and service to the poor should be stressed on the young minds.

EDUCATION AND SOCIAL AND NATIONAL INTEGRATION

Social and national integration is crucial to the creation of a strong, united country, which is an essential pre-condition for all progress. It has a varied content-economic, social, cultural and political-and its different facets are closely interconnected.

It needs:

- Confidence in the nation’s future;
- A continuous rise in the standard of living for the masses and the reduction in unemployment and in the disparities in development between different parts of the country, all of which are essential to promote a sense of equality of opportunity in political, economic and social terms;

- A deep sense of the values and obligations of citizenship and a growing identification of the people, not with sectional loyalties but with the 'nation' as a whole;
- Assurance of good and impartial administration, equal treatment for every citizen, in fact and not merely in law, based on the integrity of the public services; and
- Mutual understanding of and respect for the culture, traditions and ways of life of different sections of the nation.

To make this social and psychological revolution possible, it is necessary to deal with the short-term problems facing the country in this field, particularly in regard to the growing and dangerous symptoms of social disorganization. These express themselves as the widening gulf between the rich and the poor, the privileged and the unprivileged, the urban and the rural, the educated and the uneducated. They are reflected in the general weakening of the feeling of national solidarity under the increasing impact of local, regional, linguistic, religious and other sectional or parochial loyalties. Effective steps must be taken to bridge these dangerous gulfs and strengthen national consciousness and unity. Social and national integration is a major problem which will have to be tackled on several fronts, including education.

In our view, education can and should play a very significant role in it by:

- Introducing a common school system of public education;
- Making social and national service an integral part of education at all stages;
- Developing all modern Indian languages, and taking necessary steps to enrich Hindi as quickly as possible so that it is able to function effectively as the official language of the Union; and
- Promoting national consciousness.

COMMUNITY LIVING IN SCHOOLS AND COLLEGES

Every educational institution should try to develop a rich community life of its own and provide adequate and satisfying opportunities for students to participate in it and help in organizing it. Various opportunities for such work exist in the classroom, on the campuses of the schools and the colleges. in the hostels and on the playgrounds. For instance, instead of utilizing servants and hired labour for educational institutions and their hostels, it is possible to have much of the work done by students, not primarily to save money, but to provide valuable experience. Japan has followed this Practice in her national system, and many schools in India are also doing so to some extent. This type of work-experience would cultivate in students the habit of work and a sense of the dignity of labour.

If this is done on a nation-wide scale, it will incidentally effect some saving in expenditure which could be utilized for providing certain basic amenities for students. The training institutions for basic teachers have tried to develop such traditions of community living by eliminating the use of servants to a large extent. We recommend that this pattern of hostel life should be introduced in all schools and colleges so far as

possible. The practice of making self-help and manual work a part of the daily life and training in all types of educational institutions-as was the case in many of the old Ashrams and Academies-would yield good educational results. We cannot totally recreate the conditions of the past in our educational institutions; but we can certainly profit from the useful elements in past experience.

PARTICIPATION IN PROGRAMMES OF COMMUNITY DEVELOPMENT

In addition to these institutional programmes, it is essential to make participation in meaningful programmes of community service an integral part of all education from the primary to the undergraduate stage. Such participation can help to create positive attitudes towards social service and to develop closer ties between the educated persons and the rest of the people. It can also help in building up a sense of social purpose and self-confidence and giving students a sense of participation in community life and activities.

The organization of the programmes will naturally vary from stage to stage:

- At the primary stage, it will take the form of bringing the school closer to the community with an accent on serving the community in suitable ways.
- At the secondary stage, such programmes can be more ambitiously designed, and have a greater impact on young minds. Every secondary school should develop carefully planned programmes for promoting good school-community relations and suitable forms of service to the community. We recommend that about ten days a year may be fully devoted to such programmes. Where this is not possible, it should be obligatory for the students to participate in the Labour Service Camps for secondary school students to be organized by the State Education Departments on a district basis. The programme is discussed more fully elsewhere.
- Similarly, it should be obligatory for every college student, before he is awarded his first degree, to put in at least 60 days of national service in one to three stretches. Each college should develop its own programme, suited to its objectives, its resources and the age and competence of its students. Participation in Labour and Social Welfare Camps or the NCC should also be regarded as alternative forms of such service.

The NCC programme, which has been made compulsory at the university stage during the last three years, has shown some good results. It has possibilities of promoting national development by building up closer ties between the people and the defence services. It also helps to lessen the difference between the so-called 'martial' and other classes and inculcates the idea of the defence of freedom in all classes of people. As it has been started comparatively recently, we think that it should be continued on its present basis for some time longer, say, till the end of the Fourth Five Year Plan. We would, however, recommend that the authorities concerned explore the possibility of organizing this training, not throughout the period of three years as at present, but in a

concentrated and whole-time programme spread over about 60 days which, as stated earlier, could be completed by students in one to three stretches according to their convenience. As alternative forms of social service come into being, NCC should be made voluntary, leaving the students free to opt for it or any other form of social service provided.

EVOLUTION OF A LANGUAGE POLICY

The development of a proper language policy can also assist materially in social and national integration. Of the many problems which the country has faced since independence, the language question has been one of the most complex and intractable, and it still continues to be so. Its early and satisfactory solution is imperative for a variety of reasons- educational, cultural and political.

DEVELOPMENT OF MODERN INDIAN LANGUAGES

It is hardly necessary to emphasize that the development of the Indian languages is both urgent and essential for the development of the Indian people and as a way of bringing together the elite and the masses. It can make scientific and technical knowledge more easily accessible to the people in their own languages and thus help not only in the progress of industrialization but also in the wider dissemination of science and the scientific outlook.

Energetic action is needed to produce books and literature, particularly scientific and technical, in the regional languages. This should be regarded as a specific and imperative responsibility of the universities; and the UGC should provide general guidance and allot adequate funds for the programme. Medium of Education in Schools and Colleges. The development of the modern Indian languages is inextricably linked with the importance given to them in the educational system, specially at the university stage. The medium selected should enable students to acquire knowledge with facility, to express themselves with clarity and to think with precision and vigour. From this point of view, the claims of the mother-tongue are preeminent.

About thirty years ago, delivering the convocation address of the Calcutta University, Rabindra Nath Tagore had expressed his views in this matter in no uncertain terms:

- In no country in the world, except India, is to be seen this divorce of the language of education from the language of the pupil. Full hundred years have not elapsed since Japan took its initiation into Western culture. At the outset she had to take recourse to textbooks written in foreign languages, but from the very first, her objective had been to arrive at the stage of ranging freely over the subjects of study in the language of the country. It was because Japan had recognized the need of such studies, not as an ornament for a select section of her citizens, but for giving power and culture to all of them, that she deemed it to be of prime importance to make them universally available to her people. And in this effort of Japan to gain proficiency in the Western arts and sciences, which was

to give her the means of self-defence against the predatory cupidity of foreign powers, to qualify her to take an honoured place in the comity of nations, no trouble or expense was spared. Least of all was there the miserly folly of keeping such learning out of easy reach, within the confines of a foreign language.

- Learning through a foreign medium compels the students to concentrate on cramming instead of mastering the subject-matter. Moreover, as a matter of sound educational policy, the medium of education in school and higher education should generally be the same. Prior, the position was at least consistent. English was the medium both in the upper stages of school and college education. As we have rightly adopted the regional languages as the media of education at the school stage, it follows that we should adopt them increasingly at the higher stage also.

This proposal has also been supported strongly as a measure to promote social and national integration. The Emotional Integration Committee was of the view that the use of regional languages as media of education from the lowest to the highest stage of education was a matter of 'profound importance for national integration'.

This was supported by the National Integration Council which said:

- The change in the medium of instruction is justified not so much by cultural or political sentiments as on the very important academic consideration of facilitating grasp and understanding of the subject-matter. Further, India's university men will be unable to make their maximum possible contribution to the advancement of learning generally, and science and technology in particular, unless there is a continuous means of communication in the shape of the regional languages between its masses, its artisans and technicians and its university men. The development of the talent latent in the country will also, in the view of the Council, be retarded unless regional languages are employed as media of instruction at the university stage. We generally agree with these observations.

It has been sometimes argued that there should be a single medium of education at the university stage-English for the time being, to be ultimately replaced by Hindi-on the ground that it would promote mobility of teachers and students from one part of the country to another, provide for easy communication between academic and professional men and administrators, further intellectual cooperation amongst the universities and help in other ways in developing a corporate intellectual life in the country. We are inclined to think, on a balance of considerations, that this solution is not feasible. In practice, it will probably mean the indefinite continuance of English as the only medium of higher education, a development that we cannot support in the larger interests of the country. The adoption of Hindi as a common medium of education in all parts of India is not possible for some years to come, and in non-Hindi areas it will still have some of the disadvantages associated with the use of a foreign medium and is likely to be resisted.

It would, therefore, be unwise to strive to reverse the present trend for the adoption of the regional languages as media of education at the university stage and to insist on

the use of a common medium in higher education throughout the country. There will, however, be one important exception to this general rule, namely, all- India institutions which admit, in considerable numbers, students from different parts of the country. These now use English as the medium of education, which should continue undisturbed for the time being. A change-over to Hindi may be considered in due course provided two conditions are fulfilled. The first is the effective development of Hindi as a medium of education at this level. This is a matter which can be left to the UGC and the institutions concerned to decide.

The second is the equally important political consideration that, in such a change-over, the chances of students from non-Hindi areas should not be adversely affected and that the proposal should have the support of the non- Hindi States. The latter Principle has been already conceded by the Government of India even in the larger sphere of the use of Hindi in official communications between the States and the Centre. Simultaneously, it is necessary to make the regional languages the official languages of the regions concerned as early as possible so that higher services are not de facto barred to those who study in the regional medium. The acceptance of the regional languages as media at the university is much more likely when good employment, which now depends largely on a knowledge of English and is more easily open to students who have studied through English, becomes available to those who have studied through the regional medium.

We might also add here that, though Urdu is not a regional language in the ordinary sense of the word, it has an all-India significance since it is spoken by certain sections of the people in different parts of the country. Due encouragement must be given to it at all stages not only because of this peculiar character but also because of its close links with the official language, Hindi.

CHANNELS OF INTERNATIONAL COMMUNICATION

The introduction of the regional languages as media of education should not be interpreted to mean underrating the importance of English in the university. For a successful completion of the first degree course, a student should possess an adequate command over English, be able to express himself in it with reasonable ease and felicity, understand lectures in it and avail himself of its literature. Therefore, adequate emphasis will have to be laid on its study as a language right from the school stage. English should be the most useful 'library language' in higher education and our most significant window on the world. It is also important to encourage the study of other foreign languages on a more extensive scale for a variety of academic and practical purposes, Russian has a special significance for the study of science and technology in the present-day world.

In addition, French, German, Japanese, Spanish and Chinese are important world languages for communication and for acquiring knowledge and culture. We recommend that all universities, some selected colleges, and also a small proportion of carefully

selected schools should provide for the teaching of these languages. The knowledge of another foreign language besides English should be a requirement for a doctorate degree, and in certain subjects, even for the Master's degree. The country will need in increasing numbers a group of persons, small but extremely proficient in important foreign languages and their literature.

From this point of view, it would be necessary to establish a few schools which will begin teaching, right from an early age, the important foreign languages referred and will use them also as media of education. The admission to these schools should be on a selective basis and there should be an adequate provision of scholarships. It would be an important step towards the general development of higher education, and also towards international cooperation and understanding, if there were established a small number of institutions, at university level, with some of the important 'world languages' as media of education.

A beginning has already been made with the Institute of Russian Studies at New Delhi. It would be desirable to set up, during the Fourth Plan, institutions on somewhat similar lines for German, French, Spanish and Japanese languages. We could also establish one or two more institutions for Russian. The institutions will have to be largely residential. It will be an advantage to have them as constituent units of universities in their neighbourhood.

CHANNELS OF INTERNAL COMMUNICATION

Hindi must be greatly developed and enriched before it can attain the status of a library language, that is, a language which can serve as a vehicle for acquiring a substantial part of the current and rapidly expanding stock of world knowledge. This has to be taken into account fully in determining our language policy. This implies, as stated earlier, that every graduate will need to acquire a reasonable proficiency in a library language, which will be English for most students. It will thus serve as a link-language in higher education for academic work and intellectual inter-communication. It is, however, equally obvious that English cannot serve as the link-language for the majority of our people. It is only Hindi which can and should take this place in due course. As it is the official language of the Union and the link-language of the people, all measures should be adopted to spread it in the non-Hindi areas.

The success of this programme will largely depend on the extent to which it is voluntarily accepted by the people of these areas. We were interested to know that in the Kerala University, where students can take Hindi in place of Malayalam at the undergraduate stage, a large proportion of students choose to study Hindi. We have also seen increasing evidence on the part of non-Hindi areas to take to the study of Hindi. AR their efforts in this matter, particularly in the non-official sector, should receive encouragement. In addition to Hindi, it is essential to provide multiple channels of inter-State communication in all modern Indian languages. In every linguistic region, there should be a number of persons who know all the other modern Indian languages

and some who are familiar with their literatures and able to contribute to them. For this purpose, we recommend that there should be adequate arrangements, both in schools and colleges, for teaching different modern Indian languages. In addition, steps should be taken to establish efficient departments in some of the modern Indian languages in every university. It may also be advisable to create a Small Dumber of special institutes for the comparative study of different languages and their linguistic problems. At the B.A. and M.A. levels, it should be possible to combine two modern Indian languages. This will incidentally supply the bilingual persons needed for language teaching in schools and colleges.

PROMOTING NATIONAL CONSCIOUSNESS

India is a land of diversities of different castes, peoples, communities, languages, religions and cultures. What role can the schools and universities play in enabling their students to discover the 'unity in diversity' that India essentially is, and in fostering a sense of national solidarity transcending narrower loyalties ? Unfortunately, our school system as a whole has had no strong tradition of striving to develop a sense of national unity and national consciousness. Under the British administration, the schools were expected to teach loyalty to the British rule rather than instil love for the motherland. Consequently, the effort to develop national consciousness was made largely outside the educational system, particularly during the struggle for freedom. Its foundation was a renewed belief in the value of national culture and traditions and a pride in India's past achievements.

It recognized that, in the cultural exchange between India and the West, India also had something worth while to give in return for the gift of modern science and technology, and that Indians should strive to maintain the valuable elements in their own traditional culture while accepting all that is good in the West. This was not meant as an attempt at revivalism, though it was certainly so interpreted by some sections. It was an attempt, as Gandhiji put it, to refuse to be blown off one's feet even when all the windows of one's house were thrown open to fresh winds from every corner of the world. With the attainment of Independence, the situation has changed greatly.

The one rallying point which had kept many diverse elements together ceased to exist, and as the schools have been unable to inculcate and strengthen national consciousness, the sentiment itself seems to be wearing thin, particularly in the new generation now coming of age. We saw something similar to the old spirit reviving in recent years-in the wake of the Chinese invasion and during the recent conflict with Pakistan. Obviously, such unfortunate international conflicts cannot be considered the normal ways to awaken and strengthen the national spirit.

The only enduring solution to the problem is to place this responsibility on the educational system and to make it an effective instrument for the purpose. There is much evidence in the world to show that schools and universities can play a major and helpful part in the regeneration and advancement of a people and in generating the

psychological energy needed for the purpose. If rightly, oriented, this approach is not, and should not be incompatible with the wider loyalties to mankind as a whole which we have also emphasized.

The deepening of national consciousness can be fostered specially by two programmes:

- The understanding and reevaluation of our cultural heritage, and
- The creation of a strong driving faith in the future towards which we aspire.

The first would be promoted by well-organized teaching of the languages and literatures, philosophy, religions and history of India, and by introducing the students to Indian architecture, sculpture painting, music, dance and drama. In addition, it would be desirable to promote greater knowledge, understanding and appreciation of the different parts of India by including their study in the curricula, by the exchange of teachers wherever possible, by the development of fraternal relations between educational institutions in different parts of the country, and by the organization of holiday camps and summer schools on an inter-State basis designed to break down regional or linguistic barriers.

It would further be necessary to establish and maintain all-India institutions which will admit students from different parts of the country. Faith in the future would involve an attempt, as a part of the course in citizenship, to bring home to the students the principles of the Constitution, the great human values referred to in its preamble, the nature of the democratic and socialistic society which we desire to create and the five-year plans of national development. At the higher stages of education the students should learn to assess the value of modern movements and tendencies so that they may learn which of them can and should be usefully assimilated into our culture. We should, however, take care to avoid identifying all 'modern' with 'western' values.

EDUCATION FOR INTERNATIONAL UNDERSTANDING

There is no essential contradiction, as we have pointed out earlier, between this objective and the development of international understanding for the 'one-world' towards which we are moving. On the whole, textbooks used in national school systems perhaps contain fewer untrue or hostile remarks about other countries than they did in the past, thanks partly to the work of UNESCO, under whose auspices multi-national and bi-national efforts have been made for their revision. The sins are more often now of omission than of active commission. We should, however, guard ourselves against this also, for ignorance is often not less dangerous than hostility. Our studies in the humanities and social sciences should be so oriented that, while helping students to become good and active citizens of their own country, these will also enable them to acquire a knowledge and understanding of the essential features of the outside world, particularly of our neighbouring and Afro-Asian countries.

Indian culture has had a strong and honourable tradition of international understanding, of valuing, on the whole with an open mind, the contributions of different

countries and races to human civilization. For this purpose, the fact that she is a country with many communities but of one common citizenship should actually prove to be an advantage. The constitution Of UNESCO speaks of ‘the fruitful diversity of the cultures of mankind’.

If through the schools and universities, a person learns to appreciate the basic rightness of that phrase in his own federal and multi-community country, he is less likely to forget it on the international plane where he deals with persons belonging to different nations but enjoying equal membership in the community of mankind. The programme of social and national integration which will minimize the internal barriers and suspicions which exist today will not only be a national gain, but may even make an impact on international relations and the international situation as a whole. This was one of the great inspirations of Jawaharlal Nehru’s policy.

DEMOCRATIC VALUES

In this context, Special emphasis has to be laid on the development of values such as a scientific temper of mind, tolerance, respect for the culture of other national groups, *etc.*, which will enable us to adopt democracy, not only as a form of government, but also as a way of life. As stated earlier, the population of India consists of persons who profess different religions, speak different languages, belong to different races, castes, classes and communities. It is precisely in such a situation that democracy can make its most significant contribution. A healthy development of democratic trends will help to soften the impact of this division into social, economic and cultural groups. The task is admittedly difficult; but it can convert the differences of language, cultural pattern, religion, *etc.*, into the warp and woof of a very rich and rewarding social and cultural life. The problem of national integration is essentially one of harmonizing such differences, of enabling different elements of the population to live peacefully and cooperatively and to utilize their varied gifts for the enrichment of the national life as a whole. We have to cultivate a spirit of large- hearted tolerance, of mutual give and take, of the appreciation of ways in which people differ from one another. This is a very exacting experiment in living’ that we have launched and no education will be worthwhile if the educated mind is unable to respond to this situation with intelligence and imagination.

SOME PROBLEMS OF NATIONAL DEVELOPMENT

SELF-SUFFICIENCY IN FOOD

The first and the most important of these is food. Mahatma Gandhi said: ‘If God were to appear in India, He will have to take the form of a loaf of bread’. Even at its present level of population, the country is in short supply for food. Every five years, the

addition to the population is greater than the entire population of the U.K. During the next two decades, there will be an addition of about 230 million persons even on the assumption that the existing birth- rate of 40 per thousand persons is reduced. If it is not, this increase may be 320 million or 65 per cent of the present population. On the basis of present trends, in another 10-15 years, no country is likely to have a surplus of food to export. Even if such surpluses existed, we would have no resources to import the huge quantities of food required, or even to import the fertilizers needed. Self-sufficiency in food thus becomes not merely a desirable goal but a condition for survival.

ECONOMIC GROWTH AND FULL EMPLOYMENT

Allied to this is the colossal poverty of the masses and the large incidence of underemployment or unemployment among the people, particularly among the educated. India is one of the poorest countries of the world. Both the rate of growth as well as the level of national income is unsatisfactory. What is worse, even this income is very unevenly distributed. It has been suggested that our immediate objective should be to assure a minimum consumption of Rs. 35 per month to every citizen, sufficient to cover the balanced diet recommended by the Nutrition Advisory Committee and leave a modest allowance for other essential items. This, by itself, is no high standard. But at present, only the top 20 per cent of the population can afford it.

The lowest 30 per cent have a monthly income of less than Rs. 15 and the lowest ten per cent, of less than Rs. 10. If this minimum standard of Rs. 35 per month is to be reached, great effort will have to be made on three fronts. The first is to secure a rapid rate of economic growth, at not less than six per cent per year and if possible at about seven percent per year. The second is to distribute income more equitably so that the more deprived sections of the population could have a comparatively larger share in the total national income. The third is to control the growth of population so that the birth-rate is reduced at least by one-third and preferably by half. The least to be attempted would be to double the national income per capita in the next twenty years. Similarly, steps will have to be taken to provide full employment to the people, especially to the educated.

SOCIAL AND NATIONAL INTEGRATION

Even more important is the role of education in achieving social and national integration. Indian society is hierarchical, stratified and deficient in vertical mobility. The social distance between the different classes, particularly between the rich and the poor, the educated and the uneducated, is large and is tending to widen. Our people profess a number of different religions; and the picture becomes even more complicated because of caste, an undemocratic institution which is still powerful and which, strangely enough, seems to have extended its sphere of influence under the very democratic processes of the Constitution itself. The situation, complex as it was, has been made critical by recent developments which threaten both national unity and social progress.

As education is not rooted in the traditions of the people, the educated persons tend to be alienated from their own culture. The growth of local, regional, linguistic and state loyalties tends to make the people forget 'India'.

The old values, which held society together, are disappearing, and as there is no effective programme to replace them by a new sense of social responsibility, innumerable signs of social disorganization are evident everywhere and are continually on the increase. These include strikes, increasing lawlessness and a disregard for public property, corruption in public life, and communal tensions and troubles. Student unrest, of which so much is written, is only one, and probably a minor one, of these symptoms. Against this background, the task to be attempted—the creation of an integrated and egalitarian society—is indeed extremely difficult and challenging.

POLITICAL DEVELOPMENT

The political challenge has many aspects, but three stand out. The first is the need to strengthen democracy. In spite of all odds, Indian democracy has given a fairly good account of itself so far. But it will not be permanently viable unless its foundations are deepened by the creation of an educated electorate, and a dedicated and competent leadership, and the cultivation of essential values like self-control, tolerance, mutual goodwill and consideration for others, all of which make democracy not only a form of government, but a way of life. The second is related to the defence of the country's freedom. The overriding priority of this challenge is recognized, for one must live before one can grow. But it cannot be met only by maintaining a large and efficient army or training all able-bodied youth for military service. The defence of freedom is a national concern, to which every citizen contributes his best in order to achieve self-sufficiency in food and other essentials, to strengthen the country's economy, and to create a secular, united and strong democratic State.

The third is the growing awakening among the masses who, suppressed for centuries, have now awakened to a sense of their rights and are demanding education, equality, higher standards of living and better civic amenities. This 'explosion of expectations' has also to be met through a planned programme of national development. Internationally, the country is facing another equally significant and urgent challenge. The gap between the standards of living in India and those of industrialized nations is very large. The first scientific industrial revolution which developed in the West over the last 200 years almost passed us by. The agro-industrial revolution which is even more crucial for us has yet to begin in our country.

The world is now at the beginning of the second scientific industrial revolution of automation and cybernetics, which is likely to be in full swing before the close of the century. It is difficult to visualize the changes it will make in man's life. One thing, however, is certain: unless proper steps are taken right from now, the gap between us and the industrialized countries following this second revolution may become too wide to be bridged. There is still another aspect to the challenge. It is true that knowledge is

international and that there can be no barriers, except those of our own creation, to its free import. But India cannot for ever remain at the receiving end of the pipeline. She must make her own contribution as an intellectual and cultural equal to the eternal human endeavour to extend the frontiers of knowledge. This demands a large-scale programme for the discovery and development of talent and the creation of centres of excellence in higher education which can compare favourably with the best of their kind in the world.

The difficulty of these problems is equalled only by the complexity of the situation, the gravity and urgency of the challenge, and the magnitude of the stakes involved. The Indian situation with its federal constitution, its multi-party system of democratic government, its multireligious and mixed society consisting of highly sophisticated groups who live side by side with primitive ones, its mixed economy which includes modern factories as well as traditional agriculture, and its multiplicity of languages, presents such a complex picture that it almost resembles a miniature world. In India's attempts at national development, the welfare of one-seventh of the world's population is at stake and the future of democracy and free societies is in the balance. She is heir to an ancient and great civilization which can make a contribution to human progress by striving to create what Acharya Vinobaji has described as the 'age of science and spirituality'. She has to raise herself from her present standards of living which are amongst the lowest in the world and to take her rightful place in the comity of nations as soon as possible—a task to be accomplished within a generation at the most.

Obviously, the solution of these problems makes large demands on us, the Indian people of this generation—we need a clear focus, deeper understanding, collective discipline, hard and sustained work, and dedicated leadership. The solution of these problems also needs the cooperation and assistance of the richer industrialized nations which share India's faith in democratic socialism and sympathize with her struggle to create a new social order.

EDUCATION: THE MAIN INSTRUMENT OF CHANGE

DEVELOPMENT OF HUMAN RESOURCES

These difficult, complex, significant and urgent problems are all interdependent, and the shortest and most effective way to their solution is obviously to make a simultaneous attack on all fronts. *This will have to be attempted through two main programmes:*

- The development of physical resources through the modernization of agriculture and rapid industrialization. This requires the adoption of a science-based technology, heavy capital formation and investment, and the provision of the essential infrastructure of transport, credit, marketing and other institutions; and
- The development of human resources through a properly organized programme of education.

It is the latter programme, namely, the development of human resources through education, which is the more crucial of the two. While the development of physical resources is a means to an end, that of human resources is an end in itself; and without it, even the adequate development of physical resources is not possible. The reason for this is clear. The realization of the country's aspirations involves changes in the knowledge, skills, interests and values of the people as a whole. This is basic to every programme of social and economic betterment of which India stands in need. For instance, there can be no hope of making the country self-sufficient in food unless the farmer himself is moved out of his age-long conservatism through a science-based education, becomes interested in experimentation, and is ready to adopt techniques that increase yields.

The same is true of industry. The skilled manpower needed for the relevant research and its systematic application to agriculture, industry and other sectors of life can only come from a development of scientific and technological education. Similarly, economic growth is not merely a matter of physical resources or of training skilled workers; it needs the education of the whole population in new ways of life, thought and work. Robert Heilbroner describes the journey to economic development undertaken by a traditional society as 'the great ascent' and points out that the essential condition for its success is human 'change on a grand scale'. He observes: 'The mere laying of a core of capital equipment, indispensable as that is for further economic expansion, does not yet catalyse a tradition-bound society into a modern one. For that catalysis to take place, nothing short of a pervasive social transformation will suffice; a wholesale metamorphosis of habits, a wrenching reorientation of values concerning time, status, money, work; and an unweaving and reweaving of the fabric of daily existence itself.' These observations are applicable to advances on the social, political and cultural fronts as well.

EDUCATION AS INSTRUMENT OF CHANGE

If this 'change, on a grand scale' is to be achieved without violent revolution there is one instrument, and one instrument only, that can be used: EDUCATION. Other agencies may help, and can indeed sometimes have a more apparent impact. But the national system of education is the only instrument that can reach all the people. It is not, however, a magic wand to wave wishes into existence. It is a difficult instrument, whose effective use requires strength of Will, dedicated work and sacrifice. But it is a sure and tried instrument, which has served other countries well in their struggle for development. It can, given the will and the skill, do so for India. This emphasis on the social purposes of education, on the need to use it as a tool for the realization of national aspirations or for meeting national challenges, does not imply any underestimation of values for the individual. In a democracy, the individual is an end in himself and the primary purpose of education is to provide him with the widest opportunity to develop his potentialities to the full.

But the path to this goal lies through social reorganization and emphasis on social perspectives. In fact, one of the important principles to be emphasized in the socialistic pattern of society, which the nation desires to create, is that individual fulfillment will come, not through selfish and narrow loyalties to personal or group interests, but through the dedication of all to the wider loyalties of national development in all its parameters. This direct link between education, national development and prosperity which we have emphasized and in which we deeply believe, exists only when the national system of education is properly organized, both qualitatively and quantitatively

The naive belief that all education is necessarily good, both for the individual and for society, and that it will necessarily lead to progress, can be as harmful as it is misplaced. Quantitatively, education can be organized to promote social justice or to retard it. History shows numerous instances where small social groups and elites have used education as a prerogative of their rule and as a tool for maintaining their hegemony and perpetuating the values upon which it has rested. On the other hand, there are cases in which a social and cultural revolution has been brought about in a system where equality of educational opportunity is provided and education is deliberately used to develop more and more potential talent and to harness it to the solution of national problems. The same is even more true of the quality of education.

A system of university education which produces a high proportion of competent professional manpower is of great assistance in increasing productivity and promoting economic growth. Another system of higher education with the same total output but producing a large proportion of indifferently educated graduates of arts, many of whom remain unemployed or are even unemployable, could create social tensions and retard economic growth. It is only the right type of education, provided on an adequate scale, that can lead to national development; when these conditions are not satisfied, the opposite effect may result.

Judged from this point of view, it becomes evident that the present system of education, designed to meet the needs of an imperial administration within the limitations set by a feudal and traditional society, will need radical changes if it is to meet the purposes of a modern democratic and socialistic society—changes in objectives, in content, in teaching methods, in programmes, in the size and composition of the student body, in the selection and professional preparation of teachers, and in organization. In fact, what is needed is a revolution in education which in turn will set in motion the much desired social, economic and cultural revolution.

The main concern of this Report is to identify the major programmes that can bring about this educational revolution which has three main aspects:

- Internal transformation so as to relate it to the life, needs and aspirations of the nation;
- Qualitative improvement so that the standards achieved are adequate, keep continually rising and, at least in a few sectors, become internationally comparable; and

- Expansion of educational facilities broadly on the basis of manpower needs and with an accent on equalization of educational opportunities.

It is with the first group of the programmes, namely, the transformation of the system to reflect the needs and aspirations of the Indian people.

RELATING EDUCATION TO THE LIFE, NEEDS AND ASPIRATIONS OF THE PEOPLE

As is well known, the existing system of education is largely unrelated to life and there is a wide gulf between its content and purposes and the concerns of national development.

For instance,

- The educational system does not reflect the supreme importance of agriculture which is neglected at all stages and does not attract an adequate share of the top talent in the country; enrolment in the agricultural faculties of universities is extremely low; and agricultural colleges are comparatively weak and underdeveloped;
- The main task before the nation is to secure rapid economic development. If this is to be successfully accomplished, education must be related to productivity. The present system is too academic to be of material help in increasing national wealth;
- The schools and colleges are largely unconcerned with the great national effort at reconstruction and teachers and students generally remain uncommitted to it. They are often even unaware of its principles and very rarely have opportunities to participate in its programmes;
- Instead of promoting social and national integration and making an active effort to promote national consciousness, several features of the educational system promote divisive tendencies; caste loyalties are encouraged in a number of private educational institutions; the rich and the poor are segregated, the former attending the better type of private schools which charge fees while the latter are forced, by circumstances, to attend free government or local authority schools of poor quality; and - at a time when the need to cultivate a sense of moral and social responsibility in the rising generation is paramount, education does not emphasize character-formation and makes little or no effort to cultivate moral and spiritual values, particularly the interests, attitudes and values needed for a democratic and socialistic society.

There is hardly any need to multiply such instances. The nature of the transformation needed in our educational system is generally recognized. What we wish to emphasize is its urgency. Traditional societies which desire to modernize themselves have to transform their educational system before trying to expand it, because the greater the expansion of the traditional system of education, the more difficult and costly it becomes

to change its character. This truth has been lost sight of and, during recent years, we have greatly expanded a system which continues to have essentially the same features it had at its creation about a century ago. In our opinion, therefore, no reform is more important or more urgent than to transform education, to endeavour to relate it to the life, needs and aspirations of the people and thereby make it a powerful instrument of social, economic and cultural transformation necessary for the realization of our national goals. *This can be done if education is related to productivity;*

- Strengthens social and national integration; consolidates democracy as a form of government and helps the country to adopt it as a way of life;
- Hastens the process of modernization; and
- Strives to build character by cultivating social, moral and spiritual values.

All these aspects are interrelated and in the complex process of social change, we cannot achieve even one without striving for all.

EDUCATION AND PRODUCTIVITY

India is in transition from a society in which education is a privilege of a small minority to one in which it could be made available to the masses of the people. The immense resources needed for this programme can be generated only if education is related to productivity so that an expansion of education leads to an increase in national income which, in turn, may provide the means for a larger investment in education. Education and productivity call thus constitute a 'rising spiral' whose different parts sustain and support one another.

The link between education and productivity can be forged through the development of the following programmes which should receive high priority in the plans of educational reconstruction:

- Making science a basic component of education and culture;
- Making work-experience an integral part of general education;
- Vocationalization of education, especially at the secondary school level, to meet the needs of industry, agriculture and trade; and
- Improvement of scientific and technological education and research at the university stage with special emphasis on agriculture and allied sciences.

SCIENCE AS A BASIC COMPONENT OF EDUCATION AND CULTURE

A basic distinction between traditional and modern societies is the development and use by the latter of science-based technology which helps modernization of agriculture and the development of industries. In a traditional society, production is based largely on empirical processes, experience, and trial and error, rather than on science; in a modern society, it is basically rooted in science. The electrical industry was probably the first to become science-based; next came the chemical industry; and now, in industrialized countries, agriculture is fast becoming a branch of applied science. This close interlocking and interdependence between science and technology is a

characteristic of the contemporary world. In recent years, several countries have been able to raise their GNP very rapidly because of their investment in basic science, technology and education.

We are at a crucial stage in the process of development and transformation; and in this context the role of science is of the utmost importance. Science education must become an integral part of school education; and ultimately some study of science should become a part of all courses in the humanities and social sciences at the university stage, even as the teaching of science can be enriched by the inclusion of some elements of the humanities and social sciences. The quality of science teaching has also to be raised considerably so as to achieve its proper objectives and purposes, namely, to promote an ever deepening understanding of basic principles, to develop problem-solving and analytical skills and the ability to apply them to the problems of the material environment and social living, and to promote the spirit of enquiry and experimentation. Only then can a scientific outlook become part of our way of life and culture.

It is necessary to highlight this last point which is sometimes underestimated. Science strengthens the commitment of man to free enquiry and to the quest for truth as his highest duty and obligation. It loosens the bonds of dogmatism and acts as a powerful dispeller of fear and superstition, fatalism and passive resignation. By its emphasis on reason and free enquiry, it even helps to lessen ideological tensions which often arise because of adherence to dogma and fanaticism. Although it is largely occupied with the understanding of Nature at present, its development is tending more and more to help man to understand himself and his place in the universe. In the developments that we envisage in the future, we hope that the pursuit of mere material affluence and power would be subordinated to that of higher values and the fulfillment of the individual. This concept of the mingling of 'science and spirituality' is of special significance for Indian education.

WORK-EXPERIENCE

As another programme to relate education to life and productivity, we recommend that work-experience should be introduced as an integral part of all education-general or vocational. We define work-experience as participation in productive work in the school, in the home, in a workshop, on a farm, in a factory or in any other productive situation.

In our opinion, all good and purposeful education should consist of at least four basic elements:

- 'Literacy' or a study of languages, the humanities and the social sciences;
- 'Numeracy' or a study of mathematics and the natural sciences; work-experience; and
- Social service.

In the present educational system, most of the time is taken up with the first, although even in this limited sphere, the attainments are not appreciable. The second is still quite

weak and needs a great deal of emphasis. But the third and fourth have been almost totally absent till recently and need to be highlighted-the former mainly for relating education to productivity, and the latter as a means of social and national integration. The need to include work-experience as an integral part of education is to some extent inherent in the very nature and organization of formal education. Traditionally, an individual grew up in society through participation in its activities, and work-experience formed the bulk of his education.

While this method had several advantages, its weakness consisted in that it was not essentially dynamic and forward-looking and tended to perpetuate traditional patterns of behaviour. Formal education, on the other hand, tended to withdraw the child temporarily from participation in community activities and to train him, in an artificial environment for his anticipated future role in society. This created a cleavage between the world of work and the world of study. This defect is particularly conspicuous in our system of education which tends to strengthen the tradition of denigrating work and alienates the students, particularly the first-generation learners, from their homes and communities. The introduction of work-experience is intended to overcome, to some extent, these weaknesses and to combine the advantages of the formal and informal systems of education. Work-experience is thus a method of integrating education with work. This is not only possible but essential in modern societies which adopt science-based technology.

In all traditional societies, an antithesis between education and work is usually postulated, partly because the techniques of production are primitive and do not necessarily require formal education, special skills or high intellectual ability, and partly because the work is generally manual, low-paid, akin to drudgery and confined mostly to the uneducated 'lower' classes. As against this, education is generally the privilege of the upper classes who are interested, not in working for a living, but in the cultivation of interests which may help them in the enjoyment of life. The educated elite thus become largely parasitical in character and the real productive workers-at a low level of efficiency, generally-are the unlettered peasants and artisans.

The complex techniques of production adopted in modern societies, on the other hand, require higher forms of general or technical education and a comparatively higher level of intellectual ability. High talent is required for research in technology, and even at the lower levels of work, brains become more important than physical strength. The traditional resistance of educated persons to engage themselves in productive work tends to disappear because, with the adoption of the new technology, work in industry or on the farm becomes more productive and remunerative and ceases to be looked down upon socially.

The educated person thus becomes an important source of production and the uneducated person, an unproductive burden on society. This process, which has already started in our country, needs to be accelerated and therefore the inclusion of workexperience as an integral part of all education acquires an urgent significance. The

need to provide some such corrective to the over-academic nature of formal education has been widely recognized. In the curricula of most contemporary school systems, particularly in the socialist countries of Europe, a place is found for what is variously called 'manual work' or 'work-experience'. In our country, a revolutionary experiment was launched by Mahatma Gandhi in the form of basic education. The concept of work-experience is essentially similar. It may be described as a redefinition of his educational thinking in terms of a society on the road to industrialization. In addition to being an effective educational tool, work-experience can, in our view, serve some other important purposes. It can help to make the distinction between intellectual and manual work less marked as also the social stratification based on it.

It could make the entry of youth into the world of work and employment easier by enabling them to adjust themselves to it. It could contribute to the increase of national productivity both by helping students to develop insights into productive processes and the use of science, and by generating in them the habit of hard and responsible work. And it might help social and national integration by strengthening the links between the individual and the community and by creating bonds of understanding between the educated persons and the masses. In providing work-experience, every attempt should be made to link programmes realistically to technology, to industrialization and to the application of science to productive processes, including agriculture. This 'forward look' in work-experience is important for a country which has embarked on a programme of industrialization.

In a well-organized programme, work-experience, at least from the higher primary stage, should also result in some earning for the student-either in cash or in kind. This would meet, to some extent, the expenditure which the students have to incur on their education or on their maintenance while at study. The amount of this earning will naturally increase as the students go up the educational ladder and it becomes possible to organize work-experience in a manner that would enable them to 'earn and learn'. The ultimate objective should be to move towards a situation in which the education of a student is not held to be complete unless he participates in some type of work-experience in real-life conditions and earns some amount, however small, towards his own maintenance. This will also help to develop in him values which promote economic growth, such as appreciating the importance of productive work and manual labour, willingness and capacity for hard work, and thrift. We realize that this is no easy task. But it will pay adequate dividends in the long run.

VOCATIONALIZATION

Another programme which can bring education into closer relationship with productivity is to give a strong vocational bias to secondary education and to increase the emphasis on agricultural and technological education at the university stage. This is of special significance in the Indian situation. where, as we have pointed out, the educational system has been training young persons so far mostly for government services

and the so-called white-collar professions. The introduction of practical subjects in secondary schools so as to divert them into different walks of life was first recommended, as far back as in 1882, by the Indian Education Commission.

But little or no effective action was taken to implement the recommendations and even today the enrolment in the vocational courses at the secondary stage is only nine per cent of the total enrolment, which is among the lowest in the world. Even at the university stage, vocational education was mostly ignored throughout the last century. The Calcutta University Commission pointed out that the great majority of university students about 22,000 out of 26,000- pursue purely literary courses which do not fit them for any but administrative, clerical, teaching and legal careers. About fifty years later, we find that the overall picture has improved only slightly and the proportion of students at the university stage enrolled in all courses of professional education is only 23 per cent of the total enrolment.

Our proposals on this subject are discussed elsewhere. But it may be briefly stated here that we visualize the future trend of school education to be towards a fruitful mingling of general and vocational education-general education containing some elements of prevocational and technical education, and vocational education, in its turn, having an element of general education. In the kind of society in which as suggested, be living in the coming years, a complete separation between the two will not only be undesirable but impossible. We also expect a considerable expansion of professional education at the university stage, especially in the agricultural and technological fields.

IMPROVEMENT OF SCIENTIFIC AND TECHNOLOGICAL EDUCATION

For the planned development of the national economy we need a large-scale expansion of enrolment in engineering and agriculture, and at the postgraduate level, in pure science subjects. The increase has to be several times the present enrolments. The quality of education and research needs a radical improvement.

REORGANISATION OF EDUCATION AT DIFFERENT STAGES

EARLY CHILDHOOD CARE AND EDUCATION

The National Policy on Children specially emphasises investment in the development of the young child, particularly children from parts of the population in which first generation learners predominate. Recognising the holistic nature of child development, *viz.*, nutrition, health and social, mental, physical, moral and emotional development, Early Childhood Care and Education will receive high priority and be suitably integrated with the Integrated Child Development. Services programme, wherever possible. Day-care centres will be provided as a support service for universalisation of primary education, to enable girls engaged in taking care of siblings to attend school and as a

support service for working women belonging to poorer parts. Programmes of ECCE will be child-oriented, focussed around play and the individuality of the child. Formal methods and introduction of the 3 R's will be discouraged at this stage. The local community will be fully involved in these programmes. A full integration of child care and pre-primary education will be brought about, both as a reeder and a strengthening factor for primary education and for human resource development in general. In continuation of this stage, the School Health Programme will be strengthened.

Elementary Education

The new thrust in elementary education will emphasise two aspects:

1. Universal enrolment and universal retention of children up to 14 years of age, and
2. A substantial improvement in the quality of education.

Child-Centred Approach

A warm, welcoming and encouraging approach, in which all concerned share a solicitude for the needs of the child, is the best motivation for the child to attend school and learn. A child-centred and activity-based process of learning should be adopted at the primary stage. First generation learners should be allowed to set their own pace and be given supplementary remedial instruction. As the child grows, the component of cognitive learning will be increased and skills organised through practice. The policy of non- detention at the primary stage will be retained, making evaluation as disaggregated as feasible. Corporal punishment will be firmly excluded from the educational system and school timings as well as vacations adjusted to the convenience of children.

School Facilities

Provision will be made of essential facilities in primary schools, including at least two reasonably large rooms that are usable in all weather, and the necessary toys, blackboards, maps, charts, and other learning material. At least two teachers, one of whom a woman, should work in every school, the number increasing as early as possible to one teacher per class. A phased drive, symbolically called- OPERATION BLACKBOARD will be undertaken with immediate effect to improve Primary Schools all over the country. Government, local bodies, voluntary agencies and individuals will be fully involved. Construction of school buildings will be the first charge on NREP and. RLEGP funds.

Non-Formal Education

A large and systematic programme of non-formal education will be launched for school drop-outs, for children from habitations without schools, working children and girls who cannot attend whole- day schools. Modern technological aids will be used to

improve the learning environment of NFE centres. Talented and dedicated young men and women from the local community will be chosen to serve as instructors, and particular attention paid to their training. Steps will be taken to facilitate their entry into the formal system in deserving cases.

All necessary measures will be taken to ensure that the quality of non-formal education is comparable with formal education. Effective steps will be taken to provide a framework for the curriculum on the lines of the national core curriculum, but based on the needs of the learners and related to the local environment. Learning material of high quality will be developed and provided free of charge to all pupils. NFE programmes will provide participatory learning environment, and activities such as games and sports, cultural programmes, excursions, *etc.* Much of the work of running NFE centres will be done through voluntary agencies and panchayati raj institutions. The provision of funds to these agencies will be adequate and timely. The Government will take over-all responsibility for this vital sector.

A Resolve

The New Education Policy will give the highest priority to solving the problem of children dropping out of school and will adopt an array of meticulously formulated strategies based on micro- planning, and applied at the grass-roots level all over the country, to ensure children's retention at school. This effort will be fully coordinated with the network of non-formal education. It shall be ensured that all children who attain the age about 11 years by 1990 will have had five years of schooling, or its equivalent through the non-formal stream. Likewise, by 1995 all children will be provided free and compulsory education upto 14 years of age.

Secondary Education

Secondary education begins to expose students to the differentiated roles of science, the humanities and social sciences. This is also an appropriate stage to provide children with a sense of history and national perspective and give them opportunities to understand their constitutional duties and rights as citizens. Conscious internalisation of a healthy work ethos and of the values of a humane and composite culture will be brought about through appropriately formulated curricula. Vocationalisation through specialised institutions or through the refashioning of secondary education can, at this stage, provide valuable manpower for economic growth. Access to secondary education will be widened to cover areas unserved by it at present. In other areas, the main emphasis will be on consolidation.

Pace-Setting School

It is universally accepted that children with special talent or aptitude should be provided opportunities to proceed at a faster pace, by making good quality education available to them, irrespective of their capacity to pay for it. Pace-setting schools intended

to serve this purpose will be established in various parts of the country on a given pattern, but with full scope for innovation and experimentation. Their broad aims will be to serve the objective of excellence, coupled with equity and social justice to promote national integration by providing opportunities to talented children largely rural, from different parts of the country to live and learn together, to develop their full potential, and, most importantly, to become catalysts of a nation-wide programme of school improvement. The schools will be residential and free of charge.

Vocational

The introduction of systematic, well-planned and rigourously implemented programmes of vocational education is crucial in the proposed educational reorganisation. These elements are meant to enhance individual employability, to reduce the mis-match between the demand and supply of skilled manpower, and to provide an alternative for those pursuing higher education without particular interest or purpose. Vocational education will be a distinct stream, intended to prepare students for identified occupations spanning several areas of activity. These courses will ordinarily be provided after the secondary stage, but keeping the scheme flexible, they may also be made available after Class VIII. In the interests of integrating vocational education better with their facilities the Industrial Training Institutes will also conform to the larger vocational pattern. Health planning and health service management should optimally interlock with the education and training of appropriate categories of health manpower through health-related vocational courses. Health education at the primary and middle levels will ensure the commitment of the individual to family and community health, and lead to health-related vocational courses at the +2 stage of higher secondary education. Efforts will be made to devise similar vocational courses based on Agriculture, Marketing, Social Services, *etc.*

An emphasis in vocational education will also be on development of attitudes, knowledge, and skills for entrepreneurship and self-employment. The establishment of vocational courses or institutions will be the responsibility of the Government as well as employers in the public and private sectors; the Government will, however, take special steps to cater to the needs of women, rural and tribal students and the deprived parts of society. Appropriate programmes will also be started for the handicapped. Graduates of vocational courses- will be given opportunities, under predetermined conditions, for professional growth, career improvement and lateral entry into courses of general, technical and professional education through appropriate bridge courses. Non-formal, flexible and need-based vocational programmes will also be made available to neoliterates, youth who have completed primary education, school drop-outs, persons engaged in work and unemployed or partially employed persons.

Special attention in this regard will be given to women. Tertiary level courses will be organised for the young who graduate from the higher secondary courses of the academic stream and may also require vocational courses. It is proposed that vocational

courses cover 1 per cent of higher secondary students by 1990 and 25 per cent by 1995. Steps will be taken to see that a substantial majority of the products of vocational courses are employed or become selfemployed. Review of the courses offered would be regularly undertaken. Government will also review its recruitment policy to encourage diversification at the secondary level.

Higher Education

Higher education provides people with an opportunity to reflect on the critical social, economic, cultural, moral and spiritual issues facing humanity. It contributes to national development through dissemination of specialized knowledge and skills. It is therefore a crucial factor for survival. Being at the apex of the educational pyramid, it has also a key role in producing teachers for the education system. In the context of the unprecedented explosion of knowledge, higher education has to become dynamic as never before, constantly entering uncharted areas.

There are around 15 universities and about 5, colleges in India today. In view of the need to effect an all round improvement in these institutions, it is proposed that, in the near future, the main emphasis will be on the consolidation of, and expansion of facilities in, the existing institutions. Urgent steps will be taken to protect the system from degradation. In view of mixed experiences with the system of affiliation, autonomous colleges will be helped to develop in large numbers until the affiliating system is replaced by a freer and more creative association of universities with colleges. Similarly, the creation of autonomous departments within universities on a selective basis will be encouraged. Autonomy and freedom will be accompanied by accountability. Courses and programmes will be redesigned to meet the demands of specialisation better. Special emphasis will be laid on linguistic competence.

There will be increasing flexibility in the combination of courses. State level planning and coordination of higher education will be done through Councils of Higher Education. The UGC and these Councils will develop coordinative methods to keep a watch on standards. Provision will be made for minimum facilities and admission will be regulated according to capacity. A major effort will be directed towards the transformation of teaching methods. Audio-visual aids and electronic equipment will be introduced; development of science and technology curricula and material, research, and teacher orientation will receive attention. This will require preparation of teachers at the beginning of the service as well as continuing education thereafter.

Teachers' performance will be systematically assessed. All posts will be filled on the basis of merit. Research in the universities will be provided enhanced support and steps will be taken to ensure its high quality. Suitable mechanisms will be set up by the UGC for coordinating research in the universities, particularly in thrust areas of science and technology, with research undertaken by other agencies. An effort will be made to encourage the setting up of national research facilities within the university system, with proper forms of autonomous management. Research in Indology, the humanities

and social sciences will receive adequate support. To fulfil the need for the synthesis of knowledge, inter-disciplinary research will be encouraged. Efforts will be made to delve into India's ancient fund of knowledge and to relate it to contemporary reality. This effort will imply the development of facilities for the intensive study of Sanskrit and other Classical languages. In the interest of greater coordination and consistency in policy, sharing of facilities and developing interdisciplinary research, a national body covering higher education in general, agricultural, medical, technical, legal and other professional fields will be set up.

Open University and Distance Learning

The Open University system has been initiated in order to augment opportunities for higher education and as an instrument of democratising education. The Indira Gandhi National Open University, established in 1985 in fulfilment of these objectives, will be strengthened. This powerful instrument will have to be developed with care and extended with caution.

Delinking Degrees From Jobs

A beginning will be made in de-linking degrees from jobs in selected areas. The proposal cannot be applied to occupation-specific courses like Engineering, Medicine, Law, Teaching, *etc.* Similarly, the services of specialists with academic qualifications in the humanities, social sciences, sciences, *etc.* will continue to be required in various job positions. De-linking will be applied in services for which a university degree need not be a necessary qualification. Its implementation will lead to a re-fashioning of job-specific courses and afford greater justice to those candidates who, despite being equipped for a given job, are unable to get it because of an unnecessary preference for graduate candidates. Concomitant with de-linking, an appropriate machinery, such as a National Testing Service, will be established, in appropriate phases, to conduct tests on a voluntary basis to determine the suitability of candidates for specified jobs and to pave the way for the emergence of norms of comparable competence across the nation.

Rural University

The new pattern of the Rural University will be consolidated and developed on the lines of Mahatma Gandhi's revolutionary ideas on education so as to take up the challenges of micro-planning at grassroot levels for the transformation of rural areas. Institutions and programmes of Gandhian basic education will be supported.

TECHNICAL AND MANAGEMENT EDUCATION

Although the two streams of technical and management education are functioning separately, it is essential to look at them together, in view of their close relationship and complementary concerns. The reorganisation of Technical and Management Education should take into account the anticipated scenario by the turn of the century, with specific

reference to the likely changes in the economy, social environment, production and management processes, the rapid expansion of knowledge and the great advances in science and technology. The infrastructure and services sectors as well as the unorganised rural sector also need a greater induction of improved technologies and a supply of technical and managerial manpower.

This will be attended to by the Government. In order to improve the situation regarding manpower information, the recently set up Technical Manpower Information System will be further developed and strengthened. Continuing education, covering established as well as emerging technologies, will be promoted. As computers have become important and ubiquitous tools, a minimal exposure to computers and a training in their use will form part of professional education. Programmes of computer literacy will be organised on wide scale from the school stage.

In view of the present rigid entry requirements to formal courses restricting the access of a large segment of people to technical and managerial education, programmes through a distance-learning process, including use of the mass media, will be offered. Technical and management education programmes, including education in polytechnics, will also be on a flexible modular pattern based on credits, with provision for multi-point entry. A strong guidance and counselling service will be provided. In order to increase the relevance of management education, particularly in the non-corporate and under-managed sectors, the management education system will study and document the Indian experience and create a body of knowledge and specific educational programmes suited to these sectors. Appropriate formal and non-formal programmes of technical education will be devised for the benefit of women, the economically and socially weaker parts, and the physically handicapped. The emphasis on vocational education and its expansion will need a large number of teachers and professionals in vocational education, educational technology, curriculum development, *etc.* Programmes will be started to meet this demand. To encourage students to consider “self-employment” as a career option, training in entrepreneurship will be provided through modular or optional courses, in degree or diploma programmes. In order to meet the continuing needs of updating curriculum, renewal should systematically phase out obsolescence and introduce new technologies or disciplines.

Institutional Thrusts

Some polytechnics in the rural areas have started training weaker groups in those areas for productive occupations through a system of community polytechnics. The community polytechnic system will be appraised and appropriately strengthened to increase its quality and coverage.

Innovation, Research and Development

Research as a means of renovation and renewal of educational processes will be undertaken by all higher technical institutions. It will primarily aim at producing quality

manpower capable of taking up R and D functions. Research for development will focus on improving present technologies, developing new indigenous ones and enhancing production and productivity. A suitable system for watching and forecasting technology will be set up. The scope for cooperation, collaboration and networking relationships between institutions at various levels and with the user systems will be utilised. Proper maintenance, and an attitude of innovation and improvement will be promoted systematically.

Promoting Efficiency and Effectiveness at All Levels

As technical and management education is expensive, the following major steps will be taken for cost-effectiveness and to promote excellence:

- High priority will be given to modernisation and removal of obsolescence. However, modernisation will be undertaken to enhance functional efficiency and not for its own sake or as a status symbol:
- Institutions will be encouraged to generate resources using their capacities to provide services to the community and industry. They will be equipped with up-to-date learning resources, library and computer facilities:
- Adequate hostel accommodation will be provided, specially for girls. Facilities for sports, creative work and cultural activities will be expanded:
- More effective procedures will be adopted in the recruitment of staff. Career opportunities, service conditions, consultancy norms and other perquisites will be improved.
- Teachers will have multiple roles to perform: teaching, research, development of learning resource material, extension, and managing the institution. Initial and in-service training will be made mandatory for faculty members and adequate training reserves will be provided. Staff Development Programmes will be integrated at the State, and coordinated at Regional and National levels.
- The curricula of technical and management programmes will be targetted on, current as well as the projected needs of industry or user systems. Active interaction between technical or management institutions and industry will be promoted in programme planning and implementation, exchange of personnel, training facilities and resources, research and consultancy and other areas of mutual interest.
- Excellence in performance of institutions and individuals will be recognised and rewarded. The emergence of substandard institutions will be checked. A climate conducive to excellence and innovation will be promoted with full involvement of the faculty.
- Select institutions will be awarded academic, administrative and financial autonomy of varying degrees, building in safeguards with respect to accountability.
- Networking systems will have to be established between technical education

and industry, R D organisations, programmes of rural and community development, and with other sectors of education with complementary characteristics.

Management Functions and Change

In view of the likely emergence of changes in management systems and the need to equip students with the ability to cope with them, effective mechanisms will be devised to understand the nature and direction of change *per se* and to develop the important skill of managing change. In view of the integrated nature of the task, the Ministry of Human Resource Development will coordinate the balanced development of engineering, vocational and management education as well as the education of technicians and craftsmen. Professional societies will be encouraged and enabled to perform their due role in the advancement of technical and management education.

The All India Council for Technical Education will be vested with statutory authority for planning, formulation and the maintenance of norms and standards, accreditation, funding of priority areas, monitoring and evaluation, maintaining parity of certification and awards and ensuring the coordinated and integrated development of technical and management education. Mandatory periodic evaluation will be carried out by a duly constituted Accreditation Board. In the interests of maintaining standards and for several other valid reasons, the commercialisation of technical and professional education will be curbed. An alternative system will be devised to involve private and voluntary effort in this sector of education, in conformity with accepted norms and goals.

EDUCATION AND MODERNIZATION

We have already stated that the most distinctive feature of a modern society, in contrast with a traditional one, is in its adoption of a science-based technology. It is this which has helped such societies to increase their production so spectacularly. It may be pointed out, however, that science-based technology has other important implications for social and cultural life and it involves fundamental social and cultural changes which are broadly described as 'modernization'. We shall briefly discuss the impact of this modernization on programmes of educational reconstruction.

EXPLOSION OF KNOWLEDGE

There has been a great explosion of knowledge during the last few decades. In a traditional society, the stock of knowledge is limited and grows slowly so that the main aim of education is interpreted to be its preservation. In a modern society, on the other hand, the stock of knowledge is far greater and the pace of its growth is infinitely quicker. One of the main tasks of education in a modern society is to keep pace with this advance in knowledge. In such a society, knowledge inevitably ceases to be something to be received passively; it is something to be actively discovered. If this is rightly understood, it would involve a revolution in traditional education where 'to

know' has come to mean 'to know by heart', where respect for all inherited knowledge is assiduously cultivated and where the assimilative faculties tend to be emphasized to the neglect of the critical and creative ones. In India, as in other countries where similar conditions prevail, this would require, among other things, a new approach to the objectives and methods of education, and changes in the training of teachers. Unless they are trained in new ways of teaching and learning, the students in schools and colleges will not be able to receive the type of education needed for the new society.

RAPID SOCIAL CHANGE

Another feature of a modern society is the quick, almost breath-taking rate at which social change takes place. In a traditional society, change is so slow that the conservatism of the educational system does comparatively little harm. In a modern society, on the other hand, change is so rapid that the school must always be alert if it is to keep abreast of significant changes. There is, therefore, an imperative need for adopting a dynamic policy in such a situation. An educational system which does not continually renovate itself, becomes out of date and hampers progress because it tends to create a lag between its operative purposes and standards and the new imperatives of development, both in quality and quantity. The very aim of education has to be viewed differently-it is no longer taken as concerned primarily with the imparting of knowledge or the preparation of a finished product, but with the awakening of curiosity, the development of proper interests, attitudes and values and the building up of such essential skills as independent study and the capacity to think and judge for oneself without which it is not possible to become a responsible member of a democratic society.

NEED FOR RAPID ADVANCE

Two other aspects of modernization need emphasis. The first is that once a society launches itself upon a programme of modernization, there is no turning back, no half-way house where we can arrest the process. In the initial stages, such a change must disturb the traditional equilibrium reached and maintained over centuries which, though it had its obvious disadvantages, had some built-in redeeming factors as well. The attempt to create a new social order naturally creates a host of unexpected social, economic, cultural and political problems. But if one tinkers with the problems involved or tries to march with faltering steps, if one's commitments and convictions are half-hearted and faith is lacking, the new situation may turn out to be worse than the old one. The only solution to these transitional problems is to move rapidly forward and create a new equilibrium, based on the full implications of the process of modernization.

MODERNIZATION AND EDUCATIONAL PROGRESS

The progress of modernization will, therefore, be directly related to the pace of educational advance and the one sure way to modernize quickly is to spread education, produce educated and skilled citizens, and train an adequate and competent intelligentsia.

The Indian society of today is heir to a great culture. Unfortunately, however, it is not an adequately educated society, and unless it becomes one, it will not be able to modernize itself and to respond appropriately to the new challenges of national reconstruction or take its rightful place in the comity of nations. The proportion of persons who have so far been able to receive secondary and higher education is very small at present—less than two per cent of the entire population. This will have to be increased to at least ten per cent to make any significant impact. The composition of the intelligentsia must also be changed; it should consist of able persons, both men and women, drawn from all strata of society.

There must also be changes in the skills and fields of specialization to be cultivated. At present, the intelligentsia consists predominantly of the white-collar professions and students of the humanities, while the proportion of scientists and technical workers in its ranks is quite small. To change this, greater emphasis must be placed, as we have argued earlier, on vocational subjects, science education and research. The average level of competence is not at all satisfactory, due mainly to inadequate standards maintained in the universities. This is damaging to Indian academic life and its reputation. In order to change this situation radically, it will be necessary to establish a few ‘major’ universities in the country which attain standards comparable to the best in any part of the world, and which will gradually spread their influence to others. This is one of the basic reforms needed in our system of higher education.

Concept of Social Change

According to Davis: *“By social change is meant only such alterations as occur in social organization, that is, structure and functions of society.*

According to Gillin and Gillin: *“Social changes are variations from the accepted modes of life; whether due to alterations in geographical conditions, in cultural equipment, composition of the population, or ideologies and whether brought about by diffusion or invention within the group.”*

According to Jones: *“Social changes are a term used to describe variations in, or modifications of any aspect of social processes, social patterns, social interaction or social organization.”*

According to Fairchild: *“Social change means variations or modifications in any aspect of social processes, patterns or form.”*

According to Jenson: *“Social change may be defined as modification in the ways of doing and thinking of people.”*

According to MacIver and Page: *“Change in social structure is social change.”* Social structure is the network of relationship in a society involving social status, social roles and social norms.

According to Dawson and Gettys: *“Cultural change is social change, since culture is social in its origin.”*

According to Merrill and Eldredge: *“Social change means that large number of persons are engaged in activities that differ from those which they or their immediate forefathers engaged in sometime before. When human behaviour is in the process of modification, that is only another way of indicating that social change is occurring.”*

According to Ginsberg's view: "By social change I understand change in social structures such as the size of a society, the composition or balance of its parts or the type of its organization."

SOCIAL CHANGE AND CULTURAL CHANGE

Relation between social change and cultural change: Many of the sociologists and social thinkers like Gillin and Gillin, Malinowski, Ruth Benedict, Pigginton *etc.* are of the view that social change and cultural change are one. They believe that social changes take place as a result of cultural changes and so they think that they do not differ from one another. **Difference between social change and cultural change:** The fact is that social change and cultural change may be related but there is certain difference between the two.

Social change is a part of cultural change which is quite comprehensive affair. Social change implies change in social relation only. Cultural change is not mere social change. It also includes changes that take place in the field of art, knowledge, belief, law, science, ethics, philosophy and technology *etc.* Thus cultural change is a broader term. Secondly, society is a net work of various types of relationships and any type of change in the social relationship results into a social change. On the other hand, cultural change means change in the field or art, literature, religion *etc.* From this point of view, a social change is more dynamic as compared to cultural change because here the change takes place at a fast speed. However, social changes mainly refer to changes taking place in the non-material aspect of culture. Thus social change can be defined as the alteration or transformation in the material and non-material culture of a society.

Social change and cultural change in spite of similarity are quite different from one another. The difference between the two may be studied under the following heads:

Social relationship: Social change leads to changes in social relationship. As a result of this, some changes take place in the social structure. Cultural change brings about change in the art, science, literature *etc.* As a result of this change different aspects of the culture change take place. It does not mean that necessarily a change in the social structure or social relationship takes place.

Social change as a part of cultural change: Social change is more intimately related to change in the non-material aspect of culture whereas cultural change comprehends both material and non-material changes. Thus social change is a part of cultural change. Cultural change is more comprehensive, social change is relatively restricted or narrow. Social change is limited to changes in the structure of society and the functioning of this structure.

Causes: Social change is caused by physical causes as well as conscious efforts. The cultural change is the result of conscious and planned effort.

Speed: Social change is faster as compare to cultural change. New types of inventions and technological developments bring about changes in social structure and relationship.

as compared to these changes, changes in religion, traditions, norms, values which form part of the culture, take place at a slower pace. The two are different but they are inter-related as well. Change in one necessarily affects the change in other.

PROCESSES OF SOCIAL CHANGE

The term “Social change” itself suggests nothing as far as its direction is concerned. It is a generic term describing one of the categorical processes. It only suggests a difference through time in the object to which it is applied. Social changes are of various types and can be explained by different terms such as Growth, Progress, Evolution, Revolution, Adaptation, and Accommodation, *etc.* Here we shall consider only two terms, *i.e.*, Progress and Evolution.

THE MEANING OF EVOLUTION

Evolution is a process of differentiation and integration. The term ‘evolution’ comes from the Latin word ‘evolvere’ which means ‘to develop’ or ‘to unfold’. It is equivalent to the Sanskrit word ‘vikas’. It means more than growth. The word ‘growth’ connotes a direction of change but only of a quantitative character, *e.g.*, we say population grows. Evolution involves something more intrinsic, change not merely in size but at least in structure also, for example when we speak of biological evolution, we refer to the emergence of certain organisms from others in a kind of succession.

Evolution describes a series of related changes in a system of some kind. It is a process in which hidden or latent characters of a thing reveal themselves. It is an order of change which unfolds the variety of aspects belonging to the nature of the changing object. We cannot speak of evolution when an object or system is changed by forces acting on it from without.

The change must occur within the changing unity as the manifestation of forces operative within it. But since nothing is independent of the universe, evolution also involves a changing adaptation of the object to its environment, and after adaptation a further manifestation of its own nature. Thus, evolution is a continuous process of differentiation-cum-integration.

The concept of evolution as a process of differentiation-cum integration was first developed by the German sociologist Von Bae and subsequently by Darwin, Spencer and many others. Spence writes, “Societies show integration, both by simple increase of mass and by coalescence and re-coalescence of masses. The changes from homogeneity to heterogeneity is multitudinously exemplified; from the simple tribe, to the civilized nation full of structural and functional unlikeness in all parts. With progressive integration and heterogeneity goes increasing coherence... simultaneously comes increasing definiteness.

Social organisation is at first vague; advance brings settled arrangement which grows slowly more precise; customs pass into laws, which while gaining fixity, also become more specific in their application to variety of actions, and all institutions, at first

confused] intermingled, slowly separated at the same time that each within itself marks off more distinctly its component structures. Thus in all respects is fulfilled the formula of evolution. There is progress towards greater size, coherence, multiformity and definiteness.”

Herbert Spencer thus Prescribes four Principles of Evolution these are:

- (i) Social evolution is one cultural or human aspect of the law of cosmic evolution;
- (ii) Social evolution takes place in the same way in which cosmic evolution takes place;
- (iii) Social evolution is gradual;
- (iv) Social evolution is progressive.

Social Evolution Does not Always Proceed by Differentiation

But the point at issue is whether this process of differentiation-cum-integration is sufficient to explain the general march of society excluding thereby any other kind of interpretation. Ginsberg writes, “The notion that evolution is a movement from the simple to the complex can be, and has been, seriously disputed.” In every field where we find the forces of differentiation at work, there the opposite trends are also manifest.

Thus, in the development of languages where the process of differentiation has been stressed we have many disconcerting facts. The modern languages derived from Sanskrit like Bengali or Gujrati cannot be compared in their structure with the richness and diversity of their origin. Here the process is not towards differentiation but towards simplification.

In the development of religion too the transition from fusion to differentiation is difficult to see. The state has made inroads into the institutions once administered by the church. Many of the functions once performed by the church are now being absorbed by the state. Instead of differentiation there is fusion between state and religion. In the economic system too we find the state controlling more and more the economic activities of the people, the period of laissez-faire being over. On the whole we find that social evolution does not always proceed by differentiation, but also by simplification and synthesis.

To define, social evolution is the process by which individuals are detached from or fail to be attached to an old group norm so that ultimately a new norm is achieved. According to Hobhouse, “Social evolution is development, planned and unplanned of culture and forms of social relationships or social interaction.”

Looking to the difficulties about the version of social evolution the French sociologist, Claude-Levi-Strauss was of the opinion that “sociology should relinquish every attempt at discovering origins and forms of evolution.” However, in spite of the various difficulties the concept of evolution still retains its usefulness. MacIver too has supported the principle of social evolution. He has criticized the practice of believing social evolution to be imaginary. Social evolution is a reality. Nadel writes: “We need

the concept of evolution as it were, to satisfy our philosophical conscience; but the 'law' of evolution is of too huge a scale to help us in understanding the behaviour of Toms, Dicks and Harrys among societies and culture, which after all is our main concern. Perhaps indeed there are no particular 'laws' of evolution, but only one law', or postulate if you like, that there is evolution."

The Idea of Progress

In the earlier theories of biological evolution the idea of progress was closely associated with that of evolution. For the social evolutionists of the nineteenth century social evolution was in effect social progress. The technological advance of the same century led many philosophers and sociologists to conclude that the major trends of social phenomena made for social progress. But from what has been discussed in these pages it is clear that the idea of progress is different from that of evolution.

Differentiation Between Evolution and Progress

What, in fact, do we mean by progress is "a development or evolution in a direction which satisfies rational criteria of value" According to Ogburn, progress "is a movement towards an objective, thought to be desirable by the general group, for the visible future. According to MacIver, "By progress we simply not merely direction, but direction towards some final goal, some destination determined ideally not simply by the objective consideration at work." According to Burgess, "Any change or adaptation to an existent environment that makes it easier for a person or group of persons or other organised form of life to live may be said to represent progress." According to Lumley, "progress is change, but it is change in a desired or approved direction, not any direction."

The nature of progress depends on two factors: the nature of the end and the distance at which we are from it. Thus, when we say that we are progressing, we mean that society is flourishing both materially and morally. Evolution is merely change, the change may be for the better or the worse. When we speak of social evolution we refer to the emergence of certain institution. The emergence of the institution may or may not be welcomed by the people. The reference is to an objective condition which is not evaluated as good or bad.

But when we speak of progress we imply not merely direction, but direction towards some final goal, some destination determined ideally. Progress means change for the better, and hence implies a value judgment. It is not possible to speak of progress without reference to standards. Hobhouse writes. "By evolution I mean any sort of growth, by social progress the growth of social life in respect of those qualities to which human being can attach or can rationally attach values."

According to Mazumdar, H.T. Progress must at Least Contain Six Ingredients:

1. Enhancement of the dignity of man,
2. Respect for each human personality,
3. Ever increasing freedom for spiritual quest and for investigation of truth,

4. Freedom for creativity and for aesthetic enjoyment of the works of nature as well as of man,
5. A social order that promotes the first four values, and
6. Promotes life, liberty and the pursuit of happiness, with justice and equity to all.

Now it is easy to see why evolution cannot be progress. It is not logically necessary that evolutionary process, should always move in the direction of progress. That society has evolved, all agree. That society has progressed, all would not agree because we cannot speak of progress” without reference to standards, and standards, as we know, are eminently subjective. If the process of evolution satisfies also our sense of values and if it brings a fuller realization of the values we cherish then for us it is also progress.

Different people may look differently on the same social changes and to some they may spell progress, to others decadence. Evolutionary changes are welcomed by some and are opposed by others. Civil marriages, divorce, women’s participation in public life, free mixing of young boys and girls may appear to some to be in line with progress while, to others it may seem retrogression because they have different values.

Primitivism has always had its champions and it still has them today. Many of the conditions on which important human values such as contentment, economic security, honesty and freedom depend are not often realized more adequately in the more evolved society. Industrialization led to urbanization and urbanization led to congestion, epidemics, poor health, and more accidents on the road. Similarly, competition, rivalry, corruption and dishonesty are the other effects of industrialization.

In fact, strong indictments have been drawn against civilization on the basis of social and moral values. Clearly, therefore, we cannot associate progress with evolution. In short, no single criterion can be used as a test of progress. Societies are complexes made up of many important elements. Progress is achieved if, in a society, all aspects of social life move in a coordinated manner towards desired ends.

To Briefly put the Characteristics of Progress are the Following:

- (i) Progress is change — a change in some direction:
- (ii) Change can be called progress only when it fulfills the desired aim:
- (iii) Progress is communal *i.e.*, related to social system,
- (iv) Progress is volitional. It requires desire and volition;
- (v) The concept of progress is variable. What is considered today the symbol of progress may tomorrow be regarded as sign of regress.
- (vi) There are no limits to human progress.

Have we Progressed

To the question whether we are progressing or not or whether we are more cultured than our ancestors, no absolute answer can be given. Comte, it may be recalled, believed in the perfectibility of society, although he considered that perfection was something that men would have via science. Marx also advanced the thesis that progress was a law

of society. Nothing could prevent the coming of communism where all men would share alike and all would be content. In those days progress was regarded as a 'cultural compulsion.' Of recent, the social philosophers have changed their mood. They consider the modern civilization as a failure or as an experiment doomed to failure. Standards of morality are no respecters of technical achievement. However, the answer to whether we have progressed or not depends upon our standards of moral value.

Our parents do not share many of our moral standards, for standards are not objective. In the near past, progress was taken for granted; now in some circles, the very idea arouses indignation, and the multitudinous deficiencies in human social conduct are pointed to with something approaching triumph. The national wealth of the country has gone up, but is the acquisition of wealth progress? We have invented aeroplanes and other fast-moving mobiles, but does it bring more security of life? Our country is on the way to industrialization but does this bring health, happiness or peace of mind? Some people marvel at our material achievement but often question whether it really represents progress. Thus, there can be much difference of opinion about whether we have progressed or not. Progress in science is possible but no one is obliged to regard progress in science as a good thing in itself. Evidence of progress in morality from preliterate society to modern civilization is simply lacking. In spite of the many technological achievements, big industries and imposing dams the fact remains that in India the evils of unemployment, crime, violence and disease have not lessened.

The family bonds have loosened. More marriages break now than yesterday. The social evils like drug-addiction, dowry system, prostitution, alcoholism, child exploitation and delinquency have increased manifold. We are politically hypocrites, economically corrupt, socially dishonest and morally unfaithful. In the face of these multitudinous defects in our social conduct it would be hard to maintain that we have progressed.

Thinkers like Mahatma Gandhi and Aurobindo Ghose have warned mankind against moral degeneration. No universal standards of progress. But as stated above it is all a question of one's standard of moral value and outlook, if we think that increased scope for personal development is really better than opportunity for only a few, if we think that education makes for more enlightened judgment and further if we believe that in India more people have now scope for development than before, then we may justly say that we have progressed. Nobody would deny that we have progressed in the case of technology. Tools have become more varied and efficient.

Whether the influence of tools on society has been for human happiness or not is a question to which no definite answer can be given for there are different standards for different people to measure human happiness. Conceptions of happiness differ as to ideals of what is good for a people. In a word it is difficult to find clear and definite standards that all people would accept and to formulate definite conceptions of progress which may apply to all time and to all cultures. While general principles do serve as tools to be used in thinking out the course of action we wish to pursue, they do not

afford specific guidance. While considering social progress, it is well to note the time and place qualifications. Thus, abolition of female labour at night may be deemed a step in the direction of progress but may not be so deemed a hundred years hence.

It may be interesting to speculate on the probability of change in the future. Some thinkers are of the opinion that men have all what they need in material goods and that there is no need for further invention. However, it would be unwise to assert that further inventions be stopped because mankind has all the material goods it needs. Man's wants are limitless. Changes will continue in future also.

SOCIAL PROGRESS OF SOCIAL CHANGE

SOCIAL PROGRESS

Social progress is the idea that societies can or do improve in terms of their social, political, and economic structures. This may happen as a result of direct human action, as in social enterprise or through activism, or as a natural part of sociocultural evolution.

The concept of social progress was introduced in the early 19th century social theories, especially social evolution as described by Auguste Comte and Herbert Spencer. It was present in the Enlightenment's philosophies of history. As a goal, social progress has been advocated by varying realms of political ideologies with different theories on how it is to be achieved.

HISTORY OF SOCIAL PROGRESS AS CONCEPT

The Enlightenment (1650–1800)

The big breakthrough to a new idea in Europe was the Enlightenment, when social commentators and philosophers began to realize that people *themselves* could change society and change their way of life. Instead of being made completely by gods, there was increasing room for the idea that people themselves *made their own society* – and not only that, as Giambattista Vico argued, *because* people practically made their own society, they could also fully comprehend it. This gave rise to new sciences, or proto-sciences, which claimed to provide new scientific knowledge about what society was like, and how one may change it for the better.

In turn, this gave rise to progressive opinion, in contrast with conservational opinion. The social conservationists were skeptical about panaceas for social ills. According to conservatives, attempts to radically remake society normally make things worse. Edmund Burke was the leading exponent of this, although later-day liberals like Hayek have espoused similar views. They argue that society changes organically and naturally, and that grand plans for the remaking of society, like the French Revolution, National Socialism and Communism hurt society by removing the traditional constraints on the exercise of power.

The Notion of Freedom

This new idea implied a new concept of human freedom, *i.e.*, people independently making their own lives using their own judgment. Initially, this concept appeared rather paradoxical; thus, Jean-Jacques Rousseau wrote, “People are born free, but are everywhere in chains”. A big breakthrough was the French Revolution of 1789, which inspired a lot of new philosophical thought.

In the philosophy of the German thinker Hegel, history radically recasts itself as the continual development of humanity towards ever-greater freedom, continually extending the limits of freedom. This philosophy is still religious and mystical however, insofar as Hegel sees history as culminating in the unity of God with the world, but at the same time, Hegel also affirmed and imputed a Logos or teleology to human history, and fully recognized that both evolutionary and revolutionary transformations took place in history. This was a hopeful philosophy, which in a rational way sees real progress occurring in history. It was possible to detect human advances, as well as human regressions to an earlier state. In Hegel’s view, if something existed, it was rational. If it passed out of existence, that was because it had become irrational. This contained a very important idea, namely that history was not a fluke of fate (a *kismet*) but that it could be *rationally understood*, at least in principle.

Marxist Theory (Late 19th Century)

Marx Developed a theory of Historical Materialism. He Describes the mid-19th Century Condition in the Communist Manifesto as follows:

The bourgeoisie cannot exist without constantly revolutionizing the instruments of production, and thereby the relations of production, and with them the whole relations of society. Conservation of the old modes of production in unaltered form, was, on the contrary, the first condition of existence for all earlier industrial classes. Constant revolutionizing of production, uninterrupted disturbance of all social conditions, everlasting uncertainty, and agitation distinguish the bourgeois epoch from all earlier ones. All fixed, fast frozen relations, with their train of ancient and venerable prejudices and opinions, are swept away, all new-formed ones become antiquated before they can ossify. All that is solid melts into air, all which is holy is profaned, and man is at last compelled to face with sober senses his real condition of life and his relations with his kind. Furthermore Marx described the process of social progress which is in his opinion based on the interaction between the productive forces and the relations of production:

No social order is ever destroyed before all the productive forces for which it is sufficient have been developed, and new superior relations of production never replace older ones before the material conditions for their existence have matured within the framework of the old society.

Capitalism is thought by Marx as a process of continual change, in which the growth of markets dissolve all fixities in human life, and Marx admits that capitalism is progressive and non-reactionary. Marxism further states that capitalism, in its quest for

higher profits and new markets, will inevitably sow the seeds of its own destruction. Marxists believe that, in the future, capitalism will be replaced by socialism and eventually communism.

In Modernist Thought (20th Century)

“The reasonable man adapts himself to the world. The unreasonable man persists in trying to adapt the world to himself. Therefore, all progress depends on the unreasonable man.”

—George Bernard Shaw

Many advocates of capitalism such as Schumpeter agreed with Marx’s analysis of capitalism as a process of continual change through creative destruction, but, unlike Marx, believed and hoped that capitalism could essentially go on forever.

Thus, by the beginning of the 20th century, two opposing schools of thought – Marxism and liberalism – believed in the possibility and the desirability of continual change and improvement. Marxists strongly opposed capitalism and the liberals strongly supported it, but the one concept they could both agree on was modernism, a trend of thought which affirms the power of human beings to make, improve and reshape their society, with the aid of scientific knowledge, technology and practical experimentation.

Postmodernist Thought (Late 20th Century)

“As it will be in the future, it was at the birth of Man. There are only four things certain since Social Progress began. That the Dog returns to his Vomit and the Sow returns to her Mire.”

—Rudyard Kipling

In the postmodernist thought steadily gaining ground from the 1980s, the grandiose claims of the modernizers are steadily eroded, and the very concept of social progress is again questioned and scrutinized. In the new vision, radical modernizers like Joseph Stalin and Mao Zedong appear as totalitarian despots, whose vision of social progress is held to be totally deformed. Postmodernists question the validity of 19th century and 20th century notions of progress - both on the capitalist and the Marxist side of the spectrum. They argue that both capitalism and Marxism over-emphasize technological achievements and material prosperity while ignoring the value of inner happiness and peace of mind. Postmodernism posits that both dystopia and utopia are one and the same, overarching grand narratives with impossible conclusions.

MEASURING SOCIAL PROGRESS

The Social Progress Index is a tool developed by the International Organization Imperative Social Progress, which measures the extent to which countries cover social and environmental needs of its citizenry. There are fifty-two indicators in three areas or dimensions: Basic Human Needs, and Foundations of Wellbeing and Opportunities which show the relative performance of nations.

DIFFERENCES BETWEEN “SOCIAL CHANGE” AND “SOCIAL PROGRESS”

1. Social Change is also different from progress. According to Ogburn, “Progress is a movement towards an objective which is thought to be higher by the group.” Thus, when we say that society is progressing, we mean that it is flourishing or improving its conditions.

As such progress implies an upward change or a change for the better. But in case of social change, it may be either progressive or retrogressive. It is progressive when it leads to some improvement or betterment in society.

It is retrogressive when it leads to downfall, setback or retardation or degradation of society. Sometimes, change can be upward and sometimes downward. If social change comes through planning, it can lead to progress, but if it is brought by some destruction or war, it is always harmful for the society.

2. According to Giddings, the difference between social progress and social change is that progress is planned, while social change can be planned as well as unplanned. It is believed that progress cannot exist without substantial and proper planning.

According to Hobbhouse, Social progress consists in the realization of ethical order in society. He even believed that only moral development leads to social progress. But in case of social change, it may lead to the development of moral values or a degeneration of moral values.

3. According to R.M. MacIver, progress is a vague term and hence cannot be defined. It cannot be stated precisely as to which should be the criterion of progress. There are some social thinkers, who are of the opinion that social progress can be explained from the stand-point of economic progress. However, some others go to the extent of saying that economic progress can never be equated with social progress. In this way, we find that a change which is regarded to be progress by a few persons or by a group may be regarded as downfall of society by others.
4. In contemporary times, while some people say that society is progressing, some others say that it is going from bad to worse. Therefore, the term ‘progress’ cannot be fully defined and we cannot decide unanimously whether a society is progressing or not.

However in case of social change, this is not a big problem. Any change, whether it is regarded as bad or good constitutes a change in social relations. In other words, there is little confusion or controversy regarding the nature of social change. Moreover, social change includes social progress within itself. Some social scientist like Ward, believe that social change should be fully planned, should be equated with social progress.

RECENT SOCIAL CHANGES IN INDIA

CHANGE IN FAMILY STRUCTURE

The impact of modern social forces is evident in the change of family structure. Basically joint family system is prevalent in India in which the head of the family

exercised absolute power over its members. He distributes work among family members. It is thus authoritarian which encourages collectivism rather than individualism. But due to modern forces, joint family system is disintegrating. It is now exhibiting characteristics of urban family. Family authority and family ties are now gradually weakening. Now there is a trend towards the development of nuclear family.

Change of Caste System

Impact of modern forces is also evident in the institution of caste. Modern means of communication, growth of competitive economy and expansion of western education have shattered the traditional structure of caste. Thus is a transformation of rigid caste into modern mobile classes. Certain castes have attained the privileges of upper classes while certain castes are losing their previous status. Besides, there is secularisation for the selection of occupation. This type of change has been revealed in every aspect of social life. Among high castes there is a trend towards modernisation while among low classes there is a trend towards sanskritisation.

Changes in the Social Stratification

The original stratification of Indian society is based on the caste system which gives exclusive importance to birth rather than individual achievements. Now birth is no longer a criterion of role allocation. It is gradually being replaced by the achievement system. Due to the impact of modern forces Indian society is being transformed from feudal system to capitalist system. Therefore, the stratification based on class system is gradually emerging. The main cause for the genesis of new classes is based on the disintegration of caste control.

Increasing Mobility

In India caste system prevents the mobility of the people from one occupation to another. But the new class system has encouraged social mobility. Further, legal compulsion has also encouraged social mobility and social change. According to Prof. S.C. Dubey, *“An excellent example of social change by legal compulsion is provided by the abolition of untouchability by law in India. In a favourable social climate legislation can be a potent instrument of social change.”*

Changes in the marriage institution: In India, the institution of marriage has undergone a tremendous change. Formerly people used to observe endogamy as regards the 'caste' and exogamy as regards the gotra'. But due to the impact of modern forces, the endogamous restrictions are gradually declining. Among educated people, there are a number of instances of the inter-caste marriage.

CASTE AND CLASS

All types of societies and races have some sort of functional social stratification. The only difference is that in most cases it remains in an elementary form and elastic.

The peculiar feature of the Indian caste system is that in India the aforesaid process has crystalized into watertight compartments. And it has become permanent feature of the Hindu society.

It is very difficult to give out a dried definition of caste. We may offer fairly accurate description of caste system. 'A caste is an endogamous group or collection of groups, bearing a common name and claiming a common origin, following the same traditional occupation and occupying the position of superior and inferior rank of social esteem in comparison with other group maintaining a social exclusiveness with reference to diet, marriage and observing certain ceremonies and rituals.'

The essential feature of caste are thus food, marriage restriction, hereditary, occupation, hierarchical organisation, social exclusiveness and religious sanctions. In India social stratification has, through caste been carried to length unparalleled elsewhere. Hindu society is divided into about 3, 000 castes and each of these is segregated from every other by restriction with regard to marriage, food and sometimes even personal contact. A person's caste, his station in life, his occupation, the people with whom he may associate and among whom the caste he may marry all determined for him by the fact of his birth in a particular caste. It is pre-determined of him by the law of karma or *purva-sanchit*. The first distinguished feature of the Indian caste system is its absolute rigidity and immobility. A man dies in the same caste in which he is born. And it is the caste that determines his station of life.

The caste system Brahmin is not allowed to eat non-vegetarian food. Kshatriyas and Vaishyas can but even for them certain kind of non-vegetarian food such as of buffalo is forbidden. But Shudra can eat any type of food. Of all the features of caste system, endogamy is the most important one. The essence of the system is endogamy. Occupation for instance, though usually associated with caste, has never been an essential part thereof. Coles Brooke writes, 'daily observation shows even Brahmins exercising the menial profession of a Shudra. But the different castes are strictly endogamous.'

Each one must marry within his own caste and within the subgroup if there be any in that particular castes. Brahmins for instances, if inter-caste marriages become extremely difficult because two persons belonging to two different castes differ in their food-habits, cultural habits *etc.* Thirdly, according to the Indian caste system, the caste of jati is more often than not named after the hereditary profession and the son inherits the profession of his father. There are thus jatis such as sutar, lohar, sonar, koli according to Carpenters, Blacksmiths, Goldsmiths, Fishermen respectively.

And these occupations are for the most part hereditary and even if someone changes the occupation, the caste name remains the same and the rules of endogamy apply. The caste system however forbids shudras to *tapasya*, or to recite Vedas *etc.* Secondly both of them are the distinguishing feature of practically every society, save the primitive savage tribe. However the two differ in most important respects. The fourth one is the hierarchical occupation. There is definite social gradation in which the different castes are related as higher and lower. The Brahmins are at top of the ladder whereas; Shudras

are at the lowest rung of the ladder. Brahmins are twice born, and they alone can devote exclusively to lore and learning. This gives them a position of advantage. He is placed even higher than the kind whose duty is to rule. Agriculture and trade is in the hands of Vaishyas over whom the king rules whereas Shudras are properly speaking rightless mass of people. Finally this rigid stratification is sanctioned by the religion. The violation of the system is an offence. And the law of dharma says that the advancement of each one can be best brought by each sticking to his own station in life and doing this duty prescribed by dharma loyalty.

We have already described what the caste system is at the beginning of this answer. We shall presently note the difference between a caste and class. There is a resemblance between classes and castes and castes. In as much as both are the forms of social stratification implying a sort of consciousness of superior and inferior.

Secondly both of them are the distinguishing feature of practically every society, save the primitive savage tribe. However the two differ in most important and respects.

Classes are elastic whereas castes are rigid. One can on his merit strive for money and success in life and with wealth he can change his social status implied in the class distinction. A man may be born in a particular class, but is not predetermined for him that they must die in the same class. There is a possibility of his moving upwards or downwards.

This is well nigh impossible in case of caste system. Once a man is born in a particular caste he remains in it for his life time and makes his children suffer the same fate. Thus classes are changeable, while castes are water-tight compartments. Secondly classes are secular in origin, they are capable of adaptation to changing environment and are determined by social needs castes, however, are believe to be divinely ordained. They are founded on religious dogmas. Finally while the class distinction have served an impetus to further progress, the caste distinction have proved a great dragon on social progress.

RELIGION

In the Indian way of life religion plays an important role and the basis of our day-to-day life is religion. Political leaders right from the beginning felt that if there is any possibility of retaining unity in India, it should be by remaining secular. That is why Gandhiji had been preaching brotherhood among the different religious groups. Nehru was a strong supporter of secularism. Their efforts could not divorce religion from politics rather in politics the vested interests started exploiting caste and religion for gaining political advantage.

With the passage of time India was divided into Pakistan and Bharat only because two nations theory was accepted by the Britishers. Even after Independence, the religious fervour could not be finished because the trail of the memories of the partition haunted the minds of the people, Still India managed to keep the communal forces under check. But the opposition parties exploit religion and theocratic States established in Pakistan

and Iran encouraged fundamentalism all over the world. Recently in Punjab religion and politics are so closely interwoven that it has become difficult to separate them.

Religious places are used for political propaganda and the religious sentiments of the people are excited in order to gain political control of the State. This emergence of religion-political party has threatened the secular character of India. It is feared that if it succeeds there is a possibility that many other political parties with caste and religion as the basis may come up.

Mixing of religion with politics is a dangerous trend because religious attitude is diametrically opposed to democratic feelings. Religion encourages fanaticism and suspends our reasoning power and we repose full faith in leaders. We are prepared to make sacrifice because sacrifice will be considered martyrdom. This mental attitude is directly opposed to democratic spirit. Democracy demands open mindedness, universal brotherhood and thinking based upon reason and capable of taking its own decision. In such cases, there is no herd tendency and the person is liberal in outlook.

If religious forces are allowed to become powerful there will be disintegration of the nation and sovereignty of the State will be in danger. There are a large number of religions, castes and sub-castes in India, and unfortunately some of them are opposed to one another as far as their practices are concerned. Under such circumstances, there is no possibility of keeping them together if once there is fragmentation.

Religion is a private affair and if it is allowed to appear in public affairs it will corrupt politics. All the crimes committed in the name of religion in the past as well as in the present one cannot forget. A large number of people have been put to death in Iran only because they do not follow the Islamic religion up to the last Point. So religion makes a man blind and it will never encourage opposition.

So if we want to consolidate democracy give firm foundations to it and make its working successful, it is necessary that the people should keep religion apart from politics. It is wrong to think that with the help of laws it is possible to divorce religion from politics. Till the attitude of the people is changed, and till they rise above the petty considerations it is not possible to keep religion and politics apart. By keeping them independent of each other, we can retain democratic set-up.

LANGUAGE

It is not surprising that there should be so much confusion and conflict over the issue of language. It is assumed that it is single problem with a unique solution. One group argues that English language owes its position to British imperialism and that it is inconsistent with national self-respect to continue to use it to the exclusion of our beautiful regional languages. A second group contents that, whatever the evils of British imperialism, English has helped Indian nationalism to achieve freedom; that it has been the vehicle of higher education, technology and scientific research and that it has been the invaluable link for forging the unity of India and any attempt to discard it will result in a breakdown of our educational system and take the country back by two

centuries in all aspects of modern life. A third group is convinced that as all regional languages of India are well developed and have worth literatures, all of them should have equal status.

There is a lot of confused thinking going on in India today over the problem of finding a universally acceptable link language for the country. The debate on the subject rages 'ad infinitum' in all available forums like the parliament, the press, the platform *etc.*; but none of the many points of view being put forward every day appears to be gaining acceptance. That is so mainly because most of the thinking on the subject is supercharged with emotion and the issues involved have been greatly clouded by the import of prejudices which can be traced to extra-national loyalties.

What is precisely meant by a "link language"? The term is of comparatively recent coinage. In the pre-independence years, the same connotation was sought to be conveyed through 'national language'. But the expression underwent a change in meaning when the constitution-makers of free India conferred the status of 'national language' on fourteen regional languages (now 15) in use in the country as also on Sanskrit. In the constitution, the term which came nearest to expressing the idea was 'official language'. Over the last two or three years however, even this term has been replaced with another, *viz.*, 'link language'. The change is significant. It is symptomatic of the trend of popular thinking on the subject.

The people were prepared to accept their own regional languages as the 'official' languages in their states. But when it came to accepting another Indian language for the purposes of inter-state and state-centre communications, they were not prepared to submit to the hegemony of another Indian language. In that context, the word 'official' in the expression 'official language' snarled for imposition and they were not prepared to brook any imposition whatsoever. Hence 'official' had to yield place to 'link' which was found to be more in accord with the spirit of the times.

The context in which the problem has to be considered has radically changed since 1950 when the founding fathers laid down official policy in the matter. At that time, the problem did not appear to be so intractable. The leaders of the national movement had almost unanimously endorsed the choice of Hindi as the official language of India. They were not, however, unmindful of the difficulties involved in a sudden change-over. Therefore, they made provision that English should continue to be used for all official purposes for 15 years after the introduction of the constitution, *i.e.*, up to 1965. Even after that, the Parliament was given the option to extend the period, if necessary.

All these precautions were designed to cushion the shock of change by making it a gradual process. Even then, within a few years, the proposed change became the subject of nation-wide debate which has several times assumed the proportions of a law and order problem. In the main, it was a direct consequence of the ruling party's disproportionate pre-occupation with linguism. That led to the reorganisation of states on a linguistic basis in 1956 (and the process has continued since). The step, though taken with the object of creating conditions in which the regional languages could

flourish and come into their own, encouraged separatism, and strengthened the forces of disintegration. It gave rise to a sort of linguistic patriotism which gained precedence over nationalism in the affections of a large number of people.

The constitution-makers' decision that English should be replaced with Hindi in course of time was a sound one. Being understood by the largest number of Indians (40 per cent), Hindi was no doubt in the best position to serve as the hand maiden of all other national languages and as such, to replace English as the official language of India. But the new language consciousness injected into the body politic by the ruling party, combined with the aggressive attitude of some advocates of Hindi who insisted on calling it the national language, and a fear being at a disadvantage in the central services made the decision largely unacceptable to people whose mother tongue was not Hindi. The Dravida Munnetra Kazhagara leaders called it "a conspiracy to relegate the people of the South to the position of second class citizens."

As the deadline officially fixed for the replacement of English drew near, representatives of the 'non-Hindi speaking people' as they were called bent their energies towards making sure that it did not happen. They even put forward suggestion that the language clause in the constitution should be suspended and English retained as the official language of India for all time to come. It was to allay their fears that, the late Pt. Jawaharlal Nehru, speaking in a debate on the subject in the Lok Sabha on August 7, 1959 said that English would be retained as an alternative official language as long as those who did not know Hindi wanted it. Accordingly, the Official Languages Bill passed by the Lok Sabha in April, 1963 provided that "English may continue to be used after January 26, 1965, in addition to Hindi for all the official purposes of the Union." The measure, however, did not satisfy people belonging to regions where the language in use was not Hindi. They wanted the assurance given by Pt. Nehru to be incorporated into law.

In his life-time, Pt. Nehru had opposed the suggestion for the excellent reason that such a step would 'amount to limiting the powers of Parliament by giving only part of the members the right to decide. This notwithstanding, the demand was pressed forward by every available means-through violent agitation and otherwise. Ultimately, the government had to give in. A draft of a bill seeking to confer statutory status on Jawaharlal Nehru's assurances to non-Hindi speaking people that English would continue as the associate official language of India indefinitely was circulated to state Chief Ministers. It would have been introduced in the 1967 budget session of the Parliament, but had to be held over as comments of some Chief Ministers had not been received before the session came to an end.

From all this, what have we gained? Have we moved forward or backward in the matter? What the opponents of Hindi have managed to secure after a prolonged struggle and many sacrifices can be described as a negative gain at best. They have not been able to suggest a better alternative to Hindi from among all the national languages of India. They have been harking back to English.

How this has come about would bear scrutiny. There are at least three main factors responsible for things having come to such a pass. Most responsible for bringing about this situation has been official inaction in not preparing the country for the change-over. The Government failed to take advantage of the long period to create the necessary climate for the purpose. Its forte was inaction, and not action to support this group or that the fears entertained by people who did not speak Hindi, of being dominated by the Hindi-speaking people in the event of Hindi becoming the sole link language in the country were not totally unfounded.

The second factor which contributed to creating the impasse we are facing were the protagonists of Hindi. They proved to be the worst enemies of the language for which they sought to gain acceptance as the official language. Instead of accepting it in all humility, they crowded over the role assigned to Hindi and displayed an attitude of arrogant superiority which could not but provoke adverse reaction. The third factor which added to the confusion were the chauvinists who thrived on regional and sectarian rivalries. The language issue came handy to them and they used it to whip up popular agitations out of which they made political capital.

All these currents and cross-currents of emotions and motives naturally served to cloud the real issues which, simply stated, were; (i) the fear of domination by one linguistic group over the other; and (ii) probable effect on each region's share in the central services. For lack of a positive approach to the problems, whatever is being done to resolve these issues, *viz.*, retention of English for an indefinite period; the proposed introduction of regional languages as media for U.P.S.C. examinations and lately, their adoption as media of instruction for higher education, is bound to have a distorting influence on our political life and further undermine our already brittle sense of nationhood. The genesis of peasant movements rest in the relationship patterns of different social categories. It is essential for the unity of India that graduates of all Indian Universities and technological institutes should form a single intellectual pool. Though, in the Universities of each State, the medium of instruction may be the regional language to hold on to English at present as the necessary link language at the higher level and hope that at some time in the future, Hindi may also blossom into another such language is all that a far sighted Indian patriot can hope.

REGIONALISM

India is a large country having continental dimensions and comprising no fewer than 28 States and 7 Union Territories. It is a multi-racial, multi-lingual nation. There are scores of regional languages, various strains of culture and different loyalties, single as well as multiple. Amidst the amazing diversities, it is natural that regional feelings, regional parties, regional institutions and similar other organisations meant for voicing the aspirations of local people and providing forums for them, should emerge. Indeed, with the passage of years, the multi-faceted aspirations, which together may be described as regionalism, have gained strength.

It is not a new phenomenon. In fact, the fillip given to regionalism by the emergence of the Telegu Desam Party (TDP) in Andhra Pradesh in 1982 has a historical continuity. The growth of this trend can be traced back to the fast unto death by Potti Srivamulu over the demand for the creation of Andhra Pradesh which set in motion the reorganisation of State along linguistic lines in 1956.

In principle, regionalism need not be regarded as an unhealthy or anti-national phenomenon-unless it takes a militant, aggressive turn and encourages the growth of secessionist tendencies, (as it did in Punjab during the past five years or so). National unity is not impaired if the people of a region have a genuine pride in their language and culture. But regionalism develops into a serious threat to national unity if politicians do not go beyond their regional loyalty and claim to stand only for their regional interests if regionalism is to be regarded as an unhealthy phenomenon, decentralisation too would be objectionable, which of course it is not. So there is nothing basically contradictory between nationalism and regionalism. Nor does the growth of regional values and consolidation of regional forces as such pose a challenge to the central administration of the country.

Any attempt to counter regionalism in the erroneous belief that is not conducive to national interests, would be ill-conceived. Enforced uniformity in a huge country like India would be sheer-folly. Regional parties do not hinder national unity and integrity as long as they do not exceed their area of activity.

Several regional political parties have merged in recent years and have gained strength for obvious reasons. The handful of national parties cannot, by the very nature of things, adequately represent and pursue regional causes. Most of the national parties have even failed to live up to the people's expectations. That explains why more State-based parties have been formed in various regions and are quite successful in their aims.

Regional parties are not a new phenomenon. Several parties have been existing in the country for the last many decades. They have held power, or are still holding power, in many states such as Andhra Pradesh, Tamil Nadu, Goa, Pondicherry, Jammu and Kashmir, Assam and other States in the North-Eastern region. But never before were regional parties dubbed as anti-national or regarded as a threat to the nation's unity.

People repose confidence in regional parties because they believe that they alone can safeguard the interests of the State concerned and can fight for the legitimate rights and powers of the States without being hamstrung by their association with a national party. Regional parties naturally concentrate on safeguarding and promoting regional interests. But they do not sacrifice the larger interests of the country. It is also significant that in the Lok Sabha a regional party (Telegu Desam) now forms the largest opposition group.

Among the causes of the growth of regionalism is prolonged maladministration and neglect of an area or State by the Central Government. There has been a creeping disillusionment against Central rule. Regional symbols, regional culture, history and in many cases a common language, all promote regionalism. The Centre's indifference to

the development of certain regions has created imbalances. Some areas particularly in the North, are well developed, with adequate infrastructure while others are way behind. This explains why there is Telegu Desam in Andhra Pradesh, DMK in Tamil Nadu and the Jharkhand Movement in Bihar.

There is much concern among leaders of the Congress (I) about the growth of regionalism in the country. It is looked upon with suspicion and is even regarded as a challenge to democracy and national integration. But this concern is largely unwarranted; Regionalism will come into conflict with nationalism only when it becomes aggressive and when members of the various regional parties tend to forget that they are Indians first and last, citizens of the same country.

Non-regional conflicts are however a cause for concern. There are constitutional means to deal with regional conflicts, while the communal and caste conflicts have often to be settled in the streets. Inter-regional or centre-region disputes have never created a serious explosion whereas communal clashes frequently cause havoc.

Unfortunately, there are important differences among the regional parties themselves in the country. The differences of approach and policy have hindered the formation of an effective, durable and viable combination of regional parties so as to facilitate the emergence of a national alternative to the ruling party at the centre. The growth of regional parties in itself is not incompatible with the process of nation-building.

In a democracy, ideological options are open in the sense that any individual or group can adopt any ideology, provided, of course, it is within the legal framework. Political parties have the freedom to compete for power and pursue their respective ideologies. Since ideologies are no respecters of geographical boundaries, they also check the exclusiveness of regional identities. In fact, it has been the decline of the party system in recent years that has inflated the role of regionalism in the country.

As for the cures, three suggestions may be made. First, there should be a greater spirit of accommodation on the part of the Central authorities. This implies a reversal of the process of concentration of power which has admittedly been much in evidence in the country, causing resentment among the opposition-governed State. Power and authority must be shared on an equitable basis between the Centre and the constituent units, of the Indian Federation. Harmonious, balanced growth should be the administrations aim, not suppression of local desires and demands.

Of course, firmness is necessary when regionalism, assumes militant forms, as it has done in Punjab in the form of Operation Blue Star and Operation Thunder, where in recent years certain groups of misguided youth started running a parallel government and creating chaos. Regionalism must not be allowed to become a shield for militancy, extremism, establishing a reign of terror and carrying on other anti-national activities.

The regional parties patriotism should not be suspected, regionalism does not weaken India. The majority groups should not become arrogant or obsessed with power. They should be generous towards the minorities, religious, cultural and linguistic suppression

of regional aspirations is not the right remedy. There are some uniting factors which need to be further promoted. The emergence of a national market, the spread of communications, the influence of all India institutions, the widening transport facilities, the vast network of the electronic media, the establishment of a common structure of formal education almost throughout the country. These factors helps to counter regional tendencies.

Education and Social Theory

THE FUNCTIONS OF EDUCATION

The manifest function of education is the transmission of knowledge to students in order to help them contribute back to society, either as knowledge producers themselves (*i.e.*, scientists) or in other ways, like in business or industry. However, education also has a latent function, and it is a function that is actually quite important when it comes to financial outcomes. One of the latent functions of a mass educational system is to foster social skills. And, it turns out, those social skills literally pay-off in the long run. Students who score high on measures of sociability earn more money and get more education than equally intellectually gifted students who don't score as high in social skills.

EDUCATION AND SOCIAL INTERACTION

EDUCATION AND INEQUALITY

Gender

A recent topic of policy debate in the U.S. is the concern that there is a "boy crisis," as boys are less likely to graduate from high school and go on to college than are girls. However, recent research by Judith Kleinfeld suggests that painting this as a "crisis" exclusive to boys may be a little misleading. Boys at the primary and secondary

educational levels do have a number of problems: lower rates of literacy, lower grades, lower engagement in school, higher dropout rates, higher rates of placement in special education, higher rates of emotional disturbance, higher rates of learning disabilities, higher rates of suspension and expulsion from school, and lower rates of entrance into college. Outside of school, boys are also more likely to: commit suicide, exhibit conduct disorders, die young, and be arrested or sent to juvenile detention centres. Clearly, boys have a number of problems.

However, girls have their own share of problems that are no less serious than those facing young men, just different: higher rates of depression, higher incidence of eating disorders, higher rates of suicide attempts, lower scores on math and science tests, and less likely to be the highest achievers. Kleinfeld does conclude that the issues facing boys are probably more serious (*e.g.*, 25 per cent of American male high school graduates cannot understand a newspaper article vs. 10 per cent of females), but also reminds us that women are facing many serious issues as well.

In the U.S. in 1960, 65 per cent of all Bachelors degrees went to men; in 1982 it reached parity between men and women. In 2004, women received 58 per cent of Bachelor's degrees. The 1960s and 1970s were pivotal years in the reversal of educational disparity between men and women. According to recent research, the reversal in disparity of educational attainment between men and women is largely attributable to higher drop-out rates of men from 4-year colleges, particularly males whose fathers have just a high school education or are absent from the home.

Additionally, women experience earlier academic success than men do (in junior high and high school); the study skills and commitment that contribute to that success carries over to college where they do better than men. Additional reasons why this reversal has taken place may include: a faster rise in the value of college for females than for males (resulting in women benefiting more from completing their education) and rising divorce rates, which force women to be more independent. The increasing completion rate of women may have substantial influences on society in general, as education and income are important factors when considering who one will marry.

Class and Income

One of the primary factors contributing to educational inequality and poverty in general in the US is the funding system for public education. In most areas of the US, public schools are funded primarily through local property taxes. As a result, schools in wealthier neighbourhoods have substantially larger budgets, which translates into better facilities, better teachers, and better resources. When these monetary advantages are coupled with the advantage of having a safe, supportive, and intellectually enriching home environment that comes with wealth, it is not surprising that children who attend better funded public schools tend to be more successful than those who attend more poorly funded public schools. Additionally, within schools in the U.S., tracking is commonly employed (separating students into different classes based on their abilities

and skills). Tracking increases disparity in educational attainment rather than reducing those disparities.

There is also evidence to suggest that the lifetime educational possibilities of most kids are set by the time they are six years old. This is due to the fact that several family background characteristics are very strong predictors of future educational attainment, including: parental support, parental expectations for schooling, household income, and parental educational attainment, with the last two being the most important factors. Also surprising in this study was the finding that the lifetime educational attainment predicted using background characteristics when individuals were in the first grade were just as good at predicting educational attainment as were similar variables when individuals were in high school, and on some measures better. In real terms, this translates into children living in poorer neighbourhoods receiving less education than children who live in middle-class or upper-class neighbourhoods. What's more, the lifetime educational attainment seems to be set by the time children are starting school, not when they are finishing it.

Educational deficits resulting from inequality also affect future life trajectories. Colleges tend to draw students from a relatively advantaged background because of their high costs and stiff academic requirements for enrolment. What's more, because colleges want to maintain their rankings in various college ranking systems (*e.g.*, U.S. News and World Report), colleges favour students with higher standardized test scores and aggressively recruit them using "merit" scholarships. In 2000, affluent students, students who could otherwise afford to pay for college, received "merit" scholarships worth 82 per cent of the need-based aid received by students with the lowest family incomes. In other words, affluent students who could pay for college often do not have to because the advantages they received attending better elementary, middle, and high schools translated into higher standardized test scores, which are attractive to universities when it comes to recruiting. As a result, there is less funding available for students who actually need it. This ends up reinforcing the existing status hierarchy by keeping the affluent wealthy and the poor unable to attend college.

Some of those under-privileged youth choose another path as a result. Despite being both volunteer and high risk, the military remains attractive to a certain segment of the U.S. population, primarily young people from less advantaged families and neighbourhoods. Such individuals view the military as more promising than entry-level jobs in the workforce without a college diploma. Cognitively, young people from disadvantaged homes are about average, but are socio-economically disadvantaged, which is one of the primary reasons they turn to the military rather than college to pursue financial success, with some success.

Health and Longevity

Education is a strong and significant predictor of greater life expectancy. The more years of education a person has, the greater their likelihood of engaging in healthy

behaviours (and, inversely, the lower their likelihood of engaging in unhealthy behaviours, like smoking).

ISSUES IN EDUCATION

CHARTER SCHOOLS

Charter schools are publicly funded elementary or secondary schools in the United States that have been freed from some of the rules, regulations, and statutes that apply to other public schools in exchange for some type of accountability for producing certain results, which are set forth in each school's charter. Their founders are often teachers, parents, or activists who feel restricted by traditional public schools. State-run charter schools (schools not affiliated with local school districts) are often established by non-profit groups, universities, and some government entities.

The basic concept of charter schools is that they exercise increased autonomy in return for greater accountability. They are accountable for both academic results and fiscal practices to several groups, including the sponsor that grants them, the parents who choose them, and the public that funds them. Charter schools can theoretically be closed for failing to meet the terms set forth in their charter, but in practice, this can be difficult, divisive and controversial. One example was the 2003 revocation of the charter for a school called Urban Pioneer in the San Francisco Unified School District, which first came under scrutiny when two students died on a school wilderness outing. An auditor's report found that the school was in financial disarray and posted the lowest test scores of any school in the district except those serving entirely non-English-speakers. It was also accused of academic fraud, graduating students with far fewer than the required credits.

Opponents of charter schools argue that charter schools probably cannot fix any broader problems with the U.S. educational system. If they do anything, they probably encourage segregation while drawing money out of the public school system. Additionally, recent data indicates that children in charter schools are not doing any better on standardized achievement tests.

At best, they are doing as well as children in public schools, but when charter schools are not associated with a public school district, children in the schools score significantly less well than their public school counterparts. Charter schools appear not to be the solution to the lackluster educational system of the U.S.

High School Completion

An issue often obfuscated by confusing statistics, high school completion rates in the U.S. are often lower than reported. High school completion rates are calculated at the state level and aggregated at the federal level. The percentage of students who graduate from high school often reported by the federal government in the U.S. is around 85 per cent, but a more accurate percentage is about 71 per cent. Seventy-one per cent

is a sobering statistic when you think about its implications - slightly less than 3 out of 4 students who start high school in the U.S. finish; 1/4 of young people in America today are moving into adulthood without a high school diploma. The lack of a high school diploma contributes to lower wages and increased rates of poverty, as well as poorer general understandings of society, health, and science.

Additionally, the U.S. used to lead the world in sending high school graduates on to higher education, but in 2008 it was 5th and 16th out of 27 industrialized countries in the proportion of students who start college and complete it. We also have a disproportionate share of low-performing students; almost 1/3 of students in U.S. secondary schools don't have the minimum education to contribute to a globalized, high-tech economy.

Public vs. Private Schools

While it has long been held that private schools provide better education than public schools, current research suggests that is not the case. Public school students test higher, on average, in math than do private school students when social and economic backgrounds are controlled. Longitudinal data illustrates that, while students beginning in public and Catholic schools test at about the same level in math, by the fifth grade the public school students make significantly greater gains, equivalent to almost an extra half year of schooling. While the limited number of private schools makes it difficult to compare a wide variety of religiously affiliated schools with public schools, conservative Protestant private schools show the worst math performance of all types of schools.

Math is particularly telling in this regard as it is the subject least influenced by home environment and most greatly influenced by the effectiveness of the schooling environment. However, it should be noted that school type only explains about 5 per cent of the variation in math scores; the bulk of the difference is explained by demographic characteristics. The implication of this research is that, while private schools may provide a specific type of schooling (*i.e.*, one that includes religious instruction), there is little evidence that private schools in general provide better instruction or improve educational outcomes.

Other Issues

- Standardized testing
- Home schooling
- Tracking
- People with disabilities
- Adult education
- Teacher shortages and teacher quality.

Some of these issues, as well as others, are discussed in a Wikibook called Contemporary Educational Psychology.

THE ROLE OF STUDENTS

Education is not solely about the provision of information to students. Students also play a role in the education process. One of the ways they do so is in employing effective study techniques. Unfortunately, there are a number of myths about effective studying techniques for students.

For instance, many educators encourage elaborate study techniques, like concept-mapping, claiming that they improve the encoding of information into memory. Recent research suggests it is more beneficial for recall to practice retrieving information than using elaborate encoding techniques. Students' time is better spent practising free recall of information studied than employing complex study techniques.

CHALLENGES IN EDUCATION

The goal of education is the transference of ideas and skills from one person to another, or from one person to a group. Current education issues include which teaching method(s) are most effective, how to determine what knowledge should be taught, which knowledge is most relevant, and how well the pupil will retain incoming knowledge. Educators such as George Counts and Paulo Freire identified education as an inherently political process with inherently political outcomes. The challenge of identifying whose ideas are transferred and what goals they serve has always stood in the face of formal and informal education.

In addition to the "Three R's", reading, writing, and arithmetic, Western primary and secondary schools attempt to teach the basic knowledge of history, geography, mathematics (usually including calculus and algebra), physics, chemistry and sometimes politics, in the hope that students will retain and use this knowledge as they age or that the skills acquired will be transferrable. The current education system measures competency with tests and assignments and then assigns each student a corresponding grade. The grades usually come in the form of either a letter grade or a percentage, which are intended to represent the amount of all material presented in class that the student understood.

Educational progressives or advocates of unschooling often believe that grades do not necessarily reveal the strengths and weaknesses of a student, and that there is an unfortunate lack of youth voice in the educative process. Some feel the current grading system risks lowering students' self-confidence, as students may receive poor marks due to factors outside their control. Such factors include poverty, child abuse, and prejudiced or incompetent teachers.

By contrast, many advocates of a more traditional or "back to basics" approach believe that the direction of reform needs to be quite the opposite. Students are not sufficiently inspired or challenged to achieve success because of the dumbing down of the curriculum and the replacement of the "canon" with inferior material. Their view of self-confidence is that it arises not from removing hurdles such as grading, but by

making them fair and encouraging students to gain pride from knowing they can jump over these hurdles.

On the one hand, Albert Einstein, one of the most famous physicists of our time, credited with helping us understand the universe better, was not a model school student. He was uninterested in what was being taught, and he did not attend classes all the time. However, his gifts eventually shone through and added to the sum of human knowledge. On the other hand, for millennia those who have been challenged and well-educated in traditional schools have risen to great success and to a lifelong love of learning because their minds were made better and more powerful, as well as because of their mastery of a wide range of skills.

There are a number of highly controversial issues in education. Should some knowledge be forgotten? What should be taught, are we better off knowing how to build nuclear bombs, or is it best to let such knowledge be forgotten?

PARENTAL INVOLVEMENT

Parental involvement is an essential aspect of a child's educational development. Early parental involvement in the child's life is critical such as reading to children at an early age, teaching patterns, interpersonal communication skills, exposing them to diverse cultures and the community around them, educating them on a healthy lifestyle, *etc.* The socialisation and academic education of a child are aided by the involvement of the student, parent(s), teachers, and others in the community and extended family.

EDUCATIONAL SOCIOLOGY VS SOCIOLOGY OF EDUCATION

Educational Sociology and Sociology of Education are two branches of study that are sometimes understood as one and the same branch, but they are actually not so. They indeed show some differences between them when it comes to the subjects of their study and the nature of the branches of study.

The sociology of education is the study of how public institutions and individual experiences affect education and its outcomes. The study of the development of public schooling systems and its impact on modern industrial societies forms the subject matter of the branch of study of sociology of education. Topics such as higher education, further education, adult education and continuing education can be included in the branch of study of sociology of education.

On the other hand educational sociology is the branch of study that deals with the various methods providing better education to society through an in depth research of our culture and society. Educational sociology is a subject that has to take both the sociologists and the educationists into consideration. This makes the subject an invaluable asset to all the students and researchers of social sciences, particularly sociology and education. It is a general belief that such of those who are involved in a deep study of education will benefit more from the branch of educational sociology. In the sociology of education, education has been viewed as a basically optimistic human effort

characterized by aspirations for improvement and betterment. Education is thus a fundamental endeavour of every individual. It is interesting to note that the experts of the sociology of education opine that education is looked upon as an endeavour by which children can develop according to their needs. The potential in the children plays a great part in the role of education.

THE SOCIAL IMPORTANCE OF EDUCATION

The social importance of education is widely recognised today, especially in modern industrialised societies. In such societies education has become one of the means of acquiring social and technical skills. Education has come to be not only a way of training people of work in different fields but also a qualification for jobs in certain fields. It fits people for increasingly specialised roles. More than that, education has become an essential need today to register progress in scientific and technological fields. As such, it is a means of promoting economic prosperity. Education, as a means of bringing about social change, is no less significant.

It promotes social mobility, that is, movement of people from one social status to another. It influences social stratification. Education is often made use of in totalitarian and communist countries as an instrument to propagate some chauvinist and communist ideologies. The famous writer, Newman said that the main practical purpose of a university is to produce socially responsible people. President Truman of America, stressing the importance of education, once remarked that man with wide experience, practical vision of things, intellectual depth and capacity to take right decisions at right time should be given the reins of administration to rule the country.

Dr. S. Radhakrishnan said that the main objective of education was to give training to students to undertake occupations effectively and to become proper leaders in various social fields in which they happen to work. Dr. Kothari Commission, appointed by the Government of India to recommend suitable educational reforms declares in the very beginning of its very comprehensive report that, The destiny of India is being shaped in her classrooms today. Sociology and Education are not, however, one and the same. Sociology studies all kinds of social relationship including the educational. Hence its scope is wider. Education, on the other hand, deals with only the educational activities of man and hence has smaller field of study. Sociology is a general social science, whereas education, as a branch of knowledge, specialises itself in only one kind of human activity, that is, educational.

RELATIONSHIP DID SOCIOLOGY AND EDUCATION SHARE IN THIS MODERN WORLD?

Education, as such, is a triangular process. At one end there is the educator, on the other end is the educant and still on the other end is society. Thus, education is a social need. Without education, no society can prosper. Education is very much related to sociology. Social process goes hand in hand with Education. Educational Sociology is

very important branch of the discipline of Sociology. The meaning of educational Sociology is well explained by George Payne: "By Educational Sociology we mean the Science which describes and explains the institutions, the social groups and social processes - that is the social relationship in which or through which the individual gains and organises his experiences." Brown says, "Educational Sociology is the study of the interaction of the individual and his cultural environment, which includes other individuals, social group and patterns of behaviour".

To put in simple terms Educational Sociology utilises all that has been learned in a social and educational field but joins them in a new science called 'Educational Sociology' by applying sociological principles to the whole process of education, including subject matter and activities, method of social organisation and measurement to quote Dodson, "Educational Sociology is interested in the impact of total cultural milieu in which and through which experience is acquired and organised".

The educational system motivates, equips and even allocates individual in a society to perform different roles. In the same way the other segments of the social system have important consequences for the system of education as a whole. Generally, a given system of education performs those functions which are more or less, adequate to the "survival" and "contribution" of a society at that point of time.

WHAT IS THE RELATIONSHIP BETWEEN EDUCATION AND SOCIETY?

Sociological tendency in education made several thinkers in the field of education to discuss and classify the relationships between education and society. Education was considered as a sub-system of the larger society. The character of society and social changes taking place must influence the system of education also. It was emphasized that these social changes and emerging social needs must be reflected in the theory and practice of education also. This point was very much emphasized by Dewey in his book, "The School and Society" wherein he said that while proposing to bring about a change in education the social point of view should always be kept in mind, otherwise it will be considered merely an arbitrary fad. "Education for the society" emerged as the focus of educational thought.

EDUCATION AND POLITICS

The relationship between education and politics also was emphasized as an aspect of sociological tendency in education. Education for democracy was, particularly emphasized. Democratic aims of education, democratic methods of teaching- learning, democratic organisation: of the classroom and the school, democratic discipline, *etc.*, were several new concepts which were extensively discussed.

EDUCATION AND ECONOMIC FACTORS

Sociological tendency in education forced the educational thinkers to analyse and see how economic factors affect education. It was realised that economic factors are

important constraints of educational activities. How they affect education and how education affects the economy of the society were some of the themes which emerged for discussion. A new kind of thinking began in the field of education.

CULTURAL AIM OF EDUCATION

The sociological tendency in education resulted into a greater stress on cultural aim of education. It was emphasized that education help in the transmission of cultural heritage.

THE CURRICULUM

Sociological tendency in education resulted into a ranged approach to curriculum construction. These views took the form of sociological principles of curriculum construction. It was emphasized that the curriculum should conform to the conditions, problems and needs of the society. A functional curriculum which could serve the needs of the society was said to be the need of the hour, curriculum for international understanding, curriculum for citizenship, curriculum for vocation, *etc.*, were the ideas about curriculum which gained popularity as a result of sociological tendency in education.

THE METHOD OF TEACHING

In the context of method of teaching, the sociological tendency resulted into considering the classroom and the school as the society-in-miniature. Hence, it was emphasized that teaching should take help of the principles of group dynamics. Group methods of teaching, group discussions, social interactions, group planning, group activities, project method, *etc.*, were considered important. Thus, sociological tendency in education tried to approach education, its purposes, methods, content and other aspects from the point of view of the society and sociological forces working within it.

EVOLUTIONARY THEORY

Evolutionary theories are based on the assumption that societies gradually change from simple beginnings into even more complex forms. Early sociologists beginning with Auguste Comte believed that human societies evolve in a unilinear way- that is in one line of development. According to them social change meant progress towards something better.

They saw change as positive and beneficial. To them the evolutionary process implied that societies would necessarily reach new and higher levels of civilization. L. H Morgan believed that there were three basic stages in the process: savagery, barbarism and civilization. Auguste Comte's ideas relating to the three stages in the development of human thought and also of society namely-the theological, the metaphysical and the positive in a way represent the three basic stages of social change. This evolutionary view of social change was highly influenced by Charles Darwin's theory of Organic

Evolution. Those who were fascinated by this theory applied it to the human society and argued that societies must have evolved from the simple and primitive to that of too complex and advanced such as the western society. Herbert Spencer a British sociologist carried this analogy to its extremity.

He argued that society itself is an organism. He even applied Darwin's principle of the survival of the fittest to human societies. He said that society has been gradually progressing towards a better state. He argued that it has evolved from military society to the industrial society.

He claimed that western races, classes or societies had survived and evolved because they were better adapted to face the conditions of life. This view known as social Darwinism got widespread popularity in the late 19th century. It survived even during the first phase of the 20th century. Emile Durkheim identified the cause of societal evolution as a society's increasing moral density. Durkheim viewed societies as changing in the direction of greater differentiation, interdependence and formal control under the pressure of increasing moral density. He advocated that societies have evolved from a relatively undifferentiated social structure with minimum of division of labour and with a kind of solidarity called mechanical solidarity to a more differentiated social structure with maximum division of labour giving rise to a kind of solidarity called organic solidarity.

Despite the wide variety in the possible directions change may take, various generalisations have been set forth. Because the lot of mankind generally has improved over the long term, by far the most numerous classes of theories of the direction of change comprise various cumulative or evolutionary trends. Though varying in many ways, these theories share an important conclusion that the course of man's history is marked up 'upward' trend through time.

The notion of evolution came into social sciences from the theories of biological evolution. With the advent of Darwinian Theory of biological evolution, society and culture began to be regarded as undergoing the same changes and demonstrating the same trends. It was conceived that society and culture were subject to the same general laws of biological and organism growth. Some thinkers even identified evolution with progress and proceeded to project into the future more and more perfect and better-adapted social and cultural forms.

Charles Darwin (1859), the British biologist, who propounded the theory of biological evolution, showed that species of organisms have evolved from simpler organisms to the more complicated organisms through the processes of variations and natural selection. After Darwin, 'evolution', in fact, became the buzz word in all intellectual inquiry and Darwin and Spencer were the key names of an era in the history of thought.

Herbert Spencer (1890), who is known to be the forerunner of this evolutionary thought in sociology, took the position that sociology is "the study of evolution in its most complex form". For him, evolution is a process of differentiation and integration.

BASIC ASSUMPTIONS AND DISTINCTIVE FEATURES OF THE EVOLUTIONARY CHANGE

The basic assumption of this theory is that change is the characteristic feature of human society. The present observed condition of the society is presumed to be the result of change in the past. Another assumption is that change is inevitable or it is 'natural'. It was assumed that the change is basically the result of operation of forces within the society or culture. Underlying all theories of evolution, there exists a belief of infinite improvement in the next stage over the preceding one.

All these Assumptions can be Summarised as Under:

1. That change is inevitable and natural.
2. That change is gradual and continuous.
3. That change is sequential and in certain stages.
4. That all successive stages of change are higher over preceding stage, *i.e.*, evolution is progressive.
5. That stages of change are non-reversible.
6. That forces of change are inherent in the object.
7. That the direction of change is from simple to complex, from homogeneity to heterogeneity, from undifferentiated to the differentiated in form and function.
8. That all societies pass through same stages of development.

All thinking of early sociologists was dominated by a conception of man and society as seen progressing up definite steps of evolution leading through every greater complexity to some final stage of perfection. The notion of evolutionary principles was extremely popular with British anthropologists and sociologists of nineteenth century. Such as Morgan (1877), Tyler (1889), Spencer (1890) and Hobhouse (1906). Although evolutionary theory in sociology is attributed to Herbert Spencer, it is clear that it was taken for granted by writers as diverse as Emile Durkheim, Karl Marx, Friedrich Engels and V. Gordon Childe.

The fact that it was used by both radical and conservative theorists is indicative of the profound cultural importance of evolutionism in the nineteenth century thought. The conception of evolution was applied not only to the development of societies but also to art, literature, music, philosophy, sciences, religion, economic and political life (state) and almost every other achievement of the mind of man. Both Spencer and Durkheim employed the concept of structural differentiation to indicate that as society develops more functions, it becomes structurally more complex. This perspective has been elaborated more recently by Talcott Parsons.

The general evolutionary model of society is represented by a large number of specific theories. C.H. Saint-Simon, one of the earliest founders of sociology, along with Auguste Comte, for example, put an evolutionary idea of social development, as a sequential progression of organic societies representing increasing levels of advancement.

His three stages were later elaborated in Comte's evolutionary scheme. Comte linked developments in human knowledge, culture and society and delineated the following

three great stages through which all societies must go—those of conquest, defence and industry. Societies passed through three stages—the primitive, the intermediary and the scientific, which corresponded to the forms of human knowledge (thought).

He conceived these stages as progressing from the theological through the metaphysical to arrive ultimately at the perfection of positive reasoning. He argued all mankind inevitably passed through these stages as it developed, suggesting both unilinear direction and progress. Spencer also displayed a linear concept of evolutionary stages. He argued that the trend of human societies was from simple, undifferentiated wholes to complex and heterogeneous ones, where the parts of the whole become more specialised but remained integrated.

William Graham Sumner (1934), who has been labelled as a ‘Social Darwinist’ also used the idea of evolution, as had Spencer, to block efforts at reform and social change, arguing that social evolution must follow its own course, dictated by nature. He said: “It is the greatest folly of which a man can be capable, to sit down with a slate and pencil to plan out a new social world.”

The evolutionary approach to social development was also followed by radical thinkers, such as Marx and Engels, who were greatly influenced by the work of the anthropologist L.H. Morgan, who sought to prove that all societies went through fixed stages of development each succeeding the other, from savagery through barbarism to civilization. Marx and Engels maintained that each stage of civilization, such as feudalism, prepared the ground for the next. It contained within itself “the seeds of its own destruction”, and would inevitably be succeeded by that stage next ‘higher’ on the scale of evolution. On this basis they concluded that the next stage in social evolution after the stage of capitalism could be attained only by violent revolution. All these theories are referred to as unilinear theories of social evolution.

Durkheim’s view of the progressive division of labour in society and German sociologist Ferdinand Tonnies’ view of *gemeinschaft* and *gesellschaft* types of society to some extent also represent the evolutionary perspective but their schemes of classifying societies are less sweeping and less explicit, and are, therefore referred to as quasi-evolutionary theories. For Durkheim the most important dimension of society was the degree of specialisation, as he called it, “the division of labour”.

He believed that there was a historical trend, or evolution, from a low to a high degree of specialisation. Durkheim distinguished two main types of society on the basis of this division of labour—the first based on mechanical solidarity and the second on organic solidarity. Durkheim believed that this second type always evolved from and succeeded the first as the degree of specialisation, the division of labour, increased.

Tonnies’ *gemeinschaft* type of society corresponded quite well to Durkheim’s mechanical solidarity and the second *gesellschaft* to organic solidarity. Numerous other scholars put forth similar ideas. The scheme of the American anthropologist Robert Redfield, who elaborated on the contrast between ‘folk’ and ‘urban’ society, reiterates the same basic dichotomy of social types suggested by Durkheim and Tonnies. Modern

theorist Talcott Parsons also viewed social change as a process of 'social evolution' from simple to more complex form of society. He regards changes in adaptation as a major driving force of social evolution. The history of human society from simple hunting and gathering band to the complex nation-state represents an increase in the 'general adaptive capacity of society.

TYPES OF EVOLUTIONARY THEORY

There are three main types of evolutionary theory:

Theory of Unilinear Evolution

It postulates the straight-line, ordered or progressive nature of social change. According to this theory, change always proceeds towards a predestined goal in a unilinear fashion. There is no place of repetition of the same stage in this theory. Followers of this pattern of change argue that society gradually moves to an even higher state of civilisation which advances in a linear fashion and in the direction of improvement. The pace of this change may be swift or slow. In brief, linear hypothesis states that all aspects of society change continually in a certain direction, never faltering, never repeating themselves. Theories of Saint-Simon, Comte, Morgan, Marx and Engels, and many other anthropologists and sociologists come under the category of unilinear theories of social evolution because they are based on the assumption that each society does, indeed must, pass through a fixed and limited numbers of stages in a given sequence. Such theories long dominated the sociological scene.

Universal Theory of Evolution

It is a little bit variant form of unilinear evolution which states that every society does not necessarily go through the same fixed stages of development. It argues, rather, that the culture of mankind, taken as a whole, has followed a definite line of evolution. Spencer's views can be categorised under this perspective who said that mankind had progressed from small groups to large and from simple to compound and in more general terms, from homogenous to the heterogeneous. The anthropologist Leslie White has been a leading exponent of this conception. Similar ideas were greatly elaborated by William Ogburn, who stressed the role of invention in social change. On this basis he gave birth to the famous concept of 'cultural lag' which states that change in our non-material culture, *i.e.*, in our ideas and social arrangements, always lag behind changes in material culture, *i.e.*, in our technology and invention.

Multilinear Theory of Evolution

This brand of evolutionism has more recently developed and is more realistic than the unilinear and universal brand of evolutionary change. Multilinear evolution is a concept, which attempts to account for diversity. It essentially means identification of different sequential patterns for different culture or types of cultures. This theory holds

that change can occur in several ways and that it does not inevitably lead in the same direction. Theorists of this persuasion recognise that human culture has evolved along a number of lines. Those who share this perspective, such as Julian Steward (1960), attempt to explain neither the straight-line evolution of each society, nor the progress of mankind as a whole, but rather concentrate on much more limited sequences of development.

It does identify some social trends as merely universal: the progression from smaller to larger, simpler to more complex, rural to urban, and low technology to higher technology but it recognises that these can come about in various ways and with distinct consequences. This theory is related to what is known as episodic approach, which stresses the importance of accidents and unique historical, social and environmental circumstances that help to explain a particular course of social change. Later on, the views of Leslie White and Julian Steward were named as neo-evolutionism.

EVOLUTIONARY THEORIES OF SOCIAL CHANGE

Social Darwinism

Herbert Spencer adapted Darwinian ideas to his own ethical theories to create a theory known as Social Darwinism. The major assumptions:

- Natural selection; it is natural, normal, and proper for the strong to thrive at the expense of the weak.
- Survival of the fittest is natural, and morally correct
- The fittest are the rich/powerful/elite groups of society - since they, found the best solutions to surviving by securing sustenance by accumulating surplus, as well as acquired power to retain control over these resources.

Application to Social Development

- Social Darwinism was used to justify exploitation eg. Colonialism through Social Darwinian ethics - colonizers viewed indigenous peoples as being uncivilized, weaker, more unfit to survive. This rationalization served to justify the taking of land and resources, without the approval of the indigenous.
- It provided a justification for the more exploitative forms of capitalism in which workers were paid low wages for long hours of hard labour.
- At its most extreme form, Social Darwinism has been used to justify eugenics programmes aimed at removing “undesirable” genes from the population; such programmes were sometimes accompanied by sterilization laws directed against “unfit” individuals.

Unilineal Evolutionary Theory

This school of thought suggests that society developed in a single line of continuation. Progressing from primitive to complex.

“Social evolutionism represented an attempt to formalize social thinking along scientific lines, later influenced by the biological theory of evolution. If organisms could develop over time according to discernible, deterministic laws, then it seemed reasonable that societies could as well”.

Spencer and Comte view the society as a kind of organism subject to the process of growth - from simplicity to complexity, from chaos to order, from generalisation to specialisation, from flexibility to organisation. They asserted that growth is social progress. Each newer, more evolved society is better. Thus progressivism became one of the basic ideas underlying the theory of social evolutionism.

Their Analysis of Cross-Cultural Data was Based on three Assumptions:

1. Contemporary societies may be classified and ranked as more “primitive” or more “civilized;”
2. There are a determinate number of stages between “primitive” and “civilized” (e.g., band, tribe, chiefdom, and state),
3. All societies progress through these stages in the same sequence, but at different rates.

Criticisms:

1. The theory is ethnocentric—it makes heavy value judgements on different societies; assuming that Western civilization is the most valuable.
2. It assumes all cultures follow the same path or progression and have the same goals.
3. It equated civilization with material culture (technology, cities, etc.)
4. It equated evolution with progress or fitness, based on deep misunderstandings of evolutionary theory.
5. It is greatly contradicted by evidence. Some (but not all) supposedly primitive societies are arguably more peaceful and equitable/democratic than many modern societies.

Multilinear Theory of Development

Emerging during the early twentieth century to challenge the ethnocentric assumptions of unilinear theories of social development. This school of thought was developed by a German-born anthropologist Franz Boas. These are the basic assumptions.

Different Societies Developed in Different Ways:

- All cultures are unique in time and place.
- There is no specific evolutionary change/s that is/are experienced by all cultures universally.
- All human societies evolve or progress - however these changes are influenced by a variety of socio-economic and political factors.
- The primary mechanism for progress involved technological innovations that help to make societies more adaptable to and dominant over the environment.

THE IMPACT OF DEMOCRACY ON VARIOUS ASPECTS OF EDUCATION

In the following lines we are throwing light on the various aspects of education in a democratic set up:

DEMOCRACY AND AIMS OF EDUCATION

In a democracy, the aims of education are as under:

Development of Democratic Values

The success of democracy does not depend upon Legislative buildings and massive structures of Parliament houses, but it rests upon the quality of the citizens devoted to democratic values. As such, the prime aim of democratic education is to promote in children a sense of devotion to democratic values. No book teaching can achieve this aim unless children are provided with opportunities to practice democratic norms and standards of behaviour. In fact, a child learns to live democratically by living democratically.

Development of Vocational Efficiency

For the success of a democratic set-up, economic contentment of citizens is a must. An indigent and poor person can be a victim of all kinds of allurements, inducements and exploitation by the resourceful and the powerful. Hence, the third aim of democratic education is to develop vocational efficiency in children, so that they are able to become self-reliant and serve the nation as much as possible.

Development of Interests in Children

The third aim of democratic education is to develop useful and worthy interests in children. Interests form character and enrich a child's life. Hence, the famous educationist, Herbart has insisted upon the fullest development of diverse interests. To achieve this aim, children should be provided with various and varied opportunities to participate in diverse activities and programmes in all fields of human life. If a large number of worthy interests are developed in children, they will be happy, well- balanced and efficient as citizens.

Development of Thinking Power

The fourth aim of democratic education is to develop thinking power of children. In fact, children of today are citizens of tomorrow when they will be confronted with all kinds of problems in political, social and economic fields. Education should develop in children the capacity to think clearly and take decisions confidently.

Development of Social Outlook

Development of social outlook is the sixth important aim of democratic education. This aim emphasizes upon, the fact that children should be imbued with the sense that

they are the integral parts of society, the welfare of which should be their ideal. Not only this, they should learn to live and die for the nation. Education should develop this sense of service and sacrifice making them learn the sacredness of obligations and duties for the welfare of the nation to which they belong.

Development of Sound Habits

The fifth aim of democratic education is to develop sound habits in children. Habits are the sources of good or bad conduct. Hence, education should develop good habits in children from the very beginning to make democracy a successful venture.

Development of Leadership

The seventh aim of democratic education is to develop leadership qualities in children. For this, education should instill in children the leadership qualities from the very beginning. They are the future citizens who will have to shoulder the multifarious duties and responsibilities of their nation in all areas. Their character, strength of will, insight, courage of convictions, clarity of thinking and decision-making will be the foundations on which the national edifice will go up and up.

Development of Harmonious Personality

The eighth aim of democratic education is to develop the individuality of a child into balanced and harmonious personality. In the modern world of strife, stress and strain a balance and harmonious personality can only seek and find adjustment with the surroundings. Hence, education should develop character, dynamism social outlook for this purpose.

Development of National and International Feelings

For the success of democracy, the ninth aim of democratic education is to develop in children the sense of ardent nationalism and devotion to international brotherhood. It may be noted that the two are not contradictory. On the other hand, they are mutually complementary and supplementary. In fact a nation cannot exist in isolation. All the nations of the world are mutually inter-dependent. Hence, education should foster the sense of inter-dependence, international good-will and fellow-feeling.

Training for Citizenship

Democratic education should impart to children training in dynamic and healthy citizenship. For this, education should instill in children:

1. Capacity to understand and solve the diverse problems of the country,
2. Capacity to distinguish between propaganda and reality,
3. Capacity to think and decide about issues,
4. Economic efficiency,
5. Consciousness one's rights and duties,

6. Capacity to shoulder responsibility,
7. Development of diverse interests,
8. Sense of service and sacrifice,
9. Good use of leisure hour,
10. Development of human qualities as love, sympathy, fellow feeling, co-operation, sense of nationalism and internationalism,
11. Healthy and dynamic outlook about problems, good behaviour and respect for moral values.

FUNCTIONALIST OR DYNAMIC THEORIES

In the middle decades of the 20th century a number of American sociologists shifted their attention from social dynamics to social static or from social change to social stability. Talcott Parsons stressed the importance of cultural patterns in controlling the stability of a society. According to him society has the ability to absorb disruptive forces while maintaining overall stability. Change is not as something that disturbs the social equilibrium but as something that alters the state of equilibrium so that a qualitatively new equilibrium results. He has stated that changes may arise from two sources. They may come from outside the society through contact with other societies. They may also come from inside the society through adjustment that must be made to resolve strains within the system.

Parsons speaks of two processes that are at work in social change. In simple societies institutions are undifferentiated that is a single institution serves many functions. The family performs reproductive, educational, socialising, economic, recreational and other functions. A process of differentiation takes place when the society becomes more and more complex. Different institutions such as school, factory may take over some of the functions of a family. The new institutions must be linked together in a proper way by the process of integration. New norms must be established in order to govern the relationship between the school and the home. Further bridging institutions such as law courts must resolve conflicts between other components in the system.

CONFLICT THEORIES

Whereas the equilibrium theories emphasize the stabilising processes at work in social systems the so-called conflict theories highlight the forces producing instability, struggle and social disorganisation. According to Ralf Dahrendorf the conflict theories assume that - every society is subjected at every moment to change, hence social change is ubiquitous. Every society experiences at every moment social conflict, hence social conflict is ubiquitous. Every element in society contributes to change. Every society rests on constraint of some of its members by others.

The most famous and influential of the conflict theories is the one put forward by Karl Marx who along with Engel wrote in Communist Manifesto 'all history is the history of class conflict.' Individuals and groups with opposing interests are bound to

be at conflict. Since the two major social classes the rich and poor or capitalists and the proletariat have mutually hostile interests they are at conflict. History is the story of conflict between the exploiter and the exploited. This conflict repeats itself off and on until capitalism is overthrown by the workers and a socialist state is created. What is to be stressed here is that Marx and other conflict theorists deem society as basically dynamic and not static. They consider conflict as a normal process. They also believe that the existing conditions in any society contain the seeds of future social changes. Like Karl Marx George Simmel too stressed the importance of conflict in social change. According to him conflict is a permanent feature of society and not just a temporary event. It is a process that binds people together in interaction. Further conflict encourages people of similar interests to unite together to achieve their objectives. Continuous conflict in this way keeps society dynamic and ever changing.

FACTORS OF CHANGE

Physical Environment

Major changes in the physical environment are very compelling when they happen. The desert wastes of North Africa were once green and well populated. Climates change, soil erodes and lakes gradually turn into swamps and finally plains. A culture is greatly affected by such changes although sometimes they come about so slowly that they are largely unnoticed. Human misuse can bring very rapid changes in physical environment which in turn change the social and cultural life of a people. Deforestation brings land erosion and reduces rainfall. Much of the wasteland and desert land of the world is a testament to human ignorance and misuse. Environmental destruction has been at least a contributing factor in the fall of most great civilization. Many human groups throughout history have changed their physical environment through migration. In the primitive societies whose members are very directly dependent upon their physical environment migration to a different environment brings major changes in the culture. Civilization makes it easy to transport a culture and practice it in a new and different environment.

Population Changes

A population change is itself a social change but also becomes a casual factor in further social and cultural changes. When a thinly settled frontier fills up with people the hospitality pattern fades away, secondary group relations multiply, institutional structures grow more elaborate and many other changes follow. A stable population may be able to resist change but a rapidly growing population must migrate, improve its productivity or starve. Great historic migrations and conquests of the Huns, Vikings and many others have arisen from the pressure of a growing population upon limited resources. Migration encourages further change for it brings a group into a new environment subjects it to new social contacts and confronts it with new problems. No major population change leaves the culture unchanged.

Isolation and Contact

Societies located at world crossroads have always been centres of change. Since most new traits come through diffusion, those societies in closest contact with other societies are likely to change most rapidly. In ancient times of overland transport, the land bridge connecting Asia, Africa and Europe was the centre of civilizing change. Later sailing vessels shifted the centre to the fringes of the Mediterranean Sea and still later to the north-west coast of Europe. Areas of greatest intercultural contact are the centres of change. War and trade have always brought intercultural contact and today tourism is adding to the contacts between cultures says Greenwood. Conversely isolated areas are centres of stability, conservatism and resistance to change. The most primitive tribes have been those who were the most isolated like the polar Eskimos or the Aranda of Central Australia.

Social Structure

The structure of a society affects its rate of change in subtle and not immediately apparent ways. A society which vests great authority in the very old people as classical China did for centuries is likely to be conservative and stable. According to Ottenberg a society which stresses conformity and trains the individual to be highly responsive to the group such as the Zunis is less receptive to the change than a society like the Ileo who are highly individualistic and tolerate considerable cultural variability.

A highly centralized bureaucracy is very favourable to the promotion and diffusion of change although bureaucracy has sometimes been used in an attempt to suppress change usually with no more than temporary success. When a culture is very highly integrated so that each element is rightly interwoven with all the others in a mutually interdependent system change is difficult and costly. But when the culture is less highly integrated so that work, play, family, religion and other activities are less dependent upon one another change is easier and more frequent. A tightly structured society wherein every person's roles, duties, privileges and obligations are precisely and rigidly defined is less given to changes than a more loosely structured society wherein roles, lines of authority, privileges and obligations are more open to individual rearrangement.

Attitudes and Values

To people in developed nations and societies change is normal. Children there are socialized to anticipate and appreciate change. By contrast the Trobriand Islanders off the coast of New Guinea had no concept of change and did not even have any words in their language to express or describe change. Societies differ greatly in their general attitude towards change. People who revere the past and preoccupied with traditions and rituals will change slowly and unwillingly. When a culture has been relatively static for a long time the people are likely to assume that it should remain so indefinitely. They are intensely and unconsciously ethnocentric; they assume that their customs and techniques are correct and everlasting.

A possible change is unlikely even to be seriously considered. Any change in such a society is likely to be too gradual to be noticed. A rapidly changing society has a different attitude towards change and this attitude is both cause and effect of the changes already taking place. Rapidly changing societies are aware of the social change. They are somewhat skeptical and critical of some parts of their traditional culture and will consider and experiment with innovations.

Such attitudes powerfully stimulate the proposal and acceptance of changes by individuals within the society. Different groups within a locality or a society may show differing receptivity to change. Every changing society has its liberals and its conservatives. Literate and educated people tend to accept changes more readily than the illiterate and uneducated. Attitudes and values affect both the amount and the direction of social change. The ancient Greeks made great contributions to art and learning but contributed little to technology. No society has been equally dynamic in all aspects and its values determine in which area-art, music, warfare, technology, philosophy or religion it will be innovative.

Cultural Factor influences the direction and character of technological change Culture not only influences our social relationships, it also influences the direction and character of technological change. It is not only our beliefs and social institutions must correspond to the changes in technology but our beliefs and social institutions determine the use to which the technological inventions will be put. The tools and techniques of technology are indifferent to the use we make of them. For example the atomic energy can be used for the production of deadly war weapons or for the production of economic goods that satisfy the basic needs of man.

The factories can produce the armaments or necessities of life. Steel and iron can be used for building warships or tractors. It is a culture that decides the purpose to which a technical invention must be put. Although technology has advanced geometrically in the recent past, technology alone does not cause social change. It does not by itself even cause further advances in technology. Social values play a dominant role here. It is the complex combination of technology and social values which produces conditions that encourage further technological change. For example the belief or the idea that human life must not be sacrificed for wants of medical treatment, contributed to the advancement in medical technology. Max Weber in his *The Protestant Ethic and the spirit of Capitalism* has made a classical attempt to establish a correlation between the changes in the religious outlook, beliefs and practices of the people on the one hand and their economic behaviour on the other. He has observed capitalism could grow in the western societies to very great extent and not in the eastern countries like India and China. He has concluded that Protestantism with its practical ethics encouraged capitalism to grow in the west and hence industrial and economic advancement took place there. In the East, Hinduism, Buddhism, Judaism and Islam on the other hand did not encourage capitalism.

Thus cultural factors play a positive as well as negative role in bringing about technological change. Cultural factors such as habits, customs, traditions, conservatism,

traditional values *etc.* may resist the technological inventions. On the other hand factors such as breakdown in the unity of social values, the diversification of social institutions craving for the new thoughts, values *etc.* may contribute to technological inventions. Technological changes do not take place on their own.

They are engineered by men only. Technology is the creation of man. Men are always moved by ideas, thoughts, values, beliefs, morals and philosophies *etc.* These are the elements of culture. These sometimes decide or influence the direction in which technology undergoes change. Men are becoming more and more materialistic in their attitude. This change in the attitude and outlook is reflected in the technological field. Thus in order to lead a comfortable life and to minimize the manual labour man started inventing new techniques, machines, instruments and devices.

Technological Factors

The technological factors represent the conditions created by man which have a profound influence on his life. In the attempt to satisfy his wants, fulfil his needs and to make his life more comfortable man creates civilization. Technology is a by-product of civilization. When the scientific knowledge is applied to the problems in life it becomes technology. Technology is a systematic knowledge which is put into practice that is to use tools and run machines to serve human purpose. Science and technology go together. In utilising the products of technology man brings social change. The social effects of technology are far-reaching. According to Karl Marx even the formation of social relations and mental conceptions and attitudes are dependent upon technology. He has regarded technology as a sole explanation of social change. W.F Ogburn says technology changes society by changing our environment to which we in turn adapt.

These changes are usually in the material environment and the adjustment that we make with these changes often modifies customs and social institutions. A single invention may have innumerable social effects. Radio for example has One of the most extreme expressions of the concern over the independence of technology is found in Jacques Ellul's 'the technological society'. Ellul claims that in modern industrial societies technologism has engulfed every aspect of social existence in much the same way Catholicism did in the middle ages. The loss of human freedom and the large-scale destruction of human beings are due to the increasing use of certain types of technology which has begun to threaten the life support systems of the earth as a whole.

IMPACT OF TECHNOLOGY CHANGE

INDUSTRIALISATION

Technology has contributed to the growth of industries or to the process of industrialisation. Industrialisation is a term covering in general terms the growth in a society hitherto mainly agrarian of modern industry with all its circumstances and problems, economic and social. It describes in general term the growth of a society in

which a major role is played by manufacturing industry. The industry is characterised by heavy, fixed capital investment in plant and building by the application of science to industrial techniques and by mainly large-scale standardised production.

The Industrial Revolution of 18th century led to the unprecedented growth of industries. Industrialisation is associated with the factory system of production. The family has lost its economic importance. The factories have brought down the prices of commodities, improved their quality and maximized their output.

The whole process of production is mechanised. Consequently the traditional skills have declined and good number of artisans has lost their work. Huge factories could provide employment opportunities to thousands of people. Hence men have become workers in a very large number. The process of industrialisation has affected the nature, character and the growth of economy. It has contributed to the growth of cities or to the process of urbanisation.

Urbanisation

In many countries the growth of industries has contributed to the growth of cities. Urbanisation denotes a diffusion of the influence of urban centres to a rural hinterland. Urbanisation can be described as a process of becoming urban moving to cities changing from agriculture to other pursuits common to cities and corresponding change of behaviour patterns. Hence only when a large proportion of inhabitants in an area come to cities urbanisation is said to occur. Urbanisation has become a world phenomenon today. An unprecedented growth has taken place not only in the number of great cities but also in their size. As a result of industrialisation people have started moving towards the industrial areas in search of employment. Due to this the industrial areas developed into towns and cities.

Modernisation

Modernisation is a process which indicates the adoption of the modern ways of life and values. It refers to an attempt on the part of the people particularly those who are custom-bound to adapt themselves to the present-time, conditions, needs, styles and ways in general. It indicates a change in people's food habits, dress habits, speaking styles, tastes, choices, preferences, ideas, values, recreational activities and so on. People in the process of getting themselves modernised give more importance to science and technology. The scientific and technological inventions have modernised societies in various countries. They have brought about remarkable changes in the whole system of social relationship and installed new ideologies in the place of traditional ones.

DEVELOPMENT OF THE MEANS OF TRANSPORT AND COMMUNICATION

Development of transport and communication has led to the national and international trade on a large scale. The road transport, the train service, the ships and the aero planes have eased the movement of men and material goods. Post and telegraph, radio and

television, newspapers and magazines, telephone and wireless and the like have developed a great deal. The space research and the launching of the satellites for communication purposes have further added to these developments. They have helped the people belonging to different corners of the nation or the world to have regular contacts.

TRANSFORMATION IN THE ECONOMY AND THE EVOLUTION OF THE NEW SOCIAL CLASSES

The introduction of the factory system of production has turned the agricultural economy into industrial economy. The industrial or the capitalist economy has divided the social organisation into two predominant classes-the capitalist class and the working class. These two classes are always at conflict due to mutually opposite interest. In the course of time an intermediary class called the middle class has evolved.

Unemployment

The problem of unemployment is a concomitant feature of the rapid technological advancement. Machines not only provide employment opportunities for men but they also take away the jobs of men through labour- saving devices. This results in technological unemployment.

Technology and War

The dangerous effect of technology is evident through the modern mode of warfare. The weaponry has brought fears and anxieties to the mankind. They can easily destroy the entire human race reveal how technology could be misused. Thus greater the technological advancement the more risk for the mankind.

CHANGES IN SOCIAL INSTITUTIONS

Technology has profoundly altered our modes of life. Technology has not spared the social institutions of its effects. The institutions of family, religion, morality, marriage, state, property have been altered. Modern technology in taking away industry from the household has radically changed the family organisation. Many functions of the family have been taken away by other agencies. Marriage is losing its sanctity. It is treated as a civil contract than a sacred bond

Marriages are becoming more and more unstable. Instances of divorce, desertion and separation are increasing. Technology has elevated the status of women but it has also contributed to the stresses and strains in the relations between men and women at home. Religion is losing hold over the members. People are becoming more secular, rational and scientific but less religious in their outlook. Inventions and discoveries in science have shaken the foundations of religion. The function of the state or the field of state activity has been widened. Modern technology have made the states to perform such functions as -the protection of the aged, the weaker section and the minorities

making provision for education, health care *etc.* Transportation and communication inventions are leading to a shift of functions from local government to the central government of the whole state. The modern inventions have also strengthened nationalism. The modern governments which rule through the bureaucracy have further impersonalised the human relations.

CULTURAL LAG

To provide a law of social change comparable to the laws of physics and biology that William F. Ogburn in 1922 advanced his theory of social lag. Ogburn pointed out that social changes always originate in the invention by some individual of a new way of doing something new to do. So far he was following in the tradition established by Gabriel Tarde; but Ogburn then began to wander in the tracks of Marx. Historically, he argued, inventions occur most often in the field of material technology, if only because the advantages of an improvement in technology are self-evident.

With each development in technology there comes, however, some disturbance to the effective working of the existing social order. A strain or stress is set up between the new technique and various organisational aspects of the social system, changes in which come slowly if at all; the result, disequilibrium between new technology and old social organisation, is social lag. The core of Ogburn's theory is the idea that change first occurs in the material technology.

SOCIAL MOVEMENTS

Social Movement is one of the major forms of collective behaviour. We hear of various kinds of social movements launched for one or the other purpose. A social movement can be defined as collectively acting with some continuity to promote or resist change in the society or group of which it is a part. Horton and Hunt have defined it as a collective effort to promote or resist change. Smelser defines it as organised group effort to generate or resist social change. According to M.S.A Rao social movement includes two characteristics.

Collective Action

Social Movement involves collective action. However it takes the form of a movement only when it is sustained for a long time. This collective action need not be formally organised. But it should be able to create an interest and awakening in relatively large number of people.

Oriented towards Social Change

A social movement is generally oriented towards bringing social change. This change could either be partial or total. Though the movement is aimed at bringing about a change in the values, norms, ideologies of the existing system, efforts are also made by some other forces to resist the changes and to maintain the status quo. The counter

attempts are normally defensive and restorative rather than innovative and initiating change. They are normally the organised efforts of an already established order to maintain itself.

According to Yogendra Singh social movement is a collective mobilisation of people in a society in an organised manner under an individual or collective leadership in order to realise an ideologically defined social purpose. Social movements are characterized by a specific goal which has a collective significance ideological interpretation of the collective goal a rank of committed worker and strong leadership. Social movements have a life-cycle of their own origin, maturity and culmination. T.K Oomen observe that a study of social movements implies a study of social structure as movements originate from the contradictions which in turn emanate from social structure. He states that all social movements centre around three factors- Locality, Issues and social categories. Anthony Wallace view social movement as an attempt by local population to change the image or models they have of how their culture operates.

An important component of social movement that distinguishes it from the general category of collective mobilisation is the presence of an ideology. A student strike involves collective mobilisation and is oriented towards change. But in the absence of an ideology a student strike becomes an isolated event and not a movement. A social movement requires a minimum of organisational framework to achieve success or at least to maintain the tempo of the movement.

To make the distinction clear between the leaders and followers to make clear the purposes of the movement to persuade people to take part in it or to support it, to adopt different techniques to achieve the goals - a social movement must have some amount of organisational frame-work. A social movement may adopt its own technique or method to achieve its goal. It may follow peaceful or conflicting, violent or non-violent, compulsive or persuasive, democratic or undemocratic means or methods to reach its goal.

Social Movements Types

Reform Movements

Reform movements are organised to carry out reforms in some specific areas. The reformers endeavour to change elements of the system for better. For example: Civil Rights Movement, Women's Liberation Movement, Arya Samaj Movement, Brahmo Samaj Movement *etc.*

Revolutionary Movements

The revolutionary movements deny that the system will even work. These movements are deeply dissatisfied with the social order and work for radical change. They advocate replacing the entire existing structure. Their objective is the reorganisation of society in accordance with their own ideological blueprint. Revolutionary movements generally

become violent as they progress. Example: The Protestant Reformation Movement, the Socialist Movement, the Communist Revolution of China.

Reactionary or Revivalist Movement

Some movements are known as reactionary or regressive movements. These aims to reverse the social change. They highlight the importance and greatness of traditional values, ideologies and institutional arrangements. They strongly criticize the fast moving changes of the present.

Resistance Movement

These movements are formed to resist a change that is already taking place in society. These can be directed against social and cultural changes which are already happening in the country.

Utopian Movement

These are attempts to take the society or a section of it towards a state of perfection. These are loosely structured collectivities that envision a radically changed and blissful state, either on a large scale at some time in the future or on a smaller scale in the present. The Utopian ideal and the means of it are often vague, but many utopian movements have quite specific programmes for social change. The Hare Krishna Movement of the seventies, the movement towards the establishment of Ram Rajya and the Sangh Parivar, the Communists and Socialists pronouncement of a movement towards the classless, casteless society free from all kinds of exploitation *etc.*

Peasant Movement

Peasant movement is defined by Kathleen Gough as an attempt of a group to effect change in the face of resistance and the peasant are people who are engaged in an agricultural or related production with primitive means who surrender part of their or its equivalent to landlords or to agents of change. The history of peasant movements can be traced to colonial period when repressive economic policies, the new land revenue system, the colonial administrative and judicial system and the ruin of handicrafts leading to the overcrowding of land transformed the agrarian structure and impoverished the peasantry.

In the zamindari system peasants were left to the mercies of the Zamindars who exploited them in form of illegal dues. The British government levied heavy land revenue in the Ryotwari areas. Peasants were forced to borrow money from the moneylenders and they were reduced to the status of tenants at will, share croppers and landless laborious while their lands, crops and cattle passed into the hands to landlords, trader moneylenders and such peasants. When the peasants could take it no longer they resisted against the oppression and exploitation through uprisings. Peasant Movements occupy an important place in the history of social unrest in India though the aims and objectives

of these movements differ in nature and degree from region to region. It is in this sense that these movements also aimed at the unification of the peasants of a region, development of leadership, ideology and a peasant elite. Through these movements emerged a new power structure and peasant alliance.

The genesis of peasant movements rest in the relationship patterns of different social categories existing within the framework of feudal and semi feudal structure of our society. In the post Independence period the nature and objectives of the peasant movement have changed to getting remunerative prices for agricultural produce, to increase agricultural production, to establish parity between prices of agricultural produce and industrial goods and to get minimum wages for the agricultural laborious.

Women's Movement

The women's movement in India is a rich and vibrant movement which has taken different forms in different parts of the country. Fifty years ago when India became independent, it was widely acknowledged that the battle for freedom had been fought as much by women as by men. One of the methods M K Gandhi chose to undermine the authority of the British was for Indians to defy the law which made it illegal for them to make salt. At the time, salt-making was a monopoly and earned considerable revenues for the British. Gandhi began his campaign by going on a march - the salt march - through many villages, leading finally to the sea, where he and others broke the law by making salt.

No woman had been included by Gandhi in his chosen number of marchers. But nationalist women protested, and they forced him to allow them to participate. The first to join was Sarojini Naidu, who went on to become the first woman President of the Indian National Congress in 1925. Her presence was a signal for hundreds of other women to join, and eventually the salt protest was made successful by the many women who not only made salt, but also sat openly in marketplaces selling, and indeed, buying it. The trajectory of this movement is usually traced from the social reform movements of the 19th century when campaigns for the betterment of the conditions of women's lives were taken up, initially by men.

By the end of the century women had begun to organise themselves and gradually they took up a number of causes such as education, the conditions of women's work and so on. It was in the early part of the 20th century that women's organisations were set up, and many of the women who were active in these later became involved in the freedom movement. Independence brought many promises and dreams for women in India - the dream of an egalitarian, just, democratic society in which both men and women would have a voice.

The reality was, however, somewhat different. For all that had happened was that, despite some improvements in the status of women, patriarchy had simply taken on new and different forms. By the 1960s it was clear that many of the promises of Independence were still unfulfilled. It was thus that the 1960s and 1970s saw a spate of

movements in which women took part: campaigns against rising prices, movements for land rights, peasant movements. Women from different parts of the country came together to form groups both inside and outside political parties. Everywhere, in the different movements that were sweeping the country, women participated in large numbers. Everywhere, their participation resulted in transforming the movements from within. One of the first issues to receive countrywide attention from women's groups was violence against women, specifically in the form of rape, and 'dowry deaths'.

This was also the beginning of a process of learning for women: most protests were directed at the State. Because women were able to mobilize support, the State responded, seemingly positively, by changing the law on rape and dowry, making both more stringent. In the early campaigns, groups learnt from day to day that targeting the State was not enough and that victims also needed support. So a further level of work was needed: awareness raising so that violence against women could be prevented, rather than only dealt with after it had happened. Legal aid and counselling centres were set up, and attempts were made to establish women's shelters. Knowledge was also recognised as an important need. The women's activity was geared towards improving the conditions of women's lives. In recent years, the euphoria of the 1970s and early 1980s, symbolized by street-level protests, campaigns in which groups mobilized at a national level, has been replaced by a more considered and complex response to issues. In many parts of India, women are no longer to be seen out on the streets protesting about this or that form of injustice. This apparent lack of a visible movement has led to the accusation that the women's movement is dead or dying.

While the participation of urban, middle class women is undeniable, it is not they who make up the backbone of the movement, or of the many, different campaigns that are generally seen as comprising the movement. The anti-alcohol agitation in Andhra Pradesh and similar campaigns in other parts of India were started and sustained by poor, low-caste, often working-class women. The movement to protect the environment was begun by poor women in a village called Reni in the northern hill regions of India, and only after that did it spread to other parts of the country. One of the biggest challenges women have had to face in recent years is the growing influence of the religious right in India. Right-wing groups have built much of their support on the involvement of women: offering to help them with domestic problems, enabling them to enter the public space in a limited way, and all the while ensuring that the overall ideology within which they operate remains firmly patriarchal.

For activists too, this has posed major problems. It has forced them to confront the fact that they cannot assume solidarity as women that cuts across class, religion, caste, ethnic difference. It is important to recognise that for a country of India's magnitude, change in male-female relations and the kinds of issues the women's movement is focusing on will not come easy. For every step the movement takes forward, there will be a possible backlash, a possible regression. And it is this that makes for the contradictions, this that makes it possible for there to be women who can aspire to, and

attain, the highest political office in the country, and for women to continue to have to confront patriarchy within the home, in the workplace, throughout their lives.

Backward Caste Movement

The Backward castes have been deprived of many social, economic, political and religious privileges. These people provided manual labour and the untouchables occupied the lowest position among the caste hierarchy. They were subjected to extreme form of exploitation. The colonial power accentuated the disparities in the distribution of economic power.

The atrocities united the lower castes against the upper castes. Some of the important backward caste movement which came up was Satyashodak Samaj and Nadar Movement which consolidated the masses along the castelines. E.V Ramaswamy started Self-Respect movement against the Brahmins in South India.

The SNDP movement in Kerala was more of a reformist movement. In 1950s there was a widespread desire among the non-Brahmin castes to be categorized as Backward. Subsequently Backward Class commission was set up to look into the conditions and requirements of these classes. Mandal Commission submitted its report in 1980 recommending reservations for backward castes in educational institutions and government offices. However this move resulted in anti- Mandal Commission movement which resulted in large scale violence and many students lost their lives.

Dalit Movement

Dalits are the suppressed people at the lost rung of the cast-based hierarchy. Their inferior occupations and low levels of ascriptive status make them vulnerable for attacks at the hands of upper-caste people. The organisational efforts made by Dalit leadership for uplifting their status are known as Dalit movement. It is a protest against untouchability, casteism and discrimination faced by the dalits. Dalit movement indicates some trends of protest ideologies which entail the following -withdrawal and self organisation, high varna status and extolling of non-Aryan culture's virtues, abandoning of Hinduism and embracing other religions like Buddhism and Islam. Mahatma Gandhi in 1923 founded the All India Harijan Sevak Sangh to start education and schools for the dalits.

Another most important dalit leader Dr. Ambedkar struggled to secure the basic human dignity to the dalits. The Mahad Satyagarh for the right of water led by him was one of the outstanding movements of the dalits to win equal social rights. The role of All India Depressed Classes Association and All India Depressed Classes Federation were the principal organisations which initiated a movement to improve the conditions of the dalits.

These organisations aimed at improving their miserable conditions and to spread education among them. They worked to secure rights of admission to school, drawing water from the public wells, entering the temples and to use the roads.

THEORIES ON SOCIAL CHANGE

The major sociological theories of change can be classified in several methods. One can, for instance create a distinction flanked by evolutionary, (linear) and cyclical theories of social change. In the middle of the former, the mainly important are those of Comte, Spencer, Hobhouse and Marx. In the middle of the latter, the mainly prominent ones are those of Spengler, Pareto, and Sorokin.

In this unit, we shall briefly look at the following perspectives on change:

- The evolutionary perspective,
- The cyclical perspective,
- The structural-functional perspective, and
- The disagreement perspective.

The Evolutionary Perspective

The notion of social development was taken from the theories of biological development. Spencer propounded an analogy flanked by social and organic growth and flanked by society and an organisation. The theories of social development are composed of one or more of the following principles change, order, direction, progress and perfectibility. The principle of change states that the present system is the outcome, of more or less continuous modification from its original state. Some evolutionists add to the principles of change the notion that change necessity have an order.

Other evolutionists combine the principles of change and order with the principle of direction, thereby suggesting that there is a natural linear order of change in a social system. The evolutionary procedure of change implies, that every society goes through distinctive and successive states of subsistence and orientation. Comte, for instance, proposed a directional theory of society.

He suggested that a society evolves from a theological orientation, to a metaphysical orientation to a positivistic orientation. Durkheim classified societies into easy societies united through parallel of their members, (what he described mechanical solidarity) and intricate societies based on specialisation and functional interdependence of members (what he described organic solidarity). This also suggests a directional evolutionary pattern.

It has been pointed out that it is sometimes hard in evolutionary theory, to differentiate easy direction from progress. The general theme in much of the evolutionary literature is that society's progress in excess of time, to a point where they industrialise and develop in the path and manner of western nations. Extreme expressions of this location are contained in the notion of perfectibility. Societies continue to move towards some ideal advanced state of industrialisation.

Though, the neo-evolutionary theories that have appeared in recent years, are more tentative than the evolutionary theories of the 19th century and early 20th century. These neo-evolutionary theorists do not assert that change proceeds beside the similar path. They suggest that there is a common trend towards a more elaborate division of

labour. They take on a relativistic view, in that they recognise that dissimilar cultures have dissimilar ideas of what constitutes progress. One of the greatest troubles of older theories of development was that they too often contained untreatable, sometimes ethnocentric propositions.

Cyclical Theories

The vital premise of the cyclical theories is: cultures and civilizations pass through stages of change, starting and often ending with the similar stage. This passing through stages is described a cycle. The cycle when completed, repeats itself in excess of and in excess of again. The ancient civilizations in Greece, China and India for instance, can be explained through the principle of cycles.

Some cyclical theorists are pessimistic in that they think that decay is inevitable. Oswald Spengler whispered that every society is born, matures, decays and eventually dies. The Roman Empire rose to power and then slowly collapsed. The British empire grew strong, and then deteriorated. Spengler whispered that social change may take the form of progress or of decay, but that no society lives for ever. Pareto presented in his theory of the circulation of elites, an interpretation of history just as to which social change is brought in relation to the, through the thrash about flanked by groups for political power. His theory was inadequate in that it was based on a limited instance of the circulation of elites in ancient Rome. His conception of political change ignored the growth of democratic government in contemporary times.

More recently Sorokin has presented theories which have some characteristics of the cyclical perspective. Sorokin's theory is based on the principle of immanent socio-cultural change. This implies that any socio-cultural system alters through virtue of its own forces and properties. This principle is interlinked to another principle, namely, the principle of limited possibilities of change. There is a limit to the number of alterations that can develop in a system. For instance, there is a limit to the new shapes of change, and to new patterns of behaviour, that can emerge in a society. The system basically runs out of combinations in due time. If it does not die, it eventually starts running through the changes again.

Therefore, there is "recurrence" or "rhythm" in the histories of socio-cultural systems.] Sorokin also creates a distinction flanked by three broad kinds of culture-ideational, idealist and sensate-which he conceives as succeeding each other in cycles, in the history of societies.

Ideational culture is spiritualistic, mystical and indeterminate. Sensate culture is the realm of science and of direct sensory experiences. Idealistic culture has sure feature of both the ideational and sensate cultures. These three kinds of cultures are looked upon as three views of reality that change just as to the two principles.

Sorokin's work is especially noteworthy not only because it contains a mass of historical analogies and comments on scrupulous social transformations, but also because it saw societies as 'changing' rather than necessarily progressing or decaying.

Structural Functionalist Perspective

In the middle of modern scholars it is mainly closely associated with the work of Parsons and Merton. Structural functionalists consider that society, like the human body, is a balanced system. Each institution serves a function in maintaining society. When events outside or inside the society, disrupt the social order, social institutions create adjustments to restore stability. They also argue that change usually occurs in a gradual, adjective fashion and not in a sudden violent, radical fashion. Even changes which appear to be drastic, have not been able to create a great or lasting impact on the core elements, of the social and cultural systems.

Change just as to them comes from basically three sources:

1. Adjustment of the system to exogenous change,
2. Growth through structural and functional differentiation,
3. Innovations through members of groups within society.

The mainly significant and vital factor creation for social integration and stability, just as to this school of thought, is value consensus. The term ‘cultural lag’ is often used to describe the state of disequilibrium flanked by material and non-material characteristics of a culture. Ogburn who coined this word, explained that ‘cultural lag’ occurs when parts of a culture that were once in adjustment with each other change at dissimilar rates, and become incompatible with each other.

Ogburn pointed out how the non-material culture (values, beliefs, norms, family, and religion) often lags behind material culture (technology, means of manufacture output of the economic system). For instance, family scheduling technologies (*i.e.*, material culture) have advanced, but people take their time to accept them. Some sections of the population may reject the very thought of ‘family scheduling’ and consider in having a big family. Again, when an event such as augment in population or a depletion in natural possessions cause a strain in society, it takes some time for the society to understand and absorb the strain and alter its values and institutions to adapt to the change. But in order to function smoothly, societies adjust to uphold and restore themselves.

Critics have pointed out that the amount and type of changes that can be explained, with the help of the structural functionalist perspective is limited. This view neglects revolutionary changes which are profound and sudden. It also overlooks the possibility of a society going through extensive periods of malintegration, as throughout times of economic recession.

Disagreement Perspective

The disagreement theory takes the principle of dialectic (opposites) as central to social life. Disagreement theory also has its origins in early sociology, especially in the works of Marx. Disagreement theorists do not assume that societies smoothly evolve to higher or intricate stages. Just as to this school every pattern of action, belief and interaction tends to generate an opposing reaction. Contemporary life is full of examples.

The legalisation of abortion has provoked the anti-abortion movement. The feminist movement has stimulated a reaction from men and women. The liberalization of sexual mores has led to open denunciation. The vital premise is that one of the outcomes of disagreement in the middle of groups is social change. The greatest limitation of this approach is that it lays too much emphasis on disagreement, as the mainly significant factor of change. In more recent sociological writing, there is yet another perspective of social change described the 'development perspective'. The development perspective grew from three main sources:

- From the revise of economic growth. Economists and to a great extent other social scientists, view quantitative growth in the economic sphere of life, as an significant indicator of a country's progress. For instance, they point out that a country's prosperity can be measured in conditions of GNP (Gross National Product) or per capita income.
- From the categorisation of all societies into technologically advanced, and less technologically advanced. Sometimes, the emphasis is on industrialisation and consequently societies that are highly industrialised, are seen to be more urbanised than societies which are basically agricultural.
- From the comparison of the capitalist countries with the socialist or communist countries.

Several social scientists have compared the socialist economy and social organisation with Western capitalist economy and organisation. At this juncture we will not elaborate on this perspective, as you are going to look at it in the after that unit. The development approach to social change, brought into sharp focus, the need for formulating a broad relative perspective, which would take into explanation the intricate and diverse relationships flanked by developing countries, flanked by technologically advanced countries, and flanked by technologically advanced countries and developing nations. It can be said from the above discussion of the several perspective, that no single theory can explanation for the complexity of social change.

Social Change in Knowledge and Information Technology

An essential requirement for envisioning India's future is to recognise that the equations which determine national development have changed in recent years, opening up greater possibilities than before. The same factors continue to be at work, but their relative contribution and importance is rapidly shifting along several dimensions.

The sectoral composition of the GDP changes with economic development. The predominance of agriculture in the least developed economies is reduced by the increasing importance of manufacturing, and subsequently, services, as they move up the ladder of development. As this occurs, the rates of economic growth tend to increase. This transition is now occurring globally and is reflected in the explosive growth of the services sector, especially in the fields of financial services, information and communication technology (ICT), insurance, education and health.

India's services sector has already become the dominant contributor to GDP, accounting for 46 per cent of the total, but its share is still far below the UMI reference level of 60 per cent. The country very soon will get the opportunity to skip the long slow phase of industrialisation that the most developed nations have passed through, and transit rapidly into a predominantly service economy by 2020, creating services that meet human needs, generate employment covering the large unorganised segment of the economy, raise incomes and increase purchasing power. Even our notion of services may need to evolve further to recognise the importance of the emerging knowledge-intensive services. Knowledge has replaced capital as the most important determinant of development. In a path breaking study in mid-1950s, Nobel laureate economist Robert

Solow showed that seven-eighths of the growth of US from 1900 to 1950 was accounted for by technical progress, while only one-eighth was driven by capital. A study by Denison, of factors contributing to the growth of the US economy from 1929 to 1982, attributes 94 per cent of that growth to factors relating to knowledge generation and dissemination: 64 per cent of this is linked to advances in knowledge generation (*i.e.*, R&D) and another 30 per cent to advances in education.

Better resource management, which is an application of knowledge, is also identified as a more important factor than capital. This fact bodes well for countries whose economic planners are able to escape from their earlier faith in capital and fully tap the enormous productive potential of non-material, knowledge resources. India's Green Revolution is a dramatic example of how the input of greater knowledge in the form of improved production technologies can rapidly increase the productivity of scarce land resources.

India's IT Revolution is a striking instance of how the importance of human capital has come to acquire a higher position than that of material plant and machinery. All efforts to project India's future progress get at times blinded by the question of resources, more specifically, the financial resources needed for all plan activities. We start with the conviction that financial (capital) resources will not be the key factor that decides the course of our future progress.

If we fail, it will be mainly for want of a vision of what is possible, knowledge of how to realise it, belief in ourselves, commitment to achieve, will for the effort or skill in implementation—and not for lack of finance. The knowledge revolution is not just a short-term blip on the radar screen which peaked in 2000 with the boom in dot com companies. It is a real and profound opportunity for countries around the world to increase the pace and scope of the benefits of development. It marks a significant shift in the relative importance of different resources or factors of production in the development process. This shift from material to knowledge-based resources opens up vast opportunities for the developing countries to accelerate the pace of development. India's rate of economic growth can be substantially increased if the country becomes a superpower in knowledge and if the potentials of information and information technology are fully understood and exploited.

Thus far, the potentials have been narrowly focused on the export potentials of the IT sector. But far greater potential lies in the extension and application of IT to stimulate the development of other sectors of the domestic economy. Information is a revolutionary force in bridging the digital divide that currently separates the advantage and the disadvantaged of our nation. Apart from generating new employment opportunities, the application of IT can vastly extend access to education, health care, markets, financial services, vocational skills, administrative services and other aspects of modern society, to many more people at far lower cost.

It can dramatically reduce the cost of communications, improve access to technology and marketing capabilities for the rural poor, eliminate intermediary exploitation in the

production and distribution chains, increase government accountability and stimulate democratic participation. Therefore, rather than addressing IT as a specific sector, in this report, we have chosen to highlight its significant contributions under different headings, like employment, education, infrastructure and governance.

KNOWLEDGE RESOURCES

There are a host of non-material, knowledge-based human resources that we possess in abundance and can apply to achieve far greater results.

Knowledge Resources:

- Technology
- Organisation
- Information
- Education
- Skills.

Technology

Knowledge in the form of information technology (IT) has opened up the opportunity for India to become the premier, low-cost provider of computer software and IT-enabled services to the industrialised world. It can not only provide high paying jobs and rising exports, but also transform the way we educate our youth—increasing the speed, quality and efficiency of learning manifold.

In addition, it can and is already transforming the way we communicate among ourselves and with the rest of the world, shrinking the distances between hemispheres, providing instantaneous access to the whole world's knowledge base and customer base. Knowledge in the form of biotechnology offers not only a lucrative field for employment and economic growth, but a means for improving the health of our people and the productivity of our fields. In the form of agricultural technology, knowledge can increase crop yields from the present level, which is far below world averages, to levels two, three or four times higher. Pioneering Indian farmers have already achieved it for a variety of crops. What they have done individually, we can do as a nation. Finally, knowledge in the form of manufacturing technology will raise the competitiveness of the Indian manufactures to international standards of costs and quality.

Organisation

Technology is not the only knowledge resource now abundantly at our disposal. Today we have access to the whole world's experience in organisation. Organisation is nothing but the know-how for carrying out work most efficiently and expeditiously. India's highly successful Green Revolution and White Revolution were the results of organisational innovations as much as technology. We have the opportunity to fashion new and better forms of organisation to carry out the tasks of education, health-delivery, governance, commerce, industry and social welfare.

Information

Physical and biological reactions require the presence of catalytic agents to set them in motion and speed completion. Human social processes depend on a catalytic agent too and that catalyst is information. Free movement of information releases society from fear of uncertainties. Information about prices and market potentials spurs an entrepreneur into commercial activity. Information about scientific and technological discovery prompts a scientist or an engineer to adopt new innovations and practical applications. Widely disseminated public information about proper health care and nutrition contributes more powerfully to the general health of the community than does a hospital or medical innovation. Information about government policies enables individuals and communities to fully exercise their rights and take advantage of public programmes.

Information about distant places spurs tourism and trade. Information in all forms and all fields—administration, commerce, education, finance, health, science and technology—is the very source from which we shape our dreams, plans, decisions and actions. The more and better the quality of that information, the more enlightened, expansive, productive and effective will be our efforts at individual and social advancement. Today, the average Indian citizen has access to a wider range of timely and reliable information than had the government leaders in the world's most advanced nations a few decades ago. The fairly easy access to computers and the Internet has placed the world at our fingertips. Spread of information is further facilitated by the advancement of telecommunications technology, rapid expansion of cellular telephone networks, as well as the recent legalisation of Internet telephony, that makes live voice communication possible at a fraction of the cost, both within the country and internationally.

Education

What is true of information is true of education as well. Dissemination of useful information can be said to constitute the so-called unorganised sector of public education. The formal educational system is its organised counterpart. Education is the process whereby society passes on the accumulated knowledge and experience of past generations to its youth in a systematic and abridged form, so that the next generation can start off where past generations have ended and move on from there. Today, through education we have access not only to the knowledge of our own direct ancestors but to the accumulated experience and wisdom of people the world over.

With the development of modern media that brings sound and video images into every household, and with the advent of the Internet that enables us to reach out to sources of knowledge around the world, education offers both unprecedented richness of content and the capacity to deliver it. If only we could break free from the limitations of out-dated curriculum and out-moded delivery systems, we could utilise the opportunity to close the education gap that separates the world's most prosperous communities from their poorer cousins.

Skills

Productive skills form another component of the precious human resource that we can and must fully utilise as leverage for national development. Skill is the ability to direct human energy efficiently to achieve desirable goals. A large reserve of unskilled people may be perceived as a problem, but a large population of skilled workers is a huge asset. It takes both knowledge and skill to train people and we have these in abundance. Imparting employable skills to our entire workforce is not only highly desirable but highly achievable as well. All the resources that we have enumerated—technology, organisation, information, education and skill—are knowledge-based resources.

Knowledge-based resources differ significantly in character from material resources. While material resources are consumed when they are utilised, knowledge resources increase when shared. Material resources are costly to transport and store, whereas knowledge resources are easily transportable at rapid speed and can be stored at negligible cost. Historically, development has occurred under conditions in which access to critical resources was restricted to a relatively small portion of the population. The distinct characteristic of knowledge as a resource makes it possible, for the first time, to spread and share a resource among the entire population. The pace of India's future progress will depend to a large extent on its ability to make available the latest and most useful knowledge to vast parts of the population.

S&T CAPABILITIES

Literacy and general education form the base of the knowledge pyramid that is essential for rapid and sustained development in the 21st century. The continuous advancement of science and application of improved technology form the middle rung; and social ideals and spiritual values form the apex. Technical education, both vocational and professional, constitutes the foundation for development of science and technology. India is rightly proud of the international standing of its IITs, but a handful of world class technical institutions is not sufficient. A large number of the country's approximately 500 engineering colleges need to be upgraded to quality standards nearer to those of the IITs, and given similar autonomy. Private sector initiative and investment, whether from Indian corporates or NRIs or reputed foreign universities, need to be fully encouraged. Close links need to be fostered between technical institutions and industry. India's enormous manpower base of scientists and engineers is often coveted.

President Kalam observed that India's human resource base is one of its greatest core competencies. This is true in absolute terms, but as a percentage of the total population, we are at 1/100th of the US levels and 1/50th of the Korean level. Even China's manpower base of scientists and engineers as a percentage of its population exceeds ours by more than three times. Absolute numbers do count, but manpower alone is not sufficient to excel in the development and application of science and technology. In terms of total investment in R&D, India's expenditure is 1/60th of that

of Korea, 1/250th of that of the USA, and 1/340th of that of Japan. More significantly, atomic energy, space and defence research account for 71 per cent of all central spending on science and technology, which means that relatively little is left for investment in agriculture, energy, telecommunications and other crucial sectors within the sphere of science and technology. R&D expenditure even in India's fast-growing IT sector has been averaging around 3 per cent of sales turnover (STO), which is much lower as compared to the 14-19 per cent expended by internationally reputed software firms. These low figures reflect on our R&D performance.

India's share of global scientific output in 1998 was only 1.58 per cent of the world's total. Out of 500,000 new patent applications filed globally each year, China accounts for 96,000 and Korea accounts for 72,000, while India accounts for only 8,000. Of greater concern has been the country's inability to capitalise on our huge pool of manpower and extensive network of scientific research organisations for transferring proven technologies from the lab to the land and to the factory. Despite possessing the world's largest cadre of agricultural scientists, we have not been able to extend the momentum of Green Revolution to other regions and crops and to update the scientific practices of our farmers to levels comparable with most other nations. Crop productivity remains far below and production costs far above world averages. A similar gap exists in the application of science and technology for food processing and many other industries. High technology exports account for only 6 per cent of total manufacturing exports for India, compared to over 20 per cent for the UMI reference level.

India has a number of premier universities for scientific and technical education which produce world class scientists and engineers, but the national research system rarely produces world class results. One reason is the missing link between scientists in universities and research institutes and practitioners in agriculture and industry. Another is that non-teaching research institutes lack the continuous inflow of young talented researchers required to challenge assumptions and infiltrate new ideas and approaches. Bureaucracy, tenure and lack of accountability minimise the pressure for practical application of scientific knowledge.

Other countries have done a much better job of providing optimum incentives to their scientific community and creating close linkages between science and industry. The management of technology and the administration of government require different sets of skills and a different culture. Management of technology demands innovation, global competitiveness, and wide latitude for individual freedom, characteristics that are difficult to foster under an administrative regime. The objectives of many institutes and the mechanisms for achieving them need a radical reorientation to bring them in tune with the changing environment brought about by liberalisation of imports, technology tie-ups with foreign agencies and foreign direct investment in key sectors.

If the national R&D bodies do not reorganise themselves to face globalisation and global competition, nor take into consideration the nature of the factor endowment of the country along with many existing indigenous technologies so far neglected, then

they risk becoming irrelevant to the mainstream activities of industrial development. The nature of service offered by these institutions has to undergo a sea change. Grass-root consultancy and turnkey project consultancy, including project implementation and monitoring are increasingly in demand. Another essential step is to improve the linkages between technology development and technology application by fostering close ties between basic research and business. In the USA, a large proportion of scientific research at the university level is financed by business, and most leading universities operate technology incubators designed to promote commercialisation of new products and processes.

India's recent experiment with the Science and Technology Entrepreneur Parks (STEPs) has been successful in promoting application of technologies, new business start-ups, and employment. The STEP in combination with technology incubators need to be rapidly multiplied around the country. Effective R&D management backed by suitable incentives for commercialisation are essential for bridging the gap between the lab and the land or factory. R&D is not only essential for propelling growth in industrial fields. It can also become a growth industry on its own.

Much of the work done by Indian software companies for overseas clients and parent companies rightly fall under the category of new product development. Vast potential exists for complementing India's strengths in IT with corresponding strengthening of fields such as biotechnology, pharmaceuticals, designer-made materials, and others. India's investment in the biotechnology sector is expected to increase five-fold by the end of the current decade largely as a result of growing collaboration between multinational corporations and indigenous research efforts. Huge opportunities also exist for R&D in the field of genomics, bioinformatics, DNA technologies, clinical studies and genetically modified crops. Rationalisation of procedure, backed by effective bio-safety regulations are needed to govern testing and approval of biotech products. India has the potential of converting its strong R&D infrastructure into a global research, design and development platform.

KNOWLEDGE TYPOLOGIES

Previous organizational knowledge research has resulted in the development of several typologies. Although typologies differ slightly, they share a common perspective that knowledge is multi-dimensional and complex. Although the theoretical roots of discerning different types or characteristics of knowledge go back to ancient philosophers most recent theorizing about knowledge types and domains build upon Polyani's differentiation between explicit knowledge and tacit, or implicit, knowledge.

Indeed, most typologies in the 1990s were variations of Polyani's concepts of explicit and tacit knowledge. A sample of knowledge typologies offered by scholars for understanding the complex nature of organizational knowledge. Several typologies were developed directly from Polyani's conceptualization. Spender summarized his view of knowledge with a grid of categories that represent combinations of individual, social,

explicit, and implicit knowledge. Conscious knowledge, in this typology, is both individual and explicit, such as that knowledge one obtains from reading the newspaper. Objectified knowledge is social and explicit, such as that written in standard operating procedures and followed by organizational members. Automatic knowledge is individual and implicit, such as everyday routines one does to accomplish tasks without even noticing. Finally, collective knowledge is social and implicit, such as shared work routines practiced in an organization.

According to Spender, these categories are not rigidly separated from one another. To the contrary, he argued that boundaries are flexible between explicit and tacit knowledge, in particular. Typologies offered by Cook and Brown and Tywoniak use categories similar to those presented by Spender. Cook and Brown note that certain types of knowledge, such as explicit, have been given priority in theorizing and in organizational applications. In accordance with Spender's argument about porous boundaries between categories, Tywoniak posited that tacit, explicit, personal, and common knowledge are "interdependent dimensions of knowledge that interact dynamically."

That is, knowledge types shade into and influence each other. Cook and Brown contended that categories were distinct and should be viewed as functioning in different ways, but Spender's grid demonstrated how knowledge types involve different axes, so to speak, and so should not be viewed as mutually exclusive domains. This position seems most tenable, and is also supported by a completely different typology offered by Collins and elaborated by Blackler. Typologies clearly demonstrate that organizational knowledge is multi-faceted, grounded in various aspects of human experience, and individual as well as collective. Like Cook and Brown, Blackler criticized the tendency in research and organizations to privilege some types of knowledge over others.

However, and similar to Spender and Tywoniak, Blackler argued that more relationships might exist among types of knowledge than current conceptualizations recognize when focusing only on differences. Spender also recognized the limits of presenting typologies and matrices, pointing out that such categories do little in terms of explaining how knowledge types interact and how organizations specifically become contexts for knowledge development and application.

This limitation is somewhat addressed with Collins' conceptualization of knowledge types, which moved away from the use of the explicit/tacit divide with a typology of four images of knowledge: embrained, embodied, encultured, and encoded. Blackler extended Collins' typology with a fifth image, embedded knowledge. Embrained knowledge is knowledge grounded in conceptual skills and cognitive abilities; it is abstract knowledge "about" things.

Embodied knowledge is knowledge developed in action; it is knowledge "how" to go about activities. Encultured knowledge is grounded in shared understandings that are based in cultural meaning systems. Embedded knowledge resides in system rules and routines and is grounded in relations among system roles, technologies, and

procedures. Encoded knowledge is information that is conveyed by signs and symbols; knowledge gained through books and the like. Although Blackler does not treat knowledge as constituted in interaction, these images can be thought of as resources situated in social processes. Blackler's typology of knowledge is particularly useful within the framework of structuring activity theory used in this chapter.

One theoretical foundation of SAT, structuration theory, focuses on the knowledgeability of actors in situated contexts. Knowledgeability constitutes everyday, practical knowledge that guides action but might not be readily articulated. For example, individuals know how to conduct themselves in meetings at work, even though they might not articulate all the rules of engagement that they follow. Knowledgeability is inherent in social practices, and therefore is one form of knowledge that individuals draw upon in the policy process.

This knowledgeability used in everyday routines corresponds with encultured and embedded knowledge, in that individuals use this knowledge as they engage in everyday activities; however, it is such a part of who they are that they might not be able to articulate this knowledge. Likewise, abstract, conceptual knowledge that is embrained is an inherent part of people who are interacting regarding policy. It is a resource for individuals and groups, and likely is influential in policy interactions. Embodied knowledge that involves how to do things is only partly explicit, and might be important in policy implementation contexts where knowledge develops through practice. More explicit knowledge that individuals can articulate, such as features of policy texts, corresponds with Blackler's encoded knowledge.

Accordingly, these different knowledge types likely contribute in distinctive ways to policy-related interactions within and between social systems, and are likely involved in varying ways in policy knowledge construction across organizational boundaries. Previous policy and organizational knowledge research indicates that these knowledge types are important.

For example, Pan, Newell, Huang, and Galliers demonstrated how different phases of a new organizational process involved embrained, embodied, encultured, embedded, and encoded knowledge. Presenting only encoded knowledge outcomes, according to Pan and colleagues, would have been an inaccurate and partial portrayal of the organizational knowledge process. Furthermore, Fisher and Owen noted that scholars and practitioners in the public policy arena often dismiss the experiential, embodied, knowledge of policy implementers and the relational, encultured, knowledge that develops through policy practices.

As in other organizational arenas, research and practice in public policy tend to favour the explicit, encoded, knowledge of policy texts. This tendency clearly holds implications for who is seen as knowledgeable, and how knowledge processes are explained. Knowledge typologies have been used to consider if different types of organizations produce, use, and privilege certain types of knowledge over others. Lam and Blackler both argued that various sorts of knowledge are differentially important to

organizations based on their structures or functions. Although these two scholars differed in their interpretations of which knowledge types dominate which organizational types, they agreed on the idea that different organizational contexts depend on different knowledge types. For example, a hospital, which Lam would label a “professional bureaucracy” and Blackler would label an “expert-dependent organization,” produces, uses, and privileges distinct types of knowledge in different ways than does an industrial factory, which Lam would call a “machine bureaucracy” and Blackler would call a “knowledge-routinized organization.” Examining differences in how these types of knowledge develop, or not, in different organizational contexts constitutes a process-based empirical extension of this typology.

Such an application demonstrates the potential value of typologies, in spite of limitations inherent to any static categorization scheme. One more recent typology presented deserves mention. Boer used cultural-historical activity theory to propose six knowledge domains that he contended are used as resources for system participants to accomplish ongoing activity.

“S-knowledge” is personal knowledge one member has of another member, such as background, interests, abilities, and such; “M-knowledge,” about mediating artifacts in the system, is knowledge about what tools are available to accomplish the activity and how to use them, as well as what language to use within the system; “O-knowledge” is about the collective object of activity, knowledge about the ultimate goal and motivation for the whole system; “D-knowledge,” about the division of labour, is knowledge about how tasks are divided, chains of command, and such; “A-knowledge,” about involved actors, is knowledge about who knows what, similar to transactive memory systems and “R-knowledge” is about social rules, which is knowledge about appropriate behaviour and relationships.

Although this typology presents another alternative to the explicit/tacit approach to organizational knowledge, it is limited by its system-specific level of analysis and emphasis on targets of knowledge rather than resources for knowledge development. Identifying these domains of knowledge does little for understanding how knowledge might be constructed between members of different activity systems.

Accordingly, the project reported in this chapter adopted Blackler’s typology for examining the types of knowledge developed and used in interpreting and implementing policy in an education setting. The five images of embrained, embedded, encoded, encultured, and embodied knowledge are not limited to particular system elements, but by definition allow for analyzing the influence of such system elements on the knowledge construction process.

For instance, the current environment for education systems requires highly standardized practices driven by institutional policies and routines. Such an environment might privilege encoded or embedded knowledge, but it is important to be open to other types of knowledge being used and generated, particularly in interactions that cross organizational boundaries. As Blackler noted, the nature of modern organizations—

even governmental bureaucracies—is transforming. So, too, might be the nature of knowledge produced and used within them.

FAMILY/PROFESSIONAL KNOWLEDGE PROCESSES

When considering a public policy context, organizational knowledge processes extend beyond organizational boundaries. The above discussion of organizational knowledge in policy contexts necessarily leads to a discussion of how such knowledge is developed between organization members and members of related social systems, such as families that use or benefit from public policies. Most interest in organizational knowledge has been based on the recognition that knowledge processes are consequential for competitiveness, success, and viability of for-profit organizations.

However, I argue that interests in these processes should be expanded by recognizing that they are consequential for more than profit and competitiveness. Indeed, applying organizational knowledge theory and research to the ubiquitous public policy arena is a “next move” in knowledge theorizing. Previous research of lay–expert interactions regarding policy issues indicates that much remains to be known about communicative knowledge construction in lay–expert interactions.

For example, Meyer et al. studied welfare and child support policy knowledge among recipients, and found there were no significant associations between recipients’ level of policy knowledge and contact with welfare staff or staff knowledge. However, there were significant positive associations between knowledge level and previous experience receiving support, as well as when recipients had specifically talked with a staff member about child support and lived in a country with highly knowledgeable staff members. These somewhat contradictory findings in one study suggest that more is going on than “knowledge transfer” through contact with professionals. Additionally, as discussed above, the way in which knowledge was operationalized could limit the ability to assess actual knowledge—of varying types—possessed by recipients. Although encoded knowledge was measured, perhaps more salient to the study sample, and to the use of welfare benefits, is embedded knowledge developed through relationships with knowledgeable staff, and embodied knowledge developed through lived experiences in the welfare system.

Previous research of parent–professional interactions regarding disability and special education policy has indicated that parents rely on professionals to interpret policy and inform parents of policy provisions that relate to them. When asked about how they learned about disabilities, available services, and relevant policy provisions, parent participants in these studies uniformly identified educational professionals as their main source of information. Few parents in these studies relied on policy texts to understand provisions that impact their children or their families, which would be encoded knowledge typically dependent for its meaning on embedded and even encultured

knowledge. Rather, their knowledge development relied on the filter of organizational knowledge processes of professionals with whom they interacted. This dependence upon experts, and perceived expertise, comports with Adams contention that the public policy arena privileges expert knowledge domains over other domains, such as local family situations. These findings also point to a need to recognize and examine cross-system consequences of organizational knowledge processes. Previous research also indicates that parents tend to rely on their experiences with their children in a variety of contexts in decision-making interactions with policy implementers.

These studies suggest that policy knowledge for family members might be mediated, or influenced, by elements of their family systems, such as cultural assumptions and values, resources, roles, and priorities. In other words, constructing policy knowledge across professional and family systems cannot be separated from the embedded and encultured knowledge of policy beneficiaries.

However, research also indicates that professional policy implementers, particularly in the realm of special education policy, often do not take such knowledge types or processes into account during implementation interactions that would potentially construct policy knowledge.

For example, a related analysis from the larger project described below examined parent–professional interactions in special education decision-making meetings. That analysis revealed that there were significantly more instances of professional “logistics talk” regarding procedures, documents, and testing results in these policy implementation meetings than instances of parents providing input about students’ home behaviour and background. In sum, these studies indicate that although public policy users, such as parents of children with disabilities or chronic health problems, might develop various types of knowledge in different ways, expert policy knowledge is privileged when professional and family systems intersect regarding policy. A question remains as to how to tap different types of knowledge to benefit policy implementers as well as policy beneficiaries.

CONSTRUCTING POLICY KNOWLEDGE ACROSS SYSTEMS

PROJECT DATA

This analysis was part of a larger study of the communicative construction of special education policy knowledge conducted over a five-month period in a public elementary school district in the southwest United States. Participants included 82 educational professionals and 18 parents/guardians who interacted regarding the United States’ Individuals with Disabilities Education Act. Educational professionals included regular and special education teachers, administrators, psychologists, and speech and language pathologists. There were eight intersecting activity systems representing three types of

systems: specialists, schools, and families. Data included federal, state, and school district policy documents, transcripts of 49 audio-recorded staff and staff–parent meetings, transcripts from 24 audio-recorded semi-structured interviews, 60 district e-mail messages/threads regarding policy, and field notes of approximately 10 hours of additional observations. Meetings and e-mail messages focused on changes in policy provisions, implementation practices related to those changes, and educational decisions for students as part of policy implementation or compliance. Interviews focused on ways professionals and parents learned about policy and interactions regarding policy changes and implementation.

DATA ANALYSIS

Previous reports provided detailed explanations of the data analysis process involved in the larger project. To summarize the analysis procedures, the constant comparative method was used to code and categorize the data. The coding process resulted in 108 content codes that were organized into 11 axial codes reflecting elements of activity systems and knowledge construction. These axial codes were activity intersections, community, consensus building, dissensus building, division of labour, giving information, mediating resources, object, rules, seeking information, and subject. This analysis extends previous reports of the study by examining how the policy knowledge construction process reflected the use and development of different knowledge types.

MULTI-DIMENSIONAL POLICY KNOWLEDGE

Previous analyses of project data revealed that participants developed policy knowledge that was shaped by mediating elements of intersecting activity systems. These mediating elements included rules, such as following agendas; aspects of individual system members, such as their existing knowledge; material and language resources, such as technology and metaphors; divisions of labour, such as authority; and community, such as meeting interactions. The discussion below explains how mediating elements were integral to different types of knowledge being used and constructed in the policy process. Additionally, it was previously reported that participants developed policy knowledge with five primary communication processes: identifying priorities, expressing lack of knowledge, offering explanations and clarifications, expressing difference, and posing potential consequences. Distinction of these processes provides a foundation for the following discussion of the role of different knowledge types as participants constructed policy knowledge within and across policy-related systems.

Identifying Priorities

The communicative process of identifying priorities involved the development of primarily encoded knowledge, although the construction of encoded knowledge involved the use of different types of knowledge. Encoded knowledge is conveyed by signs and

symbols, such as in hand-outs and policy documents. This is explicit knowledge that can be articulated by members as a resource for constructing policy knowledge within and across activity systems. Identifying policy priorities by distributing informational hand-outs, using policy documents, and defining terms were ways of using and constructing encoded knowledge both within activity systems and between policy-related systems. However, individuals also drew on the embedded knowledge of particular activity systems, which is knowledgeability of the routines and relations of a system, represented by system rules and divisions of labour. For example, participants used the rule of following an agenda to identify policy priorities.

Accordingly, the knowledge constructed was mediated by the use of agendas, and therefore included participants' embedded knowledge of system rules. Identifying priorities within activity systems was also shaped by shared language and understandings of the community that reflected encultured knowledge. Members of related activity systems shared a common understanding of the legitimacy of policy to structure activity, and members of specialist systems shared common understandings of policy terms, using that encultured knowledge to identify policy priorities in their professional systems. For example, a special education administrator stated in an e-mail to a transportation administrator, "Next year we need to make sure we are providing the right amount of instructional time for our students in the self contained programmes." Although this e-mail was sent in the fall, administrators of these intersecting activity systems shared a common understanding that procedures such as school and transportation schedules take considerable time to develop, which is embedded knowledge, as well as a common understanding of terms such as "instructional time" and "self contained programmes," which is encultured knowledge, to facilitate the development of policy knowledge across systems.

The embedded and encultured knowledge drawn on and reinforced was primarily that of professional activity systems. As would be expected, parents did not identify policy priorities, but they also rarely explicitly drew on embedded and encultured knowledge of their family systems when priorities were identified in decision-making meetings with professionals. Rather, members of specialist or school systems would present policy priorities to parents, and in the process structure parent-professional interactions to comply with those priorities.

Thus, although multiple knowledge types were evident in the communication process of identifying priorities, participants clearly privileged knowledge resources of professional systems over family systems and accordingly reinforced the authority of those systems for shaping what-and whose-knowledge counts in the policy process. One consequence of the knowledge construction process, the reproduction of expert authority and of the power divide between experts and lay people.

Expressing Lack of Knowledge

Previous analyses indicated that direct and indirect expressions of lack of knowledge were important for developing requisite policy knowledge within and across activity

systems. For example, direct admissions of lack of knowledge were important for parents to develop knowledge about special education policy. Professionals leading parent–professional decision-making meetings often presented information mechanically according to the organization of policy implementation documents.

When parents openly admitted that they did not know or understand a policy feature, there was an opportunity for parents to access the expert system through the encoded document knowledge, the embrained conceptual knowledge of specialists, and the embodied expertise regarding the practice of special education that professionals possessed. Members of school and specialist activity systems also used direct expressions of lack of knowledge, especially in meetings with people perceived to have expert authority, such as the director of special education or school principals.

Accordingly, this communication process was important for both within-system and cross-system knowledge development. Participants also expressed their lack of knowledge indirectly by requesting additional information or clarification about policy features. People in authority often used informational hand-outs or policy documents to identify policy priorities. Participants would use these hand-outs as basic encoded information, but express the inadequacy of that information for developing requisite knowledge by requesting clarifications and additional information.

Many of these requests stemmed from participants' own embodied knowledge developed in organizational practice. These requests for clarifications and additional information also included questions about specific aspects of the context, demonstrating that understanding of the policy issue involved embedded knowledge of routines of the system and by encultured, shared understandings of the community.

For example, participants in several specialist activity systems discussed that there were conflicts between their school-day schedules and school bus schedules. This topic revealed that certain special education students were not getting as many hours of school as other students in the district. One specialist asked at a district meeting, "So I feel like it's not fair for everybody. So, are we wanting the same time to get out? [sic] Or they—because they are VE?" This request for clarification reflects embodied knowledge that led to a judgment about fairness, the embedded knowledge about how procedures work across related activity systems of the school district, and encultured knowledge about common terms. Accordingly, expressing lack of knowledge was mediated by elements of participants' activity systems as well as elements of related systems as they used embodied, embedded, and encultured knowledge to generate shared encoded policy knowledge across systems.

Offering Explanations and Clarifications

Expressions of lack of knowledge often led to explanations and clarifications that involved different types of knowledge. Explanations and clarifications reflected individuals' existing encoded knowledge of policy, their abstract embrained knowledge, their embedded knowledge of routines, and their shared encultured knowledge.

Sometimes participants offered their existing knowledge of policy texts as a resource for constructing common knowledge within and between activity systems, drawing on their encoded knowledge as they interacted with the community. Participants often referenced past practices as a way of explaining current policy changes or policy features, drawing on embedded and encultured knowledge of the community. For example, changes regarding service provisions to private school students involved extended discussions among members of various specialist activity systems that used past policy and procedures as a way to make sense of the new policy. These explanations and clarifications reflected knowledge that was embedded in the rules, routines, and practices of particular activity systems.

Explanations and clarifications were also used to develop policy knowledge across intersecting systems. Professionals often used examples of common classroom or playground practices to explain to parents how policy provisions would work for students. In these cases, explanations and clarifications provided connections between family, school, and specialist activity systems by involving embodied knowledge of what happens in schools, encultured knowledge of normative expectations, and embedded knowledge of common school procedures. Occasionally, parents would explain past procedures for policy implementation regarding their child's education if they had transferred from a different school. In these instances, parents reflected both embrained and encoded knowledge regarding policy that contributed to decision-making for students.

Posing Potential Consequences

Many policy issues and changes were controversial, and involved participants discussing potential consequences of decisions. Posing potential consequences involved various forms of existing knowledge. For instance, participants drew on their own abstract thinking, or embrained knowledge, to introduce potential scenarios about policy changes to the community.

Also, participants used their existing encoded knowledge based on their professional position to frame potential consequences of policy features and changes. Potential consequences reflected participants' embedded knowledge of existing system routines based on the division of labour as resources for talking about potential consequences of policies and procedures. For example, a new scholarship programme for students with disabilities raised numerous policy concerns among school psychologists. One psychologist noted the potential for legal problems stemming from the new programme, "Can you imagine how you can—you've now proven—you've been awarded a scholarship because you're dissatisfied with the programme.

Now you've opened a whole new course for due process." The knowledge that developed over time regarding connections between IDEA policy provisions and the scholarship programme was influenced by comments such as this, which reflected embrained, encoded, encultured, and embedded knowledge. Furthermore, these different

types of knowledge used as resources were shaped by mediating elements of participants' activity systems, such as their divisions of labour, community interactions, and policy text resources.

When participants discussed policy across related systems, such as during parent–professional meetings, posing potential consequences also involved several types of knowledge and mediating elements of intersecting systems. Professionals often discussed potential consequences of children entering special education. For example, one psychologist explained to a parent, “Removal for the resource room may result in him losing time with his non-disabled peers. The positive effects would hopefully be a more individualized programme where he can have more access to curriculum in ways that he best learns.” These consequences were stated based on encoded knowledge of policy as well as embedded knowledge of school systems, but also provided a connection for parents to the school activity system.

Expressing Difference

Participants drew on their existing knowledge, values, and experiences to express differences and shape policy discussions, thereby shaping knowledge about policy. Expressions of difference were mediated by individuals drawing on their existing encoded and embrained knowledge; they used their existing knowledge of mediating resources such as policy language and policy provisions to provide evaluations of policy changes, suggest alternatives, and offer alternative perspectives about issues.

This process also was shaped by individuals' perceptions of how to engage in activity, reflecting implicit knowledgeability or embodied knowledge. For example, in one school meeting a special education teacher used embodied knowledge as he noted differences among staff members about how to implement and interpret policy, “and I disagree strongly on this. I feel it's more important to serve the needs of many rather than one.” This comment led to a discussion among system members about how best to interpret policy provisions regarding a specific type of educational service. Expressions of difference also involved shared, or encultured, knowledge of the community as participants used interactions with other members of the system community in meetings to share diverse perspectives that reflected common understandings of their activity.

Expressions of difference sometimes were mediated by the division of labour, reflecting embedded knowledge, as participants disagreed with each other or offered diverse opinions based on their hierarchical position or job function, which then influenced policy interpretations for the entire activity community. Parent–professional interactions rarely involved expressions of disagreement or diverse perspectives.

Interviews with parents after decision-making meetings revealed that many parents were hesitant to “cause trouble” for teachers, and so refrained from expressing disagreements regarding policy implementation. The lack of disagreement or diversity in these intersections of professional and family activity systems is consistent with parent reports, noted earlier, that they rely on gaining information from expert

professionals. However, it also suggests that the value of knowledge parents possess goes unrecognized in policy implementation, as such interactions reproduce the privileging of expert, encoded knowledge over other ways of knowing in the policy process.

NEW ENGINEERS OR NEW INFORMATION PROFESSIONALS

Bradley's book was prepared with what she said was 'new kind of engineer in mind' in a society where ICTs were pervasive in augmenting all forms of communication processing. However, given the increasing diversity of people engaged in knowledge and information work that involves questions of design, testing, implementation and theorization about ICT artifacts and relationships, it is not just a question of a 'new type of engineer'. This is an observation that can be drawn from the critical approach developed by Hirschheim and others in their attempt to deconstruct the many complex domains and tasks which cover Information Systems design: no one person, or particular discipline, can be expected to cover the huge range of skills required for the creation of increasingly complex social-technical networks and products in a globalized world.

As Bradley observes, broad education and exposure to fundamental social theories and concepts are necessary for the creation a broader cohort of ICT professionals and practitioners, who can think beyond traditional 'engineering' problems. A broader educational model also indicates the need for a dialogue between non-technical practitioners such as community development workers and those responsible for the design of technical systems, and critically, the engagement of communities themselves in the design process.

This is where the perspectives drawn from Community Informatics and the related field of Development Informatics can help to establish richer understandings of what is needed to make more professionals conscious of the complexities of working with social-technologies for social betterment. For those unfamiliar with the literature and practice of Community Informatics, it is an increasingly well-documented approach to empowering communities with information and communication technologies which brings to bear insights from Information Systems, and Social Informatics.

Gurstein, who is widely cited in Community Informatics, recently suggested that Community Informatics involves:

- A commitment to universality of technology-enabled opportunity including to the disadvantaged; a recognition that the "lived physical community" is at the very center of individual and family wellbeing—economic, political, and cultural; a belief that this can be enhanced through the judicious use of ICT; a sophisticated user-focused understanding of Information Technology; and applied social leadership, entrepreneurship and creativity.

That is, Community Informatics is a type of social-technology theorization and practice that promotes social change and human development in conjunction with technology by engaging with local communities and even smaller groupings, such as community organizations, families and informal groups.

The study of embedded and localised, in situ interactions has led to an increasing number of studies in the IS field, in part because of the failure of traditional, positivist methods of IT research in preference for other forms of theory and research, that 'make visible' what can be so easily missed. Technologies can be constructive or disruptive to established ways of doing things, with any combination of effects between, and this is just as much the case in community settings as anywhere else.

How to recognise and work with social-technical diversity is also part of the Community Informatics domain of activity.. From the community development perspective, communities function as locales for 'solidarity and agency', in which solidarity represents deeply held bonds, brought to fruition through human agency, though of course, communities can be disruptive and non-harmonious as well. Despite this restriction, the idea of 'solidarity and agency' at the core of community life also provides a less bounded dimension to the idea of community since it transcends geographic limitations, and can incorporate more dispersed, as well as virtual affiliations.

Day and Schuler provide an additional perspective by suggesting that:

- Community and voluntary sector groups and organizations form the bedrock of community life through the planning, organization, provision, and support of community activities and services. Although usually under-resourced and over-stretched the community and voluntary sector play a significant role in building and sustaining community.

With its emphasis on empathetic understandings and action being developed with a community's engagement with technology, Community Informatics is part of the interpretive tradition of action research that values and accepts multiple, divergent, and even contradictory 'definitions of the situation', in natural and real world settings. This orientation is often predicated on theories of social change, and is associated with community development practice in both developed or developing countries.

As an example of this approach, 'technologies of care', that emphasize human communication and personal development predominate in many community settings. Studies have been based upon close studies of the meanings given to technology by workers engaged in face-to-face and sometimes, distributed human services work and community education such as that found in smallscale neighbourhood welfare and educational support organizations.

Of course, not all community organizations need have the same orientation towards the use of technology but in many situations, there is 'situated use' for people-focussed problem-solving. Technologies of care can thus be understood as an 'instrumental ensemble' of personal tools and professional techniques, particularly influenced by women's culture and practices in the helping professions.

Once again, homage should be paid to Bradley's insights on gender, and this perspective can also be filled out by a growing literature on the feminization of the workforce, including the impact of ICTs on women's work. ICTs as part of this ensemble influence the shape of communications and action, subject to human intervention and modification. However, as part of the overall technologies of care, ICTs are not pivotal as the vehicle of communication and must be conceived of as part of a process of action and agency for community development and support. They have an adjunct and subsumed function in supporting a cycle of knowledgeable and skilled practices or technologies in the area of community support. However, the personal meaning of work is not well explored in Community Informatics writing, and this is where Bradley's model becomes useful.

INCREASING TRAFFIC, REGULATION AND ONLINE SOCIAL INTERACTION

By the early 80s, the network traffic increased to the point of becoming sluggish. As a result, in 1983 ARPANET was split. MILNET was used for military sites and ARPANET was used for non-military traffic. However, it did not take long for growth to go beyond control a second time. To deal with the unceasing growth, standardized protocols were required by the Secretary of Defence.

Unknown at the time, these standardizations laid the ground work for the successful commercialization of the Internet. Utilizing these standards, the National Science Foundation funded or partially funded six supercomputer centers known as NSFNET. These centers merged into a backbone of the Internet. NSFNET permitted an easier access in order for researchers to share their ideas in a more user-friendly environment. Time could be spent developing scientific ideas rather than spending countless hours figuring out the network. A key innovation occurred with Paul Mockapetris's November 1983 memo entitled "Domain Name System."

To compensate for the unexpected increase of addresses, he proposed an international system which includes seven "top-level domains":

1. Com (Commercial)
2. Edu (Educational)
3. Gov (Government)
4. Mil (Military)
5. Net (Networking Organizations)
6. Org (Non-commercial Organizations)
7. Int (International Organizations, like the United Nations).

Three-letter county codes were added later. This was a complex political task, not an engineering problem. By 1992, the Internet had continued to grow without any controls. No single entity could claim authority. The real success of the Internet is not attributed to the various technical consortia but consistently is attributed to "cooperation" among

the users. When one person or a group violates the boundary of reasonableness, users work together to establish a position. Observances of Internet activity are an irreplaceable social psychology laboratory. Two extremely different behavioural forces on the Internet set the tone for legislative control. First, Stoll's (1989) *The Cuckoo's Egg* discusses the most cited and most exciting problem that emerged from the Internet.

Stoll describes how he stumbled into a ring of eastern block spies. The process of his discovery and cooperation has inspired the Internet community. On the other extreme and dark side of the Internet, we have the case of the Internet worm. Burstyn (1990) outlines the actions and consequences of Robert Morris, son of a scientist of the National Security Agency's Computer Security Council, and a Cornell graduate student. He developed a computer programme that could have clogged the entire Internet. Although he claimed his action was an accident and that he had no ill intentions, he received a three-year sentence. No physical damage was inflicted nor was any information lost, but many human and computer hours were spent in a network clean-up. Schmidt and Darby state that the effect of the worm was devastating while Burstyn (1990) estimates that approximately 10% of computers connected to the Internet were affected.

The most serious problem was the interruption of the Army's Ballistic Research Laboratory which stayed off the Internet for a week. The most important outcome was federal legislation concerning criminal acts on the Internet. Congress wanted to send a message to pranksters and saboteurs. Thus, Morris's action induced a precedent for legal control of the Internet, while Stoll's experiences encouraged lawmakers to refrain from over-involvement.

Congress faced a paradox. They realised the need for legislation for damage control, but desired the rich contributions made in an atmosphere of organized chaos. The history of the Internet is a history of rapid technological improvements wed to an environment of human cooperation. The technological advances and social linkages occurred in three general categories that are not necessarily mutually exclusive.

They include:

- Communication
 - E-Mail
 - InterNIC
 - Listserv
 - Chat
- Information Acquisition
 - Archie
 - Bulletin Board
 - Gopher
 - Hytelnet
 - UseNet NewsGroups
 - Veronica
 - WAIS

- World Wide Web
- Other
 - File Transfer Protocol
 - Telnet.

Perhaps the most exciting of the World Wide Web. WWW has gained more national and international attention than all the other Internet services combined. The WWW is a network of Internet sites that use a common site identification system and common language for displaying information organized by Web pages.

In 1989, the first Web page was developed under the sponsorship of the European Laboratory for Particle Research (the combined resources of 18 countries). The first Web navigation software was far from user-friendly. It was run on UNIX-based computers and little thought was given to PC-based machines. As soon as Mosaic was written in 1993 by students at the National Center for Supercomputing Applications (University of Illinois at Urbana-Champaign), a new computer milestone emerged. The WWW exploded into popularity because Mosaic is a user-friendly and intuitive software that could be adapted for non-UNIX computers.

In 1994, Mosaic dominated the use of the Web by personal computers. However, by 1996, Rafter (1996) notes that Mosaic's popularity dropped rapidly when Netscape appeared on the market. Mosaic was a "hassle to install," slow, and lacked an integrative e-mail function. However, Mann (1996) predicts that with all the improvements in Microsoft's Explorer, Netscape may soon follow the path of Mosaic. Since January 1994, Pitkow has been involved in surveying the Web. To date, at least seven comprehensive surveys have been completed.

THE INTERNET AND SOCIAL WORK: A PERSONAL PERSPECTIVE

A vast array of social work Web sites have sprung up. The first systematic social work invasion of Web service was in educational institutions. Michael McMurray (Colorado State University) was the first, most notable effort. A long-lived and well-managed example is the University of South Carolina's SWAN (Social Work Access Network). At the time of this writing well over 25% of all accredited social work academic programmes have Web sites. Clearly, Web pages that are owned and operated by large institutions are the most reliable. Nevertheless, I believe the future power of the Web lies elsewhere. Although psychotherapy on the Internet existed prior to the popularity of the Web, employing the Web as a medium of psychotherapy has mushroomed. During a recent survey, well over 1,000 psychotherapists were capitalizing on the Web.

Each time Marson and Brackin study psychotherapy on the Web, the number of therapists increases. NASW News is chronicling efforts of private practitioners in this arena. Clearly, the Web will have a profound effect on future strategies for psychotherapy

and other forms of social work practice. For example, Schein-Levi and Pollack (1997) demonstrate how social workers can employ the Web for enhancing parenting skills.

According to Anderson et al. (1995), the most widespread use of the Internet among the general public is not browsing the Web, but e-mail and listservs. Marson (1997) confirms this finding among social workers. A personal history will help explain the importance of these two uses. When I first learned of access to the Internet, I also learned about listservs and immediately searched for a listserv that addressed social work issues.

In my search, I discovered the Social Work discussion list. SOCWORK started as an idea for introducing students at the School of Social Work, University of Toronto, to computer communication technology. Bellamy (1987), and Bellamy and Mielniczuk (1988) offer historical details of how and why the list became operational on February 3, 1988. It began with 11 students and four faculty and staff. Within four months there were 20 members. By May 1989 there were approximately 75 members and a new management under the watchful eye of Harry Chaiklin at the University of Maryland. It was about this time that I discovered SOCWORK and joined the list when there were fewer than 80 subscribers.

During Harry's tenure, two major developments occurred. First, after much prodding among various members of the listserv, the Council on Social Work Education subscribed to SOCWORK. Second, staff from the National Association of Social Workers [NASW] subscribed. Participation by our two major professional organizations established an aura of legitimacy and personal subscriptions to SOCWORK surged. Later, when the subscribers grew to approximately 700 in the spring of 1995, Cindy Jones from the University of Arkansas inherited SOCWORK. Soon SOCWORK had over 1,000 subscribers and many thought the listserv was out of control. As a result, in the fall of 1996, Ogden Rogers from the University of Wisconsin at River Falls became the new manager. The number of subscribers shrank to less than 300, but SOCWORK became more manageable. Listservs are now flourishing and are being systematically evaluated for their contribution to the profession.

Berman (1996) provides an excellent example of such research in his analysis of two listservs. Noticing the congressional proposals to decentralize social services, the Consortium of State SOCWORK Listservs was founded in 1993 by Steve Marson at the University of North Carolina at Pembroke. With decentralization, each state has gained greater control over social service resources.

The direction of the services may greatly depend on the ability of state social workers to spread knowledge to grassroot organizations and to lobby the state house in a favourable direction. State social work listservs should prove to be vital for the speedy dissemination of needed information. Social workers have a long history of technology avoidance. This avoidance was first noted by Richmond (1911) when she had to encourage turn-of-the-century social workers to exploit the telephone as a tool for practice. Towards the end of this century and with the inaugural issue of a newsletter

entitled Computer Use in Social Services Network (CUSSN), Dick Schoech (1981) took Richmond's place as the nurturer of technology. Schoech's newsletter evolved into a journal entitled Computers in Human Services which continues to lay the ground work for integrating and advancing computer technology in social work. For example, it served as a platform for Hanna (1995) to report on early efforts to combine and link many existing electronic networks through the Internet. Prior to this effort, electronic and political barriers prohibited members of one network from sending e-mail or conference with members from other human services networks.

INTERNET AND KEY SOCIAL WORK ORGANIZATIONAL EVENTS

Four happenings in recent history have had an influence on social work and the Internet. In the late 80s, CSWE board members and NASW staffs became active on the Internet. A CSWE staff member coordinated a meeting of two Internet enthusiasts. Although they communicated often over the Internet, they first met face to face at the CSWE 1990 Annual Programme Meeting.

Here, they gave birth to the Baccalaureate Programme Directors Association's Committee on Information Technology and Social Work Education. In 1992, the unofficial committee offered its first workshop at the BPD annual conference. At the 1995 APM, the committee became an important part of BPD's table of organization as a standing committee. The most stirring moment in the short history of technology and social work professional organizations was at the 1992 APM. Marshall Smith requested the floor in an open meeting of CSWE's Accreditation Board.

In a very eloquent, short, but powerfully moving speech, Marshall expressed the importance of the Internet for the accreditation process. He made a plea that board members consider accepting self-studies via the Internet. Marshall was ahead of his time. Now we are wagering on which school will take the bold step to be the first to submit a self-study via electronic means. One positive intermediate step was taken to achieve this goal. Through the advocacy of BPD's Committee on Information Technology and Social Work Education, CSWE's Directory of Colleges and Universities with Accredited Social Work Degree Programmes included e-mail addresses. Soon web page locations will be published in the Directory. These actions lay the ground work for the electronic transmission of self-studies and a more efficient means of educational research.

Technical and confidentiality problems related to document transfer are on the verge of being resolved. Third, in early 1995, Rita Vandivort, of the NASW national office, established a listserv for technologically capable social workers to plan an Internet symposium at the upcoming NASW conference. As a staff member of NASW, she was able to acquire funds to put together an impressive Internet demonstration facility at the Philadelphia conference. Members of the technology committee involved in the planning were overwhelmed and profoundly impressed with the Internet resources that

Rita was able to pull together. It was a success beyond anyone's imagination. Since the Philadelphia conference, the use of an Internet lab continues to be an important feature of the NASW annual conference.

Perhaps more than any single event in the history of social work and the Internet, the lab at the NASW conference has had the greatest influence on the utilization of the Internet among seasoned social work practitioners. Fourth, in the fall of 1992, Ogden Rogers proposed that SOCWORK subscribers who were also CSWE members meet face-to-face. Ogden dubbed the group "OffLine." Subscribers/CSWE members had been communicating with each other on a wide variety of levels. Close friendships were established, accreditation consultation support was given, and in some cases enemies were made! Steve Marson organized this group's first "get-together" during the 1993 APM in New York. The second get-together in 1994 in Atlanta was also a great success. During a SOCWORK discussion, subscribers decided to formalize OffLine in order to avoid scheduling conflicts. Since the formalization of Offline, the "social" get-togethers have been well attended and led to more serious academic conferences.

TECHNOLOGY—AN EQUALIZER

Technology can be a great equalizer for individuals with disabilities that might prevent full participation in school, work, and the community. This is most evident in the case of individuals with mobility, hearing, or vision impairments, but is also true for individuals with limitations in cognition and perception. With technology, an individual physically unable to speak can communicate with spoken language. Using a portable voice synthesizer, a student can ask and respond to questions in the "regular" classroom, overcoming a physical obstacle that may have forced placement in a special segregated classroom or required a full-time instructional aide or interpreter to provide "a voice."

Improvements in sensor controls enable subtle motor movements to control mobility devices, such as electric wheelchairs, providing independent movement through the school and community. Text and graphics enhancement software can enlarge sections of a monitor enough to be seen by persons with vision impairments. Text can be read electronically by a digitized voice synthesizer for a person who is blind. For persons with hearing impairments, amplification devices can filter extraneous noise from the background or pick up an FM signal from a microphone on a teacher's lapel.

Word processing, editing, spellchecking, and grammatical tools commonly found in high-end software facilitate the inclusion of students with learning disabilities in regular classrooms by allowing them to keep up with much of the work. Not inconsequentially, the children often feel better about themselves as active learners.

Technology is providing more powerful and efficient tools to teachers who work with children with disabilities. These tools enable teachers to offer new and more effective means of learning while individualizing instruction to the broad range of student learning needs. Educators are using computers as tools to deliver and facilitate learning beyond drill and practice, to provide environments that accommodate learning, and to

ensure enhanced and equitable learning environments to all students. Access to the World Wide Web, email, listservs, and other electronic learning environments is common in many classrooms. In these environments, students around the world can interact in real time via onscreen messaging or video and audio transmissions. In most of these learning situations, a disability makes no difference at all.

Assistive Technology Defined

The definition of assistive technology applied to education is extremely broad, encompassing “any item, piece of equipment, or product system whether acquired commercially off the shelf, modified, or customized, that is used to increase, maintain, or improve functional capabilities of individuals with disabilities.”

As a result, the potential range of AT devices is incredibly large, and both “high-tech” and “low-tech” devices are included. High-tech devices may be computers, electronic equipment, or software. Although electronically operated, high-tech devices need not be expensive, a simple low-cost switch that controls a battery-operated toy can be considered a high-tech device, as can a tape recorder. Low-tech devices are manually, not electronically, operated. This group includes devices such as pencil grips, mouth sticks, and mechanical hoists.

This definition also expands the consideration of potential educational applications with its focus on devices “used to increase, maintain, or improve the functional capabilities of persons with disabilities.” As educators, we try to increase or add new academic, social, and daily living skills and knowledge to the functional capability of all children. This is a basic goal as we prepare children to take their place in society.

In the case of children with degenerative impairments, such as muscular dystrophy, educators may be working to keep children functioning at their current level. They may be striving to help students maintain their capability to function in the world. Teachers work with students to improve skills and knowledge, making existing skills and knowledge even more functional and improving fluency so that functional capabilities may be generalized into different settings.

It is critical to understand the implications of this definition to comprehend its effect on children with disabilities in our schools. It is fairly easy to understand how the definition is applied with regard to children with physical or sensory disabilities. To see a young child who had been unable to speak for her first five years say her first sentence with a speaking computer device presents an exciting and clear picture of assistive technology. The benefit of AT is also easy to comprehend when a child who cannot hear can understand his teacher’s directions because real-time captioning converts the teacher’s speech to text projected onto his laptop computer.

The definition of assistive technology also applies to the more difficult-to-gauge tools that teachers use to deliver and facilitate learning, including instructional applications of technology. These applications range from drill and practice tutorials to facilitated learner-based environments provided through the Internet or interactive

hypermedia and multimedia-based instruction. It is important to understand that virtually all applications of technology — tools for children to learn, as well as tools for teachers to provide learning opportunities — can be defined as assistive technology. This is true for individual children with disabilities whose disability has a primary impact on academic performance (*e.g.*, learning disabilities) or functional performance (*e.g.*, multiple physical and visual disabilities).

Legal and Moral Requirements

The mandate to provide assistive technology to children with special needs is grounded in the moral concerns protected by the U.S. Constitution and its amendments. *The Education for All Handicapped Children Act* (P.L. 94-142) was based on the Supreme Court's 1954 *Brown vs. Board of Education* decision that separate education was not equal education under the 14th Amendment to the Constitution. At the time the law was passed by Congress in 1975, nearly 2 million children were excluded from schools in the United States. With the legislation, the president and the Congress established a legal requirement for a "free appropriate public education in the least restrictive environment" for children with disabilities and, as a result, the field of special education began to flourish for the first time in nearly seventy-five years.

Many controversies surfaced, however, about the extent of the required educational services and the cost to society for those services. The major debates have focused on the need for a clear definition of an "appropriate" education in the least restrictive environment and the requirement to provide assistive technology devices and services to all individuals with disabilities.

'Appropriate' Education

The requirement for an "appropriate" education in the least restrictive environment has led to the development of a separate educational system designed to meet the needs of children with disabilities. Some educators contend that this is the same type of separate system that the Supreme Court found unconstitutional in 1954. These individuals suggest that all children, regardless of ability, should be educated with their neighbourhood peers in their local school.

Others in favour of the special education system argue that it is necessary to meet the educational needs of all children with disabilities, particularly in the "continuum of services" mandated by the Individuals with Disabilities Act (IDEA). In their view, children must have specific intervention designed to "mainstream" them back into regular education. Without the intervention, these individuals believe that students will be doomed to continued and more significant failure. They also note that, while the goal of mainstreaming is reasonable, some children may not benefit appropriately from a full inclusion programme.

Although there are many arguments on both sides of the issue, it is apparent that new technologies can provide the tools to bring more children with disabilities into "regular"

educational settings. In my opinion, assistive technology will certainly mainstream more and more children in wheelchairs, children who cannot physically speak, see, or hear, and children who need computers to write, organize, think, and function educationally.

The AT Requirement

The second debate centers on the requirement to provide assistive technology to all students. The initial legislation, the Education for All Handicapped Children Act, did not require schools to provide assistive technology devices and services to individuals with disabilities. The current assistive technology mandate was created by later legislation and prompted by the technological revolution resulting from the development of the microcomputer.

Subsequent legislation passed by Congress encouraged states to develop services designed to provide assistive technology to all persons with disabilities and required provision of AT as a special education service (trained special education teachers in special classes), related service (occupational, physical, speech therapies, and other services needed to access education) or supplemental service (services necessary to maintain a child in regular education classes).

Many states have not addressed the AT issue, since assistive technology devices and services were identified as requirements only recently. This may be due to a fear of “breaking” instructional budgets by purchasing high-cost equipment in already cash-short school systems. Concern also exists that the rapid evolution of technology creates the potential of costly investment in devices that may have a relatively short life span. A close look at the situation will show that these concerns are not well grounded, however. Schools already use extensive amounts of AT, and need only to identify it as such. Nearly any use of computers falls into this category, as do tape recorded instructions or homework, copies of notes from a classmate or teacher, switch-operated toys, drawing paper taped to table tops, as well as large pencils and crayons. All of these could be noted, as required, in Individual Education Plans (IEPs) and Individual Family Service Plans (IFSPs).

Assistive Computer Technology for Persons with Disabilities

The personal computer (PC) can be the backbone of independence for millions of individuals with sensory, physical, and learning disabilities. Computers and information technology can be modified with alternative input, alternative output, and other assistive technology to empower consumers who have disabilities. Computer vendors support persons with disabilities by incorporating accessibility utilities into operating systems such as Microsoft Windows, Apple Macintosh, and UNIX. PCs equipped with assistive technology permit individuals to function independently at school, work, and home, and allow access to great quantities of information from diverse sources such as compact disks, networks, electronic mail, instant messaging, the World Wide Web, and other

Internet resources. For the purposes of this discussion, the term “assistive technology” will be used to describe any hardware device or software programme that permits individuals with disabilities to operate PCs and access information technology independently. Moreover, this article uses the definition of disability as outlined in the Americans with Disabilities Act in the United States, and the Disabilities Discrimination Act in the United Kingdom—any permanent condition that impairs a major life function such as seeing, hearing, walking, or speaking.

USE OF AUDIO-VISUAL AIDS IN EDUCATION

By audio-visual aids, we usually mean the most modern or the most recently used of these methods. This is a summary identification of very old methods and very modern instruments, and one should react against it. Visual aids are far older. They correspond to a profound tendency among the immense majority of men: to materialize their thoughts in the form of graphic or sonorous images or to give their thoughts a concrete frame of reference. Plato himself took care to set the scenery of his dialogues, and he used concrete words and concrete comparisons as foundations for his most abstract ideas.

In France, the *Très riches heures du duc de Berry* bring out the importance which ‘illustration’ can take in a work which would have otherwise sunk into oblivion. Xylographic images preceded the printing press by three-quarters of a century and the first illustrated book by nearly a century. The tremendous success of the ‘images of Epinal’ in books peddled from door to door in France was only a manifestation of popular taste in a society where illiterates continued to be in a majority and where images went with oral literature.

Films, radio and television, considered as educational instruments, have merely developed – at a rapid rate – alongside older means whose importance remains considerable. Their common denominator lies in their function as aids. This is not a theoretical conclusion, for it is confirmed by the very attitude of the educator. The educator basically must contribute to the training of the individual with a view to his integration into a given society and teach new ideas, facts and techniques to a specific public. It is thus relatively easy to define the goals at which the educator aims. Achieving these goals is another task which brings him face to face every day with the basic problem of pedagogy – that of transmitting or communicating ideas or information. To solve this problem, the educator resorts to infinitely varied means, among them audiovisual aids. If our purpose, therefore, is to aid the educator, we must then offer him as complete an arsenal as possible of these means. But it is the educator and the educator alone who chooses the means which is best adapted to his subject, his audience and his circumstances. It is thus clear that audio-visual aids cannot be separated from educational materials in general.

This tendency towards the use of concrete examples has developed through a complex process. At first, graphic representation was probably only a way to enable man to capture fleeting thoughts and the sole way of transmitting thoughts, compared to oral

transmission which was subject to rapid distortion. The invention of writing, a perfect example of a visual aid at its origin, proceeded from the same necessity. It would be interesting to study, for example, in the light of Mayan writing – of the Codex Troano – how man progressed from the talking image to the letter. We can therefore conclude that ‘illustrations’ were looked upon at first, at least by the most educated persons, as a minor complement to thought.

The entire history of publishing until the end of the eighteenth century confirms this. But, in the twentieth century, powerful means of reproduction, associated with radio, cinema and television, have changed the aspect of the problem. Sound and visual ‘illustrations’ are no longer mere minor complements to thought but they directly influence the thoughts and the very conduct of millions of individuals. It was therefore inevitable that a desire should spring up to master such a powerful instrument, to discipline it for better or for worse. But this coveted mastery is still rather crude: it is often reduced to the creating of a few conditioned reflexes, satisfying the merchant but not the educator. Certain of these audio-visual aids are both means of education and media for information and propaganda, and it is not always easy to draw the line between what belongs to the educator and what is within the province of information or propaganda.

It is also probable that the child is more affected by the violence of street posters and by the shock techniques of radio and television at home than by the visual aids used in school. Should we conclude then that these means are harmful and should we condemn them? This negative attitude would be most unrealistic. The only possible conclusion is to accept the need for basic research in these fields. It should bear essentially upon psychology and upon the social sciences. We educators have already ventured forth, but timidly, onto this terrain. Systematic establishment of contacts with research workers and specialized institutes is the duty of all those who are responsible at the national level for audio-visual services.

It can be reasonably hoped that this basic research will lead to a better use of audio-visual aids and to more scientific pedagogy based upon them. It is not difficult to observe that their use is continuing and developing outside the pale of any research. As a result, a pragmatic pedagogy is taking shape and not necessarily in contradiction to the results of the most scholarly research. Establishing or stimulating closer collaboration between research workers and educators, stimulating the writing of theses or documents containing the fruits of the work of both, and publishing and distributing the results of this work should also be the common task of pedagogical and audio-visual services.

Until now, the problem of the use of audio-visual aids has been examined from an intellectual angle. It also includes important practical and technical aspects. To tell the truth, techniques cannot and should not be separated from pedagogy. We have seen that audio-visual aids cannot be separated from educational materials as a whole, this conclusion being thrust upon us by the attitude of the user when confronted by these materials. Now this same user – whether a teacher, a professor or an adult educator – does not act any differently when pedagogy and techniques are involved. He can never

be purely a pedagogue or purely a technician. It is clear, therefore, that the pedagogy of audio-visual aids cannot be separated arbitrarily from audio-visual techniques. No one can hope to achieve good results unless he is a sensitive pedagogue and a skilled technician. The problem must be solved globally.

Unfortunately, this initiation into techniques is not always carried out in the institutions where future educators are trained. In underdeveloped countries, the lack of qualified personnel is the most frequent obstacle to such an initiation. But it is not the only one because similar shortcomings are often found in more favoured countries. Routine, lack of initiative and administrative delays are the main factors responsible for educational sluggishness. There is no doubt that audio-visual aids produce their best results when they are used in connexion with active teaching methods. Here, the task of educators is to draw the attention of their governments to these methods and to the recommendations of previous seminars concerned with the introduction of an initiation into film and radio techniques into normal schools and similar institutions.

Finally, there are other questions which should be taken up in thorough and specific studies. They are related to the use of radio and television in the teaching of reading and writing and of languages. A great deal has been said about the 'singular, specific and irreplaceable services' which can be rendered, for example, by 'teaching by radio'. A great many hopes were stirred as a result of statements repeated with such warm conviction that one could have believed them to be dictated by experience. First of all, we should note the ambiguity of the term 'teaching'. In the context of the statements to which we refer, this term covers both the teaching of subjects such as science and history as well as the teaching of reading and writing. One of the greatest problems which remains to be solved is the liquidation of illiteracy. Following hasty conclusions, a belief has grown that, thanks to radio, illiteracy can be liquidated quickly, easily and cheaply. But what do we know about it objectively? Until now, the number of experiments has been limited. Some of them were frankly admitted failures. Fortunately, we will soon be in possession of an exhaustive report of the results obtained during the best known of these experiments, the one at Colombia. One of our experts has made a global study of these results from which we think that we will be able to derive valuable lessons, if not definitive conclusions.

A few complimentary remarks should be made here. The first concerns the basic difference between teaching notions of history, geography, science, *etc.*, by radio and the teaching of reading and writing. We say teaching by radio because the problem of teaching reading and writing by television is infinitely simpler and the results already acquired are sufficiently convincing. At any rate, it can never be said too often that the global use of audio-visual aids always gives the best results.

Opposing radio to television or both to films is a typical example of a false problem. In an educational campaign organized and carried out at a national level, all complementary means must be used if possible. It is also virtually certain that results are proportionate to the means employed in a geometrical, not an arithmetical,

progression. In other words, overly strict economy does not pay. There lies a source of misunderstanding. Too often, it has been believed that making an expert and \$10,000 worth of equipment available to a government was enough to solve the problem of audio-visual aids in a given Member State. Audio-visual aids do not have this magic quality. They require serious study and, first of all, serious thought about the place which they should occupy in a budget.

Pedagogical problems always end up by leading into budgetary problems – that is, in the long run, economic, social and political problems. The educator must play his pedagogical role. The tool which is offered to him – and this is the case of audio-visual tools – can multiply the activity of the educator in large proportion. Therefore, the political authorities must be convinced of the necessity of a financial effort which is often considerably large. That is a point of view which often escapes the educator: he must also educate administrators and political authorities. Many mistakes begin here. So far, we have talked about audio-visual ‘aids’ and ‘means’. Educators obviously consider them from this angle in the best of cases. But will this tremendous development of mass information media which we continue to allow itself to be domesticated?

Let us go to the heart of the matter: must we continue to consider these information media as blind forces whose unleashing – and, as far as the educator is concerned, the unleashing begins where his own control ends – would be an educational and cultural catastrophe? Or, to put things in a less impressive but equally embarrassing way, cannot audio-visual ‘means’ be allowed to play their role without the help or simply the intervention of the educator? Before issuing a condemnation without any possibility of appeal, perhaps it might be wise to remember that films, radio and television can only be arbitrarily separated from the social, economic and cultural context which gives them their means of existence.

No doubt, the study of these problems takes us a long way from modest film strips, flannelgraphs, and even traditional black-boards which are still a luxury for thousands of schools. Let there be no misunderstanding. We know the importance of pedagogy in the use of audio-visual aids. We know that the training of good educators – in this case, good users of these didactic means – is a long and difficult matter. We know that we must think about the desperate problem of training teachers. But pedagogy itself is only a means whose end is education. And education, in the long run, is only a contribution – naturally, of capital importance – to the integration of the individual into a given society. It is in this perspective – from their production to their final use – that we must look at audio-visual aids and the various questions which they raise.

TOOLS OF INFORMATION TECHNOLOGY

SYSTEM IMPLEMENTATION STRUCTURE

To implement the WBE we propose the system implementation structure. The school classrooms, office and training centers are connected through Intranet. This Intranet is

connected to the Internet by using network operating system. Firewall is introduced between Intranets and Internet in order to provide security.

SYSTEM DESCRIPTION

In initial stage of school education the students are not expert in reading and writing. Subject understanding increases if they learn the things through visualization. With the help of multimedia or Rich Media, which includes, Audio, video, graphics and Java Applets have made WBE very effective. We propose Multimedia content database scheme as shown in fig. In the primary stage students don't have good knowledge of English. Therefore the presentation should be available in their mother tongue for the better understanding, which is also helpful in improving their pronunciation. This is possible by developing Natural Language Interface to database. One important device called, as "Tech Commander" is also useful for teachers to identify students potential by viewing any students computer display on his own monitor. If he finds something that everyone should see he could set everyone's monitor to display it.

Web Based Education: Considerations and Approaches

- *Conversion of Existing Material:* In order to shift from traditional education to WBE we have to convert the existing school educational material to the Web. The important points to consider are bandwidth, design, usability, and the necessity of high quality media elements and consistency of material across the mediums.
- *Authoring for Multiple Delivery Environments:* We have to provide consistency of interface and ease of authoring and design of an effective multi platform course.
- *Using the Web for student/Teacher Interaction:* Web site can be used for posting of assignments, student work and marks, along with the ability to submit work on-line through the site, also JavaScript and JAVA applets to demonstrate course concepts interactively. This means that course delivery on the Web must be dynamic and truly interactive between the instructor and the students.
- *Faculty Support and Training:* We have to provide centralized support and training resources for training the teachers initially.

Problems to be faced while implementing WBE in school education in Indian context:

Looking at how to use Information Technology in school education, its different tools, the system structure as defined and described it is obvious that we will face some problems while implementing WBE in school education in India.

The major problems we will be facing are:

- *Intensive Training to Schoolteachers:* Schoolteachers are not introduced to the web based education. Therefore training should be given in order to create a learning environment that will itself train and spur students on the one hand to turn the learning experience into useful, practical and personal knowledge.

- **WBL awareness and Workshops:** In rural area parents are not much knowing about WBL. So the demonstration, seminars and workshops needs be conducted for society in order to understand the importance of it.
- **Bandwidth Limitations:** Limited bandwidth of Internet connection gives slower performance for sound, video and intensive graphics, causing long waits for downloads that can affect the ease of the learning process. Improved bandwidth will help the teacher to solve his problem.
- **Effect on Teachers:** WBL will lead to reduction in manpower as per as teachers are concerned. This will lead to agitations by teacher's organization.
- **Effect on Students:** Although the students will be benefited by WBL there will some part of students opposing this introduction of technology in education.
- **Infrastructure:** WBE will primarily require free access to Web to all the learners and hence government of India will have to setup nation wide Fibre Optic Cable network.
- **Access:** Every school will not have equal opportunity to information because of access issues. The schools with fewer budgets will always face this problem. This is the major problem as per as India is concerned, as there is big gap between poor and rich communities in India.
- **Download:** The learning material that appears on web needs to be downloaded will require more time. The speed depends on the transmission methods and bandwidth, which is problem as per as India is concerned.

Important Features of Web-Based Learning Environment

While designing Web-Based Learning sites the following important features should be kept in mind:

- **The Online Syllabus:** An online syllabus provides the instructor with a way to change course material easily and as per the requirements in industry, and the student will have a complete and up-to-date picture of the course requirements. Hypertext links to sample relevant disciplinary web sites may be helpful in giving students (and also prospective students) a sense of the disciplinary context for the course.
- **Personal Home Pages:** Personal home pages can be used to foster the sense that the class is not just a collection of isolated individuals but a community of learners who can profit from interacting with one another. Home pages encourage students to learn about each other so as to encourage contact and mutual interests. This helps the learners to create a group with common interest.
- **Interactivity:** Adding discussion forums and chat sessions to your online course is a common way to add an interactive component to a web-based course. There are many implementations of bulletin board and chat session software to choose from. A second method of interactivity is, of course, e-mail. It's a good practice to have an online list of the e-mail addresses of all registered students, the

professor, and teaching assistants. This is possible with an e-mail subscription mechanism included in your Online Syllabus.

- *Assignments*: The web page listings of homework assignments, upcoming events and exams can be more interactive than the familiar print counterparts. If some homework assignments, for example, are based on online materials, they can be directly linked to the class schedule. This helps the students to plan the preparations for the examination in systematic way.
- *Announcements*: To be effective, announcements need to be read; for that to happen, students need to know when a new announcement has been posted. Alert sounds or perhaps a blinking link added to a page can let students know of new announcements, or perhaps, even a mass e-mail to all students in the course. For a home page, or a long life syllabus, various software tools can be used for the subscribers announcement about page changes. All these techniques will attract the learner's attention towards announcements.
- *Testing*: Online drill or practice testing can be used to reinforce material, even if the results are not used as part of a grade. Reading comprehension questions, for example, in short answer or multiple choice formats can provide students with an assessment of their level of understanding of text. This facilitates the students to measure his level of understanding and through continuous assessment he can try to improve his performance.
- *Course Management*: Software should be available to add or delete students from the course, assign user Ids and passwords, create or edit home pages, and manage any open discussion groups. This helps to keep up to date records of students admitted for various courses.
- *Content*: Perhaps the most difficult part of developing a web-based course is creating the online contents. You can begin by transferring your basic lecture materials to the web and integrating media such as sound, images, and video. Remember, to experiment with incorporating some of the new web-based learning paradigms. And finally, come back and rebuild the lecture building its graph structure and using more html facilities.

Other Features of a Web-Based Learning Environment

- Managing cognitive load — the amount of information people can process — is essential to effective teaching or training. Bombarding learners with too much information at once, called cognitive overload, is one of the chief obstacles to learning. This indicates that we should provide only required information in order to avoid cognitive overload.
- Dividing each tutorial lesson into segments (Classroom, Quiz, Lab, *etc.*) and then further subdividing these segments into a manageable number of chunks allows users to digest new concepts and skills in a manner that prevents overload.
- *Web-based tutorial*: Users will also enjoy a great deal of flexibility in managing

their cognitive load, selecting instructional tasks from a menu of lessons, depending upon the amount and kinds of skills they bring with them, and once engaged in a lesson, selecting which portions of that particular lesson they wish to complete. This allows the students to learn the topics in proper sequence and according to his ability of understanding.

- Because the limited capacity of working memory is rapidly overwhelmed by large amounts of new information, frequent opportunities to practice are important. Rehearsal encodes or moves information into long-term memory. The practice assignments can be presented with practice opportunities throughout the classroom portion of the lesson and is also encouraged to complete the practice portion of each lesson. This allows the student to find out how much he has understood at the end of learning a particular module.
- Finally, online testing is used to reinforce material. Elaborative rehearsal involves presenting questions, which allow the user to apply knowledge in an appropriate context, thus encoding it into permanent memory.
- Quiz questions are designed to provide an authentic assessment of user skill levels by calling on the user to apply the appropriate techniques and practices from the lesson.

RECOMMENDATIONS

The profound influence of the Internet and technologies that the Internet will spawn in the future is undisputed. Social work and the entire social welfare system will not escape utilizing the Internet anymore than social workers were able to escape the use of the telephone during Richmond's era. As a result, we have an ethical obligation to systematically encourage others to employ the Internet in their daily activities. Competence and comfort with the Internet leads to and enhances one's creativity in using the Internet, which in turn leads to improved human services. Such statements are grandiose and easy to state. How does one operationalize this? Reading and recalling recent history is a good place to start.

Our short history and experience offers some simple guidelines:

- Like Richmond, professors and field instructors must shepherd their students (particularly non-traditional students) into feeling comfortable with the Internet.
- Leaders both in education and practice must share their ideas by writing, presenting documents at conferences, and offering workshops about technology. Tom Hannah, Dick Schoech and Rita Vandivort are good contemporary role models for the rest of us.
- Leaders of professional organizations must be nurtured into accepting that computer technology and the Internet have vast implications for their organization's benefit. This can only be accomplished by a one-by-one strategy and face-to-face interaction. In addition, technologically competent professionals must organize themselves to vote for leaders who recognize the importance of

the Internet. Actions like Marshall Smith's speech is a good illustration. Members of CFTSWE make efforts to support candidates that have a history of supporting technology.

- Leaders of professional organizations should be shown that technology committees made up of volunteers are a cost effective manner of spreading the advantages that the Internet and computers have for human services. The development and actions of CFTSWE provide an excellent example.
- Workshops must be freely offered in a nonintimidating manner to those professionals and educators who are technophobic. CFTSWE's annual workshops are good illustrations of an effective strategy.
- Technophobic professionals must be approached in the same manner as Richmond approached social workers at the turn of the century. They must be nurtured, reassured, and shown that such technology is not dehumanizing but improves the quality of care. Vandivort's coordination of NASW's Internet lab at annual conferences represents an excellent example.
- Technologically competent professionals should meet socially (*i.e.*, OffLine) and professionally (*i.e.*, conferences). These meetings must be used to share ideas on effective methods of spreading the good news of technology and the Internet.
- Technologically capable human service professionals need to lead the way to make Internet technology user-friendly and practical. As stated earlier, CFTSWE is developing a structure for the confidential transmission of CSWE self studies and research via the Internet.

CONCLUSIONS

The key to advancing the Internet in social work is the theme found in the history of the development of the Internet. As a social worker, two aspects of Internet history are striking. The concept of "unintended consequences" is first apparent. The present use of the Internet is nothing like the original notion expected by its designers. A vast array of commercial and non-commercial applications for social work practice now exist. We are just witnessing the tip of the iceberg.

Second, and closely related to the concept of unintended consequences, is social work values. The values to which most social workers cling are contrary to the values behind the original design of the Internet. Recall that the original intent of the Internet was to deliver weapons of mass destruction. Now, we see social workers employing it as a system to deliver social services. The success of the Internet must be attributed to the cooperative nature of it's users. I find that the massive level of cooperation found in the Internet is unprecedented in the twentieth century. In the end, social work success in employing the Internet will be limited only by the cooperative nature of those using it in practice and research.

Education as an Instrument of Social Change

The role of education as an agent or instrument of social change and social development is widely recognized today. Social change may take place – when humans need change. When the existing social system or network of social institutions fails to meet the existing human needs and when new materials suggest better ways of meeting human needs. According to Maclver social change takes place as a response to many types of changes that take place in the social and nonsocial environment. Education can initiate social changes by bringing about a change in outlook and attitude of man. It can bring about a change in the pattern of social relationships and thereby it may cause social changes.

Earlier educational institutions and teachers used to show a specific way of life to the students and education was more a means of social control than an instrument of social change. Modern educational institutions do not place much emphasis upon transmitting a way of life to the students. The traditional education was meant for an unchanging static society not marked by any change. But today education aims at imparting knowledge. Education was associated with religion. It has become secular today. It is an independent institution now. Education has been chiefly instrumental in preparing the way for the development of science and technology. Education has brought about phenomenal changes in every aspect of men's life. Francis J. Brown remarks that education is a process which brings about changes in the behaviour of society. It is a process which enables every individual to effectively participate in the activities of

society and to make positive contribution to the progress of society. Education is the most powerful instrument of social change. It is through education that the society can bring desirable changes and modernize itself. Education can transform society by providing opportunities and experiences through which the individual can cultivate himself for adjustment with the emerging needs and philosophy of the changing society. A sound social progress needs careful planning in every aspect of life— social, cultural, economic and political. Education must be planned in a manner which is in keeping with the needs and aspirations of the people as a whole.

Indian Education Commission (1964-66) observed that realization of country's aspirations involves changes in the knowledge, skills, interests and values of the people as a whole. This is basic to every programme of social and economic betterment of which India stands in need.

The functions of education in the sphere of social change are outlined as under:

- *Assistance in changing attitudes:* Education helps to change the attitudes of people in favour of modern ways of life and develops attitudes which can fight prejudices, superstitions and traditional beliefs. It can bring about a change in attitudes of people in favour of small family towards rising above orthodox values and socio-cultural barriers of caste and class and towards religion and secularism. Education interacts with the process of social development which is another name of social change.
- *Assistance in creating desire for change:* Education creates a desire for change in a society which is pre-requisite for any kind of change to come. It makes people aware of social evils like drinking, dowry, gambling, begging, bonded labour *etc.* and creates an urge to fight and change such things. Education makes under-privileged, down-trodden and backward people aware of their lot and instills a desire to improve their conditions. Thus education creates a desire for change.
- *Assistance in adopting social change:* Whenever some social change occurs, it is easily adopted by some people while others find it very difficult to adjust themselves to this change. It is the function of education to assist people in adopting good changes.
- *Overcoming resistance to change:* Certain factors create resistance in the way of accepting social change. Education helps in overcoming resistance.
- *Analysis in change:* Education invests the individual with the capacity to use his intelligence, to distinguish between right and wrong and to establish certain ideals. Education determines the values which act as a criterion for the analysis of social change.
- *Emergence of new changes:* Education initiates, guides and controls movements for social reform. Education helps in agitating public opinion towards the abolition of many social changes.
- *Leadership in social change:* Raja Ram Mohan Roy, Swami Dayanand

Saraswati, Swami Vivekananda, Mahatma Ghandhi, highly educated and enlightened Indians, made all efforts at conscious level to bring about the social changes. Mahatma Gandhi designed Basic Education to fight the ills of mass illiteracy.

- *Advances in the sphere of knowledge:* New researches and inventions depends upon education, because only the educated individuals can search for new things in every field. Thus education contributes to social change by bringing changes in knowledge.
- *Perpetualizing eternal values:* Education protects the eternal values, promotes knowledge and acceptance in such a manner that in spite of social changes, people in general keep faith in these values.
- *Transmission of culture:* Education is the creator, generator and director of all social change. In short, education is a preservative, consolidating, establishing and creative force.
- *Stabilizing democratic values:* Education is a weapon of democracy. It can help in developing democratic attitudes and values for better living. Democratic values such as liberty, equality, fraternity, justice, tolerance, mutual respect, feeling of brotherhood and faith in peaceful methods are stabilized through education in free India. These values are helpful in bringing about social change.
- *National integration:* Education can prove very useful in bringing about national integration which is the basis for unity among people.
- *Economic prosperity:* Education is the most important factor in achieving rapid economic development and technological progress and in creating a social order founded on the value of freedom, social justice and equal opportunity.
- *National development:* Education is the fundamental basis of national development. Education is the powerful instrument of economic, political, cultural, scientific and social change.

EQUALITY OF EDUCATIONAL OPPORTUNITY

Anthropology is formed with the combination of Greek words -anthropo meaning human and logy meaning science. The meaning and purpose of anthropology is the scientific study of humanity. The inherent curiosity of man about himself was the prime reason influencing the emergence of an anthropology which systematic studied mankind. Anthropology studies who and how man is evolved over the period of time, why he looks like the way and the way he talks and acts in particular manner. It is broad discipline dedicated to the comparative study of mankind, from its first appearance to its present stage of development. It is concerned with all the varieties of human population in every part of the world, both past and present.

Anthropology looks at man both in time and space. Time means the stages of development of man in the process of evolution during different periods of time and space refers to the differentiation of physical and cultural types in modern man living in

different environs all over the world. Anthropology studies the primitive, pre-literate, simple, small societies which are not studied by other disciplines. In anthropology fieldwork or the first hand study of people through personal observations, interview and interaction is very significant. It depends on the direct evidence. Anthropology is the study of man by man. It is a study which tries to examine in objective way the evidences. People are given importance in anthropological studies as it is concerned with the nature and behaviour of each and every group of individuals. According to Roger Keesing and Felix Keesing anthropology is a study of universalities and uniqueness; a study of startling contrast and surprising similarity; a study of meaning and logic in what seems bizarre. Anthropology is thus historically and geographically vast in its scope.

EDUCATION AND SOCIAL CHANGE

The role of education as an agent or instrument of social change and social development is widely recognised today. Social change may take place - when humans need change. When the existing social system or network of social institutions fails to meet the existing human needs and when new materials suggest better ways of meeting human needs. According to MacIver social change takes place as a response to many types of changes that take place in the social and non-social environment. Education can initiate social changes by bringing about a change in outlook and attitude of man. It can bring about a change in the pattern of social relationships and thereby it may cause social changes.

EDUCATION AND MODERNISATION

Modernisation is a process of socio-cultural transformation. It is a thorough going process of change involving values, norms, institutions and structures. Political dimensions of modernisation involves creation of a modern nation state and the development of key institutions -political parties, bureaucratic structures, legislative bodies and a system of elections based on universal franchise and secret ballot. Cultural modernisation involves adherence to nationalistic ideology, belief in equality, freedom and humanism, a rational and scientific outlook. Economic modernisation involves industrialisation accompanied with monetisation of economy, increasing division of labour, use of management techniques and improved technology and the expansion of service sector. Social modernisation involves universalistic values, achievement motivation, increasing mobility both social and geographic increasing literacy and urbanisation and the decline of traditional authority.

The secular and scientific education act as an important means of modernisation. It helps in the diffusion of modern values of equality, freedom and humanism. The modern school system can inculcate achievement motivation. These values can form the basis of new relations in the society and growth of rationality can enable the development of

administrative system. Diffusion of values of equality, freedom and humanism can lay the foundations of a democratic political system. The spread of modern education in the second half of the 19th century led to the emergence of modern political elite in India who provided leadership in the freedom struggle. The diffusion of scientific and technical knowledge by modern educational institutions can help in the creation of skilled manpower to play the occupational roles demanded by the industrial economy. Other values like individualism and universalistic ethics *etc.* can also be inculcated through education. Thus education can be an important means of modernisation. The importance of education can be realised from the fact that all modernising societies tend to emphasize on universalisation of education and the modernised societies have already attained it.

EDUCATION AND CULTURE

Education encompasses teaching and learning specific skills and also something less tangible but more profound: the imparting of knowledge good judgement and wisdom. Durkheim sees education as the socialisation of the younger generation. It is a continuous effort to impose on the child ways of seeing, feeling and acting which he could not have arrived at spontaneously.

Education has as one of its fundamental goals the imparting of culture from generation to generation. Culture is a growing whole. There can be no break in the continuity of culture. The cultural elements are passed on through the agents like family, school and other associations. All societies maintain themselves through their culture. Culture here refers to a set of beliefs, skills, art, literature, philosophy, religion, music *etc.* which must be learned. This social heritage must be transmitted through social organisations.

Education has this function of cultural transmission in all societies. The curriculum of a school, its extra-curricular activities and the informal relationships among students and teachers communicate social skills and values. Through various activities school imparts values such as co-operation, teamspirit, obedience, discipline *etc.* Education acts an integrative force in the society by communicating values that unites different sections of society. The school teach skills to the children which help them later to integrate within the culture of the society. Education in its formal or informal pattern has been performing this role since time immemorial. Education can be looked upon as process from this point of view also. Education has brought phenomenal changes in every aspect of man's life.

EVOLUTION AND DEVELOPMENT OF SOCIAL CHANGE

SOCIAL CHANGE, SOCIAL EVOLUTION AND SOCIAL PROGRESS

During its early phase, sociological inquiry was mainly directed towards an interpretation of the rapid and violent changes that had overtaken European societies in

the eighteenth and nineteenth centuries. The approach was connected with the philosophy of history. Early sociologists were all concerned to explain or interpret the Social and political revolutions of their age within the framework of a general theory of history.

The writings of Saint-Simon, Comte, Marx and Spencer bear witness to this. Though Max Weber did not present a theory of universal history, he was inspired by a historical concern with the origins and significance of modern Western Capitalism and with the increasing rationalisation of social life over the years. Even Durkheim, who rejected Comte's evolutionary approach, classified societies in terms of an evolutionary scheme.

In these sociological theories, social change, social evolution, social development and social progress have been often treated as synonymous terms. Even when treated differently, these were treated as logically related terms. We have to examine how far this contention is correct. That society changes is a demonstrable certainty. The notion of social evolution, which was adopted in order to describe social change, was taken directly from the theories of biological evolution which, in the nineteenth century, powerfully reinforced the influence of the philosophy of history upon sociology.

"The term 'evolution' is used to describe any form of orderly change. In biology, it stands for a particular type of orderly change, namely, that whereby new specific forms have arisen by a process of differentiation from the old. In other words, biological evolution refers to 'descent with modifications'. This means the diversification of species from a common genus.

In the social field, however, the notion of evolution in the sense of the diversification of species cannot be appropriately applied. The term 'evolution' could be used in sociology in a sense analogous to that given to it in biology, if new elements of culture can be shown to arise from the old by a process of diversification. According to Morris Ginsberg, the notion of evolution in the sense of the diversification of species can be fruitfully applied in, at least, two realms of culture, namely, in the field of language and of tools. Languages have been classified into families or stocks and their similarities and differences have been studied with the aid of a theory which can be appropriately described as 'descent with modifications'.

Comparative philosophy is concerned with establishing connections between linguistic stocks which had formerly been regarded as remote from each other in order to explain their derivation from a common source. In the case of tools, a similar approach has also proved fruitful.

As Tylor pointed out:

"The geographical distributions of these things and their transmission from region to region have to be studied as the naturalist studies the geography of his botanical and Zoological species".

Marx made a similar point when he called for a history of tools which would be to social progress what Darwinism was to organs of animal species, and he thought that such a history of human technology ought to be easier to write than the history of 'natural technology'.

The idea of evolution in the sense of diversification of species may prove to be valuable in certain elements of culture. But in many cases the idea of evolution may not be applicable at all. According to Ginsberg, this is due to the difficulty in arriving at a workable social morphology.

The concept of social evolution which holds that each society passes through the same stages of development has been almost entirely abandoned. That is, the crude application to society of such concepts of biological evolution as natural selection, adaptation, and survival of the fittest has been thoroughly discredited.

While rejecting the concept of social evolution, V. Gordon Childe observes:

“... it is essential not to lose sight of the significant distinctions between historical progress and organic evolution, between human culture and the animal’s bodily equipment, between the social heritage and the biological inheritance. Figurative language, based on the admitted analogy, is liable to mislead the unwary...Man’s equipment and defences are external to his body; he can lay them aside and don them at will. Their use is not inherited, but learned, rather slowly, from the social group to which each individual belongs. Man’s social heritage is not transmitted in the germ-cells from which he springs, but in a tradition which he begins to acquire only after he has emerged from his mother’s womb. Changes in culture and tradition can be initiated, controlled, or delayed by the conscious and deliberate choice of their human authors and executors. An invention is not an accidental mutation of the germ-plasma, but a new synthesis of the accumulated experience to which the inventor is heir by tradition only. It is well to be as clear as possible as to the sort of differences subsisting between the processes here compared.”

In view of the difficulties associated with the use of the term ‘evolution’, some sociologists have used the term ‘development’ to denote the change of society from one stage to the other. Thus, Hobhouse held the view that in the historical development of society from the primitive stage to the advanced one, four characteristic features gradually emerged, namely, expansion, increase in efficiency, mutuality and cooperativeness, and extension of individual freedom.

The question is: Whether such an advance can be characterized as ‘development’? In its application to social phenomena, the term ‘development’ is no more precise than the term ‘evolution’. Development is a process whereby that which exists ‘potentially’ becomes actual.

In this sense, we speak of the development of a child or of a disease. It is not clear if we can speak of social development in the same way by referring a particular social phenomenon to its germ. In the case of an organic body, we can speak not only of its development but also of its decay. Can we speak in the same way of the development and decay of the whole society?

The theories of social development may be expressed in the form that human development consists in the unfolding of man’s powers, individual and collective, and the use of such powers in dealing with nature and with himself.

Bottomore Elaborates this Point thus:

“There are only two (related) social processes to which it seems possible to apply the term ‘development’ with any accuracy, namely, the growth of knowledge and the growth of human control over the natural environment as shown by technological and economic efficiency. It is, indeed, these two processes which have figured most prominently in developmental or evolutionary accounts of human society”.

It is clear, therefore, that the term ‘development’ can be, applied to particular aspects or features of society, and not to the society and a whole. That is, we cannot speak in terms of ‘development’ of the society. There is another difficulty in speaking in terms of ‘development’. Nagel points out that the term has not only a backward reference in the suggestion that something latent or hidden is progressively made manifest, but also a prospective one; “it possesses a strong teleological flavour.”

This is apparent from the fact that the development of a child is related to some known characteristics of the adult human being. In the case of particular social phenomena, we can relate the development of knowledge to a condition of more extensive and exact knowledge, and the development of control over nature to such things as survival and population size.

But the development of society as such can hardly be related to any prospective condition of society except in terms of a moral ideal. In that case development becomes synonymous with progress. We have examined the difficulties in using the terms ‘social evolution’ or ‘social development’ with reference to changes that take place in society. That society changes in course of time is an undoubted fact. But is there any specific direction in which the process is moving? Can we say with any amount of certainty that all these changes had led to social progress? An answer to these questions involves a definition of the concept of progress. Progress is development or evolution in a direction which satisfies rational criteria of value.

All throughout the nineteenth century, the idea persisted that all social change marked successive stages towards social progress. A few quotations would be revealing. Tocqueville, for whom progress consisted in the movement towards equality, urged that it was a providential fact, possessing all the characteristics of a divine decree: “it is universal, it is durable, it eludes all human interference”. From a totally different point of view.

Spencer Concluded that:

“The ultimate development of the ideal man is logically certain — as certain as any conclusion in which we place the most implicit faith”.

The Popularity of the Idea of Progress was due in Great Measure to two Factors:

1. The buoyant hopefulness inspired by the triumphs of applied science,
2. The idea was given powerful impetus by its association with the biological theory of evolution.

With reference to these two developments, Marris Ginsberg observes: “The culminating point in the history of the belief in progress was reached towards the end of

the nineteenth century. Its high priests and incense bearers were.... all rationalists. It owed its wide prevalence to the optimism inspired by the triumphs of applied science, made visible in the striking advances made in the technical conveniences of life, and, on the theoretical side, by the deeply ramified influences exerted by the ideas of development and evolution on all branches of thought and inquiry”.

Though the culminating point, of the idea was reached towards the end of the nineteenth century, no law of progress had, in fact, been scientifically established. But the general idea of progress harmonized with notions of development that had become current during the period in science and philosophy. It also provided an inspiring justification for movements of social and political reform. In spite of the fact that the concept of progress was vague and imprecise, it became “part of the general mental outlook and for many it provided the basis for a working faith of great value”.

In the twentieth century, the mood is more cautious. The concept of unilinear or automatic progress is subjected to searching criticism. The devastation of World War I shattered to pieces the confident optimism of the inevitability of progress.

“In more recent times the belief in progress has been further weakened by the growing recognition that advances in technical knowledge are by no means sufficient to ensure social and moral progress and the fear that the use of scientific knowledge for destructive purposes may outpace and arrest the growth of its powers for good”.

The idea of progress is not popular with social scientists in this century for some other reasons also. In the first place, there is no universal agreement on the standards of value and, as such, we cannot tell from the mere fact that a thing has changed whether it has progressed or not. Evolution or change is an objective condition.

Progress, on the contrary, means change for the better, and hence implies a value judgment. Secondly, since the components of society are interdependent, change at any point is likely to precipitate changes elsewhere. Some of these changes may be unwanted. A social change is never received as an unmixed blessing.

In the Words of MacIver

“Every achievement has its costs, and men often differ as to whether the costs outweigh the values accrued.” Thirdly, values change over time. What is universally hailed as a progressive measure at one period of history may be condemned with equal or even greater force as a retrograde measure at some other period of time. Faced with the difficulties involved in giving clear-cut connotation to the term progress, attempts are sometimes made to settle the matter by proclaiming a set of social values that are sufficiently general to be accepted by everybody.

Some of these Generalized Social Values are:

“Thou shalt not kill, ‘Do not tell a lie’; ‘The greatest good of the greatest number’, etc. It is true that opinions will not clash on this level of abstraction. They will clash, however, when it comes to a specification of these values. For example, there might be a general consensus on the goal of social policy that it should strive to secure the greatest good of the greatest number. But controversies will arise as to what actions are to be

taken to secure these ends. People have different ideas of what is good for them. Similarly, the commandment prohibiting 'killing' is accepted by almost all cultures. But while some people look upon both war and legal hanging as necessary, even though each results in loss of life, there are some others who want both the practices to be abolished unconditionally. It is, therefore, obvious that while it may be possible to formulate conceptions of progress or valuations of change which apply to all time, the statements are too general to afford definite and sure guidance in specific situations at different times and in different cultures".

Sometimes attempts are made to avoid these difficulties by adding a qualifying adjective to the word 'progress', such as 'agricultural', 'industrial', 'economic', *etc.* It is true that progress in these specific fields can be measured. What cannot, however, be measured is the degree in which these measurable advances in specific fields 'contributes to the ill-defined and unmeasured end, which is progress itself.' Despite these limitations and many-sided attacks, the idea of progress has not lost its vitality. It is so deeply rooted in the modern mind that its critics never reject it entirely. Some examples may be given.

Professor Toynbee, who strongly upholds a cyclic theory of civilizations, ends by suggesting that "the breakdowns and disintegrations of civilizations might be the stepping stones to higher things on the religious plane" and even goes so far as to say that the next stage in this advance may lead to a new species of society "embodied in a single world-wide and enduring representative in the shape of the Christian Church".

Even the Marxists, who reject the 'bourgeois' concept of progress, retain the belief of the early rationalists 'What man can make himself. They look forward confidently to an age when the 'true realm 'of freedom will blossom out of the realm of necessity in the fully developed Communist Society of the future. The concept of progress is as old as mankind.

In all ages, people had a concept of that which would most desirable. These concepts may and do vary from age to age. Even when everything seems lost, people look forward to a better future. In all that we do, there is a consciousness that we are working to usher in a brighter and happier future. It rings through the speeches of religious preachers, social reformers, and even politicians. The faith in progress sustains us through all vicissitudes and trials and gives us the necessary strength to face the hard and cruel realities of life.

Emphasizing that Faith in Progress is Ineradicable from the Life of Man, MacIver Observes:

"To live is to act, and to act is to choose, and to choose is to evaluate. Hence as human beings we cannot get rid of the concept of progress, though we are of course entitled to deny the reality of progress. The fact that men in-inevitably differ about it, that we cannot demonstrate the validity of our concept as against theirs, only makes it indubitably ours. If one can prove it none can refute it. At the least it is a vital myth, ineradicable from the creative strivings of life".

SOCIAL CHANGES, EVOLUTION, DEVELOPMENT AND PROGRESS

It has been understood that social change as a term shall signify such changes as affect the nature and structure of social groups and institutions and the social relations between the individual, the group and the institutions in a society. 'Development', 'evolution' and 'progress' are the different modes of change and whenever we speak of social change the importance of each of these modes has to be assessed, for the changes brought about by each of these processes will have distinct impressions upon the functioning's of social phenomena.

The term 'development' means formal and structural changes in an organism. Even though society is not an entity like the living organism, the term as applied to such organism can have its valid application in social matters. Just as life grows from the simple to the complex form, society develops in the sense that its 'energy' accumulates collectively, such energy is 'organized' for functioning in a definite direction, and 'harmony' is achieved between the different social organs for the purpose of effecting an overall development.

'Social evolution' as a term has its complexities and, as has been noticed earlier, evolution in an organism means branching out from a single amoeba into different genera and species and then from the species into various forms that are caused by the process of differentiation. In the case of a society, as Gisbert puts it in his *Fundamentals of Sociology*, evolutionary change means a 'branching off of a line in various directions, which again ramify indefinitely'. A condition of simplicity changes into situations of complexity and social evolution witnesses the progressive development of social ways and customs, norms and beliefs, and associations and institutions. 'Social progress' does not mean mere development or evolution, for in either of these terms we have witnessed the change from the simple to the complex.

Though different writers have doubted whether or not changes in society can be regarded as progress and, when the concept of social progress is upheld, what factors determine such progress, the term 'progress' has been applied to social change in a few cases. Social progress definitely connotes a valuation and such valuation is made according to certain principles. If the principle according to which, the valuation is to be made can be objectively ascertained, measuring 'progress' does not become a difficult affair; but such principle cannot always remain free from subjective value-judgments. When a subjective analysis confuses 'progress' with 'happiness' or material comforts, the conclusions tend to remain on the wrong side of value-free judgments and the sociologist must always guard against such pitfalls in reasoning.

Different Theories of Social Evolution

Evolution, as understood in a living organism, necessarily stands for a process in which simple matter develops into the complex, but such development is always caused by innate qualities of such organism and not by any extraneous factor. Darwin's *Origin of Species* may have crystallized ideas about the phenomenon of development in living

organisms, but the concept of it was grasped in some inadequate form or the other by some thinkers even before that. Particularly when it concerns social evolution, the thought has been current for the past century or two; but upon the nature of such evolution there has been difference of outlook between different students of social science.

Herbert Spencer maintains that social evolution is only a part of the general process of evolutionary development in all living matter in the world. Society evolves from the simple form into the complex one as it fulfills the functions of integration and differentiation in its various organs and consequently, out of the same unit of society, different social systems come into existence. According to Spencer, there are three stages in the evolution of society; the first stage is known as 'integration', the second as 'differentiation' and the final one as 'determination'.

In the initial stages of social development, the different units of society have to be integrated and a 'system' has to be built up. For example, if the family is taken as a basic social unit, the first stage in social evolution was the bringing together of these families and their integration into a larger unit known as 'society'.

In the second stage of development institutions like division of labour grew up and this process involved itself with differentiation in the sense that different classes, castes and tribes appeared. In the ultimate stage, however, the different segments of society came together and set up a new social structure based on harmony. This was the evolutionary stage of 'determination', for in this stage an order was established for the processes of integration and differentiation, so that harmony could be achieved.

Franz Oppenheimer has asserted in his work, 'The State, that the State evolved in the process of applying certain principles for the satisfaction of man's creature and basis needs. According to him, there are two ways in which man can supply his basic needs. One of them is work, which is an economic activity, and the other is robbery, that is, exploitation as a political activity; and the State arose when the political means were organized. Early huntsmen had no state in the sense that they had not exploitative political organization. The State came into being with the Vikings and the herdsmen who learned to exploit, to divide society into classes, to hold slaves and establish the concept of the privileged class and the unprivileged people.

Oppenheimer's analysis of the development of the State is unsatisfactory; maintaining that the State only robs and exploits is a oversimplification of its functions, since the State performs other functions too, like maintaining law and order and punishing those that violate the law. Besides that, conditions as obtained in a State might have existed even before the State as it is known came into existence; and in effect it becomes difficult to go into the question of the beginnings or the origin of a State as a social organization. McIver and Page have stressed the importance of the process of differentiation in matters of social evolution. They hold that social evolution stands for an internal change within the social system itself and as a result of such change, functional differences can be brought about within the system. According to them, primitive societies did not have many distinctions observed on the basis of different functions

and, besides the differences between tribes, clans, age groups and sex groups, not much of differentiation was noted. Division of labour in these communities was an undeveloped practice, and associations and organizations did not exist in these societies. The community as such existed on the basis of a simple solidarity and McIver and Page observe that the 'undifferentiated character' of the primitive society saw in it 'the prevalence of a simple form of communism.

Thus, the community devised a system of sharing the hunter's spoils and, even with regard to matters like sex that are treated as personal and intimate by us, customs and practices prevalent in those days allowed a kind of sharing by the community. If there was any differentiation in such society, it was based on natural distinctions of sex and age, and the multitudinous aspects of differentiation as exist in modern complex society were at best latent, if not totally absent. According to McIver and Page, associations and organizations in a modern complex society are so many that they immediately strike a contrast with the simple, institution- based primitive society. Historically speaking, diffusion of ideas from the beginnings of the earliest civilizations in Mesopotamia, Persia, India and China perhaps caused the evolutionary development of human thoughts and, therefore, of human society.

But the Generic Lines of Evolutionary Development will be best Understood if Attention is Focused upon the Following Features of Such Development:

- Society has evolved from simple communal customs into various differentiated associations. Codes apply to these associations and each association is developed by an institution comprising of specialized procedure and practice. Thus, political, economic, religious and cultural usages in a community collectively develop into community practices; and from that stage, different forms of procedure evolve as differentiated communal institutions. Finally, these institutionalized procedures become embodied in differentiated associations like the State, the economic organization, the school, the cultural association and the like.
- The Associational stage in society is so radically different from the levels that obtained in the primitive society that it has introduced an entirely different form of social relationships. There are voluntary and involuntary associations, and each great association structurally and functionally differs from the other. For example, in the second stage of development as shown above, only one set of political institution exists for the whole society, or one set of religious institution is appreciated. But in the third and the associational stage, several political organizations and diverse ideas regarding the State are entertained, just as the religious organization diversifies into various forms of religious associations. The Church, for instance, began as a religious institution in the second stage of development of society; but in the final stage it has advanced into the form of different religious associations.
- When social events are studied historically, they present the earlier types of social units and the later types of the same. The differentiation between these

types helps us to classify and characterize the diverse social systems as they evolved historically as also to understand the causes in the evolutionary processes that brought about definite results.

Hence, there is a historical pattern in social evolution and, if effects are noticed in the evolutionary process, there must be causes that explain such differentiation. This particular view of McIver and Page is not totally different from the observations made by M. Ginsberg in his *Studies in Sociology*. Ginsberg says that evolution is a process of change culminating in the production of something new but exhibiting an orderly continuity in 'transition'. McIver and Page also admit that in the evolutionary process certain factors cause the change which is but an orderly expression of a new form that has culminated from an older one.

However, Ginsberg would not admit that social evolution implies the change from the simple to the complex as McIver would put it. His idea of evolution is that while the process of transition introduces something new, such novelty is but a continuity of some social element that is permanent. Evolution truly speaking means a change that comes from within but, as Ginsberg points out extraneous factors also condition social evolution, and his view is that evolutionary changes in society are best understood when the subjects of society, that is, the individuals are taken into account.

He maintains that 'the real subjects undergoing development are men of societies who speak and think and are religious'. If this view alone is taken, it would not mean that social evolution stands for substituting the simple organism by the complex one; it would rather mean that evolutions occur in thoughts and ideas and in the interaction of human social relations.

But then human interactions and social relations cannot exist in complete detachment from social customs and institutions. Every social event is the effect of complex action and reaction of human relations just as human social relations are the effect of the complex functioning of the social structure. Ginsberg says that 'if individuals make society, it is equally true that society makes individuals'. It follows, therefore, that a mere concentration upon human actions and reactions for the proper gauging of social change will be an inadequate approach; it must be accepted that social change as a concept is an evolution of the individual as well as of the society itself.

Social Change and Progress

Every event of social change cannot be regarded as progress, for progress must connote the taking of a step forward. If at the root of evolution we have the stages of integration and differentiation, progress would stand for a development in a particular direction which is regarded as a step forward according to definite criteria of value- judgments.

While evolution has no definite direction other than the one which is inherent and irresistible in itself, progress must stand for a march in a forward direction according to some accepted principle that is formulated by a particular principle of judgment. Ginsberg maintains (Idea of Progress) that progress 'is a development or evolution in a direction

which satisfies rational criteria in value'. In order to measure progress, it is necessary to apply the test of ethical advancement made by society which, of course, is an irrelevant factor so far as evolution is concerned. Writers like Comte and Spencer would maintain that any evolutionary development of society must necessarily mean that it has progressed. Herbert Spencer particularly insists that social evolution cannot have any meaning other than that of progress. But these views are not accepted now by more modern writers. McIver states in his *Society* that 'evolution is a scientific concept and progress an ethical concept'.

Even Hobhouse observes that evolution of any form does not necessarily imply that it is changing into the better form; and, therefore, we cannot conclude that evolution necessarily implies that society is progressing. According to him, progress can be made only when the individual in society strives for ethical advancement. Social progress, therefore, is not a phenomenon marked by spontaneity; it is the product of conscious efforts made by social individuals.

The concept of progress is based on the vision of an ideal society in which every individual will have the opportunity of developing his innate qualities, in which the very basis of social relations will be principles of liberty and equality, and in which the institutions will aim at comprehending the foundations of collective good. These are, however, matters of value-judgments and the concept of progress cannot be understood without applying the test of values. Evolution, as a term, does not depend upon these values. Some modern sociologists, however, feel that the science of sociology is not concerned with ethics and, therefore, the term 'social progress', which cannot be understood unless it is related to ethical values, will not be the concern of the sociologist and will consequently have no scientific value. They maintain that no scientific observation and rational conclusion shall be based on any ethical value.

If the method followed by the sociologist in the study of society is that of positive science, and if the principles of causation are to be objectively investigated into, it will be an anomaly if facts are correlated to values. However, social facts cannot be regarded as isolated phenomena; every social event has a practical side to it and another concerning values. McIver and Page observe that social facts can also be regarded as 'value-facts' since social valuation is much concerned with them.

Therefore, the authors maintain, that science appreciates value-judgments, first, in order to test the accuracy of factual evidence in support of a value-judgment and, secondly, for testing the validity of conclusions regarding the good or the bad in so far as these conclusions are supported by reasoning from statements of facts.

For example, valuations obtainable in any social institution may be studied scientifically in order to test their validity, but the sociologist shall not apply his own personal judgment to such valuations which are ingrained in the social facts themselves. In this way, a value-judgment can be objectively made in upon the term 'social progress', but the social scientist must begin his work by looking upon evolution as a value-free fact.

Thus, one may objectively determine the degree of progress made by a particular society only after one has disinterestedly studied the growth of its associations and institutions and the psychological elements in social relations between individuals in it. An objective study of social progress can be facilitated by considering the factors that hinder and obstruct advancement in material as well as psychological terms. Any rigid attitude towards scientific development of material conditions will have both material and psychological implications.

If science and technology is looked upon with suspicion and if there is a blind adherence to outmoded custom, material development in the society will not be achieved, while social mentality in general will remain unliberated. But if technology is applied to the processes of developing and utilizing natural resources, material advancement will undoubtedly take place; and at the same time, man will have enough scope of cultivating constructive thoughts about developing his families. His social and moral consciousness grows in degree and he learns to propagate the idea of integrating efforts in the direction of realizing the common good.

Hence, we can conclude that the society in which scientific development is hindered will not progress, while the one which encourages such development will have chances of making progress; and this observation about social progress can remain scientific in so far as it is based on social facts and not merely upon ethical considerations.

However, there are problems connected with the adoption of a scientific attitude towards the study of social change, whether such change speaks of evolutionary development or of mere progress. Social change as a phenomenon is so complex by itself that the analysis of no single factor can lead us to a definite conclusion.

In trying to explain any phenomenon scientifically, any of the following mistakes can be made:

- The scientist may think that any one of the many factors relating to such phenomenon is the dominant factor; and
- He may assume that social forces can be quantitatively measured.

For example, if it is to be considered why the middle classes in India grew under the British rule; it may be wrong to conclude that this class grew and developed only because it applied itself to the task of promoting the standards of British-oriented professions and occupations; besides that, there might have been other factors too which contributed to the growth of that class. Similarly, if the social scientist allows himself to conclude from a number of social facts in the same manner in which quantitative conclusions are made following a study of a number of guinea pigs, he will not acquire for himself a proper perspective of the complex pattern of social facts.

The truth is that in the primitive society, the people unified social authority with their techniques and culture, and the basic personality structure of every individual was ascertainable. The simplified study of the primitive individual may then be possible, but modern complex society has introduced so many factors for the interplay of social relations that the scientist's task does not remain simple.

Technological advancement has in modern times broken up the primitive unity of social authority, technique and culture. Culture has been detached from other bio-physical factors and within the same society different cultures and subcultures may coexist. In the modern social system, specialization and the diversity of numerous groups and associations have given to society such multifarious facts that the sociologist cannot afford either to isolate one social factor for study or derive conclusions from a number of social facts.

Thus, he must realize that several factors combine to produce social changes and, as these factors make the combination, they interrelate themselves within the larger environment of society. The study of social change must, according to McIver, take into account the attitudes that depend on a particular cultural background and the objectives to which such attitudes are related, the system of means for fulfilling such objectives, and the physical and biological conditions that prompt the changing objectives and the human nature that follows them. The scientist must first study the interrelationship of the various factors, analyse the logical order in which they fall, and finally build up an observation upon the manner in which they make up the casual process in which social change is effected.

ROLE OF STATE IN MASS EDUCATION

Democracy gives importance to equality of opportunity. It encourages vertical mobility and recognises man as an individual thus defining man on the basis of the collective identity. Education plays very important role in making democracy more meaningful and relevant. Education offers common knowledge to all the citizens of the country making them aware about their rights and duties. Equal access to education offers equal chance to every individual to compete with other individuals to have access over highly valued occupations, skills and services in society accelerating the process of mobility. Education also helps citizens to develop common virtues important for integrative role in the social life. The history of the education reveals that it has always been monopoly of the dominant class in societies. Weber in his study found that how their monopoly over education has led to consolidation of their predomination in the status structure and class structure of the society. The concept of mass education came in prominence in American society subsequently spread to France and Britain.

Margaret Archer in her study of cross-cultural comparison of mass education system finds that American war of independence led to growth of equality in American society. The melting pot culture of America demanded that all immigrant population should abandon their cultural values to go for universal educational training to excel in world of enterprise, trade and commerce therefore education became a matter of community management. Community mobilized funds to run the schools and recruit the teachers. It became mandatory to all members of the community to send their children for basic educational programme therefore communities participation led to a total change in mass education system in America.

In France mass education got momentum only after French Revolution. In pre-revolution period education was greatly elitist and was controlled by Church. Emergent leadership during revolution period demanded mass education provision and the separation between church and schools. As a result only during 1930s through different movements Church separation from the schools became complete in French society.

In case of Britain in post industrial revolution period public opinion was divided in relation to the type of schools that Britain should set up. Elites supported missionary education for the masses. Labour class demanded government sponsorship to open up schools in slums and sub-urban areas. Multiple school system made appearance in Britain. Though the curriculum in schools was greatly adjusted to meet the demands of industry, the classical converts gave importance on value-education, spoken English, articulation drawing aristocrats in these kinds of school programme

Middle classes went for public schools and rationalized individualism, competition and gratification of self-interest. State sponsored schools running in peripheries got membership from lower class that considered themselves beneficiaries and preferred to go for submission and loyalty. As a result different schools produced different classes. Political leadership emerged from the first school, technocrats and bureaucrats and entrepreneurs from the second and semiskilled and skilled labours from the third. This explains that how expansion of mass education in Britain failed to break down the traditional class hierarchy.

In case of India at the time of independence the total literacy rate was as low as 12 per cent. The 1951 census indicates that it increased to 15.99 per cent. This figure indicated that mass education was unknown to Indian society and it was monopolized by upper castes and classes who used to get various government positions and perpetuate their domination. Indian educational system was greatly cultural focused and not linked to the Indian economy which was agricultural in nature.

Gandhian system of education known as moral education gave importance on character-building, humanism than offering scope to improve their economic position by utilising skills. To give education a new thrust, Kothari Commission in its report of 1964 suggested that mass education should be left open to community participation. As a result village communities took initiative in opening the schools and for that infrastructure, teachers' recruitment was determined by community. Subsequently these schools after running for period of five years got grant-in-aid from the state. Therefore through active community participation Kothari Commission intended to give mass education a dynamic thrust.

New Education Policy of 1986 spoke about expansion of non- formal education to rural and tribal areas. This policy also emphasized the active involvement of the NGOs, PRI to advise the parents to send their children to school. Subsequently in 1988 National Literacy Mission came into prominence and 1991 saw sharp decline in drop-out rate in schools and increase in formal literacy rate was realised. During 1999 Sarva Shiksha Abhiyan started and state Govt. together with Central Govt. and NGOs tried to give

boost to mass education. To generate funds for the purpose, Rashtriya Siksha Kosh was established in 2002. It can be advocated that success of mass education is greatly instrumental for promotion of equality in social life as it generates a kind of consciousness in the masses spelling out their duties, obligations, responsibilities both in relation to work and society. The role of government is very significant for the success of mass education programme as it has both the resources and ability to take up the work on huge scale.

EDUCATION AND INEQUALITY

The distribution of education both in terms of quality and quantity is highly uneven in most societies. Inequality in opportunities for education is found not only with reference to individuals and social classes but also in terms of regions and territorial regions such as urban and rural areas. Our educational system is urban biased in the matter of location of facilities; allocation of finance *etc.* is a well-established fact. Most institutions of higher learning and good schools are concentrated in urban areas. The awareness of their existence and utility is also greater there. The students from the urban areas are favoured for admission than those from the rural areas. This acts as a barrier to the rural person that has the effect of insulating the educational facilities from being observed by the latter. As referred by Kamat in matter of cost, adjustment and complexes urban people have a decisive edge over their rural counterparts who will have in addition to surmount the problems arising from their migration to cities to pursue education.

The rural rich alone seem in these circumstances to come closer to urbanities in the matter of higher education. According to Bauer urban bias in the educational system is sometimes defended on the ground that in the potential progress of developing societies the city plays a dynamic multiple role as it is the focal point for change and progress and that such bias is not entirely absent even in the western countries. Another group advocates that technical efficiency and economic tenability necessitates educational facilities to be developed as a part of an urban -industrial complex. These arguments counter the role of education as removing the inequalities that it supposes to do.

The inevitable consequence of the urban bias in education has been the neglect of education in villages where majority of the population lives. Not only is rural education neglected, it is also suffering from serious ills in its present state.

Past experience has shown that at the time of policy planning the greatest stress is laid on universal mass education with special emphasis on the education of women, of scheduled castes and tribes and in rural areas. But at the time of execution and performance the achievements are in urban areas. There is an upward bias in the Government's method of estimation of the educated manpower requirements for economic growth and Government has been remarkably responsive to political pressures for quantitative expansion of higher education. It is seen that good education is virtually the monopoly of a select stratum of urban society. The institutions are selective as far as

socio-economic background of the students is concerned. Students are given preference that comes from high educational, occupational and income backgrounds.

While the government has taken up the cause of ameliorating the educational situation among the scheduled tribes and castes the response has been far from satisfactory. It is also seen that despite rapid expansion in their education women are still at a disadvantage compared to men. The availing of benefits of the rapid expansion in higher education by women is chiefly confined to those belonging to the higher social strata and upper caste groups in urban areas.

Inequality of educational opportunities does not take place solely at the level of higher education. In fact the initial unequal selection takes place at the school level itself and higher education merely reinforces it. A majority of students in higher education have had English medium education and that too in privately managed schools. The public or private schools contribute to the ranks of higher education. With widespread inequality in opportunities for education resulting from the operation of socio-economic factors education seems to have failed as an effective equalizing factor. However it is not that education has not contributed at all to the process of upward social mobility but the momentum has been slow considering the social fabric of the country.

Role of the Teacher in Bringing Social Change

Teacher is the central figure in any formal educational system. He must respond to the duty. He is regarded as an effective agent of social change. Any social change without teacher is a mere figment of imagination. The teacher should know well the needs of the society and the nature and direction of social change. The present democratic society expects that the teacher should really act as an agent of social change. Regarding his role, the following points need special mention:

Model of modernity: In the process of social change important role can be performed successfully by the teacher only when he makes himself a model of modernity *i.e.*, when he himself imbibes modern values and allows them to manifest in his own conduct. In other words, the following qualities should be imbued in the behaviour of the teacher: scientific attitude, rational thought, empathy, mobility, democratic style, sentiment of respect for individual personality, secularism, faith in change and transformation.

- Competent and skilled: The teacher should be competent and skilled because he is to lead the younger generation, the future citizens of tomorrow.
- Psychological climate: The teacher must prepare the psychological climate favourable to new attitudes and values.
- Reorientation: The teacher should reorientate himself to accelerate the process of social change.
- Suggestions for change: The teacher should come forward with his wise suggestions of bring about social change. He has to be ready for bringing about

an implementing social change. Apart from giving wise suggestions, the teacher should welcome the positive changes taking place. He should welcome what science and technology has bestowed on mankind and inculcate proper attitudes and values in students. He is to encourage non-formal education, technical education, population education, inclusive education, adult education, women education, education of the backward classes and participate, actively in these activities in the larger interest of social change.

- Remove obstacles: The teacher is to help remove the obstacles in the way of implementation of social change. The Government wants social change by implementing family planning. The teacher should help the Government by breaking down the resistance of people to this programme.
- Scientific attitude among students: The teacher should develop a rational, curious, thoughtful, positive, constructive and scientific attitude in his students by encouraging debate and discussion in the class-room as well as in co-curricular activities.
- Secular attitude: The teacher should have secular attitude. He should have respect for all religions. He should eschew all caste, community and class considerations in dealing with students and colleagues. If the teacher can modernize the entire atmosphere of the school, he can give powerful impetus to the process of social change because the students of today will become the pillars of society of tomorrow.
- National and international outlook: The teacher should have national and international outlook. He should try to inculcate emotional integration, national integration and international understanding in the students.
- Pupil-centered and community-centered education: The teacher should impart pupil-centered, life-centered and community-centered education. He should provide opportunities for the maximum and best development of the pupils. He should constantly strive to improve community life through the best possible use of material and human resources of the school and community.
- Use of modernized instructional technology: The teacher should make effective use of science and technology in the routine activities of the school and in this way habituate students to their use and give them practice in applying science and technology in other situations. Kit-based instruction for teaching of science, language laboratories for teaching languages, programmed learning for all subjects, wise exploitation of radio and television lessons and educational journals *etc.* to supplement class-room teaching are essential steps.

Thus, it can be concluded that the teacher through his multifarious roles and activities, can accelerate the process of social change in a very powerful way. To quote E.W. Gordon, "It is the task of the teacher to educate for change, to educate through change and to educate for orderly planned revolution, and if necessary to educate through revolutionary action.

TEACHER MOTIVATION AND PERFORMANCE

There are several factors that significantly impact on teacher performance and motivation. These include the actual amount of time the teacher spends in class teaching, the proportion of time spent in maintaining order and doing other administrative tasks, the pedagogic techniques used by them in the classroom and disciplinary policies adopted by the teacher and the school. A number of official policies and programme documents over the years have reiterated the significance of teachers in the education of children, including the Education Commission Report, the National Policy on Education 1986, and the National Policy on Education: Programme of Action 1992.

A school without adequate numbers of teachers is non-functional, but a school which has an adequate number of teachers who are neither empowered nor professionally qualified nor committed to the basic ideals of their profession, with low morale and self-esteem, are likely to do more damage than good to students. Sub-optimal use of teachers is another serious concern in most states. Generally, teacher absenteeism is high in India.

Even if teachers attend schools, their presence in classrooms is often low. This happens at least partially because of their engagement in non-teaching activities. A recent study on teacher absence in 20 states, nearly one quarter of teachers were absent from government primary schools on a typical school day. Absence rates ranged from around 15% in Maharashtra to 38% in Bihar and 42% in Jharkhand.

The absence rate was also more than 30% in Chhattisgarh, Uttarakhand, Assam, Punjab, Bihar and Jharkhand. One factor determining the motivation and performance of teachers is the status they enjoy in society. There is a general perception that in, recent years, the teaching profession has experienced a lowering of its status. Batra suggests that teaching has declined to a least favoured profession status. A recent study reveals that teaching is considered as more of a 'safe fall back option' than a formal career choice. Further, the social distance between teachers and students is one reason why many teachers have limited commitment to the educational advancement of their pupils, as well as have limited understanding of their problems. Research indicates that the attitudes of teachers can be particularly discriminatory in the case of working children. Ramachandran et al points out in a study of working children that both children and parents see teachers treating poor children differently; they do not appreciate the predicament of children who have to work before and after school hours.

This not only has a significant negative impact on the participation of those children in schooling, but also creates negative attitudes towards schooling for those who continue with their education. There is, in fact, a discernible hierarchical division of schools in India with selected urban schools with good infrastructural facilities and teaching staff catering to rich students, while schools with poor facilities and inadequate numbers of teachers tend to cater to the poor and deprived sections of society. Educational provision has in recent years become much more diverse owing to the rapid growth of private sector provision. There is now a greater range of choice for parents who can afford to

pay for their children's education, while government schools are usually the only option for those from lower economic strata who cannot exercise such choice. As a result, many teachers working in state-run schools perceive themselves to be in a 'deficit' situation.

A typical profile of a government school teacher tends to be of someone who is poorly educated, with little or no teacher education and training, limited or almost no contact with books, technology and science, and increasingly recruited from among low income backgrounds. The majority of untrained or under-trained teachers work in rural areas, where the poorest and most deprived schools are situated.

There is also a growing criticism of public schools which coincides with a decreased confidence in teachers, when in reality teachers are often not given the conditions to ensure teaching and even less learning. Teachers, who are themselves products of the poor education system, are often seen as obstacles to educational change rather than as key human resources. Consequently, teachers tend to respond defensively by overemphasizing problems outside their direct influence.

TEACHER MANAGEMENT AND TRAINING

Historically, state governments have not had total control over the management of teachers in publicly-supported schools. While state schools were large in number, a sizeable number of schools were under the management of District Boards, Taluka Boards, Municipal Corporations, and so on. In the early 1970s, many state governments took direct control of these schools, creating a mega system of primary schooling. This arrangement began to undergo transformation in mid-1990s, following a Constitutional Amendment which called for the empowerment of panchayati raj bodies. Some, but not all, state governments took this opportunity to wholly or partially decentralize the teacher management system. Over time, the unevenness in service conditions of teachers within and between states has increased.

In some states, local self-governing bodies are given a reasonably free hand to decide on teacher recruitment at a local level, and operate within a broad framework provided by the state government. While all teachers are technically state government employees, therefore, some teachers are employed by local village level bodies, whilst others are employed by block or district level bodies.

Another significant development during the 1990s, which has the potential to significantly influence the professional development of teachers, is the creation of a system of district and sub-district level teacher resource centres. Following the recommendations of the National Policy on Education 1986, every district in the country was provided with a district-specific teacher education institution, a District Institute of Education and Training. Subsequently, under the District Primary Education Programme, block resource centres and 10-15 cluster resource centres were established in each of the 42 districts which participated in the project. This is now being universalized to reach all districts of the country under SSA.

Availability and Deployment of Teachers

Mere availability of schools does not ensure children's participation. More than schools, motivated teachers play a vital role in ensuring that children attend school regularly and take an active part in the learning process. Though there has been an impressive increase in the number of school teachers in India in the 1990s, the imbalances in intra-state teacher allocation between districts and within districts, and between rural and urban areas continues to be a major concern. An increase in the numbers of upper primary teachers has taken place in rural areas in Andhra Pradesh, Gujarat, Himachal Pradesh, Karnataka, Madhya Pradesh, and Rajasthan indicating improved completion of the lower primary cycle and increased transition to upper primary classes.

Indeed, in most of states, the percentage increase in upper primary teachers has been more than that of the primary teachers. This indicates a positive trend in transition rates from lower to upper primary classes. However, the allocation of teachers particularly between rural and urban areas continues to be a major concern in many states. In almost all states, trained teachers with higher qualifications are generally concentrated in urban areas. Several factors, including political and bureaucratic interventions, non-availability of qualified local teachers in remote areas, lack of a proper database on the status of teachers at the school level and the low motivation of urban teachers to serve in rural areas, are obstacles to the rational deployment of teachers. Moreover, teacher transfer is highly influenced by political interventions, and more influential teachers are more likely to be found working in urban schools and schools located in habitations well connected with road or rail transport networks. The lack of basic facilities also act as de-motivating factors, particularly for female teachers. These factors undoubtedly influence pupil-teacher ratios with the average PTR higher in rural areas than in urban areas.

CHALLENGES IN TEACHER EDUCATION

Unprecedented expansion of teacher education institutions and programmes during the past few years characterizes the teacher education scenario of today. With increasing school enrolments and the launch of pan-Indian primary education development programmes like Operation Blackboard, District Primary Education Programme, Sarva Shiksha Abhiyan and Universalization of Elementary Education, there was a natural increase in the demand for teachers.

Added to this, the backlog of untrained teachers in the system and the essential requirement of pre-service teacher certification for appointment as a teacher led to mounting pressure on existing institutional capacity. The demand far exceeding supply, market forces have taken over unprecedented rise in the number of teacher education institutions in most parts of the country. From 3489 courses in 3199 institutions and an intake of 2,74,072 in 2004, the numbers in December, 2008 swelled to 14,523 courses in 12,200 institutions with an intake of 10,73,661 at different levels. This expansion has taken a heavy toll on quality parameters like infrastructure, faculty learning resources

and student profile. Teacher education as a whole needs urgent and comprehensive reform. There is a need to bring greater convergence between professional preparation and continuing professional development of teachers at all stages of schooling in terms of level, duration and structure. Considering the complexity and significance of teaching as a professional practice, it is imperative that the entire enterprise of teacher education should be raised to a university level and that the duration and rigour of programmes should be appropriately enhanced.

RESEARCH AND INNOVATION

There is a need to increase research that documents practices reflectively and analytically- whether it is of programmes or of individual classrooms – so that it can be included in the body of knowledge available for study to student teachers. University departments and research institutions need to undertake such research. In addition there is a need to innovate with different models of teacher education. Institutional capacity and capability to innovate and create are a pre-requisite for the pursuit of excellence. Hence in the present scenario a lot of impetus has been given to research. Many teacher educators are encouraged to take up either major or minor research projects.

INCLUSIVE EDUCATION

There are two kinds of exclusion prevalent in schools; one is the exclusion of the child with disabilities and the second is the social exclusion of children who come from socially and economically deprived backgrounds. There is a dire need to equip teachers to overcome their biases in these regards and positively handle these challenges. The Persons with Disabilities (PWD) Act of 2005 provides for free and compulsory education up to the age of 18 years for all children with disabilities. The education of socially and economically disadvantaged groups, especially the SCs, STs and minorities has remained a primary national concern of education for several years. The enrolment and retention of girls and therefore their participation has also remained behind those of boys. Teachers will have to be specially equipped if the social deprivation has to be overcome through education.

PERSPECTIVES FOR EQUITABLE AND SUSTAINABLE DEVELOPMENT

In order to develop future citizens who promote equitable and sustainable development for all sections of society and respect for all, it is necessary that they be educated through perspectives of gender equity, perspectives that develop values for peace, respect the rights of all, and that respect and value work. In the present ecological crisis promoted by extremely commercialized and competitive lifestyles, children need to be educated to change their consumption patterns and the way they look at natural resources. There is also an increasing violence and polarization both within children and between them, that is being caused by increasing stress in society. Education has a crucial role to play in promoting values of peace based on equal respect of self and

others. The NCF 2005 and subsequent development of syllabi and materials is attempting to do this as well.

OBLIGATION TO HONOUR STUDENTS RIGHTS

Sudbury model democratic schools claim that popularly based authority can maintain order more effectively than dictatorial authority for governments and schools alike. They also claim that in these schools the preservation of public order is easier and more efficient than anywhere else.

Primarily because rules and regulations are made by the community as a whole, thence the school atmosphere is one of persuasion and negotiation, rather than confrontation since there is no one to confront. Sudbury model democratic schools' experience shows that a school that has good, clear laws, fairly and democratically passed by the entire school community, and a good judicial system for enforcing these laws, is a school in which community discipline prevails, and in which an increasingly sophisticated concept of law and order develops, against other schools today, where rules are arbitrary, authority is absolute, punishment is capricious, and due process of law is unknown.

TEACHER ENTHUSIASM

Since teachers can affect how students perceive the course materials, it has been found that teachers who showed enthusiasm towards the course materials and students can affect a positive learning experience towards the course materials. On teacher/course evaluations, it was found that teachers who have a positive disposition towards the course content tend to transfer their passion to receptive students. Teachers cannot teach by rote but have to find new invigouration for the course materials on a daily basis. Teachers have to keep in mind that they are teaching new minds every term or semester. Otherwise, teachers will fall into the trap of having done this material again and start feeling bored with the subject which in turn bore the students as well. Students who had enthusiastic teachers tend to rate them higher than teachers who didn't show much enthusiasm for the course materials.

Teachers that exhibit enthusiasm can lead to students who are more likely to be engaged, interested, energetic, and curious about learning the subject matter. Recent research has found a correlation between teacher enthusiasm and students' intrinsic motivation to learn and vitality in the classroom. Controlled, experimental studies exploring intrinsic motivation of college students has shown that non-verbal expressions of enthusiasm, such as demonstrative gesturing, dramatic movements which are varied, and emotional facial expressions, result in college students reporting higher levels of intrinsic motivation to learn. Students who experienced a very enthusiastic teacher were more likely to read lecture material outside of the classroom. There are various mechanisms by which teacher enthusiasm may facilitate higher levels of intrinsic motivation.

Teacher enthusiasm may contribute to a classroom atmosphere full of energy and enthusiasm which feed student interest and excitement in learning the subject matter. Enthusiastic teachers may also lead to students becoming more self-determined in their own learning process. The concept of mere exposure indicates that the teacher's enthusiasm may contribute to the student's expectations about intrinsic motivation in the context of learning. Also, enthusiasm may act as a "motivational embellishment"; increasing a student's interest by the variety, novelty, and surprise of the enthusiastic teacher's presentation of the material.

Finally, the concept of emotional contagion, may also apply. Students may become more intrinsically motivated by catching onto the enthusiasm and energy of the teacher. Research shows that student motivation and attitudes towards school are closely linked to student-teacher relationships. Enthusiastic teachers are particularly good at creating beneficial relations with their students.

Their ability to create effective learning environments that foster student achievement depends on the kind of relationship they build with their students. Useful teacher-to-student interactions are crucial in linking academic success with personal achievement. Here, personal success is a student's internal goal of improving himself, whereas academic success includes the goals he receives from his superior. A teacher must guide his student in aligning his personal goals with his academic goals. Students who receive this positive influence show stronger self-confidence and greater personal and academic success than those without these teacher interactions. Students are likely to build stronger relations with teachers who are friendly and supportive and will show more interest in courses taught by these teachers.

Teachers that spend more time interacting and working directly with students are perceived as supportive and effective teachers. Effective teachers have been shown to invite student participation and decision making, allow humor into their classroom, and demonstrate a willingness to play. The way a teacher promotes the course they are teaching, the more the student will get out of the subject matter. The three most important aspects of teacher enthusiasm are enthusiasm about teaching, enthusiasm about the students, and enthusiasm about the subject matter. A teacher must enjoy teaching. If they do not enjoy what they are doing, the students will be able to tell. They also must enjoy being around their students. A teacher who cares for their students is going to help that individual succeed in their life in the future. The teacher also needs to be enthusiastic about the subject matter they are teaching. For example, a teacher talking about chemistry needs to enjoy the art of chemistry and show that to their students. A spark in the teacher may create a spark of excitement in the student as well. An enthusiastic teacher has the ability to be very influential in the young student's life.

MISCONDUCT

Misconduct by teachers, especially sexual misconduct, has been getting increased scrutiny from the media and the courts. A study by the American Association of University

Women reported that 0.6 per cent of students in the United States claim to have received unwanted sexual attention from an adult associated with education; be they a volunteer, bus driver, teacher, administrator or other adult; sometime during their educational career. A study in England showed a 0.3 per cent prevalence of sexual abuse by any professional, a group that included priests, religious leaders, and case workers as well as teachers. It is important to note, however, that the British study referenced above is the only one of its kind and consisted of “a random... probability sample of 2,869 young people between the ages of 18 and 24 in a computer-assisted study” and that the questions referred to “sexual abuse with a professional,” not necessarily a teacher. It is therefore logical to conclude that information on the percentage of abuses by teachers in the United Kingdom is not explicitly available and therefore not necessarily reliable.

The AAUW study, however, posed questions about fourteen types of sexual harassment and various degrees of frequency and included only abuses by teachers. “The sample was drawn from a list of 80,000 schools to create a stratified two-stage sample design of 2,065 8th to 11th grade students” Its reliability was gauged at 95 per cent with a 4 per cent margin of error. In the United States especially, several high-profile cases such as Debra LaFave, Pamela Rogers, and Mary Kay Latourneau have caused increased scrutiny on teacher misconduct. Chris Keates, the general secretary of National Association of Schoolmasters Union of Women Teachers, said that teachers who have sex with pupils over the age of consent should not be placed on the sex offenders register and that prosecution for statutory rape “is a real anomaly in the law that we are concerned about.” This has led to outrage from child protection and parental rights groups.

UNIVERSITY

A university is an institution of higher education and research, which grants academic degrees in a variety of subjects. A university is a corporation that provides both undergraduate education and postgraduate education. The word university is derived from the Latin *universitas magistrorum et scholarium*, roughly meaning “community of teachers and scholars.” The definition of a university varies widely even within some countries. For example, there is no nationally standardized definition of the term in the United States although the term has traditionally been used to designate research institutions and was once reserved for research doctorate-granting institutions. Some states, such as Massachusetts, will only grant a school “university status” if it grants at least two doctoral degrees. In the United Kingdom, an institution can only use the term if it has been granted by the Privy Council, under the terms of the Further and Higher Education Act 1992. In India, a new tag deemed universities was created a few years ago, by the cabinet minister Arjun Singh during his tenure as the Minister for Human Resource Development. Through this provision many universities sprung up in India, which are commercial in nature and have been established just to exploit the demand of higher education.

Human Development and Social Change

THE PEOPLE

Human beings are distinguished among all living species by their capacity to develop and think. Though development is often measured in terms of so many miles of roads, number of tall buildings, airports, cars, TVs, computers *etc.*, it is not things and places alone that define development. Development must be people centric. Twenty years hence, the people of India will be more numerous, better educated, healthier and more prosperous than at any other time in our long history. Having eradicated the scourge of famine that plagued the country for centuries, we still confront the challenging tasks of providing a nutritious diet to all our children, educating our teeming masses, abolishing epidemic diseases and creating employment opportunities for all our citizens. Today India is the second most populous country in the world, with about 1.04 billion people, home to a-sixth of humanity. Although it is difficult to accurately predict population growth rates 20 years to the future, we expect this number to rise by another 300-350 million, in spite of continuous efforts to reduce fertility rates. This would raise the total population to about 1330 million by 2020.

India is in the process of a demographic transition from high fertility, high mortality and stable population to low fertility, low mortality and stable population. This transition is a global phenomenon generated by the improved availability and access to modern health care that sharply reduced mortality rates and increased life expectancy. The Crude death rate has declined to one-third of its level in 1941 and the expectation of life at birth has nearly doubled during this period.

Falling mortality rates have been followed by a steady decline in birth rates, but this decline has not been as steep as the fall in death rate; even after reaching the replacement fertility rates, the population will continue to grow because of large numbers of young persons entering reproductive age. While assessing future prospects, it is necessary to take into account the vast regional differences in demographic parameters. Population growth has recently decelerated below 2 per cent for the first time in four decades, but not uniformly across the length and breadth of the country.

Several states in south India have already reached, or are about to attain, the replacement level of fertility that would ensure a zero growth rate of population in the long run. On the other hand, in many states in north India, characterised by high fertility rates, it may take several decades to reach the replacement level of fertility. There are also similar differentials in levels of mortality, especially in infant and child mortality. At the national level, two alternate scenarios for achieving population stabilisation have been considered. In the optimistic scenario, which is based on achieving the demographic goals of the National Population Policy 2000, life expectancy is assumed to rise to 71 for males and 74 for females by 2020. Under the realistic scenario, life expectancy is assumed to reach 65 for males and 69 for females by 2020.

Under either scenario India's population would exceed 1.3 billion in the year 2020. In both cases the sex ratio of population (females per 1000 males) would marginally increase from 932 in 2000 to 950 in 2020; reversing the historical trend of falling sex ratio is expected in the 21st Century. We can be more certain of the changes in the age structure of the population as given in Table.

Table. Population Projections (millions).

Year	2000	2005	2010	2015	2020
Total	1010	1093	1175	1256	1331
Under 15	361	368	370	372	373
15-64	604	673	747	819	882
65+	45	51	58	65	76

Actually, the population under-15 years is expected to increase only marginally over the next 20 years. This means that pressure for expansion of the educational system will come only from increasing enrolment and efforts to reduce drop-out rates. The largest growth of population will be in the 15-64 year age group, which will expand by about 46 per cent by 2020 (*i.e.*, annually by 1.9 per cent as against the population growth of much lower, at 1.4 per cent). It is expected to rise from 604 million in 2000 to 883 million in 2020, *i.e.*, from 60 to 66 per cent of the total population.

This rise will accentuate the need for challenge of reducing fertility and increasing employment opportunities, so that the family size comes down and incomes rise. The elderly population is also expected to rise sharply from 45 to 76 million, (*i.e.*, by 2.6 per cent per annum) and their share in the total population would rise from 4.5 to 5.7 per cent. As a consequence of these age structural changes, the age-dependency ratio of

non-working age population to working age population) is expected to fall from 67 per cent in 2000 to 46 per cent in 2020, although the percentage of elderly people to population will increase.

INFRASTRUCTURE DEVELOPMENT

URBAN DEVELOPMENT

Disparities between the social and physical infrastructure of the urban and rural areas are common to all countries. In India they are a continuing source of concern and will become further aggravated unless innovative strategies are evolved to accelerate the development of rural infrastructure. A Vision of India 2020 must assess and try to anticipate the complexion of the future urban-rural divide. Recent evidence confirms that the rate of growth of urban centres in the country is declining more rapidly than was previously anticipated, though the proportion of people living in urban areas continues to rise. The 2001 Census, 27.8 per cent of the Indian population reside in cities, compared with 25.5 per cent in 1990.

The urban population is expected to rise to around 40 per cent by 2020. As India's cities continue to swell, the challenge of improving the urban infrastructure will be magnified. For instance, only 73 per cent of India's urban population has access to improved sanitation facilities, compared to the UMI reference level of 95 per cent. The trend towards concentration of urban population in a small number of large urban centres has been taking shape over a century. While the number of urban centres doubled between 1901 and 1991, the urban population increased eight-fold, resulting in a top heavy urban hierarchy.

Future demographic and economic growth is likely to concentrate in and around 60 to 70 large cities in the country having a population of a million people or more. The demographic trends towards urbanisation are accompanied by a change in the management and financing of urban development as a result of liberalisation. Decentralisation of municipal governance has led to a substantial reduction in budgetary allocations for infrastructural development. Greater reliance is now placed on institutional financing and capital markets for resource mobilisation and on private companies for service delivery.

Disparities in infrastructure between large and small urban areas have always been prevalent, but these disparities can be expected to increase significantly in future years. The stricter fiscal disciplines imposed by government and credit rating agencies will make it increasingly difficult for all but the largest urban centres to attract finance for infrastructural development. The large cities may be expected to experience modest to high rates of growth and to absorb a large part of the incremental migration in their peripheries and neighbouring towns.

Still greater inequality may be expected in the level of basic services across urban centres of different sizes by the year 2020, unless concerted initiative is taken to reverse

the trend. The gross inadequacies of infrastructure, especially those of public transport, water supply and sanitation, demand corrective action. A strategy for developing capabilities within the urban local bodies to cope with the infrastructural deficiencies must be adopted, with genuine decentralisation of financial and administrative powers and restructuring of the entities on the basis of sound management practices.

The face of urban poverty in 2020 is unlikely to be very different from what it is today, given that the largest indicator of poverty in cities is not so much lack of income, as lack of decent housing and civic amenities. These call for a change in a number of policies, especially those relating to land regulation, zoning, and development. The activities relating to provision of health care, water supply and sanitation, education and vocational training need to be carried out with higher levels of efficiency, to help the urban poor upgrade themselves. While improving infrastructure in existing cities/towns is not to be ignored, in the next two decades there will be need to encourage growth of new townships and take up regional urban development plans where growth corridors can be identified and public-private partnerships promoted for investment in alternative nodes of development.

Such centres will have considerable impact in improving the urban profile of the country. They will, however, need to make provision for adequate supply of water and sanitation as well as other civic amenities. They will also need to chalk out more liberal policies towards education and health care in order to attract a cosmopolitan group of investors, professionals and providers of other services.

While a large part of it will cater to the affluent, through imaginative planning the new centres can serve the needs of the hinterland also by offering avenues for employment, housing and other needs of a large workforce. Urban development represents one of the great challenges for India over the next two decades. Given the socio-political reality in India, it will be difficult for the private sector to bring about changes in the pattern of investment in infrastructure without the state becoming an active partner, bringing about the required legislative and administrative changes.

A satisfying outcome will depend on the formulation of effective public policy to accelerate all-round development of smaller urban centres and to refashion the role of the state as an effective facilitator to compensate for the deficiencies of market mechanisms in the delivery of public goods. Anti-poverty programmes should primarily be directed towards creation of a community-based infrastructure.

RURAL INFRASTRUCTURE

Along with the development of urban infrastructure, simultaneous efforts are also needed for strengthening rural infrastructure. Specific aspects of the rural infrastructure are discussed elsewhere in this report under the headings of education, health care, transport, telecom, power and water. Our vision is to create a rural infrastructure which connects every village with paved roads and telecommunication facilities, provides electricity and an assured supply of safe drinking water to all rural households, offers

access to quality primary and secondary education to all children and medical services to all citizens. The rural electrification programme, launched in 1951, has succeeded in bringing electricity to more than 5 lakh villages.

However, 80,000 villages are yet to get electricity connections. Out of these, 18,000 are in remote areas where electrification through the conventional electricity grid may not be feasible. To achieve 100 per cent rural electrification, we must rely on non-conventional sources of energy, especially for the remote villages. A detailed discussion regarding the potential for such energy sources is given elsewhere in this report. But the problem of rural infrastructure cannot be viewed or tackled in terms of a composite of separate services or in isolation from changes affecting the urban landscape.

A comprehensive and integrated strategy is required. The future development of urban centres will widen the disparities in civic infrastructure that exist between urban and rural communities, thereby stimulating greater migration to the cities and rapid expansion of urban slum areas. The natural growth of urban areas will make this trend inevitable unless bold steps are taken to promote an alternative, more geographically dispersed and equitable development paradigm.

Past efforts to develop satellite townships and growth centres in the vicinity of large cities have mitigated the concentration of population in central urban business districts to some extent, but in the process it has furthered the concentration of population in mega cities. Rural livings offer several considerable advantages over their urban counterparts: However, any successful alternative approach must address the crucial issues of infrastructure and civic amenities that make urban areas so attractive.

Comparative Advantages:

- Urban
 - *Domestic services:* Piped water, electricity, telecommunications and sanitation.
 - *Primary or localised services:* Shops, dispensaries, and playgrounds.
 - *Secondary services:* Schools, police stations, railway, fire-fighting services, and hospitals, which need not be local but must be readily accessible when required.
 - *Tertiary services:* airports, wholesale markets, referral hospitals, *etc.* which are less frequently demanded.
- Rural
 - Living space is much cheaper and more abundant.
 - The environment is much healthier.
 - Commuting times between work and home are much shorter.

One promising alternative is to link clusters of ten villages together by a high speed circular highway, thereby bringing 100,000 or more people into a circular community that can be crossed within 30 minutes travel time, and promoting a balanced and well spread out development of urban services along the periphery of the ring road. This arrangement would vastly reduce the length and cost of constructing good roads between all the villages, and enable establishment of quality services at any point around the

ring to be accessible to all members of the linked community. Telecom links and sanitation systems would naturally develop around the ring at far lower cost, and better quality schools and hospitals could develop to support the larger community. Industrial parks could be established to utilise the large workforce available within the ring. These and similar models need to be tried and modified, on a priority basis, otherwise it requires little stretch of imagination to anticipate the increasing congestion and immobility of large urban areas that is bound to occur in the near future.

TELECOM

Telecom acts as a stimulus for the development process. Over the past decade, India's achievement in this sector has been quite impressive. In terms of the overall size of main telephone lines in operation, the country rose from its 14th rank in the world in 1995 to 7th in 2001. India has adopted a gradual approach towards telecom sector reforms, through selective privatisation and managed competition in different market segments. Since the early 1990s, it has introduced private competition in value-added services, and national and international long distance telephony. Two state-owned public sector incumbents with a large existing subscriber base still dominate the fixed line service, in which they own about 35 million direct exchange lines (DEL).

While the number of DELs has multiplied six-fold since 1991, demand for new lines has grown nearly as fast, resulting in persistent shortfall of supply and a current waiting list of nearly 3 million lines. India has also achieved significant quality upgradation of its network in the 1990s, achieving almost 100 per cent conversion to digital lines. At the same time there has been a significant reduction in telecom rates paid by consumers. The opening up of the cellular market has unleashed real dynamism in the marketplace. Cellular mobile telephone subscribers in India increased from 77 thousand in 1995 to 7.3 million by June 2002, rising from a 0.6 per cent of the total subscribers to 17 per cent. This level is still relatively low compared with the average level of about 25 per cent achieved by low-income countries, 42 per cent by middle-income countries, and upwards of 50 per cent by upper middle income and high-income countries. The mobile explosion is helping in improving telecom penetration, bringing along the concomitant economic benefits of enhanced telecom accessibility, while at the same time promoting a better entrepreneurial culture, generating additional employment, and fostering a shift in the investment burden from the state to the private sector and the consumers.

Table. Number of Main Telephone Lines per 100 Inhabitants.

Country	2001
USA	66.45
Germany	63.48
Japan	59.69
UK	57.78
France	57.35

Korea, Rep.	47.60
Russia	24.33
Turkey	28.52
Brazil	21.69
China	13.81
India	3.38

In spite of a significant expansion of capacity, however, India's tele-density remains appealingly low and closing the digital divide in terms of tele-density remains a daunting task. In order to attain the current network size of UMI countries by 2020, India will have to increase the number of telephone lines seven-fold in 18 years. Targeted intervention through access promotion can be an important strategy for achieving growth with greater equity. The launching of private franchises for public call offices (PCOs) handling both domestic and long distance services has resulted in an increase in the number of PCOs from 0.2 million in 1993 to 0.9 million in 2001, including 0.4 million in villages. The presence of the private sector is widespread in PCOs and Internet Kiosks in the cities and towns, but remains insignificant in rural areas which account for only about 23 per cent of the country's fixed phone lines.

The aggressive expansion of public sector telecommunications infrastructure in geographically remote parts of the country, where private industry cannot venture profitably, is essential for unleashing the latent economic energies and market forces in these regions. Looking forward from now to 2020, the rapid expansion of India's fixed line network is likely to continue. As the fixed line market matures, more and more users will cross over to mobile communications as well, spurring a mobile revolution in India. Mobile telecommunications and the Internet are going to set the contours of further technological progress over the next two decades.

The recent convergence of voice and data transmission, global satellite systems, mobile handsets and calling cards have overcome the technological barriers of distance, topography and remoteness. By 2020, third generation (3G) mobile devices with access to mobile data and voice should be within the reach of wider parts of the Indian population. The last decade also witnessed a phenomenal growth in the use of Internet. Private Internet service providers have been allowed to set up international Internet gateways, as well as both satellite and landing stations for submarine optical fibre cables. Internet telephony is also opening up.

India must strive to complete the transition into digital switching and transmission, voice over Internet protocol (VoIP), broadband and 3G by 2020. Despite differences across countries, the global trend is towards growing privatisation and competition in telecommunications. Global experience suggests that this process is likely to undergo alternating cycles of differentiation and convergence. The convergent nature of technology may dictate mergers and acquisitions between companies in certain cases. Network operators and service providers may have to merge with content developers to add value to their services.

This may create temporary conditions of oligopoly until the emergence of new firms, offering multiple services, intensifies competition. India in 2020 is likely to see competition among big firms offering innovative value-added services to capture market share. The regulatory environment should therefore foster maximum flexibility and freedom to encourage innovation and expansion, consistent with this process of evolution. Rapid flow of information is a catalyst for social development. Vision 2020 conceives of India evolving into an information society and knowledge economy built on the edifice of information and communication technology (ICT), of which telecommunications is the springboard. Telecom is improving opportunities for people across different social strata.

A whole range of information based industries and applications have come up, creating new sources of employment and earnings with welfare enhancing consequences for both the poor and the wealthy. As a meta technology, ICT has caused rapid innovations to occur in other areas of material sciences. It has improved access in the fields of education, health care, governance and business services. It has also enhanced the ability of the poor to manage risks and mitigate vulnerabilities through provision of timely information. Rapid expansion and extension of the country's fixed and mobile telecom infrastructure is essential for stimulating growth of both the ICT sector and the economy as a whole.

TRANSPORT

Development involves a continuous increase in the number of physical transactions and the speed with which they occur, both of which are highly dependent on the size and quality of the nation's transport system. India's network of roads, railways, air services and ports is one of the largest in the world, serving a land mass of 3.3 million square km and a population of over one billion. Over the last 50 years, the growth of this network and of the demand for transport services has been enormous. The volume of railway freight has increased five-fold and passenger kilometres nearly seven-fold. The length of surfaced roads has multiplied nine-fold. The number of goods vehicles has increased 40-fold, buses 20-fold and four-wheel passenger vehicles 30-fold.

The tonnage of freight handled by the nation's expanding system of major and minor ports has grown more than 16-fold. Air-freight tonnage has increased about 30-fold. Currently, about 800 billion tonne km of freight and 2,300 billion passenger km are handled by the transport system, with the total transport demand growing at around 10 per cent a year for the past decade. The modal growth rates have varied, with road transport and air services growing at a much faster rate than the railways. The relative share of railways in total freight and passenger traffic has declined dramatically over the past five decades, from 90 to 25 per cent of inland freight and from 75 to 18 per cent of inland passenger.

Meanwhile, the share of road transport has increased from 12 to 60 per cent of freight and from 32 to 80 per cent of passenger business. Growing urbanisation is a

natural phenomenon and despite all its implications has beneficial effects on economic growth. Increasing population combined with continued urbanisation are likely to raise India's urban population to around 540 million over the next two decades, resulting in the emergence of about 60 to 70 cities with population of more than a million. Lack of adequate public transport has resulted in a rapid increase in private ownership of vehicles, particularly two wheelers, with consequent adverse effects of air and noise pollution.

In most cities, two wheelers comprise more than 70 per cent of total motor vehicles. Based on current growth rates, two-wheeler ownership in cities with more than 100,000 inhabitants is likely to rise from 102 to 393 per thousand population in the next 20 years, while the number of cars increases from 14 to 48 per thousand. To contain this explosion of personal vehicle movement in cities, massive investment in mass transport services will become necessary. Public investment in transport has declined steadily over the past four decades from 23 per cent of the total plan funds in 1960 to around 12 per cent in the 1980s and 1990s. Considering the developing nature of the country and the deficiencies in its transport system, a larger proportion of resources will need to be invested in improvements to the transport infrastructure.

Among the most daunting problems faced by India's cities are the inadequate public transport infrastructure, long travel times and high transport cost for both passenger and goods traffic. The growth of cities has been haphazard, expanding around a compact central core with industrialisation and migration bringing about rapid change in peripheral land use, resulting in urban sprawl. No organised, viable and significant public transport system is available in major cities, except the four metro areas of Mumbai, Kolkata, Chennai and Delhi. Bus and rail systems are the mainstay of public transport in these cities.

Transport planning must go hand-in-hand with land use planning. Specific plans will need to be formulated by each urban authority, starting with the provision of bus services, developing intermediate public transport and identifying corridors for future growth, including reserving land for such activity. In the long run, rail-based mass transport systems appear to be the only viable solution to the problems of urban transport in India's major metropolitan areas. Where flexibility of land use is possible, electrified surface rail transport can be adopted. In the case of highly developed and congested areas, underground or fly over systems will have to be considered. On medium density corridors, electric trolley buses or CNG operated buses may be best suited.

A key component of rural development is the provision of roads for connectivity, access being essential for social and economic well-being. Families residing along side roads benefit from better health, greater educational opportunities, smaller sized families and higher ownership of assets, compared to those families living in remote villages. While almost all villages with population above 1,500 are now connected by road, more than 45 per cent of villages with less than 1,000 people still need to be connected.

There are also significant regional imbalances in the connectivity of villages. Poor road maintenance is a related problem. The Government has allocated resources to

achieve all-weather road access to all villages with population above 1,000 by 2003, and for villages with population between 500 and 1,000 by 2007. While the progress made in the transport sector has been significant, it has not been able to meet the growing demand, particularly in the last decade when economic reforms triggered an unprecedented growth of GDP. Efforts to achieve even higher GDP growth rates in future years cannot be sustained without correspondingly greater efforts to strengthen the nation's transport system.

The existing issues and concerns, the congestion and constraints in movement leading to higher transport costs, and other unfinished tasks will be some of the challenges that need to be met during the next two decades. A huge increase in demand for both railways and road transport will require not only massive investments but also modification in pricing policies, increasing customer focus, improved terminal management, enhanced safety, reduced transit times, and other changes required to provide adequate and efficient services for meeting the growth in demand and public expectations.

Based on projected GDP growth of 8 per cent per annum, the total freight traffic is likely to reach about 5,500 billion tonne km by the year 2020, five times the level in year 2000. If a 10 per cent growth is achieved, the traffic demand would exceed 10,000 billion tonne km. Assuming that railways retain a 20 per cent market share, rail freight traffic, measured in ton km, will increase three to six-fold. The proportion of manufactured products to bulk cargo will also rise, with a larger proportion of liquids being carried through the pipelines. General merchandise will travel longer distances and in much smaller consignments, favouring smaller carrying units.

The rising value of consignments will place a higher premium on the speed of transit in order to minimise inventories. The total logistical management of transportation, marketing and distribution will become commonplace for most general merchandise. These changes, together with the impact of other factors such as energy efficiency, environmental pollution, and technological changes in each mode, will result in changing preferences for alternative modes of transport.

For medium distances up to 500 kms—with four-lane highways connecting the country, expressways coming up close to the main metro cities, and car ownership increasing rapidly—the bulk of the passenger travel may start moving towards 'own vehicle travel', a phenomenon already observed in most countries. Fast inter-city rail services will have an edge over air travel for this segment of traffic. Road transport is best suited for short distance traffic, except where volumes increase to very high levels and rail-based mass rapid transport offers a cost-effective alternative.

Future Trends in Transport:

- Improvement in the length, quality and speed of transport networks.
- Increase in productivity and economic efficiency of transport.
- Higher frequency and quality of services.
- Better control of air and noise pollution.
- Increased mobility for commuters, business and recreational travel.

- Improved urban transport systems.
- Corresponding improvements in rural transport.
- Fewer transport constraints to trade.
- Reduced transport transit times in general and at initial loading and terminal points.
- Better energy efficiency.
- Improved safety with fewer accidents.

Our vision of India 2020 is of a country with a well-developed network of roads and railways and adequate capacity to handle the growth in demand for transport. The volume of road traffic will multiply about five-fold, carried over a 70,000 km network of National Highways, including 5,000 to 10,000 km of expressways, linking the golden quadrangle of Delhi-Mumbai-Chennai-Kolkata-Delhi as well as northern, southern, eastern and western portions of the country, mostly with four or more lanes. State Highways with at least two way lanes will link most districts. Rural roads will provide access to the furthest outlying villages. Technological progress will generate vehicles that are pollution free and fuel-efficient.

An efficient public transport system will lead to a reduction in the population of two-wheelers in major urban areas. The railways will have to be expanded to handle a three-fold increase in traffic. Improved customer service, comfort and safety, a reduction in freight costs and tariffs, elimination (or at least reduction) of the uneconomic services, non-paying projects and subsidies will be necessary for this. India will need airports of international standards for passenger and cargo handling and modern handling systems at major ports to reduce delays in berthing.

Institutional arrangements will need to be in place with adequate funds for proper maintenance, especially of roads. Urban transport systems will be needed in all cities with a population of a million plus. We envisage that connection of several distributaries of the rivers through a network of canals will give impetus to the development of inland water transport

Investment requirements to meet these needs will increase to levels three to four times higher than the present in real terms. While the government will continue to be a major source of funds for infrastructure, internal generation of resources by the transport services will have to increase, supported by more realistic pricing of transport services and reduction in operating costs. The private sector will need to play a more central role in the development and operation of transport systems.

FOOD SECURITY

The single most important implication of India's rapid population growth during the second half of the twentieth century was the threat it posed to national food security. That threat reached dangerous proportions in the mid-1960s, leading to the launching of the Green Revolution, achievement of food self-sufficiency, and subsequently, a growing stock of surplus food grains by the mid-1970s.

Happily, such a threat no longer exists for the country. Growth of food production has exceeded population growth for each of the past three decades. Statistics present a confusing picture of India's progress on food security. Both per capita food grain consumption and total calorific intake have declined slightly in recent years among all levels of the population. At the same time, grain surpluses have reached peak levels and real per capita expenditure on food is rising among all income groups.

The factors influencing this trend are numerous and complex, however, it can be primarily attributed to:

- Reduction in calorie requirement due to a more sedentary life style among both the rural and urban population. Bicycle and bus travel, mechanised pumps and equipments, access to telephones and newspapers have reduced the physical work to a greater or lesser extent for most Indians.
- Diversification of the Indian diet to include a larger intake of fruits, vegetables, dairy products, sugar, oil and pulses, eggs, fish and meat products, thereby reducing the required intake of calories from cereals.

However, the challenge of achieving food security for all our people remains a real one even today, and will continue to demand our attention in the coming decades. Food security depends on both availability of food and access to it. Long-term food security requires not only producing sufficient food to meet the market demand, but also ensuring that all citizens have the required purchasing power to obtain the food they need for a nutritious and healthy life.

India has won the first battle, but the second still looms large. By best estimates, nearly half of the population still suffers from chronic under-nutrition. The most vulnerable are children, women and the elderly, specially among the lower income groups. While the number of children suffering from severe malnutrition declined significantly in the 1990s, the prevalence of mild and moderate under-nutrition, especially among those in the lower 30 per cent income group, is still high. Prevalence of micro nutrient deficiencies such as anaemia is also very high.

The problem of chronic macro and micro nutrient under-nutrition cannot be addressed simply by increases in food production or the accumulation of larger food buffer stocks. Nor has the public distribution system been able to effectively target the most needy in an effective manner. Targeted food for work programmes and targeted nutrition programmes can alleviate the problem temporarily. But in the long run, the solution is to ensure employment opportunities for all citizens so that they acquire the purchasing power to meet their nutritional requirements.

Thus, employment or livelihood security becomes an essential and inseparable component of a comprehensive strategy for national food security and must be considered as one of the nation's highest priorities. Increasing food consumption alone is not a sufficient condition for overcoming malnutrition in India. It is also necessary to address the factors responsible for the high incidence of gastrointestinal and respiratory infections as well as cultural factors responsible for faulty child feeding and weaning practices.

Assured provision of safe drinking water, improved health care and education for all women are necessary elements of a comprehensive strategy to eliminate malnutrition and achieve the goal of Food for All before 2020.

India's population is still rapidly expanding. Living standards are rising and slated to rise faster than in the past. As they rise, both calorific intake and diversification of diet will increase significantly. Although a portion of this increase can and will be obtained from abroad, a fulfilling vision of India 2020 depicts this nation—with its conducive and varied climate, the largest irrigated area in the world and a vast farming population—as a major food exporter. Both as a challenge and as an opportunity, India can and must do much more to modernise and diversify its agriculture to meet the increased domestic and international demand for a wide variety of food products. Continued growth of the agriculture sector is particularly important because it plays such a vital role in generating purchasing power among the rural population. Therefore, it is essential that agricultural development strategies for the 21st Century focus on generating both higher incomes and greater on-farm and off-farm employment opportunities. India is now entering the fourth stage of agricultural transformation. The Green Revolution phase spread in the north-western and southern states from the mid-1960s to the mid-1980s.

From the early 1980s, it spread rapidly in the central, and to a lesser extent the eastern states. This, coupled with a further growth of productivity in the north-western and southern states, enabled India to achieve a 3.77 per cent annual growth rate in agricultural production during the 1980s. The overall growth slowed to 2.72 per cent in the 1990s, which was associated with a reduction in public investment in both agriculture and agricultural research, slower growth of fertilizer consumption and the area under high yielding varieties, and degradation of soils.

Stages of Agricultural Development:

- *Pre-Green Revolution:* Boost in productivity growth of coarse grains and pulses per unit of land.
- *Green Revolution:* Expansion of area and rapid growth in productivity of wheat and rice, made possible by widespread adoption of improved varieties, expansion of agricultural research, demonstration and education, and investment in irrigation, supported by establishment of a national infrastructure to produce and supply inputs and to warehouse, distribute and market outputs.
- *Post-Green Revolution* Continued growth in productivity through intensification of chemical and labour inputs, followed by a gradual deceleration in productivity growth. Expansion of area under maize, cotton, sugarcane, and oilseeds.
- *Commercialisation* Further diversification of cropping patterns from low value to high value crops such as fruits, vegetables, flowers and other horticulture crops for domestic consumption, processing and export.

There is enormous scope for accelerating growth in agriculture, through improved soil nutrition and pest management, diversification into higher value-added crops,

expansion and more efficient use of irrigation potential, rainwater harvesting, and infrastructure development for agroprocessing industries. India's productivity on major crops ranks far below the world average. This low productivity contributes to low farm incomes and wages, lower on-farm employment generation, and relatively high food prices for all Indians.

Average yields from tomato cultivation, for example, are 76 per cent higher in Mexico and more than four times higher in the USA, while average yields on seed cotton are more than four times higher in Mexico and three times higher in USA. Even within India, some progressive farmers have achieved yields comparable to their foreign counterparts till recently; but these high achievers are a rare exception. Progressive farming must become the rule and the standard.

Raising the productivity of our soil and water resources is an effective way to increase the profit margins of farmers, while at the same time reducing the cost of farm produce, so that it is more affordable to the masses. A minimum target should be to double or triple the average yields of major commercial crops. This will reduce pressure on scarce land and water resources as well as enable the country to utilise less fertile areas for other purposes. Tapping the full potential of Indian agriculture to meet the rising domestic demand and to take advantage of the liberalisation of international trade will require, first and foremost, the recognition of the vital role that agriculture can continue to play in national development.

The other necessary conditions are: greater public investment in research; expansion and development of rural infrastructure including roads, storage capacity and organised markets; improved farmer education; effective involvement of the private sector to provide technology, investment and organisational expertise for commercialisation; and modification of land regulations to achieve greater production efficiency. A suitable land use pattern will also need to be implemented, based on the principle that each region would focus on crops best suited to their agro-climatic characteristics, soil types and water resources.

Strong measures will also be needed to address the problem of land degradation that affects an estimated 45 per cent of total land area. India's objective for 2020 must not only be to produce the food its population requires but also to fully exploit the comparative advantages it possesses—agro-climatic variety, irrigation, scientific capabilities and low labour cost—to become a low cost, high profit producer for the world market.

Figure depicts the projected food production under two scenarios, together with estimates of food demand in 2020. Business-as-usual (BAU) assumes that growth of production continues at the same rates as during the 1990s.

The Best-case scenario (BCS) assumes that production grows at the higher rates achieved during the 1980s. Even under the BAU, India will be able to meet the projected demand in all five food categories. This indicates that more emphasis can be placed on diversifying production to other value added agricultural products.

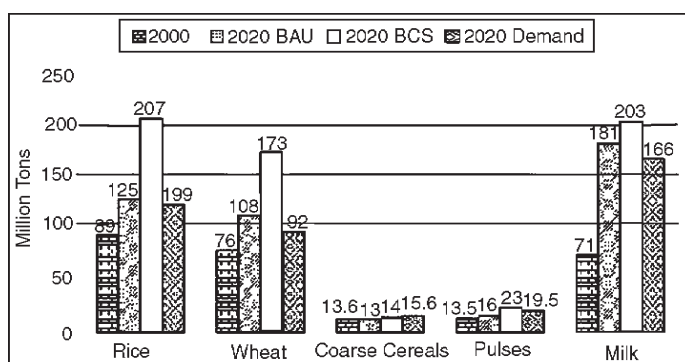


Fig. Current and Projected Food Production vs. Demand in 2020.

India needs to sustain an agricultural growth rate of 4.0 to 4.5 per cent in order to reduce food insecurity and poverty, while increasing rural purchasing power. At this growth rate, agricultural development could more rapidly diversify into horticulture, fishery, dairying, animal husbandry and other areas. It would also spur the growth of agro-processing industries in rural areas. Such an achievement is well within reach, provided there is the requisite commitment to raising crop productivity through dissemination of advanced technologies; increasing investment in irrigation, research and training; water harvesting and improved access to credit. While food production should be able to comfortably meet the total domestic demand, there will still be parts of the population that require assistance in order to meet their nutritional requirements.

In view of the high cost and inefficiency of the public food distribution system, reform should directly target the most vulnerable parts, which can be more effectively accomplished through programmes such as food vouchers or food stamps. For the vast majority of the population, food security can best be achieved by ensuring creation of gainful employment opportunities for all job seekers.

EMPLOYMENT

Population, food security, education and remunerative employment opportunities are closely interconnected. Rising levels of education and rising living standards are powerful levers for reducing birth and mortality rates. As population growth slows to replacement levels over the next two decades, India's greatest challenge will be to expand the opportunities for the growing labour force, to enrich their knowledge and skills through education, raise their living standards through gainful employment and make provisions for ensuring a good life for the aged.

India has met the challenge of producing sufficient food to feed everyone, but it has yet to meet the challenge of generating sufficient employment opportunities to ensure that all its people have the purchasing power to obtain the food they require. Gainful employment is one of the most essential conditions for food security and economic security. Conversely, food security is an essential requirement for raising the productivity

of India's workforce to international levels. India's labour force has reached 375 million approximately in 2002, and it will continue to expand over the next two decades. The actual rate of that expansion will depend on several factors including population growth, growth of the working age population, labour force participation rates, educational enrolment at higher levels and school drop-out rates. Projections based on these parameters indicate that India's labour force will expand by 7 to 8.5 million per year during the first decade of this century, and will increase by a total of about 160-170 million by 2020, *i.e.*, 2.0 per cent per annum.

Total unemployment in India has been estimated to be about 35 million persons in 2002. This figure takes into account the significant level of underemployment and seasonal variations in the availability of work. It also reflects wide variations in the rate of unemployment among different age groups and regions of the country. Approximately three-fourth of the unemployed are in rural areas and three-fifth among them are educated. The recent trends towards shedding excess labour to improve competitiveness and increasing capital intensity have further aggravated the situation. A clear consensus is now emerging that major changes in economic policy and strategy will be needed to meet the country's employment needs.

Future rates of unemployment will depend on a range of factors, including the growth rate of the labour force and changes in the structure of employment between different sectors, as well as the growth rate of the economy. Adopting the higher number takes us closer to understanding the full magnitude of the challenge the country faces for providing employment opportunities for all its people. India needs to generate on the order of 200 million additional employment opportunities over the next 20 years.

The first question that inevitably arises is whether generating nine or ten million jobs a year is feasible, and that naturally budgets the question of what rates of economic growth would be required for achieving this. However logical and inevitable it may sound, we believe this is the wrong way to approach the problem. The right question to ask is: How important is it to us as a nation to create employment opportunities for all? The answer here is simple. It is extremely important. It is as important to create job opportunities for all citizens in a market economy as it is to provide universal suffrage to all adults in a democracy. Access to employment is an essential component of freedom of economic choice. Absence of such opportunity means depriving our young not only of economic freedom but of hope as well. India's vision for 2020 must be founded on the premise of gainful Jobs for All. Access to employment should not only be a top priority of the government but a constitutionally guaranteed fundamental human right.

But can we possibly achieve such an ambitious goal and that too within the framework of a non-subsidised market economy? In order to answer this question, first of all we need to recognise that the economy we build over the next two decades and the jobs we create will be products of the decisions we make, not some irreversible logic of economic science. It is we, the nation, that have to decide what we want to accomplish and how serious we are about doing it, how willing we are to change our attitudes and to alter

our policies to achieve this desirable goal. We conclude that if the will and determination are present, the goal is achievable. Achieving full employment will require a reorientation of national priorities, technology policy and government action. Until now, planning to achieve national goals has been largely done on a sector-wise basis by respective ministries assigned with the responsibility. These parallel lines of planning need to be integrated around a central vision and set of goals, of which full employment must be one. As we have incorporated an environmental analysis into all our planning, every plan initiative needs also to be re-evaluated to consider its impact on employment.

Despite a persistent tendency to associate employment with large industry in the organised sector, an in-depth examination of employment potentials makes it evident that the largest share of new jobs will come from the unorganised sector. In evaluating the importance of developing each sector, we must include an assessment of its potential contribution to employment. Such an assessment will have to be made for the unorganised sector, which currently contributes 92 per cent of the country's employment and generates seven times greater labour intensity per unit of production, as compared to the organised sector. The public organised sector has been and will continue to shed jobs. Although the private organised sector will contribute significantly to the growth of the economy, its contribution to the overall employment generation will be quite modest, since total employment in this sector currently represents only 2.5 per cent of all jobs. Even if this sector grows by 30 per cent per annum, over five years it will contribute less than one per cent to the growth of the workforce.

International comparisons reveal that small and medium enterprises (SMEs) create the majority of jobs. In the USA, nearly half of the private workforce is employed in small firms, of which three-fifth have less than five employees. In Japan, 78 per cent of jobs are generated by small and medium enterprises. The small and medium manufacturing enterprises in Korea account for 99 per cent of all manufacturing enterprises and 69 per cent of employment in this sector. Therefore, the unorganised sector, including small and medium enterprises, must play a central role in the country's employment strategy.

This will require modification of policies and programmes to level the playing field, improve availability of credit, increase productivity, raise quality consciousness and competitiveness, and enhance job quality. Recent experiences of different countries in the context of globalisation also demonstrate that SMEs are better insulated from the pressures generated by the volatility of world trade and capital markets. They are more resistant to the stresses, and more responsive to the demands of the fast-changing technologies and entrepreneurial responses.

Indeed, they are observed to be a very important vehicle for new technology adoption and entrepreneurial development. What is true of the most developed countries today will be true for India in 2020. SMEs will play a crucial role in ensuring India's international competitiveness and its rapid assimilation of new technologies. An assessment of the different sectors reveals a vast untapped employment potential in a

wide range of fields for unskilled, semi-skilled, skilled and professionally educated workers. A list of specific sectors with the largest gross employment potential per unit of output.

High Employment Potential Sectors:

- Commercial agriculture
- Agro-industry and agri-business
- Afforestation for pulp, fuel and power
- Retail and wholesale trade
- Tourism
- Housing
- Construction
- Garment industry
- Other small scale and medium industries
- IT and IT enabled services
- Education
- Health
- Financial services
- Transport
- Communications
- Community services.

The growth of food grain production may not lead to a significant growth in on-farm employment opportunities. However, there is substantial scope for creating new jobs through watershed development programmes, expansion of the area under irrigated cultivation, raising crop yields which increases labour intensity, and diversification of cropping patterns into cash crops—especially vegetables and horticultural crops. Together, these could generate upwards of 20 to 30 million new on-farm employment opportunities during the next decade.

India processes less than 2 per cent of its fruit and vegetable products, as compared with 70 to 80 per cent in countries such as Brazil, Malaysia and Philippines. The development of downstream processing, packaging and distribution activities can generate millions of additional off-farm jobs. Policies are needed to attract greater private sector participation in terms of land development, production and processing technologies, investment, management and marketing.

A concerted effort to fully develop the potentials of agri-business could generate millions of additional jobs. The recorded forest area constitutes about 23 per cent of the geographical area (about 71 million hectares) of the country, but half of this area is degraded. Hence, it is unable to play an important role in environmental sustainability and in meeting the forest produce needs of the people, the industry and other sectors. India has more than 50 million hectares of degraded wasteland that lie outside the national forests, in addition to 30 million hectares within protected areas. In spite of this huge expanse, the country is a net importer of forest products to the extent of Rs. 2.5

billion annually. Afforestation of India's vast expanse of wastelands and depleted forest areas for production of wood pulp, timber, fuel, fodder, biomass power, edible and fuel soil, fuel, herbs and medicinal plants for exports can create millions of jobs, while reversing environmental degradation and supporting industrial development.

Every two hectares of additional area placed under plantation of forest crop such as bamboo, casuarina, eucalyptus or oil-bearing plants can generate year-round employment for one person. Development of forests can generate 10 to 15 million additional employment opportunities over the next five year plan period and increase the livelihood opportunities of low income families during the transition period, while educational levels are rising and other sectors of the economy are developing.

The contribution of primary sector employment to the total workforce has been coming down very slowly, to 56-57 per cent in 1999-2000, and should decrease to about 51 per cent by 2006-2007. This slow rate of decline is likely to continue for at least a decade, because of the hidden under-employment in the agriculture sector, partly compensated for by the vast potential for additional job creation by crop diversification and afforestation. Increasing prosperity in agriculture will naturally lead to the growth of non-farm jobs in agri-industries, agro-business and other occupations required to meet the needs of an increasingly prosperous farming community.

During the second decade of the 21st Century, increasing domestic demand for manufactured products and services, coupled with more rapid mechanisation of agriculture will draw in more and more people to non-farm occupations. By 2020, total employment in agriculture may fall to less than 45 per cent, while the share of the services sector increases proportionately. The small scale industries (SSI) sector accounts for 95 per cent of industrial units, 40 per cent of value addition, 35 per cent of exports, and 80 per cent of manufacturing employment.

Registered SSI units provide nearly 18 million jobs in the country at this time. Among manufacturing sectors, the single largest employment potential is in textiles, which is slated to generate 7 million jobs over the next five years alone. More than 40 per cent of these jobs are in garment production units in the SSI sector. A healthy and rapidly expanding small sector is essential for a vibrant growth of the Indian economy as a whole. This sector serves as the field for entrepreneurship to flourish, as an entry point for new entrepreneurs who can start small and then grow, as a vehicle for extending the regional spread of industry, as a laboratory for development of innovative products and services, and as an essential support to attract large manufacturing assembling industries from overseas.

Since SSIs are generally more employment intensive per unit of capital than large scale industry, they are also a source of much needed employment. Employment in the registered SSI sector has nearly tripled over the past 20 years. A repetition of this performance would generate an additional 36 million jobs over the next 20 years. A comprehensive package of venture capital, credit, liberalisation of controls, technical training, marketing and management measures is needed to ensure the continuous

expansion of this sector. By far, the largest number of new jobs will be in the services sector. Over the past 100 years, the US economy has more than quadrupled the total number of those employed, yet the percentage of the US workforce engaged in manufacturing is now back to the same level as it was in 1850, while agricultural employment has shrunk to less than 3 per cent of the US workforce. Today, the fastest growing job categories in the USA are in services industries such as financial services, insurance, education, health, construction and real estate.

Rapid growth is slated for a wide range of services in India, including professional, computer-related research and development, real estate, leasing, advertising, printing and packaging, marketing, telecommunications, postal and courier, audiovisual, engineering and construction, wholesale and retail distribution, all levels and types of education, environment, banking, insurance, health, travel, sports and recreation, and all categories of transport services. By the year 2020 more than 120 million jobs will come from the services sector alone.

Tourism-related occupations, including hotels and restaurants, employ 10.8 per cent of workers globally, compared to only 5.6 per cent in India. Domestic tourism will rise rapidly as living standards increase. India's domestic tourist sector is the fastest growing in the world, but with the lowest level of investment. The potential for international tourism too has not yet been exploited properly. China currently attracts more than five times as many foreign tourists, excluding visitors from Hong Kong.

Thailand, Malaysia, and Turkey attract three to four times as many. By one estimate, development of India's tourism infrastructure such as roads, airports and medium priced hotels, combined with modifications in air and hotel pricing and tax policies, could generate more than 20 million additional employment opportunities in tourist related businesses within a decade. Numerous studies have been undertaken to assess the potentials of the IT industry and to identify specific strategies needed to accelerate development of this sector. The NASSCOMMcKinsey study in 1999 projected creation of a Rs 70 billion IT industry, employing more than two million persons within 10 years. The worldwide market for IT services is expected to exceed Rs 900 billion by 2010. The recent growth of IT-enabled service businesses in India—call centres, medical transcription, technical support and back office processing, engineering and design, geographic information services, payroll and other human resource services, insurance claim processing, legal databases—is powerful evidence that the potential of IT technology and knowledge based industries extends far beyond the development of software and hardware.

The US experience shows that, as the application of information technology spreads and encompasses traditional industries, it can generate new employment opportunities ten times greater in number than those directly involved in core IT industries. IT is also a stimulant to the growth of home-based employment opportunities, especially suitable for women. The trend towards tele-working, which is growing rapidly in USA and some other countries, is just beginning in India. The lower cost and greater convenience

of home-based employment is bound to open up greater job opportunities for educated women with children. IT services will be a powerful engine of job growth, but the term 'Knowledge-based industries' includes a much wider range of commercial opportunities.

It encompasses all fields in which the application of mind, judgment and skill, rather than the application of mechanised production technology is the core resource. Education, health services, biotechnology, pharmaceuticals, insurance and financial services are among the leading industries in this category. They are also among the fastest growing industries in the world. Added to these, there is enormous scope for other knowledge intensive activities such as clinical drug trials and many other types of scientific research.

The management of all types of information is emerging as a major growth industry worldwide and India is well poised to become a global leader in this field. Increasing demand for education within the country and worldwide will create tremendous demand for qualified teachers with adequate language skills. An estimated two million additional teachers will be needed in India to support 100 per cent primary school enrolment and to reduce the teacher-student ratio for improved quality of teaching.

Another two million or more will be required to support higher rates of upper primary and secondary school education and to reduce teacher-student ratios at those levels. Teaching staff for vocational training and higher education need to be expanded as well. Demand for health services is increasing worldwide. Already nurses and medical technicians are in short supply. The USA has a doctor-nurse ratio of about twice that of India. Yet the US currently faces a shortfall of nurses, that is more than three times India's total annual production of new nurses. Physicians, nurses, medical technicians and other scientific occupations will become growth industries to rival the IT sector within the next decade. Creation of new jobs in existing industries can and should be supplemented by creation of new services industries that will in turn stimulate increased employment opportunities.

It is extremely difficult to project unemployment rates 20 years into the future, when so much doubt exists even about current levels of unemployment and underemployment. However, it is clear that rising levels of education and growth of the 60 plus age group will mitigate to some extent the growth of the labour force. Combined with the enormous opportunities for creation of new employment opportunities, the incidence of unemployment should be almost eliminated during the second decade of the century.

EDUCATION

Successful population policy is directly linked to successful education policy. Success in raising literacy rates and school enrolment rates while reducing drop-out rates, especially for women, are closely correlated with the delayed onset of marriage and child birth, improved mortality for both mothers and children, and reduction in family size. In fact, a successful education policy forms the bedrock of all fields of national development—political, economic, technical, scientific, social, and environmental.

Education is the foundation for a vibrant democracy in which informed citizens exercise their franchise to support the internal growth of the nation and its constructive role in the world community. It is the foundation for growth in productivity, incomes and employment opportunities and for the development, application and adaptation of science and technology to enhance the quality of life. Education is the foundation for access to the benefits of the information revolution that is opening up vistas on the whole world. Education is also the foundation for improved health care and nutrition.

Literacy, the basis of all education, is as essential to survival and development in modern society as food is to survival and development of the human body. Literacy rates in India have arisen dramatically from 18 per cent in 1951 to 65 per cent in 2001, but these rates are still far from the UMI reference level of 95 per cent. Literacy must be considered the minimum right and requirement of every Indian citizen. Vast differences also remain among different parts of the population.

Literacy among males is nearly 50 per cent higher than females, and it is about 50 per cent higher in urban areas as compared to the rural areas. Literacy rates range from as high as 96 per cent in some districts of Kerala to below 30 per cent in some parts of Madhya Pradesh. Rates are also significantly lower among scheduled castes and tribes than among other communities.

These differential rates of progress leave approximately 300 million illiterate adults in the country, the largest number of illiterates in the world. For these people, the traditional avenues of knowledge dissemination through education and printed information are ruled out. These are the people who are most vulnerable to the challenges of development, because they are least equipped to rapidly expand their knowledge base. Since many of them are relatively young adults who will still be active in 2020, the country cannot afford to ignore them or leave them behind. The Government has already set a goal to achieve 75 per cent literacy by the end of the Tenth Five Year Plan.

A 100 per cent literate India is of paramount importance for realising the vision for the country in 2020 as presented in this document. Even when this goal is achieved, innovative approaches will be needed to increase knowledge dissemination through TV and other means, to literate adults with little or no formal education. Literacy is an indispensable minimum condition for development, but it is not sufficient. In this increasingly complex and technologically sophisticated world, ten years of school education must also be considered as an essential prerequisite for citizens to adapt and succeed economically, avail of the social opportunities and develop their individual potentials.

Education is the primary and most effective means so far evolved for transmitting practically useful knowledge from one generation to another. India's education system has expanded exponentially over the past five decades, but its current achievements are grossly inadequate for the nation to realise its potential greatness. The net enrolment

rate in primary schools is around 77 per cent and in secondary schools it is around 60 per cent. These compare with the 99.9 per cent primary and 69 per cent secondary enrolment for the UMI reference level. The drop out rate was 40 per cent at the primary level and 55 per cent at the upper primary level in 1999-2000. These high drop out rates from both primary and secondary school, combined with low enrolment rates at the higher levels deprive tens of millions of children of their full rights as citizens.

Out of approximately 200 million children in the age group 6-14 years, only 120 million are in schools and net attendance in the primary level is only 66 per cent of the enrolment. Further, less than 7 per cent of the children ever pass the 10th standard public examination. Apart from addressing the needs of a large illiterate population, India's knowledge strategy must also develop innovative approaches to enhance knowledge acquisition among the large community of school drop-outs. Unless something is done to drastically reduce drop-out rates, by the year 2016 there will be approximately 500 million people in the country with less than five years of schooling, and another 300 million that will not have completed high school.

In other words, about two-thirds of the population will lack the minimum level of education needed to keep pace with and take advantage of the social changes occurring within the country and worldwide. Extending the primary school system to over 500,000 villages in India has brought education to the masses. Unfortunately, this huge quantitative expansion has been accompanied by a tremendous dilution in the quality of schooling. High drop out rates in rural areas is one result of single room schools, with few teaching aids and inadequate instruction both in terms of quantity and quality.

Qualitative improvement in the system can be accomplished by promoting centralised schools serving clusters of 10 or more villages, wherever distances and transportation links make that feasible. This will permit greater investment in educational infrastructure, including introduction of computers. Achieving 100 per cent enrolment of all children in the 6 to 14 year age group is an ambitious but achievable goal for 2020. This must be coupled with efforts to increase the quality and relevance of school curriculum to equip students not only with academic knowledge but also with the values and practical knowledge needed for success in life. A business-as usual scenario for primary and secondary education in 2020, based on recent trends as well as an alternative scenario designed to radically enhance the quantity and quality of school education in the country.

A tremendous expansion of schools and classrooms will be required to support a quantitative and qualitative improvement in the country's school system. In order to achieve the best-case scenario total school enrolment would have to increase by 75 million or 44 per cent. That will require a proportionate expansion in the number of classrooms. In addition, efforts to improve the quality of education by reducing the class size would require a further 20 per cent increase in the number of classrooms. Together, this will necessitate increasing the total number of classrooms by 65 per cent within 20 years.

Table. Education Scenarios in 2020.

	1980 Actual	2000 Estimated	2020 Business- as-usual	2020 Best-case Scenario
Primary enrolment (1-5)	80%	89%	100%	100%
Elementary enrolment (1-8)	77%	79%	85%	100%
Secondary enrolment (9-12)	30%	58%	75%	100%
Drop-out rate (1-5)	54%	40%	20%	0%
Drop-out rate (1-8)	73%	54%	35%	0%

An enormous increase in the number of teachers will also be required to achieve the alternative scenario, *i.e.*, eliminating primary school drop outs and reducing the teacher-pupil ratio from the present high level of 1:42 down to around 1:20, which is the UMI reference level. Together, this will require an additional three million primary school teachers, more than twice the number currently employed.

Similar increases will be required at middle and secondary school levels. The training of such large number of teachers will require the establishment of additional teacher training colleges and much larger budget allocations for teachers' salaries.

Qualitative improvements in education should reflect a change in pedagogical methods and lay emphasis on several dimensions, including:

- A shift from methods that emphasize passive learning to those that foster the active interest and ability of children to learn on their own.
- A shift from rote memorisation to development of children's capacity for critical thinking.
- A shift from traditional academic to practically relevant curriculum.
- A shift from imparting information to imparting life values such as independent thinking, self-reliance and individual initiative that are essential for success in any field of endeavour.

An important role of education is to foster in each child the attributes and values of a responsible, capable, active and healthy member of the family and society. The rigidity of curriculum, testing and teaching methods need to be relaxed so that innovative methods and new models of education can be evolved, tested and perfected. Vocational streams have to be developed and expanded to equip larger numbers of high school students with occupation-related knowledge and skills. Experimentation is needed with new methods for knowledge delivery. Television can be a very effective means for educating both school going and non-school going children and adults. It can deliver teaching materials in a more dynamic, entertaining, and interesting manner, utilising the nation's best teachers and multimedia teaching materials on each subject. A TV based curriculum can help slow learners to supplement classroom teaching, fast learners to learn at much faster rates than the rest of the class, drop outs to acquire knowledge they missed out in school, and adults to expand their level of education without returning to school.

New methods of delivery will be particularly necessary to augment access and improve delivery at higher levels of education. Although India's college and university network has expanded dramatically, it is able to accommodate only a tiny fraction of the college-age population. In addition, the quality of facilities, teaching and course materials leaves much to be desired. Recruitment and promotion are highly politicised; seniority rule precludes merit advancement; and the cost of delivery is beyond the means of the vast majority of young Indians.

Table. Growth of Higher Educational Institutions.

Years	Colleges for General Education	Colleges for Professional Education	Universities
1951	370	208	27
1998	7199	2075	229

India in 2020 must be a nation in which all those who aspire for higher education have access to college and university level courses. A national network of community colleges, similar to the highly successful American system, is needed to provide knowledge and job-oriented skills to millions of young people who lack interest in or capacity for more stringent academic studies. The advent of computer and the Internet-based educational methods offer an exciting new learning medium that can literally transform our concept of school and classroom from physical into virtual realities.

Studies in the USA project a radical reshaping of higher education over the next two decades as a result of the digital revolution. Many traditional colleges will close as more course works are delivered at a distance through alternative channels. The traditional boundaries between education and other sectors will fade, as publishers, for-profit and non-profit organisations, offer accredited, multimedia-enhanced courses directly to students, by-passing the university.

The traditional classroom type of education, which is most useful for students that require personal attention and assistance and for subjects that involve hands-on experimentation, will no longer be the predominant model of education. For all other purposes, it is very costly and not very efficient in the way it uses the time of both teachers and students. Experience shows that computer-based educational methods can lead to much faster rates and higher quality of learning, which is more interactive and motivating for students at all levels of education from pre-school to post-graduation. It is extremely effective for enhancing reading and language skills and general knowledge among the very young and even for some sophisticated professional courses such as medicine and engineering.

Given the huge number of young students that will quest for all levels of higher education in the coming decades and the severe shortage of qualified instructors, and in the light of India's outstanding expertise in the IT industry, the country needs to embark on a massive programme to convert the entire higher educational curriculum into a multi-media, web-based format and to establish accreditation standards for recognition

of the distance education so imparted. Our vision of India in 2020 is predicated on the belief that human resources are the most important determinants of overall development.

As India's IT revolution has been fuelled by the availability of a very large reservoir of well-trained engineers, its future development in many different spheres will depend on commensurate development of sufficient and surplus capabilities. Full development of India's enormous human potential will require a shift in national priorities, to commit a greater portion of the country's financial resources to the education sector. India currently invests 3.2-4.4 per cent of GNP on education. This compares unfavourably with the UMI reference level of 4.9 per cent, especially with countries such as South Africa, which invests 7.9 per cent of GNP on education. A near doubling of investments in education is the soundest policy for quadrupling the country's GDP per capita.

S&T CAPABILITIES

Literacy and general education form the base of the knowledge pyramid that is essential for rapid and sustained development in the 21st century. The continuous advancement of science and application of improved technology form the middle rung; and social ideals and spiritual values form the apex. Technical education, both vocational and professional, constitutes the foundation for development of science and technology. India is rightly proud of the international standing of its IITs, but a handful of world class technical institutions is not sufficient. A large number of the country's approximately 500 engineering colleges need to be upgraded to quality standards nearer to those of the IITs, and given similar autonomy.

Private sector initiative and investment, whether from Indian corporates or NRIs or reputed foreign universities, need to be fully encouraged. Close links need to be fostered between technical institutions and industry. India's enormous manpower base of scientists and engineers is often coveted. President Kalam observed that India's human resource base is one of its greatest core competencies. This is true in absolute terms, but as a percentage of the total population, we are at 1/100th of the US levels and 1/50th of the Korean level. Even China's manpower base of scientists and engineers as a percentage of its population exceeds ours by more than three times. Absolute numbers do count, but manpower alone is not sufficient to excel in the development and application of science and technology.

In terms of total investment in R&D, India's expenditure is 1/60th of that of Korea, 1/250th of that of the USA, and 1/340th of that of Japan. More significantly, atomic energy, space and defence research account for 71 per cent of all central spending on science and technology, which means that relatively little is left for investment in agriculture, energy, telecommunications and other crucial sectors within the sphere of science and technology. R&D expenditure even in India's fast-growing IT sector has been averaging around 3 per cent of sales turnover (STO), which is much lower as compared to the 14-19 per cent expended by internationally reputed software firms. These low figures reflect on our R&D performance. India's share of global scientific

output in 1998 was only 1.58 per cent of the world's total. Out of 500,000 new patent applications filed globally each year, China accounts for 96,000 and Korea accounts for 72,000, while India accounts for only 8,000. Of greater concern has been the country's inability to capitalise on our huge pool of manpower and extensive network of scientific research organisations for transferring proven technologies from the lab to the land and to the factory.

Despite possessing the world's largest cadre of agricultural scientists, we have not been able to extend the momentum of Green Revolution to other regions and crops and to update the scientific practices of our farmers to levels comparable with most other nations. Crop productivity remains far below and production costs far above world averages. A similar gap exists in the application of science and technology for food processing and many other industries. High technology exports account for only 6 per cent of total manufacturing exports for India, compared to over 20 per cent for the UMI reference level. India has a number of premier universities for scientific and technical education which produce world class scientists and engineers, but the national research system rarely produces world class results. One reason is the missing link between scientists in universities and research institutes and practitioners in agriculture and industry.

Another is that non-teaching research institutes lack the continuous inflow of young talented researchers required to challenge assumptions and infiltrate new ideas and approaches. Bureaucracy, tenure and lack of accountability minimise the pressure for practical application of scientific knowledge. Other countries have done a much better job of providing optimum incentives to their scientific community and creating close linkages between science and industry. The management of technology and the administration of government require different sets of skills and a different culture.

Management of technology demands innovation, global competitiveness, and wide latitude for individual freedom, characteristics that are difficult to foster under an administrative regime. The objectives of many institutes and the mechanisms for achieving them need a radical reorientation to bring them in tune with the changing environment brought about by liberalisation of imports, technology tie-ups with foreign agencies and foreign direct investment in key sectors. If the national R&D bodies do not reorganise themselves to face globalisation and global competition, nor take into consideration the nature of the factor endowment of the country along with many existing indigenous technologies so far neglected, then they risk becoming irrelevant to the mainstream activities of industrial development.

The nature of service offered by these institutions has to undergo a sea change. Grass-root consultancy and turnkey project consultancy, including project implementation and monitoring are increasingly in demand. Another essential step is to improve the linkages between technology development and technology application by fostering close ties between basic research and business. In the USA, a large proportion of scientific research at the university level is financed by business, and most leading

universities operate technology incubators designed to promote commercialisation of new products and processes. India's recent experiment with the Science and Technology Entrepreneur Parks (STEPs) has been successful in promoting application of technologies, new business start-ups, and employment.

The STEPAs in combination with technology incubators need to be rapidly multiplied around the country. Effective R&D management backed by suitable incentives for commercialisation are essential for bridging the gap between the lab and the land or factory. R&D is not only essential for propelling growth in industrial fields. It can also become a growth industry on its own. Much of the work done by Indian software companies for overseas clients and parent companies rightly fall under the category of new product development. Vast potential exists for complementing India's strengths in IT with corresponding strengthening of fields such as biotechnology, pharmaceuticals, designer-made materials, and others.

India's investment in the biotechnology sector is expected to increase five-fold by the end of the current decade largely as a result of growing collaboration between multinational corporations and indigenous research efforts. Huge opportunities also exist for R&D in the field of genomics, bioinformatics, DNA technologies, clinical studies and genetically modified crops. Rationalisation of procedure, backed by effective bio-safety regulations are needed to govern testing and approval of biotech products. India has the potential of converting its strong R&D infrastructure into a global research, design and development platform.

VOCATIONAL TRAINING

The knowledge and skill of our workforce will be a major determinant of India's future rate of economic growth as well as the type and number of jobs we create. The greater that knowledge and skill, the higher will be the productivity, the better the quality, and the lower the cost of the products and services we generate. Similarly, the better the quality and lower the cost, greater will be the comparative advantage and market potential. Currently only 5 per cent of the country's labour force in the 20-24 age category have undergone formal vocational training, compared with 28 per cent in Mexico, 60 to 80 per cent in most industrialised nations, and as much as 96 per cent in Korea. A strategy to achieve full employment must include as an important component, a strategy to ensure that all new entrants to the workforce are equipped with the knowledge and skill needed for high productivity and high quality. India has over 4,200 industrial training institutes (ITI) imparting education and training in 43 engineering and 24 non-engineering trades. Of these, 1,654 are government run ITIs, while 2,620 are private.

The total seating capacity in these ITIs is 6.28 lakh. Most of this training is conducted in classroom style in the form of one to two year diploma courses. In addition, about 1.65 lakh persons undergo apprenticeship vocational training every year in state-run enterprises. If a wider definition of applied courses is taken that includes agriculture,

engineering and other professional subjects, the total number receiving job related training is about 17 lakh per annum, which still represents only 14 per cent of new entrants to the workforce. The nature of vocational skills makes it impossible for vocational schools to fully address the nation's needs.

The variety of skills needed by the workforce is far too great. The changes in technology and work processes are too rapid for training courses and their instructors to stay up-to-date. The cost of training is also relatively high as it often demands full time enrolment for a prolonged period. Some vocational fields do not lend themselves to classroom or laboratory study at all. A comprehensive strategy is thus needed to enhance the nation's employable skills. It must begin by preparing a catalogue of the entire range of vocational skills needed to support the development of the country. The network of vocational training institutes and the range of vocational skills taught needs to be expanded substantially to impart those skills for which institutional training is most suitable.

The private sector, which promoted the rapid proliferation of computer training institutes throughout the country, should be encouraged to recognise the commercial potential of vocational training in many other fields. In addition, it is essential to fashion more effective and efficient mechanisms for disseminating useful knowledge and skills, especially through the TV media and through computerised vocational training. The importance of computer has been widely recognised as a means to improve efficiency in business, government and formal education, but its application in vocational training is not fully appreciated.

Use of Computers in Vocational Training:

- Rates of learning on computer for both academic as well as vocational or skill-based subjects are four to ten times faster than they are in a classroom setting, and learning retention is likely to be much higher.
- Computers can provide multimedia, interactive, customised and individually paced learning with instantaneous feedback and testing.
- It eliminates the need for producing and deputing a large number of highly skilled instructors. Course contents can be rapidly modified to reflect changing needs.
- For many types of vocational skills, computerised training also offers specific advantages over the live delivery of skills in a classroom.
- Training can be delivered wherever computers are available. A nationwide network of 50,000 computerised vocational centres, run as private self-employed businesses similar to the STD booths and Internet cafes, can deliver low-cost, high-quality training to 10 million workers every year—more than five times the total number covered by existing programmes.

Although 58 per cent of Indians are engaged in agriculture, vocational training for farmers is one of the weakest links in the Indian educational system. Agricultural universities cater to the nation's need for agricultural scientists and extension officers.

An extensive network of more than 300 Krishi Vignan Kendras (KVK) offers short and medium term courses for farmers on specialised subjects. The KVK network provides training to several hundred thousand farmers each year, but it can cater to the needs of only a miniscule percentage of all farmers.

Agricultural education can be moved from the campus into the village by establishing a national network of farm schools, offering practical demonstration and training on lands leased from farmers in the local community. The prime objective of the schools would be to impart knowledge and skills designed to double yields on important commercial crops. Educated farmers can be trained as self-employed instructors to operate the farm schools as private enterprises on a commercial basis.

HEALTH FOR ALL

The health of a nation is difficult to define in terms of a single set of measures. At best, we can assess the health of the population by taking into account indicators like infant mortality and maternal mortality rates, life expectancy and nutrition, along with the incidence of communicable and non-communicable diseases. The health of the Indian population has improved dramatically over the past fifty years. Life expectancy has risen from 33 years to 64 years.

The infant mortality rate (IMR) has fallen from 148 to 71 per 1000. The crude birth rate (CBR) has declined from 41 to 25 and the crude death rate (CDR) has fallen from 25 to under 9. The couple protection rate (CPR) and total fertility rate (TFR) have also improved substantially. Despite these achievements, wide disparities persist between different income groups, between rural and urban communities, and between different states and even districts within states. The infant mortality rate among the poorest quintet of the population is 2.5 times higher than that among the richest. Maternal mortality remains very high. More than one lakh women die each year due to pregnancy-related complications.

Like population growth and economic growth, the health of a nation is a product of many factors and forces that combine and interact with each other. Economic growth, per capita income, employment, levels of literacy and education—especially among females—age of marriage, birth rates, availability of information regarding health care and nutrition, access to safe drinking water, public and private health care infrastructure, access to preventive health care and medical care, health insurance, public hygiene, road safety, and environmental pollution are among the factors that contribute directly to the health of the nation.

Communicable diseases such as malaria, kalaazar, tuberculosis and HIV infection remain the major causes of illness in India. During the next five to ten years, existing programmes are likely to eliminate polio and leprosy and substantially reduce the prevalence of kalaazar and filariasis. However, TB, malaria and AIDS will continue to remain major public health problems. India has about 1.5 million identified cases of TB that are responsible for more than 3,00,000 deaths annually. Improved diagnostic

services and treatment can reduce the prevalence and incidence of TB by 2020. About 2 million cases of malaria are reported in India each year. Restructuring the “malaria workforce” and strengthening health infrastructure can reduce the incidence of this disease by up to 50 per cent within a decade.

Assessing the impact of HIV epidemic is more difficult; there are about 4 million persons infected with HIV. The National Health Policy aims at achieving a plateau in the prevalence of HIV infection by 2007. Childhood diarrhoea, another major cause of illness, is largely preventable through simple community action and public education, and deaths due to diarrhoea can be eliminated by 2010. Childhood under-nutrition, the other major area of concern, can be addressed by targeting children of low birth weights and employing low-cost screening procedures to detect under-nutrition at an early age. Given the projected improvement in living standards, food security, educational levels and access to health care among all levels of the general population, substantial progress can be made in reducing the prevalence of severe under-nutrition in children by 2020.

China’s remarkable success in combating disease over the past two decades is proof that a determined commitment to improving public health can dramatically reduce the incidence of infectious diseases within one or two decades. With the demographic and epidemiological transition taking place in the coming years, non-communicable diseases are also likely to emerge as major public health problems. Modernisation of life styles will further aggravate health problems. The rapid proliferation of two and four wheel motor vehicles, increasing congestion on city roads and intercity highways have all contributed to an increasing number of deaths and serious injuries from traffic related accidents.

Greater emphasis on education and enforcement of road safety rules by both drivers and pedestrians is an urgent need of the hour. As already noted, there will be a massive increase in population in the 15-64 age group. Reproductive and Child Health care programmes must meet the needs of this rapidly growing clientele. The population in this age group will be more literate and have greater access to information. They will have greater awareness and expectations about access to quality services for maternal and child health, contraceptive care, management of gynaecological problems, *etc.*

A major focus of vision 2020 must be on improving access to health services to meet the health care needs of women and children. India’s significant achievements in the field of health have been made possible by the establishment of a huge rural health infrastructure, along with the formation of a massive health care manpower consisting of over five lakh trained doctors working under plural systems of medicine, and a vast frontline of over seven lakh nurses and other health care workers; 25,000 primary and community health care centres; and 1.6 lakh sub-centres, complemented by 22,000 dispensaries and 2,800 hospitals practicing Indian systems of medicine and homeopathy.

This infrastructure remains under-equipped, under-manned and under-financed to cope with the challenge of eradicating major threats to human life. The inadequacy of the current health care system is starkly showed by the fact that only 35 per cent of the

population have access to essential drugs, while the UMI reference level is above 82 per cent. Infant immunisation against measles and DPT for children under 12 years is only 60 per cent and 78 per cent compared to the UMI level of over 90 per cent for both diseases. As a larger proportion of the population reaches middle class standards of living, an increasing number of people will turn to private health care providers.

This development is welcome, because it will permit the public health care system to concentrate more resources on meeting the needs of the poorer parts. But at the same time, the level of public expenditure on health care needs to rise about four-fold from the current level of 0.8 per cent of GDP to reach the UMI reference level of 3.4 per cent. Rapid growth of the private health care system, however, requires the formulation of competence and quality standards to check and balance the increasing emphasis on health care as a business.

Criteria for a More Equitable and Effective Health Care System:

- Universal access and access to an adequate level of health care without financial burden.
- Fair distribution of financial costs for access and fair distribution of burden in rational care and capacity.
- Ensuring that providers have the competence, empathy and accountability for delivering quality care and for effective use of relevant research.
- Special attention to vulnerable groups such as women, children, the disabled and the aged.

Development plans for India's health care systems need to place greater emphasis on public health education and prevention. The wide dissemination of health and nutrition related information through traditional channels should be supplemented by an ambitious and persistent programme of public health education through the print, television, radio and electronic media. Health insurance can play an invaluable role in improving the overall health care system. The insurable population in India has been assessed at 250 million and this number will increase rapidly in the coming two decades.

This should be supplemented by innovative insurance products and programmes by panchayats with reinsurance backup by companies and government to extend coverage to much larger parts of the population. The life expectancy of the Indian population is expected to reach above 65 years in 2020, which compares favourably with the UMI reference level of 69 years. Mortality rates for infants is expected to decline to about 20 per 1000 in 2020, which compares favourably with the UMI reference level of 22.5.

VULNERABILITY

Economic growth, rising levels of education among the young, expansion of employment opportunities for the working age population, slower population growth, and declining infant mortality, however, will not eliminate and may even aggravate, inequalities between different age groups, the sexes, income groups, communities and regions, unless specific corrective steps are taken for levelling the different degrees of

capacity and opportunity. The Vision 2020 must have special focus on bridging the existing gaps in the various levels of development and endeavour its best to fulfil the Constitutional commitment of raising the status of the vulnerable groups vis-a-vis the rest of the society.

During the next 20 years, the aged population in India will nearly double, placing much greater demand on the infrastructure of hospitals and nursing homes, while at the same time shifting the profile of health disorders from problems of the young to those of the aged. The population above 65 years of age will increase from 45 to 76 million persons by 2020. Reference has already been made to the very high incidence of malnutrition among this group and the growing incidence of diseases associated with aging, such as cancer and cardiovascular diseases.

A disproportionate number of the aged population are illiterate and living below the poverty line. While majority of these people live in rural areas where the extended family system remains prevalent, increasing urbanisation and mobility will destabilise their situation in future. The problem of coping with a larger aged population will be partially solved by a big surge in the size of the working age labour force and a reduction in the dependency ratio, meaning that there will be a larger proportion of workers to support the aged. Specific strategies will be needed to provide targeted assistance for the most vulnerable members of this group, including research on the disabilities of the aged, greater development of geriatric medicine, training of minders for the aged, and establishment of innovative support systems such as the highly successful Japanese model.

Like the aged, the disabled persons also form another subgroup that is particularly vulnerable and unlikely to benefit directly from the general advance of society, unless specific provisions are made to address their particular needs, including a special system of schools, clinics and homebased learning programmes, combined with therapeutic and institutional care. Increasing gender equity is an important challenge of the coming years. Literacy rates for females are 40 per cent lower than for males. Females represent only 43 per cent of all students at primary level, 37 per cent at higher secondary level, and 35 per cent of those in higher education. Drop out rates are also higher for girls. The UNDP's gender development index ranks India 108th among 174 countries in terms of gender equity.

It is no coincidence therefore that countries ranking highest on this index are among the most prosperous in the world. Gender equity and social development are inseparably interlinked. Reducing the disparity in nutritional, literacy and educational levels between the sexes is essential for realisation of the country's full potentials. Children being the supreme assets of the country, the 'Rights' based approach will continue to play an important role in ensuring their 'survival', 'protection' and 'development', with special attention to the girl child. Our vision for 2020 in this regard is to see a nation free from all forms of child labour. However, complete elimination of child labour through legal means is not only difficult to implement but may also prove to be counter productive in

protecting the interest of these very children. While legal enforcement can contain the demand for child labour, it may leave the poor households supplying child labour more vulnerable rather than addressing the underlying socio-economic compulsions generating the supply of such labour.

Indeed, poverty eradication combined with educational reforms to provide free (or affordable) access to quality education with an interesting, innovative and job-oriented curriculum for all can effectively eliminate child labour once and for all. The other categories of the vulnerable groups include the scheduled castes (SC), scheduled tribes (ST), other backward classes (OBC) and minorities, constituting nearly three-fourths of the country's total population. They require special attention in order to narrow down the disparities between them and the more privileged part of the population. Literacy rates for SCs and STs are 25 per cent lower than for other communities. Education enrolment rates among this group, especially for females, lag significantly behind that of other communities. Low levels of education lead to lower paying employment opportunities and lower incomes.

Thus, it becomes exceedingly difficult for these communities to come out of the vicious circle. Large number of people in other communities also suffer from similar disparities. Regional disparities in rates of development are of similar concern. Regardless of the community or the region, the overall progress of the nation will depend to a large extent on its ability to provide increasing opportunities for the disadvantaged to take initiative for their self development. Otherwise, these disadvantaged will exact an increasing toll on the stability, well-being and development of the country as a whole.

It is notable that rising levels of violence and crime in society are not so much associated with overall low levels of development, as they are with wide disparities between levels of development, which create frustration and resentment. The progress of the whole will ultimately depend on the progress of its weakest links. India's vision of 2020 must be one in which all levels and parts of the population and all parts of the country march forward together into a more secure and prosperous future.

URBAN DEVELOPMENT AND SOCIAL CHANGE

Disparities between the social and physical infrastructure of the urban and rural areas are common to all countries. In India they are a continuing source of concern and will become further aggravated unless innovative strategies are evolved to accelerate the development of rural infrastructure. A Vision of India 2020 must assess and try to anticipate the complexion of the future urban-rural divide.

Recent evidence confirms that the rate of growth of urban centres in the country is declining more rapidly than was previously anticipated, though the proportion of people living in urban areas continues to rise. The 2001 Census, 27.8 per cent of the Indian population reside in cities, compared with 25.5 per cent in 1990. The urban population is expected to rise to around 40 per cent by 2020. As India's cities continue to swell, the challenge of improving the urban infrastructure will be magnified. For instance, only

73 per cent of India's urban population has access to improved sanitation facilities, compared to the UMI reference level of 95 per cent. The trend towards concentration of urban population in a small number of large urban centres has been taking shape over a century. While the number of urban centres doubled between 1901 and 1991, the urban population increased eight-fold, resulting in a top heavy urban hierarchy.

Future demographic and economic growth is likely to concentrate in and around 60 to 70 large cities in the country having a population of a million people or more. The demographic trends towards urbanisation are accompanied by a change in the management and financing of urban development as a result of liberalisation. Decentralisation of municipal governance has led to a substantial reduction in budgetary allocations for infrastructural development. Greater reliance is now placed on institutional financing and capital markets for resource mobilisation and on private companies for service delivery. Disparities in infrastructure between large and small urban areas have always been prevalent, but these disparities can be expected to increase significantly in future years. The stricter fiscal disciplines imposed by government and credit rating agencies will make it increasingly difficult for all but the largest urban centres to attract finance for infrastructural development. The large cities may be expected to experience modest to high rates of growth and to absorb a large part of the incremental migration in their peripheries and neighbouring towns.

Being linked to the national and global economy, these large areas are likely to attract investment from the corporate sector and experience a stable demographic growth, while small and medium towns, particularly in backward regions, attract little industrial and infrastructural investment and report low and unstable demographic growth. This almost exclusive focus on improving infrastructure and basic amenities in the large cities is misplaced.

The face of urban poverty in 2020 is unlikely to be very different from what it is today, given that the largest indicator of poverty in cities is not so much lack of income, as lack of decent housing and civic amenities. These call for a change in a number of policies, especially those relating to land regulation, zoning, and development. The activities relating to provision of health care, water supply and sanitation, education and vocational training need to be carried out with higher levels of efficiency, to help the urban poor upgrade themselves. While improving infrastructure in existing cities/towns is not to be ignored, in the next two decades there will be need to encourage growth of new townships and take up regional urban development plans where growth corridors can be identified and public-private partnerships promoted for investment in alternative nodes of development.

Such centres will have considerable impact in improving the urban profile of the country. They will, however, need to make provision for adequate supply of water and sanitation as well as other civic amenities. They will also need to chalk out more liberal policies towards education and health care in order to attract a cosmopolitan group of investors, professionals and providers of other services. While a large part of it will

cater to the affluent, through imaginative planning the new centres can serve the needs of the hinterland also by offering avenues for employment, housing and other needs of a large workforce.

Urban development represents one of the great challenges for India over the next two decades. Given the socio-political reality in India, it will be difficult for the private sector to bring about changes in the pattern of investment in infrastructure without the state becoming an active partner, bringing about the required legislative and administrative changes. A satisfying outcome will depend on the formulation of effective public policy to accelerate all-round development of smaller urban centres and to refashion the role of the state as an effective facilitator to compensate for the deficiencies of market mechanisms in the delivery of public goods. Anti-poverty programmes should primarily be directed towards creation of a community-based infrastructure.

HUMAN DEVELOPMENT – CONCEPT; SELECTED INDICATORS, INTERNATIONAL COMPARISONS

The literature and theories on human development are broad and deep and offer an array of explanations and possibilities for knowing how we are unique and how we are the same, and what development and change look like. The classic debate of nature vs. nurture continues to test developmentalists – two opposing worldviews. Is an individual's behaviour influenced by external factors or is it, in fact, innate and heritable? An emerging world view questions both, and attempts to deconstruct modern social science theory and establish a new way of thinking about human development, from a contextualist perspective

Human development is nature and nurture, framed within an historical context, and follows no set direction or path. Politics and history play an important role in development and the methodologies used to support this world view are largely non-scientific – “there is no such things as independent, objective knowledge”, so state the theorists representing this world view. All knowledge is subjective, from the lived experiences of individuals, in their own voice. The process of inquiry, scientific or non-scientific, now looks very different, and the debate and theorizing continues.

As a way of beginning I will examine three worldviews on human development—mechanistic, organismic and contextualists—and discuss a theorist associated with each perspective. The worldview framework was first introduced by philosopher Stephen Pepper (1961) and is viewed as providing the most complete explanation differentiating three worldviews (Goldhauber, 2000). Pepper compares these three world views by asking three questions: (1) Is the data on human development an accurate reflection of development for all times and in all places (universal), or is development so situation-specific that it is impossible to generalize across time and place? (2) What causes us to be the way we are, and what causes us to change? (3) How do causes relate to one another? Is it possible to separate causes (reductionistic) or do they interact with each

other (holistic)? Methodology is the primary reason these three worldviews are kept separate. One cannot accept the others' methods and types of evidence as a valid test of a given theory (Goldhauber, 2000). What Pepper provides is a way of grouping theories of development into "families of theories" with a core set of beliefs, helping us to better understand and further the study of human development.

My examination will include key characteristics of each "family of theories" or world view, key contributions made by a major theorists to support that world view, and my personal observations and questions as a result of the examination. It is my intent to develop and discuss my own emerging worldview, as a result of this examination, and share insights on my own development.

MAN AS MECHANISM: A MECHANISTIC GLOBE VISION

The preeminent theorists associated with the origins of learning theory (also referred to as stimulus-response theory, behaviour theory, and social learning theory) and it is learning theory that is the dominant theory within the mechanistic worldview. From a mechanistic viewpoint, human development and behaviour are naturally occurring, universal, behaviour changes that are measurable and observable, and are therefore predictable. Watson indicated that developmental methods require the continuous observation and analysis of the "stream of activity beginning when the egg is fertilized and ever becoming more complex as age increases" (Watson, 1926). For Watson the developmental approach was "the fundamental point of view of the behaviourist – viz. that in order to understand man you have to understand the life history of his activities. It shows, too, most convincingly, that psychology is a natural science – a definite part of biology" (Cairns quoting Watson, 1926).

Theorists in this family emphasize studying both efficient (external factors) and material (genes) causes as the best way of understanding behaviour change over the life span. A single behaviour is separated and dissected so that each variable influencing that behaviour can be examined independent of every other variable – reductionism. These theories are highly testable. As Positivists, this family of theorists, or scientists, accepts only knowledge verified by the empirical sciences. Therefore, to the scientist with this worldview, behaviour is all that is directly present, factual, and observable. Any debates that exist are due to the interpretation of the observable data.

The Living Organism

A developmentalist within an organismic perspective explains rather than predicts; human development is not an exact science and is best understood using a qualitative process. Jean Piaget, William Damon, Carl Jung, Freud, Erikson and Abraham Maslow spent their lives researching and studying the human psyche and intellectual development, largely in childhood. Their work has had a profound impact on the fields of psychology, sociology, and education. From the organismic worldview, human development is a holistic, sequential process of structural changes that lead to increasingly more effective modes of adaptation,

primarily for maintaining a sense of equilibration – to exist in harmony with the environment (Piaget, 1950). We develop, or change, because of our effort to stay the same or maintain equilibrium. This change is driven internally and is unidirectional and universal in nature. Just as a living organism adapts and changes as influences outside itself impact its environment, so do we as we progressively learn and grow, building on what we know, each stage better than the previous stage, moving towards a definitive end point.

POPULATION AND DEVELOPMENT

People are central to the development process and an integral element in all development strategies. There are many different and often conflicting views on the meaning of development. The most appropriate strategies need to be followed at different points of time and in space. Large size population has been viewed by many as a negative factor in the development. However, much depends on its quality.

Relationship between population and food supply has been a subject of study ever since Malthus projected a grim future for humanity, if population continued to rise faster than food production. Given the uneven rate of population growth and technological breakthroughs in food production, there are difficulties in forecasting rates of increase in food supply or how consumption will vary. The fact, however, remains that over use or misuse of the land with a view to increase food production has its serious implications for environment and thus indirectly for food security.

The vital questions to explore are: Are these regional differences in the rate of population growth in consonance with the regional disparities in the supporting capacity of the areas? If not, how far are these differences in growth rate responsible for creating imbalance between the population and the resources? Growth of population, thus becomes a vital element in any assessment of population resource balance. But we can not ignore the fact that high growth of population or the deficiency of the resources alone are not responsible for the imbalance. The nature of social structure, the stage of technological advancement, the characteristics of distribution system and the public policies are the elements that influence the balance between the people and the resources. Thus, the number that a given piece of land can support does not merely depend upon its physical resources, but also upon a set of social, economic, technological and political conditions. Therefore, in any assessment of balance between the population and means of subsistence all these constitute important elements of the system. This complex relationship has been presented by different scholars/thinkers as models and theories. Presently, we will get introduced to the concept of Human Development which provides an alternative to the view of development equated exclusively with economic development.

HUMAN DEVELOPMENT INDEX (HDI)

The Human Development Report 1990, introduced by the United Nations Development Programme (UNDP), argues that development is not merely an expansion

of income and wealth, but a process of enhancing human functionings and capabilities. This perspective of development is termed 'human development'. The concept is defined as "the process of widening people's choices and the level of well-being they achieve". The report states that regardless of the level of development, the three choices for the people are: to lead a long and healthy life, to be knowledgeable, and to have access to the resources needed for a decent standard of living.

These are represented by three indicators: longevity as measured by life expectancy at birth, educational attainment, as measured by a combination of adult literacy (two-thirds weight) and the combined gross primary, secondary and tertiary enrolment ratio (one-third weight); and standard of living as measured by GDP per capita (Productivity Per Person US\$). The Human Development Index (HDI), constructed every year since 1990 by the United Nations, measures average achievements in basic human development in one simple composite index and produces a ranking of countries.

Computing the HDI

To construct the Index, fixed minimum and maximum values have been established for each of the indicators:

- Life expectancy at birth: 25 years and 85 years;
- General literacy rate: 0 per cent and 100 per cent;
- Real GDP per capita (PPP\$): PPP\$ 100 and PPP\$ 40,000.

Individual indices are computed first on the basis of a given formula. HDI is a simple average of these three indices and is derived by dividing the sum of these three indices by 3. With normalisation of the values of the variables that make up the HDI, its value ranges from 0 to 1. The HDI value for a country or a region shows the distance that it has to travel to reach the maximum possible value of 1 and also allows intercountry comparisons. A challenge for every country is to find ways to reduce its short fall.

International Comparisons

Of the 174 countries for which HDI was constructed in 2000 AD, 46 were in the high human development category (with HDI index more than 0.8), 93 in the medium (0.5 to 0.79) and 35 in the low category (below 0.5). Twenty countries have experienced reversal of HD since 1990 as a result of HIV/AIDS, particularly in Sub-Saharan Africa, Eastern Europe and the CIS (former USSR):

- Canada, Norway and United States rank at the top on the HDI, while Sierra Leone, Niger and Burkina Faso at the bottom.
- Disparities between regions are significant, with some having more ground to cover in making up shortfalls than others. Sub-Saharan Africa has more than twice the distance to cover as Latin America and the Caribbean, South Asia nearly three times as much as East Asia without China.
- Disparities within regions can also be substantial. In SE Asia and the Pacific HDI values range from 0.484 in the Lao Peoples Democratic Republic to 0.881

in Singapore. Among the Arab States they range from 0.447 in Djibouti to 0.836 in Kuwait.

- The link between economic prosperity and human development is neither automatic nor obvious. Of the 174 countries 97 rank higher on the HDI than on GDP per capita, suggesting that they have converted income into human development very effectively. For 69 countries, the HDI rank is lower than the GDP per capita rank. These countries have been less successful in translating economic prosperity into better lives for their people

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